

2nd
EDITION

ME404 **INDUSTRIAL**
ENGINEERING

KAILAS SREE CHANDRAN

SEMESTER 8
MECHANICAL ENGG

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HIGHLIGHTS:

- **Based on revised syllabus of APJ Abdul Kalam Technological University Kerala (KTU).**
- **Covers complete syllabus.**
- **Includes important questions at the end of each module.**
- **Model question paper also included.**
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INDUSTRIAL ENGINEERING

Based on revised syllabus of APJ Abdul Kalam Technological
University Kerala (KTU)
Subject code: ME404
Semester 8
B.Tech Mechanical Engineering
Scheme: 2015

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INDUSTRIAL ENGINEERING
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PREFACE

This textbook is prepared based on the syllabus of the subject **Industrial Engineering (ME404)** for Semester 8, B.Tech in Mechanical Engineering, 2015 scheme of APJ Abdul Kalam Technological University (KTU). Notes for all six modules have been included in this book. Students are advised to refer prescribed text books for understanding the subject thoroughly. This textbook can be used as an additional reference for improving the knowledge on the topic.

This edition (2nd) is a revised version. First edition was published on 2017 based on the syllabus prescribed by Kerala University for the same titled subject. Based on the feedback from students, important questions are listed at the end of each module. This revised edition also includes one model question paper. Previous question papers of KTU and other universities can be found from the link scanning the QR code at the back cover of this textbook. Please send your valuable suggestions and feedback about this textbook to the below mentioned email address.

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Course code	Course Name	L-T-P-Credits	Year of Introduction
ME404	INDUSTRIAL ENGINEERING	3-0-0-3	2016

Prerequisite: Nil

Course Objectives:

- To impart theoretical knowledge about various tools and techniques of Industrial Engineering.
- To create awareness about various safety procedures to be followed in carrying out different types of projects.
- To get acquainted with the Inventory management Principles and Techniques.
- To equip with the theoretical knowledge on Quality control practices and testing methods.

Syllabus

Introduction to Industrial Engineering, Plant layout and Material handling, Methods engineering, Industrial relations, Production planning and control, Quality control and Inspection

Expected outcomes:

The students will be able to

- Know various tools and techniques in industrial Engineering.
- Develop work procedure applying the principles of work study.
- Apply inventory control techniques in materials management.
- Formulate replacement and purchase decisions and arrive at conclusions

Text Books:

1. B. Kumar, Industrial Engineering Khanna Publishers, 2013
2. M Mahajan, Industrial Engineering & Production Management, Dhanpat Rai, 2005
3. Martand Telsang, Industrial Engineering & Production Management, S. Chand, 2006
4. O. P. Khanna, Industrial Engineering and Management, Dhanpat Rai, 2010

References:

1. E. S. Buffa, Modern Production management, John Wiley, 1983
2. Grant and Ieven Worth, Statistical Quality Control, McGraw Hill, 2000
3. Introduction to work study – ILO, Oxford And IBH Publishing, 2008
4. Ralph M Barnes, Motion and Time Study, Wiley, 1980

Course			
Module		Hours	End Sem. Exam Marks
I	Introduction to Industrial Engineering - Evolution of modern Concepts in Industrial Engineering - Functions of Industrial Engineering - Field of application of Industrial Engineering Product Development and research- Design function - Objectives of design, - Manufacturing vs purchase- Economic aspects- C-V-P analysis – simple problems-Development of designs- prototype, production and testing - Human factors in design- Value Engineering .	7	15%
II	Plant layout and Material handling- principles of material handling, Types of material handling equipments, Selection and application. Preventive and break- down maintenance - Replacement policy-- Methods of replacement analysis-Method of providing for depreciation- Determination of economic life - Simple problems.	7	15%

FIRST INTERNAL EXAM			
III	Methods engineering: Analysis of work methods using different types of process chart and flow diagrams- Critical examination-Micro motion study and therbligs- Principles of motion economy – Work measurement-Performance rating.-Determination of allowances and standard time. - Job evaluation and merit rating - Objectives and principles of job evaluation--Wages and Incentives-Primary wage systems- Wage incentive plans.	7	15%
IV	Industrial relations- Psychological attitudes to work and working conditions - fatigue- Methods of eliminating fatigue- Effect of Communication in Industry-Industrial safety-personal protective devices-, causes and effects of industrial disputes- Collective bargaining- Trade union - Workers participation in management.	7	15%
SECOND INTERNAL EXAM			
V	Production planning and control- Importance of planning - job, batch and mass production-Introduction and need for a new product-product life cycle. - Functions of production control - Routing , Scheduling, dispatching and follow up- Gantt charts. Inventory Control, Inventory models -Determination of EOQ and reorder level-simple problems- Selective inventory control techniques.	7	20%
VI	Quality control and Inspection- Destructive and non-destructive testing methods- process capability- Statistical quality control – causes of variation in quality- control charts for X and R. Reliability-causes of failures- Bath tub curve.-System reliability- life testing-Introduction to concepts of, TQM, ISO, Six Sigma and Quality circles (Brief description only).	7	20%
END SEMESTER EXAM			

Question paper pattern

Maximum marks: 100

Time: 3 hrs

The question paper should consist of three parts

Part A

There should be 2 questions each from module I and II. Each question carries 10 marks. Students will have to answer any three questions out of 4 (3x10 marks =30 marks)

Part B

There should be 2 questions each from module III and IV. Each question carries 10 marks. Students will have to answer any three questions out of 4 (3x10 marks =30 marks)

Part C

There should be 3 questions each from module V and VI. Each question carries 10 marks. Students will have to answer any four questions out of 6 (4x10 marks =40 marks)

Note: Each question can have a maximum of four sub questions, if needed.

MODULE 1

1.1 INTRODUCTION TO INDUSTRIAL ENGINEERING

Industrial engineering is concerned with bringing together and effective utilization of various resources or inputs to facilitate efficient production/operation. In other words, we may say that industrial engineering is an engineering approach to detailed analysis of the use and cost of the various resources of an organization.

The main resources are people/men, materials, methods, equipment and money which are to be used in right manner so that they constitute an integrated combination to achieve the production objectives (to improve productivity or profits etc.) and policies of the organization. Efficient production/operation systems are those systems which score high on actual performance standards.

It is important to note that industrial engineering is related not merely with the system of equipment, processes and materials but also with people who interact with this system. Thus industrial engineering techniques go beyond the mechanical cost factor.

These are associated with organizational structure, administrative techniques and human problems; and at the same time relationship between the efficiency and consent of the working group is also considered for better coordination.

In simple words, it is related with the improvement of whatever has been designed and/or evaluated. If it is human work, the industrial engineer would try to make it more efficient, more productive, less tiring, and causing minimum wastage of materials, time and energy thus least costing.

If it is concerned with handling task, he would try to reduce the amount of movement involved through change in lot size and rearrangement of the layout used.

If it is a manufacturing task it would try either by redesigning or by using different input materials so that better or newer production techniques could be utilized and provide better integration and flow among processing steps. Thus, it may be concluded that industrial engineering is not related merely with the system of materials, methods and equipment but also with the men who interact with this system.

The impact of people on the operating system is both from within and outside. Men who operate a machine, who handle material or who plan the method/process are within the operating system. Thus work study time and motion study wage incentive plans, motivation and ergonomics are integral part of industrial engineering.

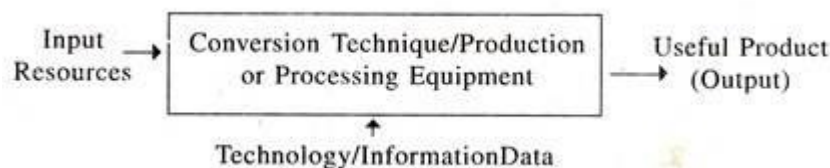
Similarly the people who are outside the operation system also play important role. Consumers outside the production system determine the demand of product. This demand is dependent on product design, quality and sales price etc. Thus industrial engineers are required to work for product and process improvement.

Another important aspect of industrial engineering is its generality. Industrial engineering is not confined only to manufacturing activities. It includes service sectors like banking, transportation, waste management, distribution and health care etc. Thus, the scope of industrial engineering is quite big and certainly not constrained within the boundary of industrial unit.

Concept of Industrial Engineering:

This term Industrial Engineering is composed of two words which give the basic concept of Industrial Engineering (I.E.). Industrial means related to industry, which means a process of production or a complete process of converting input resources into useful products or services (depending upon nature of industry).

As a system, it may be represented as follows:



Engineering may be defined as:

The profession in which knowledge of Mathematical, Physical and Social Sciences obtained by experience and practice is utilized economically for the Conversion of materials into useful products for the benefit of mankind.

Had it been the manufacturing of a product, the application of principles of Industrial Engineering (IE) would attempt to redesign or to utilize different materials in order to use newer or better production methods and provide better integration between various processing steps. In service sector like hospitals, teaching Industrial Engineering leads to productivity improvement.

1.2 FUNCTIONS OR ACTIVITIES OF INDUSTRIAL ENGINEERING

The field of Industrial Engineering (IE) represents an area that is today undergoing rapid expansion and diversification. Every activity of an industrial system involves an element that bring coordination of various activities and without which all acts would be ineffective and non-productive. This element of coordination which influences plans and objectives may be termed management. Therefore, management consists of getting the things done by others and a manager is one who achieves goals or accomplishes objectives by directing the efforts of others.

Basically, the industrial managers have to perform following:

1. Planning
2. Organizing
3. Directing
4. Controlling and
5. Staffing.

A plan is an organized scheme of doing something which is directed to achieve industrial objectives under certain constraints. An homogeneous working environment is developed to bring coordination among the various elements of plan for the execution of organized scheme.

Some activities are listed below:

1. To aid in the development of sound site selection and physical layout for the execution of organized work so as to achieve effective cost and budgetary control.
2. To analyse and plan production schedules and inventories.
3. Devise techniques to improve productivity and moral of men at work.
4. To study the equipment replacement feasibility.
5. To aid in the determination of economic lot size and the work in process requirements for each phase of operation.
6. To aid in the preparation of detailed specifications for every work and to evaluate them.
7. To aid in the development of quality specifications and quantity needed for various inputs materials.
8. To diagnose and rectify causes of poor product quality.
9. To evaluate the effectiveness of marketing, distribution and advertising policies of the organization
10. To design, develop and evaluate the complex man-machine systems involved in production process.
11. To aid in the development of non-financial incentives to maintain interest and enthusiasm among the various people of the organization.
12. To evolve better methods of production by using operation research techniques.
13. To aid in analysis and schedule of big projects by utilizing CPM and PERT techniques.
14. To aid in development and maintenance of standards governing administrative and sales budget
15. To aid in the development of training programmes for training of supervisors and workers to help them in proper performance of their duties.

Functions of Industrial Engineering:

The functions covered by industrial engineering are many and involve almost every activity connected with industrial management on scientific basis.

(i) Organization and Management Methods:

For any activity to be carried out in a systematic manner, it is essential that fundamental concepts about the activity are crystalized. Subsequently, these concepts should be transferred into an organized function so that it is carried out with maximum efficiency i.e. reduction in production cost and time. It is also now recognized modern thinking that an industrial or business activity does not exist merely for financial gains.

The social aspect i.e. the utility of industrial or business activity to the society in all respects such as consumer satisfaction, employment opportunities, welfare schemes and creation of ancillary units etc. has to be fully considered while planning and running the organization.

Organization is thus an arrangement for internal administration of the enterprise. It describes the role that each individual is required to play in the operation of the unit/enterprise.

It takes full measure to ensure that concerned effort of all of them shall be conducive to the achievement of the objectives/goals of the plant/enterprise. It is the methodology of the management by which a group of individuals of the enterprise will club its efforts for the achievement of the objectives.

This can be done by clearly demarcating the responsibility, the authority, duty, definition and division of activities by providing the channels of communication and coordination.

(ii) Plant Location and Plant Layout:

This is the most important decision, the management has to take.

The plant location depends upon various factors such as:

- (i) Industrial policies of concerned State and Central Governments.
- (ii) Availability of input raw materials
- (iii) Proximity to markets
- (iv) Availability of efficient and cheap labour/man power
- (v) Service and ancillary industries etc.

It should however be evaluated so that the proposed plant (site) should be operationally effective and cost effective. The aim should be the determination of a location which should provide minimum delivery to consumer cost of the final product.

Plant layout is an integration of manufacturing facilities and supporting facilities with the system's organization. Thus it is a technique for locating different machines/equipment and plant services within the factory.

The purpose of this technique is that each operation can' be performed at the point of greatest convenience and maximum possible output of high/optimum quality at minimum possible total cost, can be achieved.

(iii) Material Handling Methods:

Material handling is a specialised activity for modern manufacturing industries. A properly designed material handling system may save considerable floor space and also provide storage in transit. Similarly other handling equipment may save labour and effort as lot of production time is consumed in handling materials before, during and after the manufacture cycle.

Therefore, by proper selection, operation and maintenance of the material handling equipment we can improve the level of output and product quality, speed upto the deliveries and therefore, reduce the cost of production.

(iv) Methods Engineering:

It is a systematic and critical study of existing methods of doing a task with a view to evolve the most efficient method for better utilization of available resources. This is the heart of industrial engineering functions.

The six features of method analysis are:

- (a) WHAT
- (b) WHY
- (c) HOW
- (d) WHERE
- (e) WHEN
- (f) WHO



In light of these six features, methods analysis or study is basically conducted to simplify the work and working methods so that the productivity can be improved. Methods can assess the quantity of materials, time, energy and money being consumed.

So methods become the core where one can attempt to decrease the consumption of input resources thereby minimize the cost per unit output through proper methods development and design.

Thus Method Engineering is a technique which analyses each operation of a given task or piece of work, very closely in order to eliminate unnecessary operations and to develop the quickest and easiest way of performing each necessary operation. It includes the standardization of machines/equipment, method and working conditions plus training of operators to follow the standardized method.

(v) Time Study:

After a proper method has been established, the next step is to determine the time required to perform a given task. Time study is concerned with the investigation, reduction or elimination of ineffective time. Basically work measurement provides reliable and consistent standards of performance.

The usual practice is to take actual time study by stop watch and to level and rate the actual performance to obtain a normal or standard performance by also adding the usual allowances for fatigue, personal needs and unavoidable delays etc.

(vi) Production Planning and Control:

The production planning and control function is a vital management technique to assist in modern mass manufacturing operations, where the supply of correct quality and quantity at proper place and at right time has to be ensured for achieving the production goals/targets.

Production means a sequence of operations that transforms the given material in to desired shape. This transformation may be achieved by adopting different methods or techniques (i.e. Primary processes

and secondary processes). There are four factors (i.e. Quantity, Quality, Time and Cost) which govern the production system.

Planning means an analysis of data (inputs) related with the variable future so that a scheme can be chalked out for the utilization of resources in the optimized manner and the desired goal of the system may economically be achieved.

It is considered as the deterministic phase of the management and may include manufacturing planning, factory planning and production planning. Production control is the process of maintaining a balance between various activities evolved during production planning thus providing the most effective and efficient resource utilization.

Production planning and control may be summarily defined as the direction and coordination of the firm's resources towards the achievement of pre-decided production goal in the most efficient way. Thus the principles of production planning and control lies in the statement "First plan your work then work your plan".

(vii) Statistical Quality Control:

Product quality is the most important and significant factor for capturing market because it is closely allied to the cost of product and purchaser requirements. Since the quality of any product is regarded as the degree of perfection so it plays a big role in consumer's decision. Thus it is an important Industrial Engineering function.

SQC is a technique of applying statistical methods based upon the theory of probability to establish quality standards and to maintain it in most economical way. These methods and principles aim to assess not only the magnitude of "chance cause variation" but also detect the "assignable cause variation".

Statistical quality control is employed to ascertain whether the variation in quality of product is due to chance cause or due to assignable cause only. Otherwise the presence of assignable cause is detected and some corrective action is taken to improve the quality of the product.

(viii) Inventory Control:

It may be defined as scientific way of finding out the quantities to be kept in stock to meet the production requirements and the systematic location, storage and recordings of goods in such a way that desired degree of service can be provided at minimum ultimate cost.

(ix) Cost Control:

The entire industrial engineering functions are devoted to improve the efficiency of the system and reducing the cost by eliminating waste in all forms.

Cost control means the procedures and measures with the help of which the cost of carrying out an activity is kept under check.

The objective of cost control is two-fold:

- (i) To ensure that cost do not exceed beyond a certain pre-decided level.
- (ii) Therefore, as a further step, it must follow/adopt such procedures and measures by which further cost reduction is possible.

Cost control leads to efficiency regarding use of materials, machinery and manpower. Cost control provides a basis for minimization of cost which is essential to fight competition in a specific product line in the market.

(x) Job Evaluation:

The function of job evaluation is to provide both the employees/workers and management with an objective yardstick for establishing the entire wage structure which equitably reflect the value of work being done by each worker.

Job evaluation is an order by a systematic process of determining the wages worth for the job in relation to the job. It is a process of rewarding the job after it has been analysed. Each job should be paid according to its real value i.e., a high value job should receive higher wages and low value job should receive lower wages.

It may be defined as “an effort to determine the relative value of every job in a plant to determine what the fair basic wage for such a job should be”. Thus job evaluation is a method of determining wage structure of different jobs. It helps in fixing the remuneration to be paid for undertaking a particular job.

(xi) Wage Incentives:

In an under developed country like India, financial incentives are still very important to increase the output. Wage incentive is some kind of monetary reward which is directly connected with the performance of a worker that is the wages increase corresponding to an increase in output or improvement in productivity.

(xii) Operation Research:

Operation research is the organized application of modern science, mathematics and computer techniques to problems arising from operations of large complex business or industrial organization.

The purpose is to provide the top level administrators or management with explicit quantitative understanding and assessment of complex situations with a sound basis for arriving at best decisions, or seeks the optimum state in all spheres of activities and thus is able to provide optimum solution to all organization's problems.

1.3 EVOLUTION OF INDUSTRIAL ENGINEERING (HISTORY)

In any productive environment, whether it is an industrial plant, an institution such as a hospital, restaurant, office, etc. there is a need to improve the quality of work. This means that a given task should be carried out efficiently and accurately in terms of time and effort spent.

During the industrial revolution of the 18th and 19th century, many small ownership based businesses grew into larger enterprises in which a number of manual tasks were performed by mechanical and steam operated machines. At that time there weren't adequate tools or working conditions and there was considerable exploitation of labor. As a result, there were wide variations in output from different workers and different factories making the same product.

Frederick Taylor, a mechanical engineer was who observed that better methods could be established even for a simple task as handling iron ore and coal for a blast in a steel plant. He pursued the task of establishing a norm for the weight and size of the shovel for scooping and transferring material. He observed and proved that instead of using the largest shovel to move the maximum material in a day, it was better to design a shovel which could be comfortably used by the workers on a repetitive basis without tiring or injuring them at the end of the day. When his plan was implemented, he reduced the manpower by over 25%. Taylor was also considered to be the father of scientific management because he was a pioneer in improving methods and establishing the incentive system for workers with the benefit of higher productivity to the owners and higher wages for the workers.

Frank Gilbreth, started working on motion studies soon after Taylor began his work. Gilbreth was a construction contractor, who noticed that the industry lacked standardization of methods. Gilbreth and his wife Lilian, devoted their lives to motion studies.

Gilbreth's famous discovery took place when he was doing apprenticeship as a bricklayer; he observed that there were no two men, who could lay bricks the same way. Consequently, their quality and quantity of output varied. He improved the method of laying bricks by making a number of changes. He provided a platform whose height could be adjusted, so that bricklayer is always at the same height in relationship to the bricks laid. A shelf for bricks and mortar was built to save workers from bending down to pick up their material. He had bricks pre-stacked with the best side facing in the direction of the workers to avoid workers from having to turn the brick several times to find its best side just before laying.

These changes significantly reduced the number of motions in laying bricks, and resulted in higher production with lower fatigue for the workers. Lilian Gilbreth, Frank's wife, joined him in his pursuit for promoting scientific management by conducting research and application work in studies of motion and methods.

Gilbreth's work continued in motions using motion pictures for studying tasks and workers. He developed micro motion study, a breakdown of work into fundamental elements called "therbligs" (baed on Gilbreth spelled backwards).

Taylor's concept of work element was broad, and based on time study requirements like "get tool". Whereas Gilbreth's work was based on breaking down the elements further into basic individual therbligs, such as "reach for tool and pick up tool" instead of "get tool".

There were many followers of Taylor and Gilbreth in the 1900s. They believed that Taylor's work emphasized motion measurement whereas Gilbreth emphasized motion analysis. With time, it became clear that both approaches were necessary and were essentially interdependent. Ultimately what is important is the best of both of these efforts, namely, the right motions with the minimum of time. Thus evolved a term "Methods engineering" which is an important function of industrial engineering.

The concepts of time and motion studies developed by Taylor and Gilbreth are still the basis for industrial engineering. Even today, over fifty percent of industrial engineering activities are related to this concept.

1.4 TECHNIQUES OR AREAS OR APPLICATIONS OF INDUSTRIAL ENGINEERING

The main aim of tools and techniques of industrial engineering is to improve the productivity of the organization by optimum utilization of organizations resources: men, materials, and machines. The major tools and techniques used in industrial engineering are:

- | | |
|--|---|
| 1) Production planning and control. | 11) Sequencing and transportation models. |
| 2) Inventory control. | 12) Games theory and dynamic programming. |
| 3) Job evaluation. | 13) Group technology. |
| 4) Facilitates planning and material handling. | 14) Statistical techniques. |
| 5) System analysis. | 15) Quality control. |
| 6) Linear programming. | 17) Decision making theory. |
| 7) Simulation. | 18) Replacement models. |
| 8) Network analysis (PERT, CPM). | 19) Assembly line balancing. |
| 9) Queuing models. | 20) MRP-JIT-ISO-TQM |
| 10) Assignment. | |

Fields of Industrial Engineering:

Industrial Engineering rather its managerial functions have become very important in view of the fact that today's industry has become a complex system and is facing tough competition in the market.

Therefore, efforts are to be made to improve the methods of production and cut down the cost of products. It is also essential for industries to establish good working conditions and harmony between the workers and management to improve productivity.

In order to execute the functions of industrial engineering the managers, must have got a thorough knowledge of the following fields of study:

1. A good knowledge of the methods/processes involved in the production system and the various interrelated activities that work within the system.

2. Multiple feasible solutions are likely to exist at each stage of the process, so timely decisions are to be taken under multiplicity of pros and cons. One should therefore, have thorough understanding of the interaction between the production decisions and the activities of other sectors involved in similar works.

3. The industrial managers must develop sufficient skill so that one can predict the external and internal forces effecting the economy and harmony of the organization.

Thus, it appears that the industrial managers must be well conversed with the various techniques of industrial engineering. It is therefore, extremely necessary to look into various activities of Industrial Engineering (IE) and its growth. In 1943, the Work Standard Committee on Management Division of ASME tried to define the specialized field which industrial engineering should cover.

They are as follows:

(i) Manufacturing Engineering:

- (a) Equipment Selection and Specialization and
- (b) Design of Tools, Jigs and Fixtures.

(ii) Methods Engineering:

Standardization of Methods with the help of operations and motion analysis:

- (a) Work Measurement
- (b) Determination of Allowances
- (c) Computation of Standard Time

(iii) Production Planning and Control:

- (a) Materials Management (Raw materials, finished and semi-finished parts and components)
- (b) Machines and Manpower
- (c) Routing
- (d) Estimating
- (e) Scheduling
- (f) Dispatching
- (g) Expediting
- (h) Evaluation.
- (iv) Plant layout and material handling
- (v) Inventory control
- (vi) Organization management
- (vii) Inspection and quality control
- (viii) Wages and incentives
- (ix) Budgetary control and cost control.



Applications of Industrial Engineering:

Industrial Engineering is utilized in the following areas of industrial enterprises:

(i) Top Management:

It assists management in designing strategic long range plans for decision making also through the development of mathematical and statistical models and use of computers. Alternative courses of actions can be simulated and their evaluation is done.

(ii) Process and Work Design:

It aids in conducting research of all operating problems for improving productivity of the system; increase production, minimize cost, improve quality and maximize profits.

(iii) Technical Areas:

It works with technical groups to ensure that instrumentation, system design human engineering and work analysis concepts are followed in factories and machinery design.

(iv) Marketing Analysis:

It provides assistance in the analysis of complex marketing systems and pricing through the use of statistics and computer to develop alternative course of actions and their evaluation.

(v) Finance:

It helps to design improved financial information flow system which can work in conjunction with programmed computers. So the system can be developed which is used to control materials, labour and other expenses.

(vi) Procurement of Input Materials:

It helps in establishing the economic order quantities and reorder points, developing methods and procedures to provide best service for minimum cost.

(vii) Transportation and Distribution:

It provides assistance in the evaluation of alternative patterns of distribution in order to determine the optimum plans in meeting consumer's requirements particularly where advance techniques are feasible.

(viii) Production and Quality Assurance:

It aids to provide assistance to achieve maximum utilization of equipment, materials and labour to achieve optimum performance and high quality standards and assure excellent consumer service.

(ix) Maintenance and Replacement:

It works closely with maintenance department to design procedures and techniques to reduce maintenance cost and thus improve production output.

(x) Industrial Relations:

Matters concerning working conditions, training, incentives, compensation, and design changes if any require careful coordination. Industrial engineering helps in keeping employees informed on matters that effect employees relations.

1.5 PRODUCT DEVELOPMENT

In general, the Product Development can be defined as *"creating, innovating, or developing entirely a new product , or presenting an existing product with enhanced utility, improved features, more appealing design, better quality and reliability to satisfy the requirements of its end-users."*

Meaning of Product Development

Product means a good, service, idea or object created as a result of a process and offered to serve a need or satisfy a want. **Development** means the act or process of growing, progressing, or developing.

Product Development is a process of improving the existing product or to introduce a new product in the market. It is also referred as New Product Development. The functions of product development are as follows :-

1. Creation of an entirely new product or upgrading an existing product,
2. Innovation of a new or an existing product to deliver better and enhanced services,
3. Enhancing the utility and improving the features of an existing product,
4. Continuous improvement of a product to satisfy rapidly changing customer needs and wants.

Product development and design are considered the preliminary stages of production planning. When a new product is projected, the designer has to bear in mind the available resources of the enterprise plant and the possible implications of the unit required to acquire, modify or substitute the existing equipment and machines or subcontract various components (parts to ancillary) units or other suppliers.

Thus it is obvious that product development and design is the case of the development and growth of the manufacturing limbs i.e. departments. This is why product design is one of the fundamental elements of the management policy.

The idea for new or improved products may be available from many sources such as:

- (i) Consumer suggestions and complaints
- (ii) Other competitor products in the market
- (iii) R & D department of the enterprise.

After the new idea has been conceived the technical and commercial viability of the product is checked and if found fit, it is developed. The next phase is how the product should be made.

Manufacturing of a new or modified product will need the services of the following departments of the enterprise/company:

- (i) Marketing distribution and sales.
- (ii) R & D.
- (iii) Design and development of the product.
- (iv) Product manufacture.
- (v) Personnel.
- (vi) Accounts.

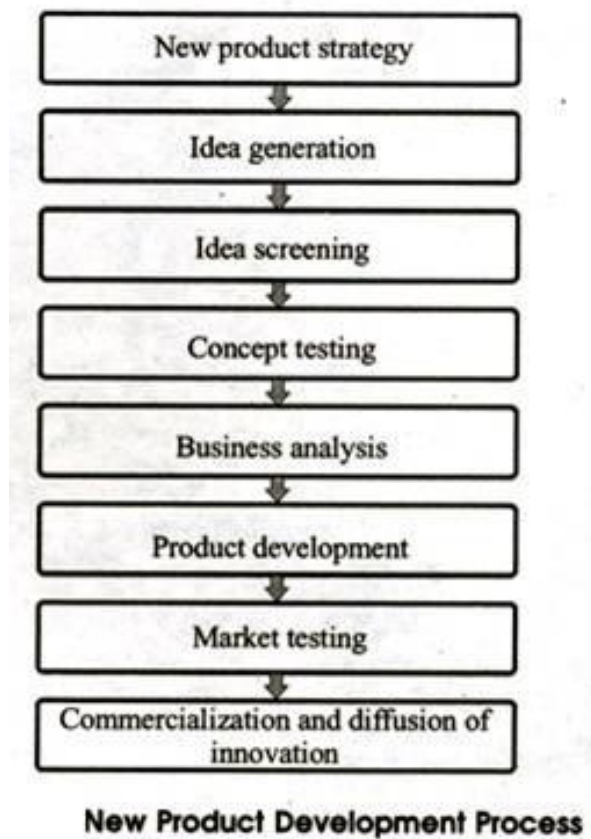
Product Development Process

Product development process is a crucial process for the success and survival of any business. Today, businesses are operating in a highly dynamic and competitive environment. Business organisations have to continuously update their products to conform to current trends. The product development process starts from idea generation and ends with product development and commercialisation. Following are the steps in the process of product development.

1. **Idea Generation** - The first step of product development is Idea Generation that is identification of new products required to be developed considering consumer needs and demands. Idea generation is done through research of market sources like consumer liking, disliking, and competitor policies. Various methods are available for idea generation like - Brain Storming, Delphi Method, or Focus Group.
2. **Idea Screening** - The second step in the process of product development is Idea Screening that is selecting the best idea among the ideas generated at the first step. As the resources are limited, so all the ideas are not converted to products. Most promising idea is kept for the next stage.
3. **Concept Development** - At this step the selected idea is moved into development process. For the selected idea different product concepts are developed. Out of several product concepts the most suitable concept is selected and introduced to a focus group of customers to understand their reaction. For **example** - in auto expos different concept cars are presented, these models are not the actual product, they are just to describe the concept say electric, hybrid, sport, fuel efficient, environment friendly, etc.
4. **Market Strategy Development** - At this step the market strategies are developed to evaluate market size, product demand, growth potential, and profit estimation for initial years. Further it includes launch of product, selection of distribution channel, budgetary requirements, etc.
5. **Business Analysis** - At this step business analysis for the new product is done. Business analysis includes - estimation of sales, frequency of purchases, nature of business, production and distribution related costs and expenses, and estimation of profit.
6. **Product Development** - At this step the concept moves to production of finalised product. Decisions are taken from operational point of view whether the product is technically and commercially feasible to produce. Here the research and development department develop a physical product.
7. **Test Marketing** - Now the product is ready to be launched in market with brand name, packaging, and pricing. Initially the product is launched in a test market. Before full scale launching the

product is exposed to a carefully chosen sample of the population, called test market. If the product is found acceptable in test market the product is ready to be launched in target market.

8. **Commercialisation** - Here the product is launched across target market with a proper market strategy and plan. This is called commercialisation phase of product development.



Objectives:

Any product of which ever type it may be must have two broad objectives i.e. immediate and ultimate objectives.

Immediate objectives of a product will be to provide a new look, offer new advantages, utilize the existing man power and machines thus stimulate the sales function and satisfy the immediate needs of the consumers. But the ultimate objectives of a product which are of more importance are to monopolise the market, tie the consumer only to branded product, to make its production possible on quantity basis and hence to cut down/reduce the cost of production and hence give its benefit to the consumers.

Objectives of a new product may be as follows:

- a) Satisfying consumer requirements.
- b) To incorporate changes in shape, form and aestheticism.
- c) To utilize tangible resources men, machines, materials and money.
- d) Stimulating sales function.
- e) To monopolize the market.
- f) Achieving low cost of production.

- g) To make production on quantity basis.
- h) To form a basis of future products.
- i) Standardisation of products.
- j) Simplification of products.

1.6 MANUFACTURING VS PURCHASE (MAKE OR BUY DECISION)

The make-or-buy decision is the act of making a strategic choice between producing an item internally (in-house) or buying it externally (from an outside supplier). The buy side of the decision also is referred to as outsourcing. Make-or-buy decisions usually arise when a firm that has developed a product or part—or significantly modified a product or part—is having trouble with current suppliers, or has diminishing capacity or changing demand.

Make-or-buy analysis is conducted at the strategic and operational level. Obviously, the strategic level is the more long-range of the two. Variables considered at the strategic level include analysis of the future, as well as the current environment. Issues like government regulation, competing firms, and market trends all have a strategic impact on the make-or-buy decision. Of course, firms should make items that reinforce or are in-line with their core competencies. These are areas in which the firm is strongest and which give the firm a competitive advantage.

Factors that may influence firms to make a part internally include:

- Cost considerations (less expensive to make the part)
- Desire to integrate plant operations
- Productive use of excess plant capacity to help absorb fixed overhead (using existing idle capacity)
- Need to exert direct control over production and/or quality
- Better quality control
- Design secrecy is required to protect proprietary technology
- Unreliable suppliers
- No competent suppliers
- Desire to maintain a stable workforce (in periods of declining sales)
- Quantity too small to interest a supplier
- Control of lead time, transportation, and warehousing costs
- Greater assurance of continual supply
- Provision of a second source
- Political, social or environmental reasons (union pressure)
- Emotion (e.g., pride)

Factors that may influence firms to buy a part externally include:

- Lack of expertise
- Suppliers' research and specialized know-how exceeds that of the buyer
- cost considerations (less expensive to buy the item)
- Small-volume requirements
- Limited production facilities or insufficient capacity
- Desire to maintain a multiple-source policy

- Indirect managerial control considerations
- Procurement and inventory considerations
- Brand preference
- Item not essential to the firm's strategy

The two most important factors to consider in a make-or-buy decision are cost and the availability of production capacity. Obviously, the buying firm will compare production and purchase costs. Burt, Dobler, and Starling provide the major elements included in this comparison.

Elements of the "make" analysis include:

- Incremental inventory-carrying costs
- Direct labor costs
- Incremental factory overhead costs
- Delivered purchased material costs
- Incremental managerial costs
- Any follow-on costs stemming from quality and related problems
- Incremental purchasing costs
- Incremental capital costs

Cost considerations for the "buy" analysis include:

- Purchase price of the part
- Transportation costs
- Receiving and inspection costs
- Incremental purchasing costs
- Any follow-on costs related to quality or service

One will note that six of the costs to consider are incremental. By definition, incremental costs would not be incurred if the part were purchased from an outside source. If a firm does not currently have the capacity to make the part, incremental costs will include variable costs plus the full portion of fixed overhead allocable to the part's manufacture. If the firm has excess capacity that can be used to produce the part in question, only the variable overhead caused by production of the parts are considered incremental. That is, fixed costs, under conditions of sufficient idle capacity, are not incremental and should not be considered as part of the cost to make the part.

While cost is seldom the only criterion used in a make-or-buy decision, simple break-even analysis can be an effective way to quickly surmise the cost implications within a decision. Suppose that a firm can purchase equipment for in-house use for ₹250,000 and produce the needed parts for ₹10 each. Alternatively, a supplier could produce and ship the part for ₹15 each. Ignoring the cost of negotiating a contract with the supplier, the simple break-even point could easily be computed:

$$₹250,000 + ₹10Q = ₹15Q$$

$$₹250,000 = ₹15Q - ₹10Q$$

$$₹250,000 = ₹5Q$$

$$50,000 = Q$$

Therefore, it would be more cost effective for a firm to buy the part if demand is less than 50,000 units, and make the part if demand exceeds 50,000 units. However, if the firm had enough idle capacity to produce the parts, the fixed cost of ₹250,000 would not be incurred (meaning it is not an incremental cost), making the prospect of making the part too cost efficient to ignore.

Example 1:

A detailed simple cost analysis is as follows:

<i>Factors</i>	<i>To buy in Rs.</i>	<i>Rs.</i>	<i>To make in Rs.</i>	<i>Rs.</i>
Material Cost				
1. Raw material	475.00		500.00	
2. Finished raw material needed	400.00		520.00	
3. Indirect material	25.00		80.00	
Total	900.00		1100.00	
10% Overhead	90.00		110.00	
Unit material	990.00	990.00	1210.00	1210.00
Labour cost				
(i) Direct cost				
1. Manufacturing	100.00		150.00	
2. Assembly	50.00		75.00	
3. Inspection	10.00		10.00	
4. Testing	40.00		90.00	
(ii) Indirect cost				
1. Production control staff	10.00		15.00	
2. Quality control staff	15.00		10.00	
3. Office staff	5.00		40.00	
4. Others	10.00		10.00	
Total	240.00		400.00	
Overhead	(100%)240.00	(80%)	320.00	
Unit labour cost	480.00	480.00	720.00	720.00
Total cost of tools		130.00		170.00
Packing etc.		10.00		10.00
Unit production cost		1610.00		2110.00
Number of units		100.00		100.00
Final cost of 100 units		161000.00		211000.00

On the basis of above cost analysis it is quite clear that the decision must be in favour of buying the part.

Economic Analysis:

(a) Buying:

As we know that economic purchase (ordering) quantity is equal to $\sqrt{2AP/C}$ or $Q = \sqrt{2AP/C}$

and also we know total cost = Procurement cost/year + Inventory cost per year

$$EOQ \text{ or } Q = \sqrt{\frac{2AP}{C}}$$

Total cost = $A \times P/Q + Q \times C/2$

This can further be modified by adding purchase cost/year.

Thus Total cost = $A \times I + A \times P/Q + Q \times C/2$

where A = Total demand per year

P = Procurement cost per order

C = Annual carrying cost per item/year

Q = Economical purchase (ordering) quantity

I = Purchase price of one unit.

(ii) Making:

Further, it can be found out that economical production quantity is equal to

$$Q_1 = \sqrt{\frac{2AS}{C \left(1 - \frac{A}{R}\right)}}$$

and Total cost = $A \times I + A \times S/Q_1 + C(R - A) Q_1/2R$

where A = Total demand per year

S = Set up cost per lot

C = Annual carrying cost per item/year

Q₁ = Economical production quantity

I = Production cost per unit



R = Production rate.

By using above mentioned formulas, an economic analysis can be made in respect of make or buy decision.

Example 2:

Suppose a supplier has the following unit cost of a part:

Direct material	Re. 0.20
Direct labour	Re. 0.10
Variable overhead	Re. 0.12
∴ Direct cost	Re. 0.42
Fixed overhead	Re. 0.08
∴ Manufacturing cost	Re. 0.50
Sales expenses	Re. 0.02
Administrative expenses	Re. 0.01
∴ Total cost	Re. 0.53
Profit	Re. 0.06
∴ Supplier's Price	Re. 0.59
Transport and packing charges	Re. 0.01
Buyer's cost	Re. 0.60

If the company likes to make this item rather than buy it could reduce the cost of part from Re. 0.60 to Re. 0.42. If the company has idle equipment that can be used to make the part and have maximum overhead expense of Re. 0.12 per unit. Further, if the company invests on additional facilities to make the part with the normal overhead, in that case if the cost of unit is equal to Re. 0.54 then it is worthwhile to make the part.

Example 3:

Suppose an item has yearly demand of 1000 units.

The different costs regarding make and buy are as follows:

	<i>Buy</i>	<i>Make</i>
<i>Item cost / unit</i>	<i>Rs. 6.00</i>	<i>Rs. 5.90</i>
<i>Procurement cost / order</i>	<i>Rs. 10.00</i>	<i>—</i>
<i>Set up cost / lot</i>	<i>—</i>	<i>Rs. 50.00</i>
<i>Annual carrying cost / item / year (22% of item cost)</i>	<i>Rs. 1.32</i>	<i>Rs. 1.30</i>
<i>Production rate equal to 6000 units per year.</i>		
<i>Indicate which of the alternative is economical</i>		

Solution:

(a) Buying:

Economical purchase quantity = Q

$$= \sqrt{\frac{2AP}{C}} = \sqrt{\frac{2 \times 1000 \times 10}{1.32}} = 123 \text{ Units}$$

Now, calculating the total cost with the help of formula,

$$\text{Total cost} = A \times I + A \times P/Q + Q \times C/2$$

For different purchase quantities, total cost is calculated, and shown in, Table.

Table 37.1

<i>Purchase Quantity</i>	<i>Total Cost in Rs.</i>
50	6233
100	6166
125	6163
132	6162
150	6165
200	6182
300	6231
400	6289
600	6413

(b) Making:

Similarly while making we have

$$Q_1 = \sqrt{\frac{2AS}{C(1-\frac{A}{R})}} = Q_1 = \sqrt{\frac{2 \times 1000 \times 50}{1.30(1-\frac{A}{R})}} = 302 \text{ units}$$

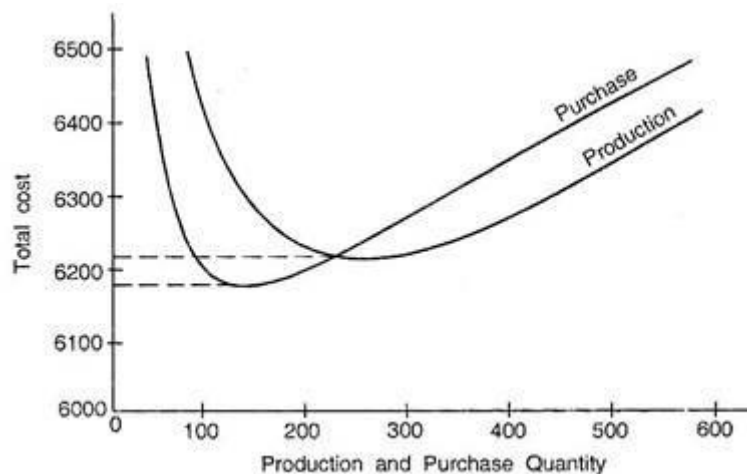
and total cost table is prepared with the help of formula

$$\text{Total cost} = A \times I + A \times S/Q_1 + C(R - A) Q_1/2R$$

<i>Purchase Quantity</i>	<i>Total Cost in Rs.</i>
100	6454
150	6314
200	6258
300	6229
302	6228
400	6241
500	6270
600	6307

Now plotting the two tables on a graph as shown below.

The decision to buy or make may be made by analysing the minimum cost for each alternative. In this example, to buy will be the choice as by choosing this we can have saving of Rs. 66 (Rs. 6228 – Rs. 6162) per year.



1.7 COST – VOLUME – PROFIT ANALYSIS (BREAK-EVEN ANALYSIS)

Break-even analysis is of vital importance in determining the practical application of cost functions. It is a function of three factors, i.e. sales volume, cost and profit. It aims at classifying the dynamic relationship existing between total cost and sale volume of a company.

Hence it is also known as “cost-volume-profit analysis”. It helps to know the operating condition that exists when a company ‘breaks-even’, that is when sales reach a point equal to all expenses incurred in attaining that level of sales. The break-even point may be defined as that level of sales in which total revenues equal total costs and net income is equal to zero. This is also known as no-profit no-loss point. This concept has been proved highly useful to the company executives in profit forecasting and planning and also in examining the effect of alternative business management decisions.

1. Break-Even Point:

The break-even point (B.E.P.) of a firm can be found out in two ways. It may be determined in terms of physical units, i.e., volume of output or it may be determined in terms of money value, i.e., value of sales.

BEP in terms of Physical Units:

This method is convenient for a firm producing a product. The BEP is the number of units of a product that should be sold to earn enough revenue just to cover all the expenses of production, both fixed and variable. The firm does not earn any profit, nor does it incur any loss. It is the meeting point of total revenue and total cost curve of the firm.

The break-even point is illustrated by means of Table 1:

Some assumptions are made in illustrating the BEP. The price of the commodity is kept constant at Rs. 4 per unit, i.e., perfect competition is assumed. Therefore, the total revenue is increasing proportionately to the output. All the units of the output are sold out. The total fixed cost is kept constant at Rs. 150 at all levels of output.

The total variable cost is assumed to be increasing by a given amount throughout. From the Table we can see that when the output is zero, the firm incurs only fixed cost. When the output is 50, the total cost is Rs. 300. The total revenue is Rs. 200. The firm incurs a loss of Rs. 100.

Table 1: Total Revenue and Total Cost and BEP

Output in units	Total Revenue	Total Fixed Cost	Total Variable Cost	Total Cost
0	0	150	0	150
50	200	150	150	300
100	400	150	300	450
1150	600	150	450	600 BEP
200	800	150	600	750
250	1000	150	750	900
300	1200	150	900	1050

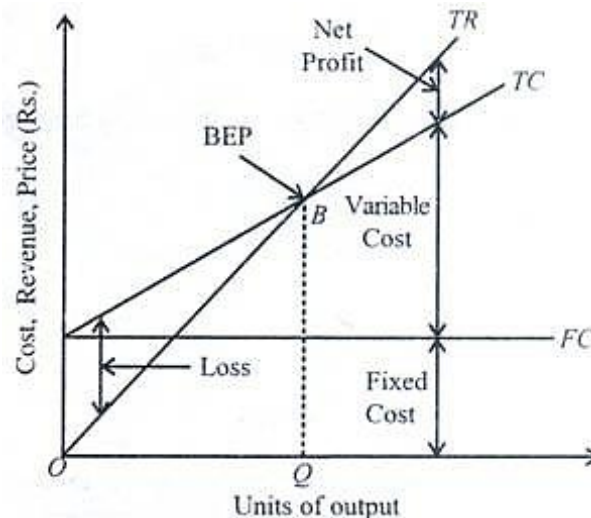
Similarly when the output is 100 the firm incurs a loss of Rs. 50. At the level of output 150 units, the total revenue is equal to the total cost. At this level, the firm is working at a point where there is no profit or loss. From the level of output of 200, the firm is making profit

Break-Even Chart:

Break-Even charts are being used in recent years by the managerial economists, company executives and government agencies in order to find out the break-even point. In the break-even charts, the concepts like total fixed cost, total variable cost, and the total cost and total revenue are shown separately. The break even chart shows the extent of profit or loss to the firm at different levels of activity. The following Fig. illustrates the typical break-even chart.

In this diagram output is shown on the horizontal axis and costs and revenue on vertical axis. Total revenue (TR) curve is shown as linear, as it is assumed that the price is constant, irrespective of the output. This assumption is appropriate only if the firm is operating under perfectly competitive conditions. Linearity of the total cost (TC) curve results from the assumption of constant variable cost.

It should also be noted that the TR curve is drawn as a straight line through the origin (i.e., every unit of the output contributes a constant amount to total revenue), while the TC curve is a straight line originating from the vertical axis because total cost comprises constant / fixed cost plus variable cost which rise linearly. In the figure, B is the break-even point at OQ level of output.



2. Determination of Break-even Point:

The formula for calculating the break-even point is

$$\text{BEP} = \text{Total Fixed Cost} / \text{Contribution Margin Per Unit}$$

Contribution margin per unit can be found out by deducting the average variable cost from the selling price. So the formula will be

$$\text{BEP} = \text{Total Fixed Cost} / \text{Selling Price} - \text{AVC}$$

Example:

Suppose the fixed cost of a factory is Rs. 10,000, the selling price is Rs. 4 and the average variable cost is Rs. 2, so the break-even point would be

$$\text{BEP} = 10,000 / (4 - 2) = 5,000 \text{ units.}$$

It means if the company makes the sales of 5,000 units, it would make neither loss nor profit. This can be seen in the analysis.

$$\text{Sales} = \text{Rs. } 20,000$$

Cost of goods sold:

$$(a) \text{ Variable cost at Rs. } 2 = \text{Rs. } 10,000$$

$$(b) \text{ Fixed costs} = \text{Rs. } 10,000$$

$$\text{Total Cost} = \text{Rs. } 20,000$$

$$\text{Net Profit} = \text{Nil}$$

BEP in term of Sales Value:

Multi-product firms are not in a position to measure the break-even point in terms of any common unit of product. They find it convenient to determine the break-even point in terms of total rupee sales. Here again the break-even point would be where the contribution margin (sales value—variable costs) would be equal to fixed costs. The contribution margin however, is expressed as a ratio to sales. The formula for calculating the break-even point is

BEP = Fixed Cost/Contribution Ratio

Contribution Ratio (CR) = Total Revenue (TR)-Total Variable Cost (TVC)/Total Revenue (TR)

For example, if TR is Rs. 600 and TVC is Rs. 450, then the contribution ratio is

$$CR = 600 - 450/600/600 = 150/600 = 0.25$$

The Contribution Ratio is 0.25

BEP = Total Fixed Cost /Contribution Ratio

$$= 150/0.25 = 600$$

The firm achieves its BEP when its sales are Rs. 600

Total Revenue = Rs.600

Total Cost = Rs.600

Net Profit/loss = Nil

Assumptions of Break-Even Analysis:

The break-even analysis is based on the following set of assumptions:

- (i) The total costs may be classified into fixed and variable costs. It ignores semi-variable cost.
- (ii) The cost and revenue functions remain linear.
- (iii) The price of the product is assumed to be constant.
- (iv) The volume of sales and volume of production are equal.
- (v) The fixed costs remain constant over the volume under consideration.
- (vi) It assumes constant rate of increase in variable cost.
- (vii) It assumes constant technology and no improvement in labour efficiency.
- (viii) The price of the product is assumed to be constant.
- (ix) The factor price remains unaltered.
- (x) Changes in input prices are ruled out.
- (xi) In the case of multi-product firm, the product mix is stable.

3. Managerial Uses of Break-Even Analysis:

To the management, the utility of break-even analysis lies in the fact that it presents a microscopic picture of the profit structure of a business enterprise. The break-even analysis not only highlights the area of economic strength and weakness in the firm but also sharpens the focus on certain leverages which can be operated upon to enhance its profitability.

It guides the management to take effective decision in the context of changes in government policies of taxation and subsidies.

The break-even analysis can be used for the following purposes:

(i) Safety Margin:

The break-even chart helps the management to know at a glance the profits generated at the various levels of sales. The safety margin refers to the extent to which the firm can afford a decline before it starts incurring losses.

The formula to determine the sales safety margin is:

$$\text{Safety Margin} = (\text{Sales} - \text{BEP}) / \text{Sales} \times 100$$

From the numerical example at the level of 250 units of output and sales, the firm is earning profit, the safety margin can be found out by applying the formula

$$\text{Safety Margin} = 250 - 150 / 250 \times 100 = 40\%$$

This means that the firm which is now selling 250 units of the product can afford to decline sales upto 40 per cent. The margin of safety may be negative as well, if the firm is incurring any loss. In that case, the percentage tells the extent of sales that should be increased in order to reach the point where there will be no loss.

(ii) Target Profit:

The break-even analysis can be utilised for the purpose of calculating the volume of sales necessary to achieve a target profit.

When a firm has some target profit, this analysis will help in finding out the extent of increase in sales by using the following formula:

$$\text{Target Sales Volume} = \text{Fixed Cost} + \text{Target Profit} / \text{Contribution Margin per unit}$$

By way of illustration, we can take Table 1 given above. Suppose the firm fixes the profit as Rs. 100, then the volume of output and sales should be 250 units. Only at this level, it gets a profit of Rs. 100. By using the formula, the same result will be obtained.

(iii) Change in Price:

The management is often faced with a problem of whether to reduce prices or not. Before taking a decision on this question, the management will have to consider a profit. A reduction in price leads to a reduction in the contribution margin.

This means that the volume of sales will have to be increased even to maintain the previous level of profit. The higher the reduction in the contribution margin, the higher is the increase in sales needed to ensure the previous profit.

The formula for determining the new volume of sales to maintain the same profit, given a reduction in price, will be as follows:

$$\text{New Sales Volume} = \frac{\text{Total Fixed Cost} + \text{Total Profit}}{\text{New Selling price} - \text{Average Variable Cost}}$$

For example, suppose a firm has a fixed cost of Rs. 8,000 and the profit target is Rs.20, 000. If the sales price is Rs.8 and the average variable cost is Rs. 4, then the total volume of sales should be 7,000 units on the basis of the formula given under target price.

Suppose the firm decides to reduce the selling price from Rs.8 to Rs. 7, then the new sales volume should be on the basis of the above formula:

$$\text{New Sales Volume} = 8,000 + 20,000/7-4 = 9,300$$

From this, we can infer that by reducing the price from Rs. 8 to Rs. 7, the firm has to increase the sales from Rs. 7,000 to Rs 9,330 if it wants to maintain the target profit of Rs. 20,000. In the same way, the sales executive can calculate the new volume of sales if it increases the price.

(iv) Change in Costs:

When costs undergo change, the selling price and the quantity produced and sold also undergo changes.

Changes in cost can be in two ways:

- (i) Change in variable cost, and
- (ii) Change in fixed cost.

(i) Variable Cost Change:

An increase in variable costs leads to a reduction in the contribution margin. This reduction in the contribution margin will shift the break-even point downward. Conversely, with the fall in the proportion of variable costs, contribution margins increase and break-even point moves upwards.

Under conditions of changing variable costs, the formula to determine the new quantity or the new selling price is:

(a) New Quantity or Sales Volume = Contribution to Margin/ Present Selling Price – New Variable Cost Per Unit

(b) New Selling Price = Present Sale Price +New Variable Cost-Present Variable Cost

Example:

The contribution margin is Rs. 64,000, the present sale price is Rs.10 and the present variable cost is Rs.6. If the variable cost per unit goes up from Rs.6 to Rs. 7, what will be the new sales volume and price?

New Sales Volume = $64,000 / 10 - 7 = 64,000 / 3 = 21,300$ units

New Sales Price = $(10 + 7 - 6) = \text{Rs. } 11$.

(ii) Fixed Cost Change:

An increase in fixed cost of a firm may be caused either due to a tax on assets or due to an increase in remuneration of management, etc. It will increase the contribution margin and thus push the break-even point upwards. Again to maintain the earlier level of profits, a new level of sales volume or new price has to be found out.

**New Sales Volume = Present Sale Volume +
(New Fixed Cost + Present Fixed Costs)/ (Present Selling Price-Present Variable Cost)**

**New Sale Price = Present Sale Price +
(New Fixed Costs – Present Fixed Costs)/ Present Sale Volume**

Example:

The fixed cost of a firm increases from Rs. 5,000 to Rs. 6,000. The variable cost is Rs. 5 and the sale price is Rs. 10 and the firm sells 1,000 units of the product

New Sales Volume = $1,000 + 6,000 - 5,000 / 10 - 5 = 1,000 + 1,000 / 5 = 1,000 + 200 = 1,200$ units

New Sale Price = $10 + 6,000 - 5,000 / 1,000 = 10 + 1,000 / 1,000 = \text{Rs. } 10 + \text{Rs. } 1$

= Rs. 11

(v) Decision on Choice of Technique of Production:

A firm has to decide about the most economical production process both at the planning and expansion stages. There are many techniques available to produce a product. These techniques will differ in terms of capacity and costs. The breakeven analysis is the most simple and helpful in the case of decision on a choice of technique of production.

For example, for low levels of output, some conventional methods may be most probable as they require minimum fixed cost. For high levels of output, only automatic machines may be most profitable. By showing the cost of different alternative techniques at different levels of output, the break-even analysis helps the decision of the choice among these techniques.

(vi) Make or Buy Decision:

Firms often have the option of making certain components or for purchasing them from outside the concern. Break-even analysis can enable the firm to decide whether to make or buy.

Example:

A manufacturer of car buys a certain components at Rs. 20 each. In case he makes it himself, his fixed and variable cost would be Rs. 24,000 and Rs.8 per component respectively.

$$\text{BEP} = \text{Fixed Cost} / \text{Purchase Price} - \text{Variable Cost}$$

$$= 24,000 / 20 - 8 = 24,000 / 12 = 2,000 \text{ units}$$

From this, we can infer that the manufacturer can produce the parts himself if he needs more than 2,000 units per year. However, certain considerations need to be taken account of in a buying decision, such as

- (i) Is the required quality of the product available?
- (ii) Is the supply from the market certain and timely?
- (iii) Do the suppliers of the components try to take any monopoly advantage?

(vii) Plant Expansion Decisions:

The break-even analysis may be adopted to reveal the effect of an actual or proposed change in operation condition. This may be illustrated by showing the impact of a proposed plant on expansion on costs, volume and profits. Through the break-even analysis, it would be possible to examine the various implications of this proposal.

Example:

A company has the capacity to produce goods worth of Rs. 40 crores a year. For this has incurred a fixed cost of Rs 20 crores, the variable costs being 60% of the sales revenue. Now company is planning to incur an additional Rs. 6 crores in fixed costs to expand its production capacity from Rs. 40 crores to Rs.60 crores. The survey shows that the firm's sales can be increased from Rs. 40 crores to Rs. 50 crores. Should the firm go in for expansion?

$$\text{BEP at present capacity} = \text{Fixed cost} / \text{Margin Contribution\%} = \text{Rs. } 20 \text{ crores} / 40\% = \text{Rs. } 50 \text{ crores}$$

$$\text{BEP at the proposed capacity} = \text{Rs } 26 \text{ crores} / 40\% = \text{Rs } 65 \text{ crores.}$$

Increase in break-even point = Rs 40 crores-Rs. 25 crores = Rs. 15 crores.

Thus we can infer that the firm should go in for expansion only if its sales expand by more than Rs. 15 crores from its earlier level of Rs. 40 crores.

(viii) Plant Shut Down Decisions:

In the shut down decisions, a distinction should be made between out of pocket and sunk costs. Out of pocket costs include all the variable costs plus the fixed cost which do not vary with output. Sunk fixed costs are the expenditures previously made but from which benefits still remain to be obtained e.g. depreciation.

(ix) Advertising and Promotion Mix Decisions:

The main objective of advertisement is to stimulate or increase sales to all customers-former, present and future. If there is keen to undertake vigorous campaign of advertisement. The management has to examine those marketing activities that stimulate consumer purchasing and dealer effectiveness.

The break-even point concept helps the management to know about the circumstances. It enables him not only to take appropriate decision but by showing how these additional fixed cost would influence BEPs. The advertisement pushes up the total cost curve by the amount of advertisement expenditure.

(x) Decision Regarding Addition or Deletion of Product Line:

If a product has outlived utility in the market immediately, the production must be abandoned by the management and examined what would be its consequent effect on revenue and cost. Alternatively, the management may like to add a product to its existing product line because it expects the product as a potential profit spinner. The break-even analysis helps in such a decision.

Example:

A fan manufacturer possesses the following data regarding his firm:

Total Fixed Cost = Rs. 1, 50,000

Volume of Sales = 5, 00,000 units

Type of Product	Price per unit	Variable Cost Per Unit	Share in Sales Volume (%)
	Rs.	Rs.	
Ordinary fan	360	240	50
Exhaust fan	600	360	30
Heaters	800	480	20

The manufacturer is considering whether or not to drop heaters from its product line and replace it with a fancy kind of fan.

He knows that if he takes the decision of dropping heaters and replaces it with fancy fans his output and cost data would be:

Total Fixed Cost = Rs. 1, 50,000

Likely Volume of Sales = Rs. 5, 00,000

Product	Price Per Unit Rs.	Variable Cost Per Unit Rs.	Share in Sales Volume (%)
Ordinary fan	360	240	50
Exhaust fan	600	360	20
Fancy fan	850	450	30

To find out the impact of proposed change, we need to compare profits in the two situations. Firstly, we have to find out the contribution ratio of each product.

$$\begin{aligned}
 \text{Contribution Ratio of a Product} &= \frac{\text{Price} - \text{Average Variable Cost}}{\text{Price}} \times \text{Share in Total Sales} \\
 \text{Contribution Ratio of Ordinary Fans} &= \frac{360 - 240}{360} \times 50\% = 0.167 \\
 \text{Contribution Ratio of Exhaust Fans} &= \frac{600 - 360}{600} \times 30\% = 0.12 \\
 \text{Contribution Ratio of Heaters} &= \frac{800 - 480}{800} \times 20\% = 0.08
 \end{aligned}$$

Therefore, the contribution ratio of the entire product line = $0.167 + 0.12 + 0.08 = 0.367$

Total Contribution = $\text{Rs. } 5,00,00 \times 0.367 = \text{Rs. } 1,83,500$

Profit = Total Contribution – Total Fixed Cost

= $\text{Rs. } 1,83,500 - \text{Rs. } 33,500$.

We have to follow the similar analysis for the second situation:

Contribution Ratio of Ordinary Fans = $\frac{360 - 240}{360} \times 50\% = 0.167$

Contribution Ratio of Exhaust Fans = $\frac{600 - 360}{600} \times 20\% = 0.08$

Contribution Ratio of Fancy Fans = $\frac{850 - 450}{850} \times 30\% = 0.141$

Thus the contribution ratio of the entire product line = $0.167 + 0.08 + 0.141 = 0.388$.

Total Contribution = $\text{Rs. } 5,00,000 \times 0.388 = \text{Rs. } 1,94,000$

Profit = Rs. 1, 94,000 — Rs. 1, 50,000 = Rs 44,000

From the above analysis, we can infer that the manufacturer should drop heaters from his product line and add fancy pens to his product line so as to earn more profit.

Limitations:

We may now mention some important limitations which ought to be kept in mind while using break-even analysis:

1. In the break-even analysis, we keep everything constant. The selling price is assumed to be constant and the cost function is linear. In practice, it will not be so.
2. In the break-even analysis since we keep the function constant, we project the future with the help of past functions. This is not correct.
3. The assumption that the cost-revenue-output relationship is linear is true only over a small range of output. It is not an effective tool for long-range use.
4. Profits are a function of not only output, but also of other factors like technological change, improvement in the art of management, etc., which have been overlooked in this analysis.
5. When break-even analysis is based on accounting data, as it usually happens, it may suffer from various limitations of such data as neglect of imputed costs, arbitrary depreciation estimates and inappropriate allocation of overheads. It can be sound and useful only if the firm in question maintains a good accounting system.
6. Selling costs are specially difficult to handle break-even analysis. This is because changes in selling costs are a cause and not a result of changes in output and sales.
7. The simple form of a break-even chart makes no provisions for taxes, particularly corporate income tax.
8. It usually assumes that the price of the output is given. In other words, it assumes a horizontal demand curve that is realistic under the conditions of perfect competition.
9. Matching cost with output imposes another limitation on break-even analysis. Cost in a particular period need not be the result of the output in that period.
10. Because of so many restrictive assumptions underlying the technique, computation of a breakeven point is considered an approximation rather than a reality.

Some Important Definitions:

(i) Angle of incidence:

It is the angle at which income line or sales line cuts the total cost line. If the angle is large, it is an indication that profits are being made at a high rate, on the other hand, if the angle is small, it indicates that less profits are being made and are achieved under less favourable conditions.

(ii) Margin of safety:

It is the output at full capacity minus the output at “Break-Even Point”. It is expressed as percentage of output at full capacity. If the margin of safety is small, a small drop in production capacity will reduce the profit greatly.

It can also be expressed as:

$$\text{Margin of safety} = \frac{(\text{Sales at full capacity} - \text{Sales at B.E.P.}) \times 100}{\text{Sales at full capacity}}$$

(iii) Contribution:

It is the difference between sales and variable cost (marginal cost). It is also called as Marginal Profit or Gross Margin. The marginal profit provides the contribution towards fixed cost and profit.

Contribution = (Sales – Variable cost) which in turn will be equal to fixed cost + Profit.

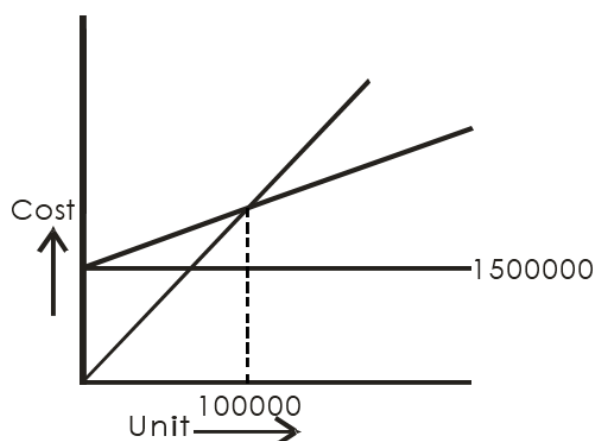
Question 1: A company is faced with a situation where it can either produce some item by adding additional infrastructure which will cost them Rs. 15,00,000/- but unit cost of production will be Rs. 5/- each. Alternatively it can buy the same item from a vendor at a rate of Rs. 20/- each. When should the company add to its capacity in terms of demand of items per annum? Draw the diagram to show the BEP.

Solution: Let the capacity is x when company will meet its demand, so

$$1500000 + 5x = 20x$$

$$15x = 1500000$$

$$\Rightarrow x = 10000$$



Question 2: The following data refers to a manufacturing unit

Fixed cost = Rs. 100000/-

Variable cost = 100/- per unit

Selling price = Rs. 200/- per unit

- (i) Calculate the BEP
 (ii) Calculate the number of component needed to be product to get a profit of Rs. 20000/-

Solution:

- (i) At break even point

$$F + Q.V = S \times Q$$

$$Q = \frac{F}{S-V} = \frac{100000}{200-100} = 1000 \text{ pieces}$$

- (ii) For fixed profit Rs. 20000/-

$$F + Q.V + P = S \times Q$$

$$Q = \frac{F+P}{S-V} = \frac{100000+20000}{200-100} = 1200 \text{ pieces}$$

Question 3: The following data from a plant is available:

Product	Price Rs.	Variable cost Rs.	Percentage sales units
A	10	6	20
B	6	4	30
C	20	12	50

Fixed cost per year = Rs. 7500

Total sales from three products = Rs. 25,000

It is decided by the management to drop the product A and add product D. To find whether the decision is profitable or not?

Assume sales for products B, C and D as Rs. 26,000.

Product	Price Rs.	Variable cost Rs.	Percentage sales units
D	16	6	10
B	6	4	50
C	20	12	40

Solution:

Total fixed cost per year = Rs. 7500.

Sales for the current year = 25,000

Product A contribution = Rs. (10 – 6) = Rs. 4

∴ Contribution for 20% sales of product A = $10 - 6/10 \times 0.20 = 0.08$

Similarly, for product B, = $6 - 4/6 \times 0.30 = 0.10$

and for product C = $20 - 12/20 \times 0.5 = 0.20$

Thus contribution ratio for product A, B and C

$$= 0.08 + 0.10 + 0.20 = 0.38$$

$$\therefore \text{Total contribution of sales} = 0.38 \times 25,000 = \text{Rs. } 9500$$

$$\therefore \text{Profit} = \text{Contribution} - \text{Fixed costs} = 9500 - 7500 = \text{Rs. } 2000$$

Similarly, the profit or contribution for new product line of products D, B, C

$$D = 16 - 6/16 \times 10\% = 0.06$$

$$B = 6 - 4/6 \times 0.50 = 0.17$$

$$C = 20 - 12/20 \times 0.40 = 0.16$$

$$\therefore \text{Total contribution ratio} = 0.06 + 0.17 + 0.16 = 0.39$$

$$\therefore \text{Total contribution from sales} = 26,000 \times 0.39 = \text{Rs. } 10,140$$

$$\therefore \text{Profit} = 10,140 - 7500 = \text{Rs. } 2640.$$

Hence proposed decision is profitable to accept.

1.8 HUMAN FACTORS IN DESIGN

Human factors and ergonomics (commonly referred to as **HF&E**), also known as comfort design, functional design, and systems, is the practice of designing products, systems, or processes to take proper account of the interaction between them and the people who use them.

The field has seen some contributions from numerous disciplines, such as psychology, engineering, biomechanics, industrial design, physiology, and anthropometry. In essence, it is the study of designing equipment, devices and processes that fit the human body and its cognitive abilities. The two terms "human factors" and "ergonomics" are essentially synonymous.

The International Ergonomics Association defines ergonomics or human factors as follows:

Ergonomics (or human factors) is the scientific discipline concerned with the understanding of interactions among humans and other elements of a system, and the profession that applies theory, principles, data and methods to design in order to optimize human well-being and overall system performance.

HF&E is employed to fulfill the goals of occupational health and safety and productivity. It is relevant in the design of such things as safe furniture and easy-to-use interfaces to machines and equipment. Proper ergonomic design is necessary to prevent repetitive strain injuries and other musculoskeletal disorders, which can develop over time and can lead to long-term disability.

To assess the fit between a person and the used technology, human factors specialists or ergonomists consider the job (activity) being done and the demands on the user; the equipment used (its size, shape, and how appropriate it is for the task), and the information used (how it is presented, accessed, and changed).

Methods

- **Ethnographic analysis:** Using methods derived from ethnography, this process focuses on observing the uses of technology in a practical environment. It is a qualitative and observational method that focuses on "real-world" experience and pressures, and the usage of technology or environments in the workplace. The process is best used early in the design process.
- **Focus Groups** are another form of qualitative research in which one individual will facilitate discussion and elicit opinions about the technology or process under investigation. This can be on a one-to-one interview basis, or in a group session. Can be used to gain a large quantity of deep qualitative data, though due to the small sample size, can be subject to a higher degree of individual bias. Can be used at any point in the design process, as it is largely dependent on the exact questions to be pursued, and the structure of the group. Can be extremely costly.
- **Iterative design:** Also known as prototyping, the iterative design process seeks to involve users at several stages of design, in order to correct problems as they emerge. As prototypes emerge from the design process, these are subjected to other forms of analysis as outlined in this article, and the results are then taken and incorporated into the new design. Trends amongst users are analyzed, and products redesigned. This can become a costly process, and needs to be done as soon as possible in the design process before designs become too concrete.
- **Meta-analysis:** A supplementary technique used to examine a wide body of already existing data or literature in order to derive trends or form hypotheses in order to aid design decisions. As part of a literature survey, a meta-analysis can be performed in order to discern a collective trend from individual variables.
- **Subjects-in-tandem:** Two subjects are asked to work concurrently on a series of tasks while vocalizing their analytical observations. The technique is also known as "Co-Discovery" as participants tend to feed off of each other's comments to generate a richer set of observations than is often possible with the participants separately. This is observed by the researcher, and can be used to discover usability difficulties. This process is usually recorded.
- **Surveys and Questionnaires:** A commonly used technique outside of Human Factors as well, surveys and questionnaires have an advantage in that they can be administered to a large group of people for relatively low cost, enabling the researcher to gain a large amount of data. The validity of the data obtained is, however, always in question, as the questions must be written and interpreted correctly, and are, by definition, subjective. Those who actually respond are in effect self-selecting as well, widening the gap between the sample and the population further.
- **Task analysis:** A process with roots in activity theory, task analysis is a way of systematically describing human interaction with a system or process to understand how to match the demands of the system or process to human capabilities. The complexity of this process is generally proportional to the complexity of the task being analyzed, and so can vary in cost and time involvement. It is a qualitative and observational process. Best used early in the design process.
- **Think aloud protocol:** Also known as "concurrent verbal protocol", this is the process of asking a user to execute a series of tasks or use technology, while continuously verbalizing their thoughts so

that a researcher can gain insights as to the users' analytical process. Can be useful for finding design flaws that do not affect task performance, but may have a negative cognitive affect on the user. Also useful for utilizing experts in order to better understand procedural knowledge of the task in question. Less expensive than focus groups, but tends to be more specific and subjective.

- **User analysis:** This process is based around designing for the attributes of the intended user or operator, establishing the characteristics that define them, creating a persona for the user. Best done at the outset of the design process, a user analysis will attempt to predict the most common users, and the characteristics that they would be assumed to have in common. This can be problematic if the design concept does not match the actual user, or if the identified are too vague to make clear design decisions from. This process is, however, usually quite inexpensive, and commonly used.
- **"Wizard of Oz":** This is a comparatively uncommon technique but has seen some use in mobile devices. Based upon the Wizard of Oz experiment, this technique involves an operator who remotely controls the operation of a device in order to imitate the response of an actual computer program. It has the advantage of producing a highly changeable set of reactions, but can be quite costly and difficult to undertake.
- **Methods Analysis** is the process of studying the tasks a worker completes using a step-by-step investigation. Each task is broken down into smaller steps until each motion the worker performs is described. Doing so enables you to see exactly where repetitive or straining tasks occur.
- **Time studies** determine the time required for a worker to complete each task. Time studies are often used to analyze cyclical jobs. They are considered "event based" studies because time measurements are triggered by the occurrence of predetermined events.
- **Work sampling** is a method in which the job is sampled at random intervals to determine the proportion of total time spent on a particular task. It provides insight into how often workers are performing tasks which might cause strain on their bodies.
- **Predetermined time systems** are methods for analyzing the time spent by workers on a particular task. One of the most widely used predetermined time system is called Methods-Time-Measurement (MTM).
- **Cognitive Walkthrough:** This method is a usability inspection method in which the evaluators can apply user perspective to task scenarios to identify design problems. As applied to macroergonomics, evaluators are able to analyze the usability of work system designs to identify how well a work system is organized and how well the workflow is integrated.
- **Kansei Method:** This is a method that transforms consumer's responses to new products into design specifications. As applied to macroergonomics, this method can translate employee's responses to changes to a work system into design specifications.
- **High Integration of Technology, Organization, and People (HITOP):** This is a manual procedure done step-by-step to apply technological change to the workplace. It allows managers to be more aware of the human and organizational aspects of their technology plans, allowing them to efficiently integrate technology in these contexts.
- **Top Modeler:** This model helps manufacturing companies identify the organizational changes needed when new technologies are being considered for their process.
- **Computer-integrated Manufacturing, Organization, and People System Design (CIMOP):** This model allows for evaluating computer-integrated manufacturing, organization, and people system design based on knowledge of the system.

- **Anthropotechnology:** This method considers analysis and design modification of systems for the efficient transfer of technology from one culture to another.
- **Systems Analysis Tool (SAT):** This is a method to conduct systematic trade-off evaluations of work-system intervention alternatives.
- **Macroergonomic Analysis of Structure (MAS):** This method analyzes the structure of work systems according to their compatibility with unique sociotechnical aspects.
- **Macroergonomic Analysis and Design (MEAD):** This method assesses work-system processes by using a ten-step process.
- **Virtual Manufacturing and Response Surface Methodology (VMRSM):** This method uses computerized tools and statistical analysis for workstation design.

Limitations

Problems related to measures of usability include the fact that measures of learning and retention of how to use an interface are rarely employed and some studies treat measures of how users interact with interfaces as synonymous with quality-in-use, despite an unclear relation.

Although field methods can be extremely useful because they are conducted in the users' natural environment, they have some major limitations to consider. The limitations include:

1. Usually take more time and resources than other methods
2. Very high effort in planning, recruiting, and executing compared with other methods
3. Much longer study periods and therefore requires much goodwill among the participants
4. Studies are longitudinal in nature, therefore, attrition can become a problem.

1.9 VALUE ENGINEERING

Value engineering (VE) is a systematic method to improve the "value" of goods or products and services by using an examination of function. Value, as defined, is the ratio of function to cost. Value can therefore be increased by either improving the function or reducing the cost. It is a primary tenet of value engineering that basic functions be preserved and not be reduced as a consequence of pursuing value improvements.

The reasoning behind value engineering is as follows: if marketers expect a product to become practically or stylistically obsolete within a specific length of time, they can design it to only last for that specific lifetime. The products could be built with higher-grade components, but with value engineering they are not because this would impose an unnecessary cost on the manufacturer, and to a limited extent also an increased cost on the purchaser. Value engineering will reduce these costs. A company will typically use the least expensive components that satisfy the product's lifetime projections.

Meaning of Value Engineering:

The society of American Value Engineers define Value Engineering as—the systematic application of recognised techniques which identify the function of a product or service, establish a monetary value for that function, and provides the necessary function reliably at the lowest overall cost.

The value engineering can be defined simply as: It is a systematic, step-by-step approach, to achieve the desired functions of a system, service, process or product at an overall minimum cost, without deteriorating the quality, performance, reliability, safety or environment.

Job Plan

Value engineering is often done by systematically following a multi-stage job plan. Larry Miles' original system was a six-step procedure which he called the "value analysis job plan." Others have varied the job plan to fit their constraints. Depending on the application, there may be four, five, six, or more stages. One modern version has the following eight steps:

1. Preparation
2. Information
3. Analysis
4. Creation
5. Evaluation
6. Development
7. Presentation
8. Follow-up



Four basic steps in the job plan are:

- Information gathering - This asks what the requirements are for the object. Function analysis, an important technique in value engineering, is usually done in this initial stage. It tries to determine what functions or performance characteristics are important. It asks questions like; What does the object do? What must it do? What should it do? What could it do? What must it not do?
- Alternative generation (creation) - In this stage value engineers ask; What are the various alternative ways of meeting requirements? What else will perform the desired function?
- Evaluation - In this stage all the alternatives are assessed by evaluating how well they meet the required functions and how great the cost savings will be.
- Presentation - In the final stage, the best alternative will be chosen and presented to the client for final decision.

Value Engineering Process

Gather information

1. What is being done now?
 - Who is doing it?
 - What could it do?
 - What must it not do?

Measure

2. How will the alternatives be measured?
 - What are the alternate ways of meeting requirements?
 - What else can perform the desired function?

Analyze

3. What must be done?

What does it cost?

Generate

4. What else will do the job?

Evaluate

5. Which Ideas are the best?

Develop and expand ideas

6. What are the impacts?

7. What is the cost?

8. What is the performance?

Present ideas

9. Alternative options are presented to the client, a recommendation is made for the best alternative and presented to the client for a decision.

Effectiveness of Value Engineering:

It assures cost effectiveness:

- It is cost avoidance instead of cost reduction.
- It locates all frills and gold plating's.
- It segregates the necessary from unnecessary.
- It is an organised study of functions and cost.
- It is not a crash cost reduction method.
- It is a combined effort of several departments.
- It provides necessary functions and avoids unnecessary functions.
- It is not a sacrifice of quality.
- It is not a mere criticism of existing system, method, design or process but an appraisal of practical alternatives.

Qualitative Advantages of V.E.:

- Better understanding between department and individuals and so better co-operation.
- Continuous updating of data base management.
- For techno-economic excellence, introspection by individuals and departments possible.
- Tendency to think functions rather than parts i.e. the hardware,
- Improved capability and preparedness to cope with changes in demand and quality.
- Since value management involves individual's participation in decision making, there is improvement in morale and commitment of the individuals.
- Atmosphere of creativity in the organisation. Individuals are encouraged to put forward suggestion which lead to ultimate cost reduction and better functioning.
- It leads to job satisfaction as there is professional autonomy to face technical challenges.
- It gives opportunities to individuals to prove their skill, ingenuity and creativity.
- It acts as a motivator towards higher productivity of the organisation.
- It encourages joint decision making so shouldering of responsibility by all.
- Improved system, procedures and communications.

Qualitative Advantages of V.E.:

- Higher productivity.
- Simplified manufacturing process.
- Overall cost reduction.
- Better performance,
- Higher reliability.
- Reduction in lead time.
- Better quality.
- Easy maintainability.
- Improved appearance.
- Simplified design.
- Reduced rejections.
- Less close and rigid tolerance.
- Higher market share.
- Higher profit.
- Less after-sales service requirements.
- Reduced down time of machine or process.
- Decision on to-make-or-to-buy, easy and correct.
- Better packaging.
- Improved logistics.
- Application of group technology possible.
- Accommodating customer's various requirements possible, hence greater customer's satisfaction.
- Export probability enhances.
- Enhanced value of product/service.
- Optimum utilisation of resources, materials, machines, facilities manpower and money.

Commandants of Value Analysis:

The followings are the 10 basic commandants of value analysis:

1. Do not use a component or part that does not contribute to the value of the product.
2. Do not use a component or part whose cost is not proportionate to its usefulness.
3. Do not provide any features to the component or the finished product that are not absolutely necessary.
4. Accept the change if the part of required quality can be made out of superior or inferior quality material where the overall cost is less.
5. If a part of required quality can be made by a method or process costing less, use the alternative.
6. Replace a non-standard part with a standard part wherever possible. It will cost less.
7. Use proper tooling and manufacturing methods considering the quantities required.
8. Cost of a component shall be proportional to the material used and the labour, overhead and profit allowed.
9. Where possible, use some other material or part better suited for the purpose.
10. If a dependable supplier can provide a part for less or a better part for the same, then do not make it yourself.

IMPORTANT QUESTIONS

1. Explain the functions and different applications of Industrial Engineering.
2. Explain Manufacturing vs Purchase.
3. What is meant by Break-Even analysis. Explain with the help of a figure, Break-even point. What are the assumptions in Break-even analysis? What is Angle of incidence & Margin of safety.
4. Explain the principles of Value Engineering.
5. Explain human factors in design.
6. Explain different steps in New Product development.
7. What are the advantages of Value engineering?



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MODULE 2

2.1 PLANT LAYOUT AND MATERIAL HANDLING

Raw materials form a critical part of manufacturing as well as service organization. In any organization, a considerable amount of material handling is done in one form or the other. This movement is either done manually or through an automated process. Throughout material, handling processes significant safety and health; challenges are presented to workers as well as management. Therefore, manual material handling is of prime concern for health and safety professional, and they must determine practical ways of reducing health risk to the workers.

Material Handling

Manual material handling ranges from movement of raw material, work in progress, finished goods, rejected, scraps, packing material, etc. These materials are of different shape and sizes as well as weight. Material handling is a systematic and scientific method of moving, packing and storing of material in appropriate and suitable location. The main objectives of material handling are as follows:

- It should be able determine appropriate distance to be covered.
- Facilitate the reduction in material damage as to improve quality.
- Reducing overall manufacturing time by designing efficient material movement
- Improve material flow control
- Creation and encouragement of safe and hazard-free work condition
- Improve productivity and efficiency
- Better utilization of time and equipment

It is critical for manufacturing organization to identify importance of material handling principle as the critical step in promoting the job improvement process. Manual material handling significantly increases health hazard for the workers in from lower back injuries. In the current competitive and globalized environment, it is important to control cost and reduce time in material handling. An efficient material handling process promotes:

- Design of proper facility layout
- Promotes development of method which improves and simplifies the work process
- It improves overall production activity.
- Efficient material handling reduces total cost of production.

2.2 PRINCIPLES OF MATERIAL HANDLING

Material handling principles are as follows:

- **Orientation Principle:** It encourages study of all available system relationships before moving towards preliminary planning. The study includes looking at existing methods, problems, etc.
- **Planning Principle:** It establishes a plan which includes basic requirements, desirable alternates and planning for contingency.
- **Systems Principle:** It integrates handling and storage activities, which is cost effective into integrated system design.
- **Unit Load Principle:** Handle product in a unit load as large as possible

- **Space Utilization Principle:** Encourage effective utilization of all the space available
- **Standardization Principle:** It encourages standardization of handling methods and equipment.
- **Ergonomic Principle:** It recognizes human capabilities and limitation by design effective handling equipment.
- **Energy Principle:** It considers consumption of energy during material handling.
- **Ecology Principle:** It encourages minimum impact upon the environment during material handling.
- **Mechanization Principle:** It encourages mechanization of handling process wherever possible as to encourage efficiency.
- **Flexibility Principle:** Encourages of methods and equipment which are possible to utilize in all types of condition.
- **Simplification Principle:** Encourage simplification of methods and process by removing unnecessary movements
- **Gravity Principle:** Encourages usage of gravity principle in movement of goods.
- **Safety Principle:** Encourages provision for safe handling equipment according to safety rules and regulation
- **Computerization Principle:** Encourages of computerization of material handling and storage systems
- **System Flow Principle:** Encourages integration of data flow with physical material flow
- **Layout Principle:** Encourages preparation of operational sequence of all systems available
- **Cost Principle:** Encourages cost benefit analysis of all solutions available
- **Maintenance Principle:** Encourages preparation of plan for preventive maintenance and scheduled repairs
- **Obsolescence Principle:** Encourage preparation of equipment policy as to enjoy appropriate economic advantage.

Factors affecting the selection of material handling equipment:

- (i) Nature of product and its portability.
- (ii) Value of production.
- (iii) Shape and size of products.
- (iv) Methods of production.
- (v) Sequence of operations.
- (vi) The production rate of the industrial unit.
- (vii) Space availability and type of layout used.
- (viii) Distance to be covered by the material.
- (ix) Power availability.
- (x) Initial cost of installation, operation and maintenance costs.
- (xi) Depreciation Costs.
- (xii) Availability and wages of unskilled labour.
- (xiii) Design of material handling equipment its capacity.



It is clear from above that the selection of material handling equipment's depend on so many factors and it is difficult to make any recommendation without taking into consideration the practical aspects of the problem. So at this stage, it is relevant to define the basic requirements of material handling equipment's.

Requirements of a good material handling equipment:

- (i) It must be able to perform the basic function of material handling like storage and transportation.
- (ii) It must facilitate production planning, inspection and process control activities.
- (iii) It should be able to reduce the work cycle time i.e. minimizing the unproductive material handling time.
- (iv) It should improve the capacity utilization of plant.
- (v) It should minimize the work in process or the total inventory requirements.
- (vi) It should be able to reduce the workers mutual and physical fatigue. This factor will ultimately improve satisfaction and safety level of workers.

2.3 DIFFERENT TYPES OF MATERIAL HANDLING EQUIPMENT

There are 4 main categories of material handling equipment:

- Bulk material handling
- Engineered systems
- Industrial trucks
- Storage and handling equipment

Bulk Handling Material Equipment

The term 'bulk handling equipment' refers to the storage, control and transportation of materials by bulk, and in loose form. There are plenty of examples of this application – the handling of food, beverages, liquids, metal items (such as screws and nails) and minerals are just a few.

In general terms, these pieces of equipment primarily handle these items when they are loose. One example is a conveyor belt that is used to move items from one part of the production process to another. Drums and hoppers may be used as well, to 'funnel' these loose items into a stage where they can be more easily manipulated, or packaged.

Conveyor belts are used for horizontal transportation. For vertical transportation, elevators are most commonly used.

Types of bulk handling material equipment

- Bucket and grain elevators
- Conveyor belts
- Hoppers and silos
- Reclaimers
- Stackers

Engineered Systems

An engineered system is one that is typically automated. Such systems are also usually created from a variety of units. When combined, they work to enable both storage and transportation.

An ‘Automated Storage and Retrieval System’ (or a AS/RS for short) is one example of a system that is engineered. This is a large, automated device that comes complete with racks, shelves and aisles. These storage solutions are accessed by a ‘shuttle’ – a mechanized device that’s similar to a cherry picker. This device can be used by the system operator to manually select the items as needed, or the entire system can be computerized and automated.

An AS/RS can be integrated with a production facility’s existing computer network to keep on top of stock control, plus other logistical systems. It can also be integrated with other stages of the production process, so that as much automation can be offered as possible.

Types of engineered systems

- Automated guided vehicles (AGVs)
- Conveyor systems
- Robotic delivery systems

Industrial Trucks

This term is another broad definition that can be applied to many different types of equipment. Such pieces of equipment do have one thing in common, though – they all provide transportation.

The scope of this term can include both small, hand-operated devices, and large-scale motorized vehicles. Some items can be driven, while others – such as pallet trucks – simply add mobility to the materials that are being handled.

Many of these types of trucks have useful characteristics such as forks or a flat surface that can be inserted under pallets or other types of storage platforms. Other trucks need a separate item of equipment to use for lifting.

Trucks have the capability to lift via powered or manual means, and can be ridden upon in a driver’s cab, or simply power-assisted when pushed. Such trucks can also be steered by human intervention, or can be completely automated, following a pre-defined track on the production floor, sunken or raised tracks, or colored strips that are laid out and sensed by optical sensors. Such automated industrial trucks also have anti-collision technology that senses when an employee or other obstacle is near.

Stacking trucks are used to stack items, while a non-stacking truck is just used for transportation, and not for product loading.

Types of industrial trucks:

- Automated guided vehicles (AGVs)
- Hand, platform and pallet trucks
- Order pickers
- Pallet jacks
- Side-loaders
- Walking stackers

Storage and Handling Equipment

Equipment that is used for storage usually only encompasses items that are not automated. Storage and handling equipment that is automated falls under the term ‘engineered systems’

Storage equipment is equipment that is used to hold products and materials when they are not being used, or when they are waiting to enter or leave the production process. These periods could be long-term, or short-term in order to allow a suitable build-up of stock or finished items.

Most items that can be described as storage and handling equipment refers to pallets, racking or shelves. Materials are stored in a neat and convenient manner to await transportation, or their entry into the production process if necessary.

Having suitable storage equipment will add to any company’s production efficiency. The efficiency of any production system is maximized by the ease at which each stage of the entire system operates. Any inefficient section creates a bottleneck that will have an effect on all other sections of the system further down the production line.

Space is also at a premium if you run a production environment. The better utilized your available space is, the more items you’ll be able to store. This means you can keep your workflow in operation for much longer without worrying about re-stocking. This helps further increase your efficiency.

Types of storage and handling equipment

- Drive-through or drive-in racks
- Pallet racks
- Push-back racks
- Shelving
- Sliding racks
- Stacking frames

Devices used for Mechanical Handling of Materials (With Diagram)

1. Transportation Equipment or Devices:

These devices are useful only for horizontal movement of materials. These devices include trucks and other similar vehicles. These vehicles are powered by hand, gasoline or electric power and have the capability of transporting material and manpower in a horizontal direction.

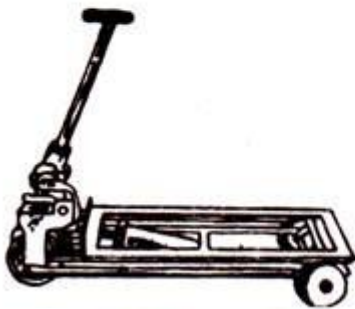
These also include variable path equipment's and can be utilized so long as travelling surfaces are available and the route is obstruction free. Thus these vehicles occupy the space intermittently and as soon as the work is over the space is free for some other operation.

The simplest among these are wheel barrows and hand trucks. But these devices need large amount of manpower for relatively small load. These involve easy portability, greater flexibility and low cost. When movement from one workstation to other is required.

Tractors and trailers are the other popular modes of horizontal transportation.



16.1: Heavy duty both Side Turn Table Platform Trailer with Solid Rubber Tyred Wheels



16.2: Elevating Truck having a Rigid Steel Structured Frame



16.3: Portable Hydraulic Stacker

Great flexibility is provided by these methods. Trailers can be left loaded and can be picked up later by other different tractors. This is considered as one of the most convenient and important methods of material handling inside the plant. Skids can be used with lift trucks.

These are the improvements over wheel barrows and hand trucks. For horizontal transportation of commodities like natural gas, oil and water etc. pipelines and pumps can also be used.





**4 Wheeled Hydraulic
Tripping Trailer**



**4 Wheeled Heavy Duty Cargo
Trailer with 3 Side Drop Down Arrangements**



Floor Mobile Life, Trolley Type



Box Trolley for Carrying Finished Products



Floor Crane. Trolley Type



Double Wheeled Barrow



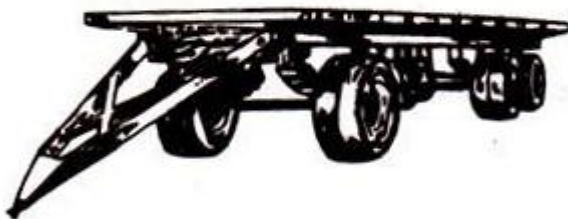
Heavy Duty Low Bed Flat Deck Trailer



Hydraulic Hand Lift Truck



Heavy Duty Goose-Neck Type Trailer



Heavy Duty Platform Trailer



Hydraulic Tipping Box Trolley



Ladder Trolley With Castor Wheels
Suitable for Use in the Stores Godown



Lifting Truck with Mechanical and
Hydraulic Lifting and Lowering Arrangements
Fitted with a Detachable Type Iron Mesh Box
Suitable for Carrying Finished Goods



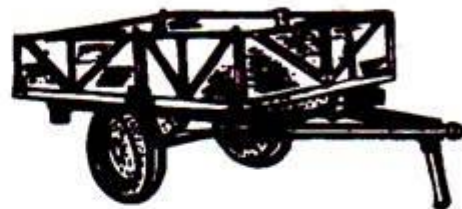
Single Wheel Barrow



**Refuse Trolley, 2 Wheeled,
All, Steel Construction**



**Platform Trolley with Single
Tubular and Swivelling Castor Wheels**



Wheeled Open Top Tractor/Trailer



Four Wheeled Sack Truck



Heavy Duty Sliding Wheeled Trolley



**2 Wheeled Open Top
Platform Trolley with Tubular Frame**



Hand Truck



2 Wheeled Hand Trolley Suitable for Carrying Finished Goods Raw Materials Complete with Iron Nett



4 Wheeled Platform Trailer with Solid Rubber Tyred Wheels



Double Wheeled Tipping Barrow



Sheet Metal Carrying Trolley



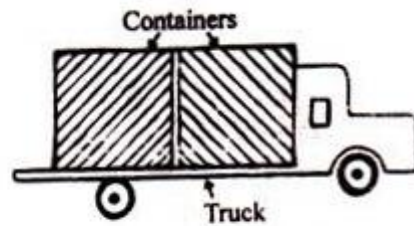
Platform Truck (high-lift type)



Fork Truck (counterbalance type)



Industrial Tractor



Truck with Containers

2. Lifting and Lowering Equipment or Devices:

These are meant for vertical transportation of material. Block and Tackle arrangement of lifting loads through vertical distance is one of the oldest and simplest device. It is still used today by moving men in hoisting machinery into position. Winch is another device used to lift loads vertically by winding rope or cable on a drum. Cranes and Hoist are the common means of vertical movement.

The equipment is able to move material vertically and laterally in a space of limited length width as well as height. If mounted on carriers like trucks and tractors etc. they can also move from one location to the other. Hoists are power driven devices, often operated between fixed guide rails.

Two types of cranes i.e., pillar and overhead cranes are generally used for light duty jobs and in various workshops (such as foundries and power houses and chemical plants etc.) respectively. Hydraulic and electric operated elevators also fall under this category of material handling devices. These are shown in figures.



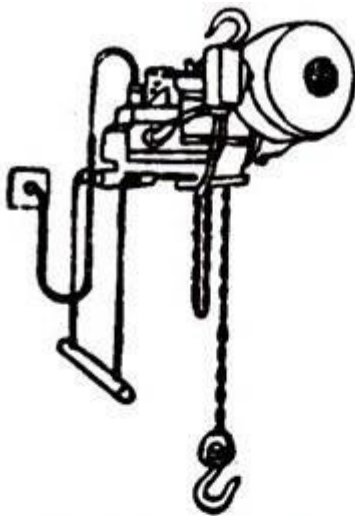
Chain Hoist



Electrical Hoist



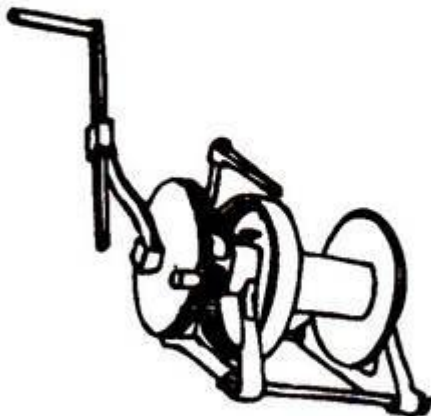
Air Hoist



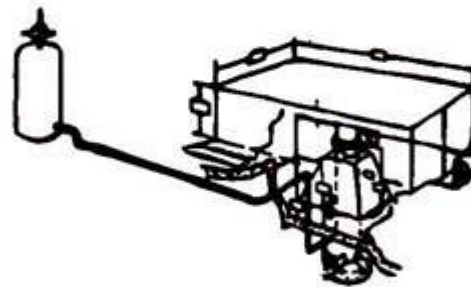
Power Hoise



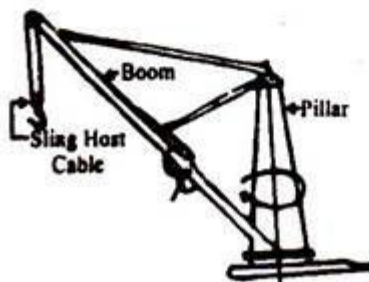
Block and Tackle



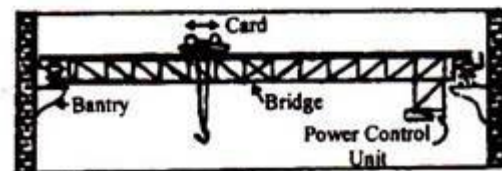
Winch



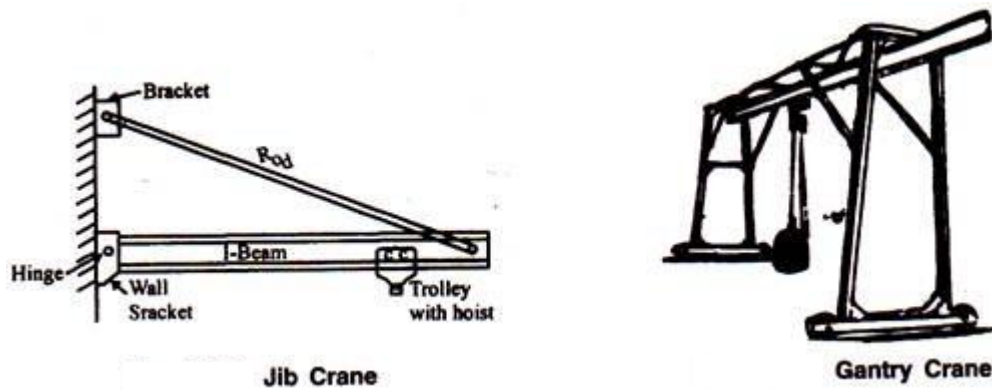
Hydraulic Elevator



: Pillar Crane



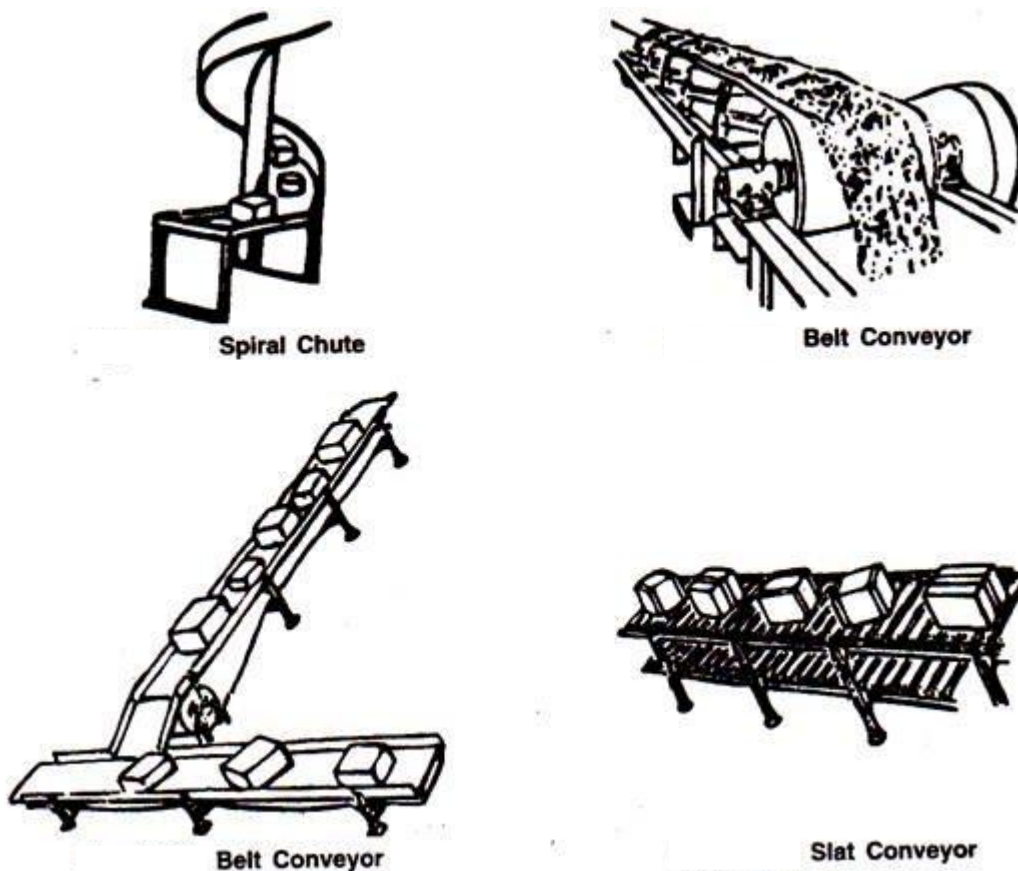
Overhead Crane



3. Combination of Transportation and Lifting Plus Lowering Devices:

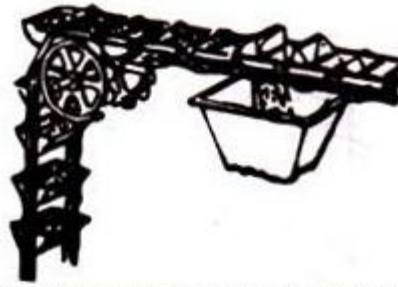
Simplest among these devices are conveyors which move materials or people in either vertical or horizontal direction between two fixed points. In conveyors, the transportation is affected by friction between materials being transported by the belt.

These conveyors have the advantage that they largely save labour cost but have disadvantage that take up considerable space and are relatively fixed and in majority of cases the investment is high. Other devices under this category of combination devices are Chute, and various types of conveyors. These are shown in Figs.

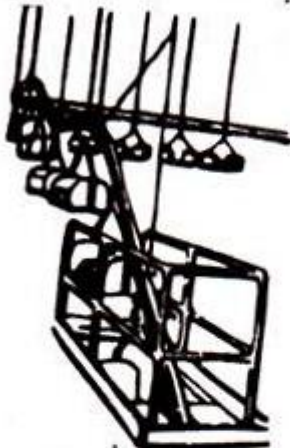




Roller Conveyor



Pivoted Bucket Conveyor



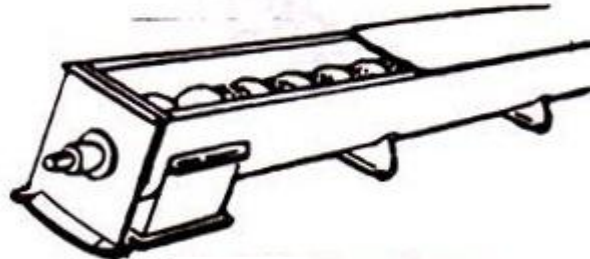
Trolley Conveyor



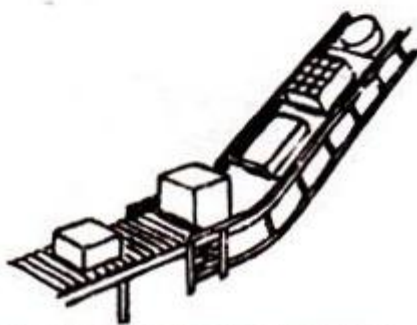
Wheel Conveyor



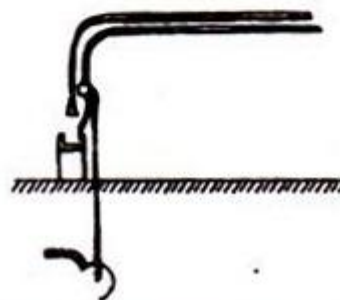
Apron Conveyor



Screw Conveyor



Pusher Bar Conveyor



Pneumatic Conveyor

Relation between Plant Layout and Material Handling

The material handling technique to be used definitely effects the plant layout and the factory building. A sound low cost method can be designed and installed only if material handling is considered an integral part of plant layout.

A well considered arrangement of production equipment, the proper location of different departments, a logical sequence of operation within the department and convenient location of store areas, tool cribs and similar activity centres is required for a good material handling arrangement. The efficient and economical material handling system can be designed and selected for installations only after the floor plan has been adequately organized.

In all types of plant layout provisions for the receiving and shipping of materials by various possible means (like trucks or train etc.) should be made. If it is required to move the materials by hand operated or power operated trucks, sufficient passage should be provided.

If the building is multi storied, lift, elevators and conveyors of different types must be utilized to enable efficient material handling. The location of items in the store room should provide for minimum handling of materials to the point of issue, accessibility and efficient space utilization.

Material Handling in Product Type of Layout:

The product type of layout require some direct means of transportation between various operations, Material handling methods and machines like the layout itself tend to be special purpose in nature for this type of layout where the nature of the product allows it, gravity chutes can be effectively utilized.

Conveyors to fit all different applications of size, shape and weight of part are commercially available. These conveyors may fit in product type of layout well. Many processes such as cleaning, painting drying and weighing as in case of certain liquid materials may take place while the material moves.

The special purpose nature of product layout design often makes it worthwhile to design special handling equipment which is integrated with processing so completely that the entire line works as a single integrated machine.

Material Handling in Functional Layout:

The basic requirement of material handling techniques used in functional layout is flexibility i.e. flexibility of size, weight and shape of load and flexibility of path. The types of material handling equipment which suit this requirement are in general, mobile trucks, tractor, trains, fork, lift, trucks and cranes etc.

The important characteristic of an efficient material handling equipment is the time required to pick up or set down load. Therefore, quick pickup system have developed around skids and pallets. Material can be directly loaded on a skid or pallet by the worker as his operation on the part is completed.

In this way without further handling, a skid truck, fork lift truck or pallet truck can pick up the entire load very quickly and move it to its goal. Therefore one important consideration in developing the details of layout is to allow easy access to all operations to be performed by efficient material handling equipment. Supplementary cranes are often needed at the work stations to handle heavy jobs to and from the machines. Overhead cranes are utilized to transport and position large heavy jobs within a fixed area.

It is common to find the material handling equipment unutilized in functional type of layout and at the same time complaints are that material can never be moved when wanted. Therefore much material handling equipment is required in this type of layout. This has always been a different question in functional type of layout in view of the demand for transporting capacity is on random basis.

2.4 FACTORS AFFECTING THE SELECTION OF MATERIAL HANDLING EQUIPMENT

The following factors are to be taken into account while selecting material handling equipment.

1. *Properties of the material*

Whether it is solid, liquid or gas, and in what size, shape and weight it is to be moved, are important considerations and can already lead to a preliminary elimination from the range of available equipment under review. Similarly, if a material is fragile, corrosive or toxic this will imply that certain handling methods and containers will be preferable to others.

2. *Layout and characteristics of the building*

Another restricting factor is the availability of space for handling. Low-level ceiling may preclude the use of hoists or cranes, and the presence of supporting columns in awkward places can limit the size of the material-handling equipment. If the building is multi-storied, chutes or ramps for industrial trucks may be used. Layout itself will indicate the type of production operation (continuous, intermittent, fixed position or group) and can indicate some items of equipment that will be more suitable than others. Floor capacity also helps in selecting the best material handling equipment.

3. *Production flow*

If the flow is fairly constant between two fixed positions that are not likely to change, fixed equipment such as conveyors or chutes can be successfully used. If, on the other hand, the flow is not constant and the direction changes occasionally from one point to another because several products are being produced simultaneously, moving equipment such as trucks would be preferable.

4. *Cost consideration*

This is one of the most important considerations. The above factors can help to narrow the range of suitable equipment, while costing can help in taking a final decision. Several cost elements need to be taken into consideration when comparisons are made between various items of equipment that are all capable of handling the same load. Initial investment and operating and maintenance costs are the major cost to be considered. By calculating and comparing the total cost for each of the items of equipment under consideration, a more rational decision can be reached on the most appropriate choice.

5. *Nature of operations*

Selection of equipment also depends on nature of operations like whether handling is temporary or permanent, whether the flow is continuous or intermittent and material flow pattern-vertical or horizontal.

6. Engineering factors

Selection of equipment also depends on engineering factors like door and ceiling dimensions, floor space, floor conditions and structural strength.

7. Equipment reliability

Reliability of the equipment and supplier reputation and the after sale service also plays an important role in selecting material handling equipments.

2.5 PREVENTIVE AND BREAK DOWN MAINTENANCE

Meaning of Preventive Maintenance:

Preventive maintenance attempts to prevent any probable failures/breakdowns resulting in production stoppages. It is said that Preventive maintenance is a stitch in time that saves time. So it follows a slogan that “prevention is better than cure”.

Preventive maintenance refers to maintenance action performed to keep or retain a machine/equipment or asset in a satisfactory operating condition through periodic inspections, lubrication, calibration, replacements and overhauls.

Preventive Maintenance Involves:

- (i) Periodic inspection of equipment/machinery to uncover condition that lead to production breakdown and harmful depreciation. Upkeeps of plant machinery to correct such conditions while they are still in a minor stage.
- (ii) The key to all good preventive maintenance programmes, however is inspection.
- (iii) Regular cleaning, greasing and oiling of moving parts.
- (iv) Replacement of worn out parts before they fail to operate,
- (v) Periodic overhauling of the entire machine.
- (vi) Machines or equipment's which are liable to sudden failures should be installed in duplicate e.g. motors, pumps, transformers and compressors etc.

Features of Preventive Maintenance:

A well-conceived preventive maintenance programme should possess the following features:

- (1) Proper identification of all items to be included in the maintenance programme.
- (2) Adequate records covering, volume of work, associated costs etc.
- (3) Inspection with a definite schedule with standing order on specific assignments.
- (4) Use of checklists by inspectors.
- (5) An inspection frequency schedule
- (6) A crew of well qualified inspectors with competency of simple repairs, as and when small trouble is noticed.
- (7) Administrative procedures which provide necessary fulfilment as well as follow up on programme.

Objectives of Preventive Maintenance:

- (1) To minimize the possibility of unanticipated production interruption or major breakdown by uncovering any condition which may lead to it?
- (2) To make plant, equipment and machinery always available and ready for use.
- (3) To maintain the value of equipment and machinery by periodic inspections, repairs, overhauls etc.
- (4) To reduce the work content of maintenance jobs.
- (5) To maintain the optimum productive efficiency of the plant equipment and machinery.
- (6) To maintain the operational accuracy of the plant equipment.
- (7) To achieve maximum production at minimum repair cost.
- (8) To ensure safety of life and limbs of the workmen along with plant equipment and machines etc.
- (9) To maintain the operational ability of the plant as a whole.

Procedure of Preventive Maintenance:

There is no readymade, on the shelf, preventive maintenance procedure for any industry or enterprise involved in manufacturing activities. In view of the fact that all industries differ in size, location, layout, construction, resources, machinery and its age so as to suit the requirements of an individual industrial plant, the preventive maintenance programmes are specifically framed.

The essential requirements for a sound preventive maintenance can be listed as follows:

(1) Proper identification of machines/equipment's and tools:

Every item must be uniquely identified by a prominent serial/identity number.

(2) Adequate past records must be available for all equipment's being utilized. It should furnish complete details regarding previous maintenance operations/activities.

(3) Breakdown/Failures Data:

Sufficient breakdown information regarding criticality and frequency of failures must be available for all machines. This would be needed for the purpose of failure identification, failure diagnostics, analysis as well as final rectification.

(4) Secondary data:

In fact, it is a sort of experienced data for similar equipment being utilized.

(5) Manufacturer's utilization recommendations:

Regarding the use of a particular machine i.e. how to utilize and provide P.M.

(6) Service manuals, instruction and maintenance sheets.

(7) Consumables and replacable parts/components should be available as and when needed.

- (8) Availability of requisite skilled manpower may be engineers, inspectors and technicians.
- (9) Availability/provision of test rigs/equipment's i.e., test rigs, sensors etc.
- (10) Clear instructions with a check list regarding preventive and corrective measures must be available to ensure proper functioning of the system.

(11) Users feedback and cooperation:

The user of the equipment/machine must provide feedback to the manufacturer regarding actual functioning of the equipment.

(12) Management Support:

For establishing a preventive maintenance system, the commitment of top management is very essential for the implementation of preventive maintenance policy of the organization.

Advantages of Preventive Maintenance:

- (1) Reduction in breakdown time and associated breakdown elements.
- (2) Reduces the odd time repairs and over time to the maintenance staff.
- (3) Fewer large scale and repetitive repairs.
- (4) Less member of standby equipment and spare parts required.
- (5) Greater safety to work force/employees due to reduced breakdowns.
- (6) Increased life of equipment and machines.
- (7) The work load of the maintenance staff can be properly planned.
- (8) It improves the availability of facilities.
- (9) Optimum production efficiency can be achieved by employing preventive maintenance.
- (10) Maintenance and repair cost reduce heavily.
- (11) It improves the quality of product and reduces rejections.
- (12) Production cost goes down by adopting RM.
- (13) Regular planned servicing and adjustment maintains and provides a high level of plant output, better equipment performance and better product quality.
- (14) Healthy, hygienic, safe and an accident free work environment can be achieved with the application of scientific preventive maintenance. This would promote industrial relations since workers do not loose any type of incentive due to breakdowns or accidents.
- (15) Reduction in inventory of spare parts.

Limitations of Preventive Maintenance:

- (1) When the cost of failure prevention is always greater than cost of failure rectification the process of P.M is very costly e.g., batch production-bridge construction.
- (2) The type of maintenance requires extra facilities and lead to under/poor utilization of basic facilities for RM.
- (3) For small scale manufacturing units which are mainly undertaking job and batch production, the P.M system is not suited and economically justified.

Breakdown Maintenance (Corrective Maintenance):

Breakdown Maintenance is the restoration of a facility to a condition substantially equivalent to its original or designed capacity and efficiency by replacement of parts, overhaul or reprocessing of materials after deterioration. Breakdown Maintenance is the most expensive since in addition to the cost of repair, there is often a loss of business or production as a result. This type of maintenance results when a piece of equipment or a structure component completely fails. There is essentially no leeway in the scheduling of this maintenance. The equipment must be repaired immediately or the operation will be shut down.

Definition of Corrective Maintenance:

- (i) A maintenance work carried out to restore the equipment/machine to a satisfactory condition after the failure has occurred.
- (ii) It may be defined as a specific maintenance operation which is well organized and performed with some prior planning, forethoughts and control activity.

Actually corrective maintenance is a special type of maintenance activity carried out to restore an item/machine when it has failed to meet an acceptable condition. Further it is basically a rectification process which is always adopted after the occurrence of a breakdown.

It may be executed in the form of:

- (a) Repair may be minor or major.
- (b) Replacement may be partial or total.

In fact, down time due to breakdown may consist not only of time taken to complete the repair work but also delays caused by lack of resources or information.

Repair time is a function of management technique so, engineering techniques and maintainability. Corrective or planned maintenance is required not only when the asset/machine item fails but also when indicated by condition based criteria.

Characteristics of Corrective Maintenance:

- (1) A corrective maintenance activity is generally planned.
- (2) A planned or unplanned corrective maintenance operation depends on the nature of breakdown and type of equipment/machine.
- (3) The maintenance work is taken up after the occurrence of a breakdown and with some permissible time lag.
- (4) Breakdown maintenance should not be very serious in nature as far as production losses, down time, loss of human life etc. are concerned.
- (5) Breakdown of individual equipment should not effect considerably the overall production loss.

In general breakdown maintenances are predictable and expected failures and hence they may be rectified over a long period of time without any time constraints.

Objectives of Corrective Maintenance:

- (1) To get equipment/machine back into operation as quickly as possible in order to minimize the interruption to production. These objectives are directly related with production capacity, costs of production, product quality and consumer satisfaction.
- (2) To control the cost of the operation of repair shops.
- (3) To keep the cost of repair crew under control, including regular and overtime of labour costs.
- (4) To control the investment in replacement of parts/components that are used/required when machines are repaired.
- (5) To control the investment required for back up machines. These replace manufacturing machines are needed until the repairs are completed.
- (6) To perform the appropriate amount of repairs at each malfunction of the asset/equipment.
- (7) To restore an asset in working order.
- (8) To maintain the operation availability of the plant and infrastructural facilities
- (9) To avoid any sudden and heavy failure (breakdown) in future.

Typical Causes of Equipment Breakdown:

The causes of equipment breakdown may be as follows:

- (i) Failure to replace worn out components/parts.
- (ii) Lack of lubrication.
- (iii) Neglected cooling arrangement/system.
- (iv) Indifference towards minor faults.
- (v) External factors such as wrong fuel, too low or too high line voltage etc.
- (vi) Indifference towards equipment vibrations, unusual sounds coming out of the rotating parts and equipment getting too much heated up.



Advantages of Corrective Maintenance:

- (1) Emergency maintenance requirements are reduced.
- (2) Heavy down time losses are reduced.
- (3) Plant availability is increased.
- (4) Results in better utilization of plant facilities.
- (6) Provides sufficient information concerning the maintenance replacement and repair.

Limitations of Breakdown Maintenance:

- (1) Breakdowns generally occur at inappropriate times. It may lead to a poor hurried maintenance and excessive delays in production schedules.
- (2) It involves prolonged down time due to non-availability of requisite manpower and spare parts, they may lead to overtime practice also.
- (3) It becomes impossible to plan workload and distribution of maintenance workforce for balanced and proper attention of all equipment's.

- (4) Reduction in production output.
- (5) There are increased chances of accidents and less safety for workforce.
- (6) It leads to faster plant deterioration.
- (7) Corrective maintenance cannot be employed for those industrial plants/enterprises which are regulated by statutory provisions for example boilers and cranes.
- (8) The maintenance of product quality is difficult.
- (9) Loss of direct profits.

2.6 REPLACEMENT OF EQUIPMENTS

The replacement of productive equipment is important strategic decisions faced by both manufacturing and service firms because purchasing a new piece of equipment often involves more cost and can affect the productivity and effectiveness of the firm. Currently, this issue is highlighted in fast changing technologies and good equipment purchase can soon become obsolete. Under these situations, the driving motivation to take replacement decisions is likely to be technological outmodedness instead of physical deterioration, of the existing equipment. This situation is typical of microcomputers, computerized numerically controlled machines, and other electronics technologies.

The replacement problems are associated with the issues that develops when the performance of an item decreases, failure or breakdown occurs. The decline in performance or breakdown may be gradual or sometimes sudden. There is a need for replacement of items when;

1. The existing item or system has become inefficient or require more maintenance.
2. The existing equipment has failed due to accident or otherwise and does not work at all.
3. The existing equipment is expected to fail shortly.
4. The existing equipment has become obsolete due to the availability of equipment with latest technology and better design. The solution to replacement problem is to device best policy that determines the time at which the replacement is most economical instead of continuing at an increased maintenance cost.

The Main objective of replacement policy is to guide the organization in many situations so that it can take correct decision such in a situations when waiting for complete failure of item or to replace earlier at the expense of higher cost of the item, whether to replace the underperforming equipment with the similar kind of item or by different kind of item. The problem of replacement occurs in the case of both men and machines.

There are numerous reasons for equipment replacement. The first reason is the equipment is depleted of function. Second reason for replacing equipment is if the equipment becomes obsolete. For example, older computers are much slower and have fewer features than their modern counterparts. In addition, older computers are harder to maintain because replacement parts and qualified technicians are much harder to find. Another reason for replacement is deterioration due to aging. Equipment is inadequate and does not meet needs, increased demand. Then it is replaced with a larger asset.

There are many types of failure in equipment:

1. **Gradual Failure:** In this, the failure mechanism is progressive. As an equipment becomes old, its performance deteriorates. This results in increased operating cost, decreased productivity of the item and decrease in resale value of item.
2. **Sudden Failure:** This type of failure occurs in equipment that do not deteriorate gradually with age but which fail suddenly after some period of service. The time period between installations and failure will not be constant for any particular equipment. However the failure pattern will follow certain frequency distribution that may be progressive, retrogressive or random in nature.
3. **Progressive failure:** progressive failure occurs when probability of failure increases with the age of an item.
4. **Retrogressive failure:** Certain items will have more probability of failure in the early years of their life and with the increase in the life of an item the chances of failure become less. That is, the ability of the item to survive in the initial years of life increases its expected life.
5. **Random failure:** Random failure occurs when continuous probability of failure is related with equipment that fails because of random causes such as physical shocks that are independent of age. In the case of random failure, virtually all items fail before aging has any effect.

The replacement situations are categorized into the following four types:

1. Replacement of capital equipment whose performance decreases with time.
2. **Group replacement items that fail completely:** Some system usually composed of a large number of low cost items that are prone to failure with age such as failure of a resistor in television, radio, computer etc. In some cases the failure of a component may cause the complete failure of the system. In such cases, the cost of overall failure will be quite higher than the cost of component itself. In such situations, two types of replacement procedures must be considered. First is Individual replacement. In this policy, an item is replaced immediately after its failure. Secondly, Group replacement in which, decision is about the age when all the items should be replaced, irrespective of whether the items have failed or not. In this policy, the items that fail before the optimal time, will be replaced individually.
3. Problem of mortality and staffing.
4. Miscellaneous problems.

Reasons for Replacement of Equipment's:

Equipment are generally considered for replacement for the following reasons:

(i) Deterioration:

It is the decline in performance due to wear and tear or misalignment indicated by;

- (i) Increase in maintenance costs.
- (ii) Reduction in product quality and rate of production.
- (iii) Increase in labour costs, and
- (iv) Loss of operating time due to breakdowns.

(ii) Obsolescence:

Technology is progressing fast, newer and better equipment are being developed and produced every year.

The equipment gets obsolete due to advancement in technology and the unwarranted manufacturing costs arising from such obsolete equipment will:

- (i) Reduce profits.
- (ii) Impair competition.
- (iii) Cause loss in value of machinery.

(iii) Inadequacy:

When the existing equipment becomes inadequate to meet the demand or it is not able to increase the production rate to desired level, the question of replacement arises.

(iv) Working Conditions:

It may be thought of replacing the old equipment and machinery which creates unpleasantness i.e. give rise to unsafe conditions for workers and leads to accidents, making the environment noisy and smoky etc.

(v) Economy:

The existing units/equipment have outlived their effective life and it is not economical to continue with them.

Factors Necessary for Replacement of Equipment:

The factors which necessitate the replacement of machinery and equipment can be classified as:

- (i) Technical Factors.
- (ii) Financial or Cost Factors.
- (iii) Tangible Factors.

(i) Technical Factors:

- (i) Whether the present equipment has become obsolete due to technological developments,
- (ii) If the present equipment is inadequate in meeting increased product demand.
- (iii) Whether the present equipment has deteriorated due to wear and tear. It may be indicated by increase in maintenance costs, reduction in product quality, rate of output, and increase in labour cost and down time etc.
- (iv) Reduced safety as compared to new machine available/developed.
- (v) Can the present equipment provide desired surface finish?
- (vi) If the present equipment is polluting or spoiling working condition of the industry.
- (vii) Possibility of performing additional operations by new machine.

- (viii) Does the present equipment make noise and vibrations and thus causing diversion of the workers.
- (ix) How often the present equipment requires maintenance and repairs.

(ii) Financial/Cost Factors:

- (i) High repair and maintenance cost of the existing equipment/machinery).
- (ii) Possibility of combining some operations and resulting increase in productivity by challenger (new machine).
- (iii) The initial cost of challenger.
- (iv) Salvage value of existing equipment and challenger at the end of its useful life.
- (v) Improvement in productivity and quality by use of challenger.
- (vi) Saving in space by use of new machine.
- (vii) Reduction in scrap and waste by use of new machine.
- (viii) Down time cost of present machine.
- (ix) Reduction in cost of jigs and fixtures by using challenger.
- (x) Effect on consumption of power by replacing the existing machine by new machine.

(iii) Tangible Factors:

These factors involve sociological and humanitarian considerations with far reaching effects:

- (i) Like replacing the existing machine which causes unpleasantness (may be noise and smoke pollution) and unsafe working conditions leading to accidents.
- (ii) Replacement may cause displacement of workers.

At the time of replacement a well-designed replacement policy should be adopted, rather than considering only the factors pertaining to the particular equipment involved, should compare thoroughly all the existing equipment with its possible replacement.

For the purpose of sound economic comparison all factors should be converted into cost and possible increase in revenue. Break even analysis can be utilized for the purpose of taking replacement decision or selection of investment alternatives.

Problems in Replacement of Equipment:

The problem of equipment replacement is a routine phenomenon of industrial enterprises. Normally, it is experienced in systems where machines, individuals or the capital assets are the main job performing units. It is the common phenomena that performance or efficiency of an item in a system deteriorates with the passage of time.

The remedy is either to adopt maintenance measures to resort to the requisite level of performance or to replace the item with some new items. Thus it is required to formulate a most economic replacement policy which is in the best interest of the enterprise or system.

The various types of replacement problems can be expressed broadly in the following situations:

(i) Replacement of Equipment/Machine/Item which Deteriorate with Time:

This situation arises when the efficiency is measured as the discounted value of all future costs associated with each alternative. The simplest replacement model in such cases is one where the deterioration rate is predictable in terms of increasing maintenance costs and decreasing salvage value with time.

The maintenance cost of the machine/assets/equipment always increases with time and a stage comes when maintenance cost is so high that it is more economical to replace it by new one.

In such cases the decision may not be to re-palace the equipment if the next year maintenance cost is less than the average cost of the previous year and replace the equipment if the next year's maintenance expenses is more than the average cost of the previous year.

There are two methods to find the appropriate solution in this case i.e.:

(i) Annual cost method.

(ii) Present worth method.

(ii) Replacements of Items that Fail Completely are Expensive to be Replaced:

In general it is a common characteristic that the probability of failure of any item in a system increases with the period of use or passage of time. A machine or equipment consisting of a number of parts/items may be considered as a system.

The system may be such that the whole system may result in breakdown with the failure of any item. This break down implies loss in production, idle labour, idle inventory and other units of the system.

It is possible that the nature of item which requires replacement may be such that immediate replacement is not available or possible. Thus there is necessity of formulating some appropriate replacement policy in such cases.

There are two possible solutions:

(a) Individual Replacement Policy:

Whenever any item fails, it should be immediately replaced.

(b) Group Replacement Policy:

All the items/parts are replaced after a certain period of time T inspite of these being in working condition, with a provision that if any item fails before this time T it can be replaced immediately. This approach decreases the probability of breakdown in the system. This approach is essential if the sudden breakdown of the equipment/machine is hazardous.

Such policy requires two fold considerations namely:

- (i) The rate of individual replacement during the period.
- (ii) The total cost incurred on individual and group replacements during the selected/chosen period of replacement.

The period for which the total cost is minimum is considered as optimal.

The following information is required to take decision in this procedure in such cases:

- (A) Probability of failure at different periods of time
- (B) Loss incurred due to these failures
- (C) Cost of individual replacement and
- (D) Cost of group replacement.

2.7 METHODS OF PROVIDING DEPRECIATION

Meaning of Depreciation:

“The term Depreciations means a fall in the value of an asset with use and passage of time for the fixed assets like plant, machinery, building, furniture fixtures etc. there is a decline in the value due to use and it is termed as Depreciation.”

Most of the fixed assets are worn out while in use over a period of time. This wear and tear is bound to occur but it can be minimized up-to some extent by proper care and maintenance.

The efficiency of these assets also reduces with the passage of time and at one time it becomes uneconomical to be used further and requires replacement conventionally, this money required for replacement is charged as depreciation.

The term depreciation means a fall in the value of an asset i.e., with the use over a period of time, there is a decline in the fixed assets. These fixed assets include plant and machinery, building furniture fixtures and tools accessories etc. Their value falls due to use. From the accounting point of view, the concept of depreciation is that of sluggish or amortized cost.

Some of the important reasons for decline in the value of the assets may be:

- (i) Lapse of time
- (ii) Wear and tear due to use over a period of time
- (iii) Lack of proper repair and maintenance
- (iv) Mishandling and accidents and
- (v) Introduction of new improved equipment in the market (obsolescence).

Purpose of Calculating Depreciation:

In the light of above discussion some money must be set aside yearly from the profits, so that when equipment goes uneconomical, it can be replaced by the new machines. Therefore, we want a system of accounting in which we distribute the initial cost of the equipment machines or assets plus installation charges less the salvage value (if any) over the estimated useful life of the assets.

In the absence of provision for any depreciation, the assets may be worn out in parts every year and thus it may swallow the profits unnecessarily. Thus for this purpose depreciation accounts for the complete plant or individual machines/equipment is opened in the organization's account books known as "Depreciation Fund".

In this way the depreciation amount is distributed scientifically over the useful life of the asset and is deducted yearly from the profits chargeable to overheads. The money so collected in Depreciation Fund is utilized for replacement of the assets at the end of useful life.

Thus the main objectives providing depreciation accounting may be as follows:

- (i) To keep annual depreciation of various assets in operation.
- (ii) To provide reserve fund for the proper repair and maintenance of plant and equipment.
- (iii) To provide depreciation data for policy formulation.
- (iv) To provide depreciation cost to get
 - (a) Estimated life of the equipment/asset in years.
 - (b) Estimated scrap value of the asset.
 - (c) Cost the equipment required for replacement.

Types of Depreciation:

Depreciation may be classified under the following types:

- (1) Physical depreciation.
- (2) Functional depreciation.
- (3) Accidental depreciation.

(1) Physical Depreciation:

Depreciation resulting from physical impairment of an equipment or assets is called physical depreciation. It manifests itself in such clear and definite ways as the wearing of metal particles from the bearing surfaces, corrosion of tubes in boilers or heat exchangers, chemical decomposition etc.

The primary causes of physical depreciation may be due to three main classes of causes as follows:

Depreciation Due to Wear and Tear:

Wearing of the equipment/machine or asset due to abrasion, shock, vibration or impact etc., which are caused due to use over a period of time. So the cost of replacement due to this cause is the value of depreciation due to wear and tear.

Depreciation Due to “Physical Decay”:

Wearing of assets/items by such ways which are independent of use e.g. rotting of wood, corrosion of pipes, decay of electric cables, chemicals etc. This reduction is depreciation due to physical decay.

Depreciation Due to Deferred Maintenance and Neglect:

If the instructions of maintenance recommended by the manufacturer are not followed properly by the user the value of asset may be reduced and depreciation in value because of this reduction is known as depreciation due to deferred maintenance and neglect.

(2) Functional Depreciation:

This type of depreciation results not from a deterioration in the assets ability to perform duties of its intended purpose but from a change in demand for the services it can render.

Such a change may occur because of the fact that the asset is no more profitable to use. Depreciation resulting from a change in the need for the service of an asset/equipment may be result of technological developments these developments cause obsolescence of the existing assets.

The two reasons are:

- (i) Obsolescence owing to the development of superior units and existing unit cannot serve the desired goal (service).
- (ii) Inability to meet the desired demand so change of existing asset by superior unit available.

(3) Accidental Depreciation:

Accidents incur a very rapid loss of value of the equipment/asset and are unpredictable. Nowadays the practice is to insure against accidental losses. These insurable losses may be of huge magnitude and are not treated as depreciation. However the premium paid may be charged from depreciation fund.

Methods for Calculating Depreciation

1. Straight Line Method:

This method assumes that depreciation is a function of time rather than use. This method is based on the assumption that each accounting period receives same a benefits from using the assets. It allocates an equal amount of depreciation in each accounting periods of the service life of the assets. Therefore, it is called Straight Line Method.

The formula for calculating depreciation charge for each accounting period is:

$$\text{Annual Depreciation} = \frac{\text{Acquisition cost} - \text{Estimated scrap value}}{\text{Estimated Life in Years}}$$

Advantages:

- (i) It is simple in use.
- (ii) It realistically matches cost and revenue and determine income of each period easily.
- (iii) There is no change either in the rate or the amount of depreciation over the useful life of the assets. Such a procedure provides for improved comparability.

Disadvantages:

- (i) It ignores the cost of capital.
- (ii) It is based on the wrong assumption of equal utility of the assets during the useful life.
- (iii) It is also wrong to consider depreciation as a function of time rather than use.
- (iv) The maintenance of asset is generally costly in the later years with the result that deductions from the revenue would be greater in later years than in the earlier years.

The journal entry for recording depreciation is

Depreciation A/c Dr.

To Asset A/c

Illustration 1:

ABC Ltd. acquires a machine on 1st July, 2007 at a cost of Rs. 2, 80,000 and spent Rs. 20,000 on its installation. The firm writes off depreciation at 10% of the original cost every year. The books are closed on 31st December every year. Prepare machinery A/c for 3 years.

Solution:			Machinery A/c		
Date	Particulars	(₹)	Date	Particulars	(₹)
2006 July 1	To Bank	2,80,000	2006 Dec. 31	By Depreciation A/c	15,000
	To Bank (installation Exps.)	20,000		By Balance c/d	2,85,000
		3,00,000			3,00,000
2007 Jan. 1	To balance b/d	2,85,000	2007 Dec. 31	By Depreciation A/c	30,000
		2,85,000		By balance c/d	2,55,000
2008 Jan. 1	To balance b/d	2,55,000	2008 Dec. 31	By Depreciation A/c	30,000
		2,55,000		By balance c/d	2,25,000
					2,55,000

Sale of Asset:

If the asset is disposed of in the middle of the year, the amount realized should be credited to the asset a/c for the time it has been in use. Any balance left in the account of asset will be profit or loss and should be transferred to P&L A/c.

Illustration 2:

On 1st Jan. 2009, a company purchased a machine costing Rs. 5, 00,000. Its estimated working life is 20 years at the end of which it will fetch Rs. 20,000. Additions are made on 1 January, 2010 and 1 July, 2011 to the value of Rs. 80,000 (scrap value Rs. 4000) and Rs. 40,000 (scrap value Rs. 2,000) respectively. The life of both the new machines is 20 years. Show machine a/c for first four years.

Solution:			Machine A/c		
2009	Particulars	(₹)	2009	Particulars	(₹)
Jan. 1	To Bank	5,00,000	Dec. 31	By Depreciation	24,000
			Dec. 31	By Balance c/d	4,76,000
		5,00,000			5,00,000
2010			2010		
Jan. 1	To balance b/d	4,76,000	Dec. 31	By Depreciation	27,800
Jan. 1	To Bank	80,000	Dec. 31	By balance c/d	5,28,200
		5,56,000			5,56,000
2011			2011		
Jan. 1	To balance b/d	5,28,200	Dec. 1	By Depreciation- (24,000 + 3800 + 950)	28,750
July 1	To Bank	40,000	Dec. 1	By Balance c/d	5,39,450
		5,68,200			5,68,200

2009	Particulars	(₹)	2009	Particulars	(₹)
2012			2012		
Jan. 1	To Bal b/d	5,39,450	Dec. 31	By Depreciation (24,000 + 3800 + 1900)	29,700
			Dec. 31	By Balance c/d	5,09,750
		5,39,450			5,39,450

Note:

In case, no method of depreciation has been stated in an examination problem, it is appropriate to use straight line method.

2. Diminishing Balance Method:

Under this method, a fixed percentage is applied to book value of the assets (cost of assets). In other words, the depreciation is calculated on the reducing balance (assets cost-depreciation) and not on the original cost. The procedure is that depreciation calculated is deducted from the cost of assets and balance known as the written down value. The written down value at the end of the estimated useful life of the assets should equal the estimated salvage or scrap value.

The formula for determining the rate of depreciation in diminishing balance method is as follows:

$$\text{Rate of depreciation} = 1 - \sqrt[n]{s/c}$$

Advantages:

1. It is easy to use.
2. There is same weight-age, in totality, on profit and loss A/c, of depreciation.

3. It facilitates replacement of fixed assets as it makes more funds available at an early stage.
4. This method is permissible under the Income Tax Act.
5. The higher depreciation is charged in the earlier years when the machine is most efficient compared to later years.

Disadvantages:

1. Under this method, value of asset can never be zero.
2. It is difficult to calculate proper rate of depreciation.
3. There is no provision of interest on capital invested in use of assets.

Illustration 3:

Y Ltd. Co. purchased a machine costing Rs. 3, 00,000 on 1st January, 2007. The depreciation is to be charged at 20% p.a. on Diminishing Balance method.

Write up Machinery A/c for first four years.

Solution:			Machinery A/c		
2007	Particulars	(₹)	2007	Particulars	(₹)
Jan. 1	To Bank	3,00,000	Dec. 31	By Depreciation	60,000
			Dec. 31	By Balance c/d	2,40,000
		3,00,000			3,00,000
2008			2008		
Jan. 1	To balance	2,40,000	Dec. 31	By Depreciation	48,000
			Dec. 31	By balance c/d	1,92,000
		2,40,000			2,40,000
2009			2009		
Jan. 1	To balance b/d	1,92,000	Dec. 31	By Depreciation	38,400
			Dec. 31	By balance c/d	1,53,600
		1,92,000			1,92,000
2010			2010		
Jan. 1	To balance b/d	1,53,600	Dec. 31	By Depreciation	30,720
			Dec. 31	By Bal. c/d	1,22,880
		1,53,600			1,53,600

Illustration 4:

X Ltd. bought a machinery for Rs. 30,000 on 1st April, 2007. One more machine was purchased on Oct. 1st, 2007 costing Rs. 20,000. On July 1st, 2008, a new machinery for Rs. 10,000 was added to the existing machinery. On 1st January, 2009, one third of the machinery which was installed on April 1st, 2007 was sold for Rs. 3,000. Show the machinery A/c in the books of X Ltd. The rate of depreciation is 10% on reducing balance method.

Solution:**Machinery A/c**

Date	Particulars	(₹)	Date	Particulars	(₹)
2007			2007		
April 1	To Bank	30,000	Dec. 31	By Depreciation	2,750
Oct. 1	To Bank	20,000		By Balance c/d	47,250
		50,000			50,000
2008			2009		
Jan. 1	To Balance b/d	47,250	Dec. 31	By Depreciation	5,225
July 1	To Bank	10,000	Dec. 31	By Balance c/d	52,025
		57,250			57,250

Date	Particulars	(₹)	Date	Particulars	(₹)
2009			2009		
Jan. 1	To balance b/d	52,025	Jan. 1	By Depreciation	3,000
			Jan. 1	By P&L A/c (loss)	5,325
			Dec. 31	By Depreciation	4,370
			Dec. 31	By balance c/d	39,330
		52,025			52,025

Illustration 5:

A machine is purchased for Rs. 3, 00,000 and its estimated useful life is 3 years with scrap value Rs. 30,000. It is decided to depreciate the machine by diminishing balance method. Find out the rate of depreciation p.a.

Solution:

$$S = ₹ 30,000$$

$$C = ₹ 3,00,000$$

$$n = 3$$

$$r = 1 - n\sqrt[n]{\frac{S}{C}} = 1 - 3\sqrt[3]{\frac{30,000}{3,00,000}}$$

$$= 1 - 3\sqrt[3]{\frac{1}{10}}$$

$$= \frac{1 - (1)^{1/3}}{10}$$

$$= 1.0 - .464$$

$$= .536$$

$$= 53.6\%$$

3. Sum of the Digits Method

Under this method, amount of the depreciation to be written off each year is calculated by the following formula:

= Remaining Life of the Asset (including the Current Year) / Sum of all the Digits of the Life of the Asset in Years x Cost of the Asset

Suppose the life of an asset costing Rs.50,000 is 10 years. The sum of all the digits from 1 to 10 comes to 55 i.e., 10+9+8+7+6+5+4+3+2+1 = 55. The depreciation to be provided in the first year will be:

$$= 10 / 55 \times 50,000 \text{ or Rs.9,091}$$

In the second year, it will be:

$$9 / 55 \times 50,000 \text{ or Rs.8,181}$$

Illustration 6:

Manoj Ltd. purchased a Computer Costing Rs. 80,000 on January 2000. The expected life is 5 years and scrap value Rs. 5000. Find the annual depreciation for 5 years by sum-of- years-digit methods.

Solution:		
$C = 80,000$	$S = 5000$	$N = 5$
Annual Depreciation = $\frac{N(C - S)}{\text{Sum}}$		
Where Sum = $\frac{N(N + 1)}{2}$		
Sum = $\frac{5(5 + 1)}{2} = 15$		
Year	Calculation	Depreciation
1.	$\frac{5(75,000)}{15}$	₹ 25,000
2.	$\frac{4(75,000)}{15}$	₹ 20,000
3.	$\frac{3(75,000)}{15}$	₹ 15,000
4.	$\frac{2(75,000)}{15}$	₹ 10,000
5.	$\frac{1(75,000)}{15}$	₹ 5,000

Illustration 7:

A bus was purchased by a Transport Company for Rs. 10, 00,000 on 1st January, 2000. The scrap value is estimated at the end of 5 years Rs. 1, 00,000. The rate of depreciation is 20% p.a. Calculate the annual Depreciation by Double Declining method.

Solution:

The rate of Depreciation = $20 \times 2 = 40\%$ p.a.

End of Year	Asset Cost at Beginning of Year	Amount of Depreciation
1	1,00,000	4,00,000
2	6,00,000	2,40,000
3	3,60,000	1,44,000
4	2,16,000	86,400
5	1,29,600	51,840

4. Unit of Production Method:

Under this method depreciation expense is allocated in proportion to the degree of asset used for production because estimation of useful life is done in terms of units of output or services hour and not a calendar time period. A machine can produce a limited number of quantity or run for a limited hour. As a result of this limitation, the depreciation is related to usage and not to time.

Under this method, the following is calculated:

Depreciation Per annum = Dep. expense Per unit x No. of units produced during the year

Or

Dep. expense per hour x No. of hours used during the year

Dep. per unit = $\text{Cost} - \text{Scrap value} / \text{Total estimated unit of output or service hrs.}$

Illustration 8:

X Ltd., bought a machine for Rs. 4, 00,000 expected to run 72,000 hrs. Its salvage value is estimated at Rs. 40,000. Calculate depreciation for an accounting year in which it was used for 20,000 hrs.

Solution:

Dep. per hour = $4, 00,000 - 40,000 / 72,000$

= $3, 60,000 / 72,000 = \text{Rs. } 5$

Dep. Amount for Accounting period

= $2, 0000 \times 5 = \text{Rs. } 1, 00,000$

5. Annuity Method:

The main disadvantages of the different methods are that they do not consider the interest on the capital invested on fixed assets. Annuity method deals with the effect of cost of capital in depreciation calculation. It makes treatment more explicit by showing the interest payment in the P&L A/c.

Under this system the capital sum is assumed to earn a certain rate of interest. The assets are, therefore, charged with interest along with actual payment; interest is calculated on the debit balance of the assets on the commencement of the year.

The following journal entry is made:

Assets A/c Dr.

To Interest A/c

At the same time, fixed amount is charged as a depreciation expense so that it eliminates the asset or brings down its salvage value, as the case may be. The important point to be noted is that the amount of depreciation to be charged every year must be calculated as to reduce the assets together with interest accumulated thereon to its salvage value at the end of the useful life of the assets.

Illustration 9:

X Ltd. acquires a lease costing Rs. 2, 00,000 on April 1st, 1997 for a term of 4 years. You find from annuity tables that in order to write off lease on the annuity method at 6% p.a. interest, the amount to be written off annually works out to be Rs. 2,88,591 for every rupee. Prepare lease A/c for 4 years. Books are closed on 31st March every year.

Solution:			Lease A/c		
Date	Particulars	Amount (₹)	Date	Particulars	Amount (₹)
1997 April 1	To Bank	2,00,000	1998 March 31	By Dep.	57,718
	To Interest (6% of ₹ 20,000)	12,000		By Bal. c/d	1,54,282
		2,12,000			2,12,000
1998 April 1	To Bal. b/d	1,54,282	1999 March 31	By Dep.	57,718
	To Interest (6% of ₹ 1,54,282)	9,527		By Bal. c/d	1,05,821
		1,63,539			1,63,539

Date	Particulars	Amount (₹)	Date	Particulars	Amount (₹)
1999 April 1	To Bal. b/d	1,05,821	2000 Mar. 31	By Dep.	57,718
	To Interest	6,349		By Bal. c/d	54,452
	(6% of ₹ 1,05,821)	1,12,170			1,12,170
2000 April 1	To Balance b/d	54,452	2000 Mar. 31	By depreciation	57,718
	To Interest (Balancing fig.)	3,266			57,718
		57,718			

Working:

Depreciation = Rs. 2, 00,000 x .288591

= Rs. 57,718.2

6. Sinking Fund Method:

Under this method, the amount of depreciation goes on accumulating till the asset is completely worn out. This method provides necessary cash to replace the asset at end of its useful life. The amount of depreciation is fixed and remains same for every year and is charged to P&L A/c.

The amount of depreciation is invested outside the business every year. The process of investing amount of depreciation and interest goes on till the time of replacement of asset. At the time of replacement all the investments are sold out and cash received, and the new asset purchased.

Journal Entries:

(a) At the end of first year.

(i) For setting aside amount of Depreciation

Depreciation A/c Dr.

To Sinking Fund A/c

(ii) For transfer of Depreciation A/c to P&L A/c

P&L A/c Dr.

To Depreciation A/c

(iii) For Investment of Amount

Sinking Fund Investment A/c Dr.

To Bank A/c

(b) Subsequent Year

(i) For receiving interest

Bank A/c Dr.

To sinking fund A/c

(ii) For setting aside amount of Depreciation

(iii) Depreciation A/c Dr.

To Sinking fund A/c



(iv) For transferring Depreciation A/c to P&L A/c

P&L A/c Dr.

To Depreciation A/c

(v) For investing amount of depreciation + interest received

Sinking fund investment A/c Dr.

To Bank A/c

(c) Last Year

(i), (ii) & (iii) entry remain same as in (b), but amount is not invested.

(iv) On sale of investment

Bank A/c Dr.

Sinking Fund A/c (if loss) Dr.

To Sinking Fund Investment A/c

To Sinking fund (in profit)

(v) On Sale of Asset

Bank A/c Dr.

To Asset A/c

(vi) Sinking Fund A/c balance is transferred to asset a/c and any balance left in the asset a/c is transferred to P&L A/c.

Illustration 10:

X Ltd. Bought a machinery for Rs. 2, 00,000 on 1st January, 2000. The useful life was estimated at 3 years with a scrap value of Rs. 20,000. The Company uses Sinking Fund Investment Method, investment to carry 5% p.a. Interest. The sinking Fund Table shows that 31720 at 5% p.a. will in 3 years accumulate to Re. 1. Prepare Machinery A/c, Sinking Fund A/c & S.F. Investment.

Solution:**Machinery A/c**

Date	Particulars	Amount (₹)	Date	Particulars	Amount (₹)
2002 Jan. 1	To Bal. b/d	2,00,000	2002 Dec. 31	By Sinking Fund A/c	1,80,000
		2,00,000		By cash	20,000
					2,00,000

Sinking Fund A/c

Date	Particulars	Amount (₹)	Date	Particulars	Amount (₹)
2000, Jan. 31	To Bal. c/d	57,097	2000, Dec. 31	By Dep. A/c	57,097
2001, Dec. 31	To Bal. c/d	1,17,053	2001, Jan. 31	By Bal. b/d	57,097
			Dec. 31	By Dep.	57,097
			Dec. 31	By Interest	2,859
		1,17,053			1,17,053
2002 Dec. 1	To Machinery A/c	1,80,000	2002 Jan., 31	By Bal. c/d	1,17,053
			Dec. 31	By Dep.	57,097
				By Interest	5,850
		1,80,000			1,80,000

Sinking Fund Investments A/c

Date	Particulars	Amount (₹)	Date	Particulars	Amount (₹)
2000, Jan. 1	To Bank	57097	2000, Dec. 31	By Bal. c/d	57,097
2002, Jan. 1	To Balance b/d	57097	2002, Dec. 31	By Bank	1,17,053
Jan. 1	To Bank	59,956			1,17,053
		1,17,053			

7. Written Down Value Method

Under this method, the rate or percentage of depreciation is fixed, but it applies to the value at which the asset stands in the books in the beginning of the year. In other words, under this method, a fixed percentage is written off every year on the reduced balance of the asset. Thus, the percentage of depreciation is not applied to the original cost but only to the balance, which remains after charging depreciation in the beginning of a year. The percentage of depreciation remains fixed for all the years of the working life of an asset but the actual amount of depreciation written off every year goes on decreasing with the reduction in the value of the asset.

8. Revaluation Method

This method is used only in case of small items like cattle (Livestock), or loose tools where it may be too much to maintain an account of each single item. The amount of depreciation to be written off is determined by comparing the value at the end of the year (valuation being done by some one having expert knowledge of the valuation of the asset) with the value in the beginning.

Suppose on 1st April 2,007 the value of loose tools was Rs.10,000 and during the year Rs.30,000 worth of tools were purchased. Now if at the end of the year, the loose tools are considered to be worth only Rs.25,000 the depreciation comes to Rs.15,000 i.e. $\text{Rs.10,000} + \text{Rs.30,000} - \text{Rs.25,000}$.

9. Depletion Method

The depletion method is used in case of mines, quarries, etc., where an estimate of total quantity of output likely to be available should be available. Depreciation is calculated per ton of output.

For example, if a mine is purchased for Rs.20,00,000 and it is estimated that the total quantity of mineral in the mine is 5,00,000 ton, the depreciation per ton of output comes to $= 20,00,000 / 5,00,000 = \text{Rs. } 4$.

If the output in the first year is 30,000 then the depreciation will be $30,000 \times \text{Rs.4} = \text{Rs. } 1,20,000$, in the second year, the output may be 50,000 ton; the depreciation to be written off will be Rs.2,00,000 i.e., $50,000 \times \text{Rs.4}$.

10. Machine Hour Rate Method

This is more or less like the depreciation method. Instead of the usual method of estimating the life of a machine in years, it is estimated in hours. Then, an accurate record is kept recording the number of hours each machine is run and depreciation is calculated accordingly.

For example, the effective life of a machine may be 30,000 hours. If the cost of the machine is Rs.4,50,000, the hourly depreciation is

$$= 4,50,000 / 30,000 = \text{Rs. } 15.$$

The depreciation for a particular year during which the machine runs for 2,500 hours will be $2,500 \times \text{Rs.15} = \text{Rs.37,500}$.

11. Insurance Policy Method or Capital Redemption Policy Method

Under this method the business takes a policy from an insurance company. The amount of the policy is such that it is sufficient to replace the asset when it is worn out. Cash, which is equal to the amount of depreciation, is paid by way of premium every year. The amount goes on accumulating with the insurance company at a certain rate of interest and is paid back to the insured at the maturity of the policy. The amount so made available by the insurance company is used for purchasing a new asset. This method to a great extent is similar to sinking fund method, but no doubt the procedure is a little different. In this method, instead of buying securities, the insurance policy is taken and premium is paid every year. Company, that receives premium, allows a small interest on compound basis.

This method is a more suitable device for ensuring the availability of cash to replace the asset. The advantage under this system is that the company need not worry whether the investments as under the Depreciation Fund Method, will be sold at best prices or not.

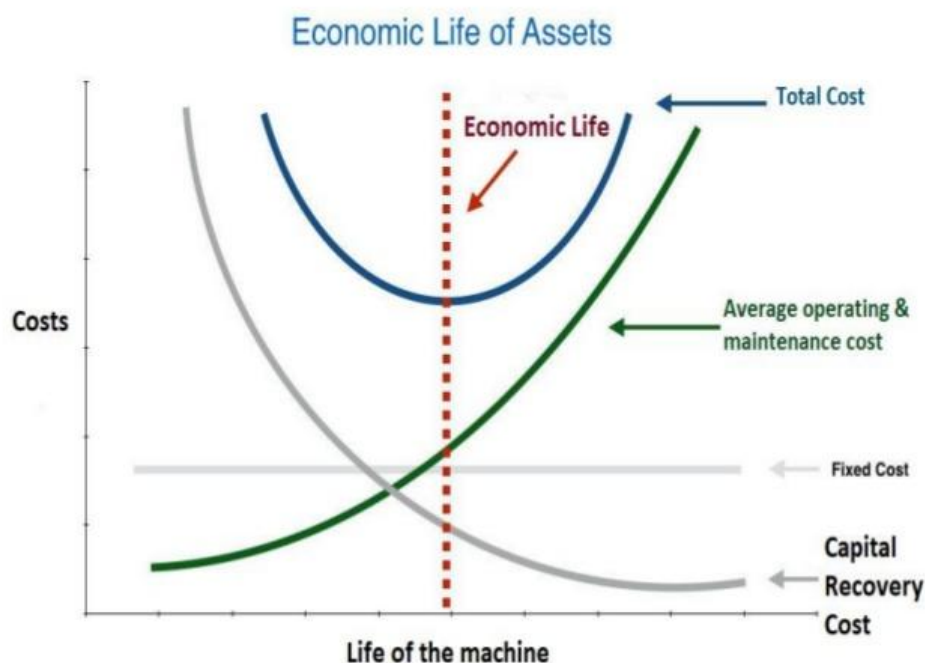
If an insurance policy is taken, it serves two purposes. Firstly, it insures the asset. Secondly, the insurance company will pay the stipulated amount to enable the company to replace assets. This method is more expensive as the insurance company has to keep its margin of profits. It is suitable for losses where the life of the asset is definitely known. It yields a very low rate of interest. It makes no adjustments for price-level changes.

2.8 DETERMINATION OF ECONOMIC LIFE

Any asset will have the following cost components:

- ☞ ⌚ Capital recovery cost (average first cost), computed from the first cost (purchase price) of the machine.
- ☞ ⌚ Average operating and maintenance cost (O & M cost)

Total cost which is the sum of capital recovery cost (average first cost) and average maintenance cost.



Example

A firm is considering replacement of an equipment, whose first cost is Rs. 4,000 and the scrap value is negligible at the end of any year. Based on experience, it was found that the maintenance cost is zero during the first year and it increases by Rs. 200 every year thereafter.

- (a) When should the equipment be replaced if $i = 0\%$?
- (b) When should the equipment be replaced if $i = 12\%$?
- (a) When $i = 0\%$. In this problem
 - (i) First cost = Rs. 4,000
 - (ii) Maintenance cost is Rs. 0 during the first year and it increases by Rs. 200 every year thereafter.

This is summarized in column B of Table

Column C summarizes the summation of maintenance costs for each replacement period. The value corresponding to any end of year in this column represents the total maintenance cost of using the equipment till the end of that particular year.

$$\text{Average total cost} = [\text{First cost (FC)} + \text{Summation of maintenance cost}] / \text{Replacement period}$$

End of year (n)	Maintenance cost at end of year	Summation of maintenance costs	Average cost of maintenance through year given	Average first cost if replaced at year end given	Average total cost through year given
		B	C/A	4,000/A	D + E
A	B (Rs.)	C (Rs.)	D (Rs.)	E (Rs.)	F (Rs.)
1	0	0	0	4,000.00	4,000.00
2	200	200	100	2,000.00	2,100.00
3	400	600	200	1,333.33	1,533.33
4	600	1,200	300	1,000.00	1,300.00
5	800	2,000	400	800.00	1,200.00
6	1,000	3,000	500	666.67	1,166.67*
7	1,200	4,200	600	571.43	1,171.43

*Economic life of the machine = 6 years

$$\text{Average total cost} = \frac{\text{First cost (FC)} + \text{Summation of maintenance cost}}{\text{Replacement period}}$$

$$= \frac{\text{First cost}}{n} + \frac{\text{Summation of maintenance cost}}{n}$$

Average first cost for the given period + Average maintenance cost for the given period

$$\text{Column F} = \text{Column E} + \text{Column D}$$

The value corresponding to any end of year (n) in Column F represents the average total cost of using the equipment till the end of that particular year.

For this problem, the average total cost decreases till the end of year 6 and then it increases. Therefore, the optimal replacement period is six years, i.e. economic life of the equipment is six years.

(b) When interest rate, $i = 12\%$. When the interest rate is more than 0% , the steps to be taken for getting the economic life are summarized with reference to Table

Table Calculations to Determine Economic Life (First cost = Rs. 4,000, Interest = 12%)

End of year (n)	Maintenance cost at end of year	P/F, 12% , n	Present worth as of beginning of year 1 of maintenance costs	Summation of present worth of maintenance costs through year given	Present worth of cumulative maintenance cost & first cost	A/P, 12% , n	Annual equivalent total cost through year given
		(B)	(C)	D	E + Rs. 4,000		
A	B (Rs.)	C	D (Rs.)	E (Rs.)	F (Rs.)	G	H (Rs.)
1	0	0.8929	0.00	0.00	4,000.00	1.1200	4,480.00
2	200	0.7972	159.44	159.44	4,159.44	0.5917	2,461.14
3	400	0.7118	284.72	444.16	4,444.16	0.4163	1,850.10
4	600	0.6355	381.30	825.46	4,825.46	0.3292	1,588.54
5	800	0.5674	453.92	1,279.38	5,279.38	0.2774	1,464.50
6	1,000	0.5066	506.60	1,785.98	5,785.98	0.2432	1,407.15
7	1,200	0.4524	542.88	2,328.86	6,328.86	0.2191	1,386.65*
8	1,400	0.4039	565.46	2,894.32	6,894.32	0.2013	1,387.83
9	1,600	0.3606	576.96	3,471.28	7,471.28	0.1877	1,402.36
10	1,800	0.3220	579.60	4,050.88	8,050.88	0.1770	1,425.00

*Economic life of the machine = 7 years

The steps are summarized now:

1. Discount the maintenance costs to the beginning of year 1.

Column D = Column B

$$1 / (1 + i)^n$$

= Column B $(P/F, i, n)$ = Column B $\times$ Column C.

2. Find the summation of present worth of maintenance costs through the year given (Column E = Column D).

3. Find Column F by adding the first cost of Rs. 4,000 to Column E.

4. Find the annual equivalent total cost through the years given.

$$\text{Column H} = \text{Column F} \times i(1 + i)^n / (1 + i)^n - 1$$

= Column F $(A/P, 12\%, n)$ = Column F $\times$ Column G

5. Identify the end of year for which the annual equivalent total cost is minimum.

For this problem, the annual equivalent total cost is minimum at the end of year 7. Therefore, the economic life of the equipment is seven years.

IMPORTANT QUESTIONS

1. Explain the principles of Material Handling.
2. Explain different types of material handling devices used in a production plant.
3. Explain the factors which affects the selection of Material handling devices.
4. What is the relation between Plant layout and material handling? How will you select a material handling device suitable for a workplace?
5. Compare Preventive and Breakdown maintenance.
6. Explain different methods of Replacement.
7. How will you determine the economic life of a product?
8. Explain different methods for providing depreciation.
9. List the advantages and limitations of preventive maintenance.
10. List the advantages and limitations of breakdown maintenance.



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MODULE 3

3.1 METHODS ENGINEERING

Methods engineering is a branch of industrial engineering specializing in the analysis of methods and the improvement and standardization of methods, equipment, and working conditions.

Alternatively it can be described as the design of the productive process in which a person is involved. The task of the Methods engineer is to decide where humans will be utilized in the process of converting raw materials to finished products and how workers can most effectively perform their assigned tasks. The terms operation analysis, work design and simplification, and methods engineering and corporate re-engineering are frequently used interchangeably.

Lowering costs and increasing reliability and productivity are the objectives of methods engineering. These objectives are met in a five step sequence as follows: Project selection, data acquisition and presentation, data analysis, development of an ideal method based on the data analysis and, finally, presentation and implementation of the method.

WORK STUDY

Work study is the investigation, by means of a consistent system of the work done in an organization in order to attain the best utilisation of resources i.e. Materials, Machines, Men and Money. All the technologies and management systems are related with productivity.

Work study is one of the basic techniques of improving productivity. In order to resolve this aspect, work study aims:

- (i) To analyse the work in order to achieve work simplification and thereby improving productivity of the system.
- (ii) To have optimum utilization of resources i.e., 4 Ms.
- (iii) To evaluate the work content through work measurement.
- (iv) To set time standards for various jobs.

In nutshell work study is mainly concerned with the examination of human work. In fact planning is not possible unless one knows how long it will take to do a particular job. Thus time is very important to the manufacturer who must keep to promise, to estimate quantities and to other industrial and business arrangements or organizations.

Work study is not a theoretical concept but essentially a practical one and deals with human beings who have their own attitude and style of working. So the success of work study is dependent upon the relations between the labour/employees and the management.

Work study involves lot of changes in various working methods. Since the manpower in general does not like changes but prefers to continue as already doing, so there will always be a tendency to resist any modification or new method suggested by work study people (officers/workers) and the manpower and

the workers have confidence in the ability, integrity and fair-mindedness of work study man, there is a good chance that sound proposals will be accepted willingly by the manpower.

Definition and Concept of Work Study:

Generally work study is used to describe a complete set of techniques with the help of which work can be simplified, standardized and measured.

When it is possible to simplify the existing work or new methods are designed and introduced such that the task/activity becomes simpler than following advantages are possible:

- (i) More production with less effort so goods/products are available at cheaper rates.
- (ii) Better equipment utilization shall lead to marked increase in the total production without addition of new resources, thus productivity may improve.

These advantages are possible through the willing cooperation of the people engaged in production work. In view of these far reaching benefits, work study has become an important tool of management.

In industries work study is considered as a tool of improving productivity by way of:

- (i) Resource utilization to a satisfactory level.
- (ii) Capital investment to introduce latest technology.
- (iii) Better management of the system.

I. British Standard Institution defines work study as a generic term for those techniques particularly “Method study” and “Work Measurement” which are used in the examination of work in all its contexts and which leads systematically to the investigation of all the factors which affect the efficiency and economy of the situation being reviewed in order to incorporate improvements at various levels.

II. Work study may be defined as “The systematic critical, objective and imaginative examination of all factors governing the operational efficiency of any specific activity in order to achieve/ effect improvement.”

Thus work study is the investigation by means of a consistent system of the work done in an organization in order to achieve the best possible utilization of resources i.e. man, machines and materials available. Every organization tries to achieve best quality production of various products in the minimum possible time.

The time required manufacturing a product/item depends on the manufacturing procedure. One phase of work study called Method Study or Motion Study aims at determining the best possible manufacturing procedure which involves least time and cause minimum fatigue to workers employed for performing the work.

In short method study is concerned with determination of the best way of doing a job. Method study may be defined as the systematic investigation of the existing method of doing a job in order to develop and

install an easy, quick, effective, efficient and comparatively less tiring method or procedure for doing the same job at cheaper rates.

This is generally achieved by eliminating unnecessary movements or motions involved in a certain procedure or by changing the sequence of the operation of the process itself

The role of time study or work measurement starts with the determination of the time allowed to complete the job by the developed improved procedure for doing that job. Thus work measurement may be defined as the application of different techniques to measure and establish the time required to complete the job by a qualified worker at a defined level of performance.

Need For Work Study:

Principles of work study have been used since long to identify the improvements to be incorporated, when industrial set up was simple and involved lesser problems. The industries of today with increased complexities and modernisation naturally demand a more systematic approach like work study in its present form for solution of various problems.

Work study is most valuable tool of management because:

- (i) It is a direct means of improving productivity of the system involving very less or no cost.
- (ii) The approach is simple, systematic, consistent and based on handling of facts. Thus the part played by opinions in taking decisions is minimized.
- (iii) No factor affecting the efficiency of operation is overlooked in this approach.
- (iv) It provides most accurate means of setting standards of performance which are helpful in the process of production planning and control.
- (v) Application of work study result in immediate savings.
- (vi) It is a universal tool for management.
- (vii) It is a most penetrating tool of investigation available to the management of the industrial unit.

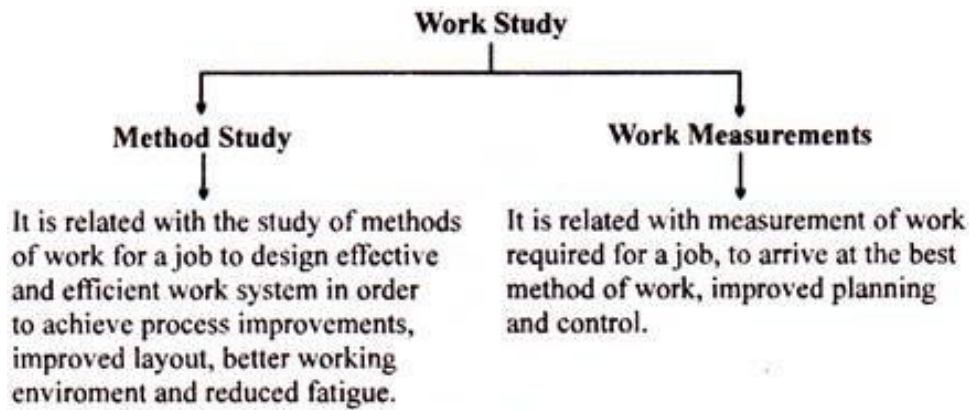
Techniques of Work Study:

Basically there are two techniques:

Work study is the term used to embrace the techniques of Method Study and Work Measurement which are used to ensure the best utilization of manpower and material resources in carrying out specified activity.

The sequential order of the correct procedure to be adopted for having effective or purpose oriented results of method study include the following:

- (i) Select the work/procedure to be analysed.
- (ii) Record all the relevant information related with the existing work system with the help of various recording devices or techniques.
- (iii) Make critical examination of collected data/facts.



(iv) Develop and improve the method which is economical and practical after giving due consideration to the alternative method possible.

(v) Install the new selected method with proper instructions.

(vi) Maintain the latest standards of methods through periodic verification etc.

Similarly, the logical order of correct procedure to be adopted for having effective results of work measurement includes the followings:

(i) Divide the selected procedure into small work elements.

(ii) By direct observations record the relevant information regarding the various work elements.

(iii) In the light of relevant information examine the work critically.

(iv) Measure the work content in the terms of time of the work elements involved in method being adopted.

(v) Define and design the new selected method.

(vi) Finally convert the work content time in standard time.

Objectives of Method Study:

- (1) Better design of plant equipment and buildings.
- (2) Less fatigue of workers by avoiding unnecessary movements of manpower.
- (3) Better working conditions and environment for workers/employees.
- (4) To have more effective utilisation of materials, machines and manpower and money.
- (5) Better Product quality.
- (6) Efficient and fast material handling equipment.
- (7) Leads to standardisation, rationalisation, simplification and specialisation.
- (8) Efficient planning of the section.
- (9) Streamlined working procedures.

Objectives of Work Measurement:

Effective planning and control of production, distribution and administration can not be achieved unless these are based on facts. One of the most important of these facts is the time required for job completion which can be obtained by motion study or work measurement.

Other objectives of work measurement are as follows:

- (1) When two alternative methods which seem equally advantageous, one which requires less time for completion of the job (i.e. faster) can be established.
- (2) The man power required for a job or new plant can be determined on the basis of accurate knowledge of the amount of work that has to be done so it helps in manpower economy.
- (3) Work measurement data provides reliable basis to decide equipment.
- (4) Provides requisite information for effective production planning and improving efficiency of the system.
- (5) Provides basis for fair and reliable incentive schemes.
- (6) Decides realistic labour budgeting and provides a basis for standard costing system.

Advantages of Work Study:

- (1) It is direct means of improving productivity.
- (2) It results in uniform and improved production flow.
- (3) It reduces the manufacturing cost.
- (4) With its help fast and accurate delivery dates are possible.
- (5) It provides better service and consumer satisfaction.
- (6) It improves employee-employer relations.
- (7) It provides job satisfaction and job security to workers.
- (8) Better working conditions are possible for workers.
- (9) It is most important tool of analysis and can help in providing better wages to workers on scientific basis.
- (10) Most accurate method and yet provides a sound basis for production planning, control and incentives for man power.
- (11) Everyone concerned with industries is benefited from it such as worker, consumer and management of the unit.

Scope of Method Study

The scope of method study is not restricted to only manufacturing industries. Method study techniques can be applied effectively in service sector as well. It can be applied in offices, hospitals, banks and other service organizations. The areas to which method study can be applied successfully in manufacturing are:

1. To improve work methods and procedures.
2. To determine the best sequence of doing work.
3. To smoothen material flow with minimum of back tracking and to improve layout.
4. To improve the working conditions and hence to improve labor efficiency.
5. To reduce monotony in the work.
6. To improve plant utilization and material utilization.
7. Elimination of waste and unproductive operations.
8. To reduce the manufacturing costs through reducing cycle time of operations.

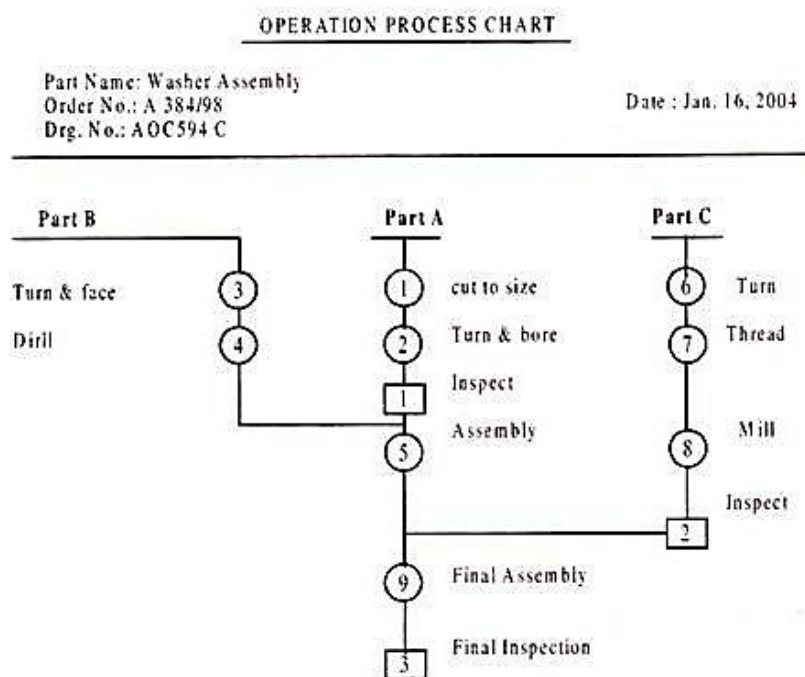
Steps or Procedure Involved in Methods Study

The basic approach to method study consists of the following eight steps. The detailed procedure for conducting the method study is shown in the following figure.

1. **SELECT** the work to be studied and define its boundaries.
2. **RECORD** the relevant facts about the job by direct observation and collect such additional data as may be needed from appropriate sources.
3. **EXAMINE** the way the job is being performed and challenge its purpose, place sequence and method of performance.
4. **DEVELOP** the most practical, economic and effective method, drawing on the contributions of those concerned.
5. **EVALUATE** different alternatives to developing a new improved method comparing the cost-effectiveness of the selected new method with the current method with the current method of performance.
6. **DEFINE** the new method, as a result, in a clear manner and present it to those concerned, *i.e.*, management, supervisors and workers.
7. **INSTALL** the new method as a standard practice and train the persons involved in applying it.
8. **MAINTAIN** the new method and introduce control procedures to prevent a drifting back to the previous method of work.

Procedure for Method Study

Method study can be performed in following six steps with the help of operation process chart as shown in Fig.



1. Selection of Work/Job to be Studied:

This is one of the most important tasks to select appropriate job, activity or situation for method analysis and mainly it is a managerial responsibility. Method study and its implementation will cost money.

Practically any activity is a potential project for improvement but only those jobs should be selected when there are some valid reasons for method study (like job/activity is unpopular or considered dirty by workers). Maximum cost benefit is the normal objective.

2. Collection and Recording of Necessary Information:

Obtain the facts about the present method of doing the job and record them. In order to improve an activity or procedure the factual information is collected from the place where the job under study is being executed by direct visual observation.

This can be done by:

- (i) Recording movement of machines or materials.
- (ii) Recording the critical procedure in the job.
- (iii) Recording the operator's performance.
- (iv) Recording the path of movement of workers w.r.t. machines and allied operations, (v) Recording the path of movement to improve workplace layout.
- (v) Recording the scrap generated.

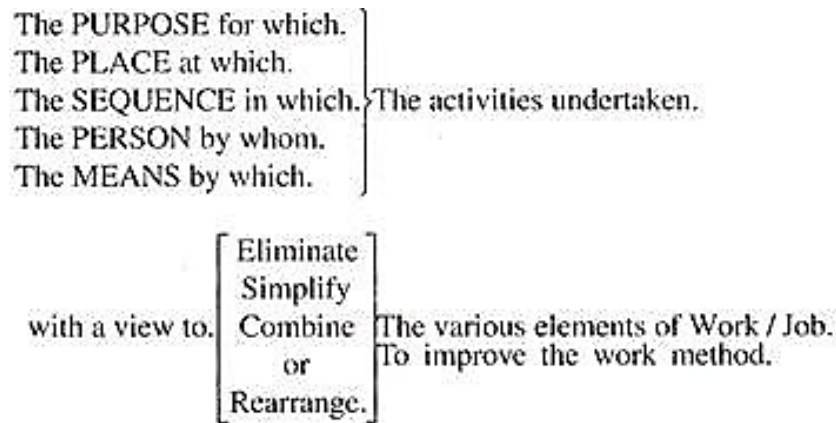
There are many standardized techniques (i.e. charts, diagrams, graphs etc.) available which may be used for proper recording and presentation for further analysis. The choice of technique depends on the type of information which is to be recorded.

This is an important step since the improvement in existing method or development of new method depends on how exactly the facts about the existing method have been recorded.

3. Critical Examination of the Existing Method:

Now examine the facts critically. This is a key step in the whole study. The information available in the form of chart and graphs after step II can now be thoroughly studied and analysed in order to detect the production stages where improvement in the method is possible. This is achieved by questioning the different activities of the process in a systematic logical and objective manner.

The questions sequence follows a well-established pattern examines:



These questions can help in the development of new method by getting the answers for:

- (i) What should be done?
- (ii) When should be done?
- (iii) Who should do it?
- (iv) Where should it be done?
- (v) How should it be done?

The following improvements are possible on the basic of answers to these questions:

- (i) Improvement in material flow.
- (ii) Improvement in working conditions.
- (iii) Improvement in manufacturing techniques.
- (iv) Improvement to clerical routine.
- (v) Reduction in material waste.

In this way a well conducted examination will determine the purpose, place, sequence of these activities, the person performing them and the resources by which it can be performed so as to select the best/optimum out of the possible alternatives.

4. Develop the New Improved Method:

The analysis of existing work method can provide a starting point for a synthesis of suggested improvements in job performance. The technique specially designed for improving work method is called the process improvement formula.

The four steps of the formula are:

- (i) Eliminate the unnecessary activities
- (ii) Combine two or more activities
- (iii) Sequence the various activities properly and
- (iv) Simplify the activities.

Complete elimination of unwanted activities is the most important step in developing an improved method. If elimination is not possible then possibility of combining the different activities should be explored. The next strategy in development stage is to identify scope of changes in the sequence of operations or activities.

The last and most important plus expensive step in process of method development is the simplification of activities in order to allow the operator/worker to complete the job more quickly and easily by reducing the number of operations, reducing or eliminating the delays and storage etc.

Finally after selecting the work method to be adopted, certain amount of experimentation will be required to locate and eliminate snags if any.

Specify the improved method. It is essential that the selected method be described fully and in a systematic manner.

Specifications of the selected method accomplish several purposes like:

- (i) Communication of the method to the management for its approval.
- (ii) Communication to those concerned with its implementation like instructions to workers and so far as the machines and layout are concerned.
- (iii) Provision of an official record of the method.

The new method should possess the following aspects:

- (a) Full use of worker's body,
- (b) Arrangements of workplace and
- (c) Design of tools and Equipment.

The acceptance of new method will mainly depend upon the cost benefit, trade unions approach, working hours and security requirements etc.

5. Install the Improved Method:

This involves training of those who are going to utilize the new method. Cooperation from both supervisory staff and operators is essential to the successful installation of any proposed and selected method. So the installation of the new method should then be entrusted to some responsible person.

The method implemented can be further improved by way of continuous observations and discussions. The trial runs may be carried out during which minor modifications may be made to facilitate working.

Any method selected is ultimately to be utilized by the employees in an environment created by the management. So employees and management should be subjected to intensive analysis so that the proposed method is operated in the best possible manner.

Installation phase of the method study is complete as soon as the newly installed method starts working smoothly and satisfactorily and provides encouraging result (such as time saving and scrap reduction etc.).

6. Maintain the Improved Method:

Even after the workers have been trained and requisite changes in machines and layout have been incorporated, there is no guarantee that the method will be used the way it was designed.

The proper functioning of the installed method is ensured by periodic checks and verifications. The purpose of checks and reverse is to determine if the method being adopted and practised is the same or it has some deviation from the selected one.

Reasons for deviation if any should be explored and the required changes may be incorporated in the procedure being practised so as to revert back to the authorized or selected one.

Views of persons concerned (like operators and supervisors) with the installed method can be of much help in exploring further improvements in the system. Therefore the method should be reviewed at intervals to provide allowances for any changes.

3.2 RECORDING TECHNIQUES FOR METHOD STUDY

The next step in basic procedure, after selecting the work to be studied is to record all facts relating to the existing method. In order that the activities selected for investigation may be visualized in their entirety and in order to improve them through subsequent critical examination, it is essential to have some means of placing on record all the necessary facts about the existing method. Records are very much useful to make before and after comparison to assess the effectiveness of the proposed improved method.

The recording techniques are designed to simplify and standardize the recording work. For this purpose charts and diagrams are used.

Symbols Used in Method Study

Graphical method of recording was originated by Gilberth, in order to make the presentation of the facts clearly without any ambiguity and to enable to grasp them quickly and clearly. It is useful to use symbols instead of written description.

- O OPERATION
- INSPECTION
- TRANSPORTATION
- D DELAY
- ▽ STORAGE

Activity	Symbol	Summary
Operation	O	Whenever the hand grasps, releases or assemble materials.
Transport	⇒	Whenever hand moves from one point to other.
Delay	D	Whenever the hand is idle or it is not performing any activity.
Hold	▽	The symbol is same, as we used for storage. But new, term hold is used, whenever hand holds an object in order to perform an operation on object by other hand.

Operation

An operation occurs when an object is intentionally changed in one or more of its characteristics (physical or chemical). This indicates the main steps in a process, method or procedure.

An operation always takes the object one stage ahead towards completion.

Examples of operation are:

- Turning, drilling, milling, etc.
- A chemical reaction.
- Welding, brazing and riveting.
- Lifting, loading, unloading.
- Getting instructions from supervisor.



Inspection

An inspection occurs when an object is examined and compared with standard for quality and quantity.

The inspection examples are:

- Visual observations for finish.
- Count of quantity of incoming material.
- Checking the dimensions.

Transportation

A transport indicates the movement of workers, materials or equipment from one place to another.

Example: Movement of materials from one work station to another.

Workers travelling to bring tools.

Delay D: Delay (Temporary Storage)

A delay occurs when the immediate performance of the next planned thing does not take place.

Example:

- Work waiting between consecutive operations.
- Workers waiting at tool cribs.
- Operators waiting for instructions from supervisor.

Storage

Storage occurs when the object is kept in an authorized custody and is protected against unauthorized removal. For example, materials kept in stores to be distributed to various work.

Charts used in methods study

This is the most popular method of recording the facts. The activities comprising the jobs are recorded using method study symbols. A great care is to be taken in preparing the charts so that the information it

shows is easily understood and recognized. The following information should be given in the chart. These charts are used to measure the movement of operator or work (i.e., in motion study).

- Adequate description of the activities.
- Whether the charting is for present or proposed method.
- Specific reference to when the activities will begin and end.
- Time and distance scales used wherever necessary.
- The date of charting and the name of the person who does charting.

Types of Charts

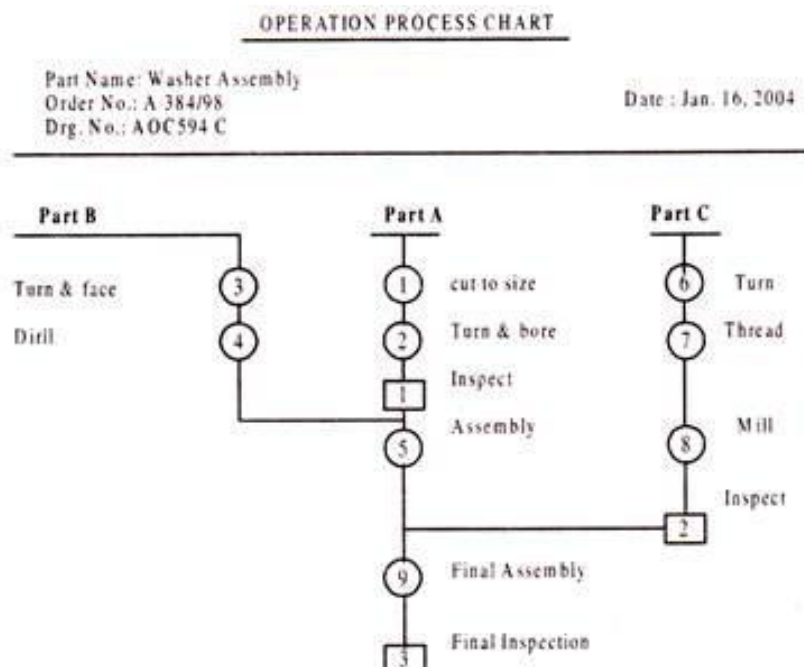
It can be broadly divided into (A) Macro motion charts and (B) Micro motion charts. Macro motion charts are used for macro motion study and micro motion charts are used for micro motion study. Macro motion study is one which can be measured through 'stop watch' and micro motion study is one which cannot be measured through stop watch.

Macro motion charts

Following four charts are used under this type:

1. Operation Process Chart

It is also called outline process chart. An operation process chart gives the bird's eye view of the whole process by recording only the major activities and inspections involved in the process. Operation process chart uses only two symbols, i.e., operation and inspection. Operation, process chart is helpful to:



- Visualize the complete sequence of the operations and inspections in the process.
- Know where the operation selected for detailed study fits into the entire process.

- c. In operation process chart, the graphic representation of the points at which materials are introduced into the process and what operations and inspections are carried on them are shown.

2. Flow Process Chart

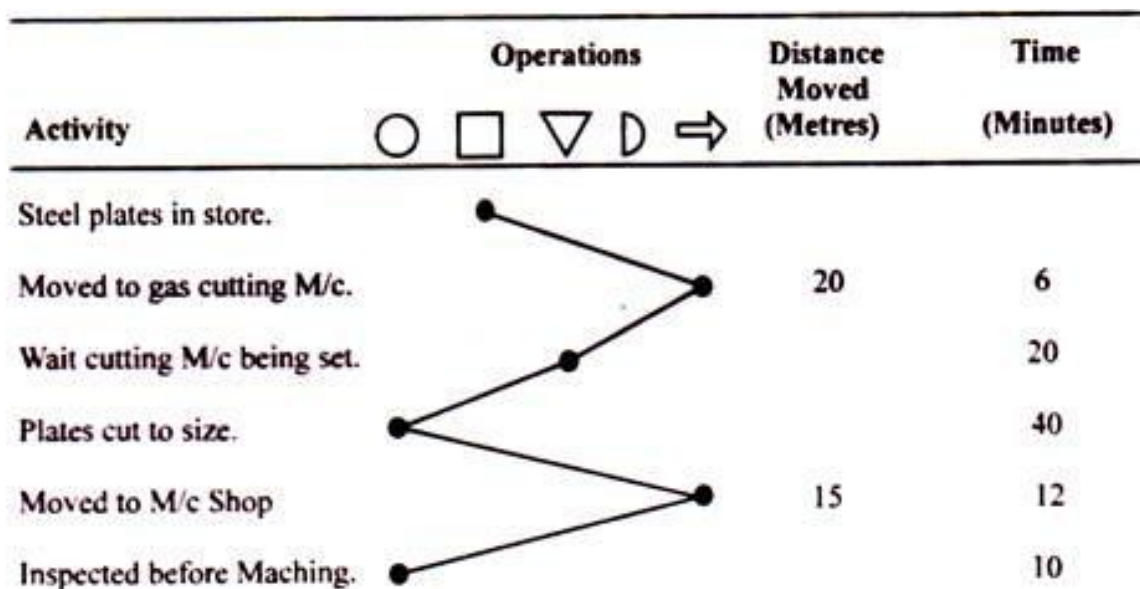
Flow process chart gives the sequence of flow of work of a product or any part of it through the work centre or the department recording the events using appropriate symbols. It is the amplification of the operation process chart in which operations; inspection, storage, delay and transportation are represented. However, process charts are of three types:

- Material type— which shows the events that occur to the materials.
- Man type—Activities performed by the man.
- Equipment type— how equipment is used.

The flow process chart is useful:

- to reduce the distance travelled by men (or materials).
- to avoid waiting time and unnecessary delays.
- to reduce the cycle time by combining or eliminating operations.
- to fix up the sequence of operations.
- to relocate the inspection stages.

Like operation process chart, flow process chart is constructed by placing symbols one below another as per the occurrence of the activities and are joined by a vertical line. A brief description of the activity is written on the right hand side of the activity symbol and time or distance is given on the left hand side.



A portion of the flow process chart (material type) showing flow of material

2. Two Handed Process Chart

A two handed (operator process chart) is the most detailed type of flow chart in which the activities of the workers hands are recorded in relation to one another. The two handed process chart is normally confined to work carried out at a single workplace. This also gives synchronized and graphical representation of the sequence of manual activities of the worker. The application of this charts are:

Left Hand Description	Symbol (L.H.)	Symbol (R.H.)	Right Hand Description
1. Move to bolt	⇒	⇒	Move to first washer.
2. Grasp the bolt	○	○	Pickup/grasp the washer.
3. Move to position	⇒	⇒	To position
4. Hold	▽	○	Assembled to bolt.
5.		⇒	Move to second washer.
6.		○	Pickup/grasp the washer.
7.		⇒	To position
8.		○	Assembled to bolt.
9.		⇒	Move to nut.
10.		○	Pick up the nut.
11. Move to box	⇒	⇒	To position.
12. Aside to box	○	○	Assembled to bolt

- To visualize the complete sequence of activities in a repetitive task.
- To study the work station layout.

3. Multiple Activity Chart

It is a chart where activities of more than subject (worker or equipment) are each recorded on a common time scale to show their inter-relationship. Multiple activity chart is made:

- to study idle time of the man and machines,
- to determine number of machines handled by one operator, and
- to determine number of operators required in teamwork to perform the given job.

The type of multiple activity charts to be drawn depends upon the type of job, important types are:

- (1) Man-Machine Activity Chart-when one operator is working on one machine.
- (2) Multi-man Activity Chart-when a group of workers are working on a machine.
- (3) Man- Multi machine activity chart- when a single operator is working on a number of machines.
- (4) Multi man- Machine chart- a group of operators working on a common central machine.

This type of chart is generally used for enabling maintenance and the similar work to be organized so that the time expensive equipment out of commission can be reduced to minimum. Obviously, the complex processes can be recorded in a simpler manner. It finds its applications in planning team work and construction jobs also.

Multiple Activity Chart (Man - Machine type)

Time (in Sec.)	Operation	Man	Machine
0	Man removes the rubber band.	Work	Idle
3	Man Picks up weight from hopper.	Work	Idle
5	Deck is placed in hopper.	Work	Idle
7	Man replaces weight in deck.	Work	Idle
9	Man Pushes start button.	Idle	Work
12		Idle	Work
20	Man picks up deck from output stacker.	Work	Idle
23	Rubber band is replaced on the deck.	Work	Idle

Key :

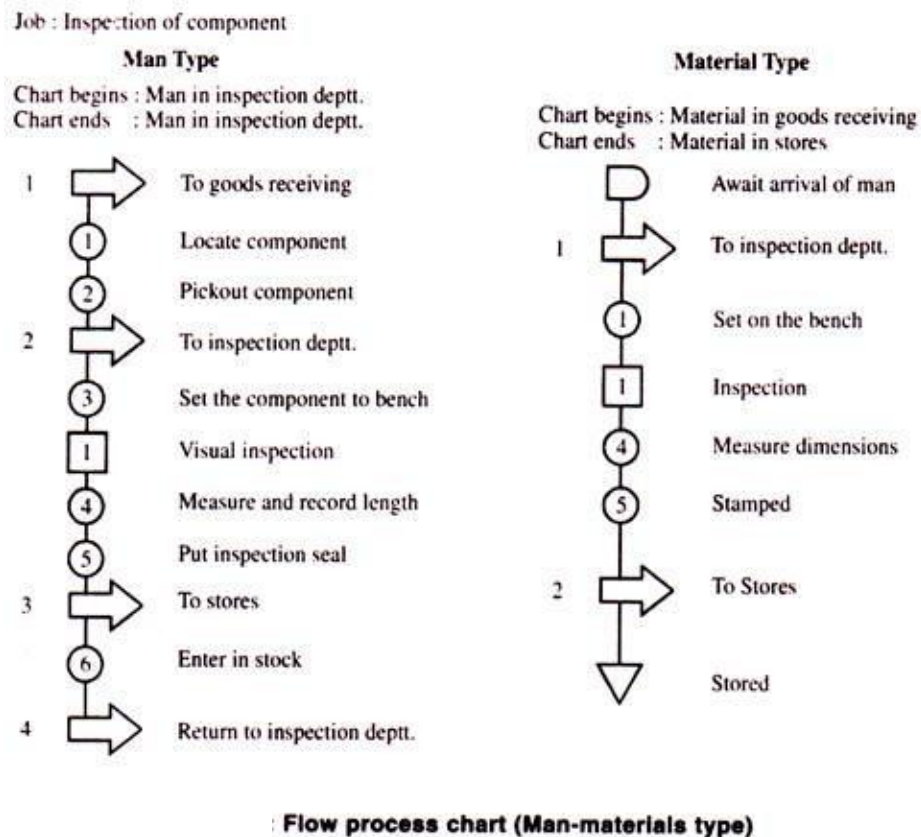
	Work
	Idle

Flow Process Chart (M-Type)		
Job Watering the garden		Chart begins : Sunil sitting in Porch Chart ends : Sunil begins watering the garden
Travel in m	Symbol	Description
3.5 m		To garage door
		Open door
3.5 m		To locker in garage
		Removes hose from locker
3.5 m		To rear garage door
		Opens door
3.5 m		To faucet at rear of garage
		Attach hose to faucet and open faucet
		Waters the garden

Diagrams Used in Method Study

The flow process chart shows the sequence and nature of movement but it does not clearly show the path of movements. In the paths of movements, there are often undesirable features such as congestion, back tracking and unnecessary long movements. To record these unnecessary features, representation of the working area in the form of flow diagrams, string diagrams can be made:

1. To study the different layout plans and thereby; select the most optimal layout.
2. To study traffic and frequency over different routes of the plant.
3. Identification of back tracking and obstacles during movements.



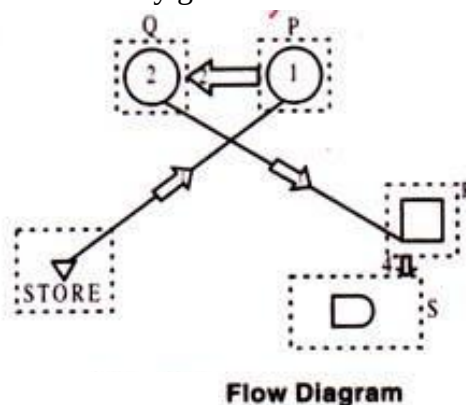
Diagrams are of two types:

1. Flow diagram and
2. String diagram.

1. Flow diagram

A flow diagram is a pictorial layout of buildings and floors on which locations of all the activities as given on the flow lines representing the movement of men and materials are shown. Such pictorial layout is valuable in incorporating improvements in the method.

The number of movements if minimized result viz. a lot of saving both in cost as well as efforts required to perform a job Fig. shows a simple flow diagram in which raw material from the store moves to work station P where an operation is performed the semi-finished product moves to work place Q where another operation is carried out hence it moves to work place R for inspection and is further sent to location S where it waits for a short period and ultimately goes out of the unit.



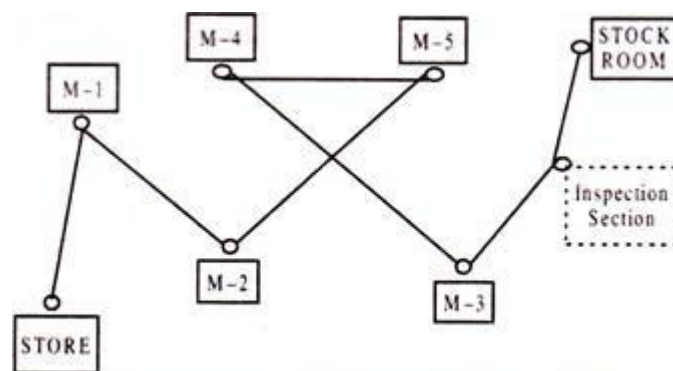
2. String diagram

The string diagram is a scale layout drawing on which, length of a string is used to record the extent as well as the pattern of movement of a worker working within a limited area during a certain period of time. The primary function of a string diagram is to produce a record of a existing set of conditions so that the job of seeing what is actually taking place is made as simple as possible.

One of the most valuable features of the string diagram is the actual distance travelled during the period of study to be calculated by relating the length of the thread used to the scale of drawing. Thus, it helps to make a very effective comparison between different layouts or methods of doing job in terms of the travelling involved. The main advantages of string diagram compared to flow diagram is that respective movements between work stations which are difficult to be traced on the flow diagram can be conveniently shown on string diagram.

Following are the procedures to draw string diagram:

1. A layout of the work place of factory is drawn to scale on the soft board.
2. Pins are fixed into boards to mark the locations of work stations, pins are also driven at the turning points of the routes.
3. A measured length of the thread is taken to trace the movements (path).
4. The distance covered by the object is obtained by measuring the remaining part of the thread and subtracting it from original length.



String Diagram (M-1 means Machine)

3.3 CRITICAL EXAMINATION OF WORK METHOD STUDY

In method study, recording is always followed by critical examination of that recorded data. Examination should be critical but impartial. So examination is just a technique to check the significance or bottlenecks of a process. But which activities to be examined first and now, are the points to be kept in mind while examination.

So, first we will categorize the various activities in two types:

- (a) Activities are which material is worked upon, moved or examined.

(b) Activities in which material are not touched, being either in storage or in a delay operation. So, the first activities to be examined must be those which are obviously non- productive or activities like delay, storage etc.

There are two stages of examining or questioning the activities -primary Questions and secondary Questions Sessions. Primary questions are based on the purpose, Place, sequence, person and means.

So, the primary questions are:

- (1) What is done and what is its significance. (Purpose).
- (2) Where the process is carried out and why at that place? (Place).
- (3) In which sequence, it is done> (sequence)
- (4) Who is performing that process and why he is performing? (Person).
- (5) By what means, process is carried out and why? (Means).

Secondary Questions takes into account the second stage of questioning technique. Here, the answers for the primary questions are further subjected to questions.

Secondary questions are:

- (1) What should be done and what else we could have done (Purpose).
- (2) Where should the process be carried out and where else we could have performed it (Place).
- (3) Which sequence is perfect for this process and what else sequence is possible? (Sequence).
- (4) Who should perform it and who else might do it? (Person).
- (5) How should it be performed and how else might it be performed? (Means).

These answers form the basis of the proposals to the improved method. In the evaluation phase, work study man considers all the possibilities with respect to the four factors namely. Safety, Economy, Work Quality and Human Factors. Economic factors are most important in most situations.

Economic factors to any proposal refer to determination of “How much this will cost”? Safety considerations ensure that the proposed method shall not make the work less safe. Quality considerations determine or check whether the purposed method will make for better product quality or quality control. Human considerations endures that the proposed method will be easy to learn, safe, less monotonous and with minimum fatigue.

3. 4 MICRO MOTION STUDY AND THERBLIGS

Micro-motion study technique is best suited for those operations or activities which are of short duration and which are repeated hundreds of time. These are the operations or motions which require very small time and it is quite difficult to measure time for these motions accurately and the time required by these motions cannot be neglected due to repetitive operations.

In such, activities it is interesting to go into greater details in order to find out which movement and effort can be avoided. All this is done to develop the best possible pattern of movement so that the operator can perform the operations repeatedly with a minimum effort and fatigue.

Definition of Micro-Motion Study:

“Thus micro-motion study is the technique of recording and analysing the timing of basic elements of a operation with the objective of achieving the best method of performing the operation.”

Such respective short duration activities involve quick movement of limbs which cannot be accurately studied and timed using two handed process charts. This is due to the fact that such record microscopic details such as different operation, inspection and transport etc. Study of such microscopic movements in short cycle repetitive jobs is not sufficient.

Short cycle operations require to be studied for microscopic motions e.g. operation of picking up a nut from bin and its fixing consists of three hand motions namely reach for the nut, grasp nut and move hand back to assembly position. Such detailed analysis help to develop the best possible pattern of movements and hence enabling the operator to perform various operations repeatedly with minimum effort and fatigue.

Micro-motion study is one of the most accurate techniques of work analysis used for work improvement. It makes use of motion pictures of the different activities or movement, so with the help of camera. Very small time upto 0.0005 minute can be measured and recorded by this system.

When picture camera is utilized, the procedure is known as “MICROMOTION STUDY”. The motion time data from the film is transferred to simo-chart. The simo chart data can be further analysed for the purpose of work place layout; or method improvement.

This technique was developed by Fran Gilbreth who considered that an operation consists of minute elements which may be repetitive or non-repetitive. He termed these elements THERBLIG (after his name Gilbreth if spelt back word is Therblig).

Purpose of Micro-Motion Study:

1. To study the nature and path of movements for obtaining the elements of an operation.
2. To study the activities of the machine and the operator.
3. To impart training to the workers or operators regarding motion; economy so that unnecessary movement by the workers may be avoided.
4. To study the relationship between the activities of operator and the machine.
5. To keep permanent record of the most efficient way of performing a task for future reference.
6. To obtain motion time data for developing synthetic time standards for various elements.
7. For carrying out research in the field of method and time study.

Advantages of Micro-Motion Study:

1. It provides a permanent record of motion study on films.
2. A large number of operators can see the procedure at any time even after the completion of motion study work.
3. Films can easily reveal the difference between the present and the proposed technique.
4. Films can be demonstrated to large workforce at any desired speed.
5. It provides very accurate time for each operation or motion in comparison to stop watch time study.
6. It helps in making detailed and accurate analysis of the prevailing technique.

Micro-motion study chart

Micro-motion study provides a technique for recording and timing an activity. It is a set of techniques intended to divide the human activities in a groups of movements or micro-motions (called Therbligs) and the study of such movements helps to find for an operator one best pattern of movements that consumes less time and requires less effort to accomplish the task. Therbligs were suggested by Frank O. Gilbreth, the founder of motion study. Micro-motion study was mainly employed for the job analysis. Its other applications include:

1. As an aid in studying the activities of two or more persons on a group work?
2. As an aid in studying the relationship of the activities of the operator and the machine as a means of timing operations.
3. As an aid in obtaining motion time data for time standards.
4. Acts as permanent record of the method and time of activities of the operator and the machine.

SIMO chart symbols and Therbligs

The micro-motion group of techniques is based on the idea of dividing human activities into division of movements or groups of movements (Therbligs) according to purpose for which they are made. Gilbreth differentiated 17 fundamental hand or hand and eye motions. Each Therbligs has a specific color, symbol and letter for recording purposes. The Therbligs are micro-motion study involves the following steps:

1. Filming the operation to be studied.
2. Analysis of the data from the film.

The recording of the data through **SIMO chart** is done as micro motion chart.

Therbligs

One of the most interesting of the Gilbreths' contributions was the analysis and breakdown of a task into its basic elements of motion, which they called "therbligs." This word, invented by Frank Gilbreth as the result of a suggestion from his wife, can in some respects be considered as a forerunner of Serutan. Reading therblig backward gives the name Gilbreth except that the t and h are reversed. The therbligs are very useful; each has a name and a symbol, as shown in figure.

Sl. No.	Code	Name	Description	Colour
1.	SH	SEARCH	Locate and article	Black
2.	F	FIND	Mental reaction at end of search	Gray
3.	ST	SELECT	Selection from a member	Light Gray
4.	G	GRASP	Taking Hold	Red
5.	H	HOLD	Prolonged group	Gold Ochre
6.	TL	TRANSPORTED LOADED	Moving an article	Green
7.	P	POSITION	Placing in a definite location	Blue
8.	A	ASSEMBLE	Putting parts together	Violet
9.	U	USE	Causing a device to perform its function	Purple
10.	DA	DISASSEMBLE	Separating parts	Light Violet
11.	I	INSPECT	Examine or test	Burnt Ochre
12.	PP	PREPOSITION	Placing an article ready for use	Pale Blue
13.	RL	RELEASE LOAD	Release an article	Carmine red
14.	TE	TRANSPORT EMPTY	Movement of a body member	Olive Green
15.	R	REST	Pause to overcome fatigue	Orange
16.	JD	UNAVOIDABLE DELAY	Idle-outside persons control	Yellow
17.	PN	PLAN	Mental plan for future action	—

 Search

 Use

 Find

 Disassemble

 Select

 Inspect

 Grasp

 Preposition

 Hold

 Release Load

 Transport Loaded

 Unavoidable Delay

 Transport Empty

 Avoidable Delay

 Position

 Plan

 Assemble

 Rest

Therbligs

Every job involves a number of these therbligs. By way of illustration, consider signing one's name. A man usually keeps his fountain pen in his inside coat pocket. In therblig terms he must search, find, select (assuming he has more than one pen), transport loaded, position, disassemble, position, use, assemble, transfer, search, find, position. A lot of work for such a simple task that is done in so short a time.

SIMO Chart

Simultaneous motion cycle chart (SIMO chart) is a recording technique for micro-motion study. A SIMO chart is a chart based on the film analysis, used to record simultaneously on a common time scale the Therbligs or a group of Therbligs performed by different parts of the body of one or more operators. It presents graphically the separable steps of each pertinent limb of the operator under study. It is an extremely detailed left and right hand operation chart.

It is the micro-motion form of the man type flow process chart. To prepare SIMO chart, an elaborate procedure and use of expensive equipment are required and this study is justified when the saving resulting from study will be very high.

It records simultaneously the different therbligs performed by different parts of the body of one more operators on a common time scale. The movements are recorded against time measured in "Winks" (1 wink = $\frac{1}{2000}$ minute). These are recorded by a "Wink Counter" positioned in such a location that it can be seen rotating during filming process.

Improving the present method:

Therbligs in the SIMO chart are critically examined to explore the possibility of:

- (i) Removing or eliminating the unproductive Therbligs such as find, select and avoidable delay etc.
- (ii) Resequencing of the productive therbligs such as assemble and disassemble etc.

A micro motion study analysis sheet or a left hand-right chart is drawn in Fig. for the following information's.

Operation: Finish hand filing copper work piece.
Time for searching, lifting and holding work piece to the vice.
Table by left hand = 0.2 min
Time for opening the vice by right hand = 0.2 min
Holding the work piece in the vice with both hands = 0.4 min
Time taken for lifting and holding file by right hand = 0.2 min
Time taken for hand filing with both hands = 1.00 min
Time for bringing the micrometer by right hand = 0.2 min
Time taken in checking dimension with both hands = 0.8 min
Time for opening the vice by right hand = 0.2 min
Time taken in removing work piece with left hand = 0.2 min



Micro motion Study

Dept..... Film No.....

Analysis Sheet

Operation: Finish hand fillings

Charted By.....

Date..... Operator.....

S.No.	Left hand description	Therblig	Time	Therblig	Right hand
1.	Searching and lifting	SH,H	0.2		
2.			0.4	U	
3.	Clamping workpiece	PP	0.8	PP	Opening the vice clamping work piece in the vice piece in the vice.
4.			1.0	TL	Take the file
5.	Do the hand filling operation.	U	2.0	U	Do the hand filing Operation.
6.			2.2	TL	Taking the micrometer
7.	Check the dimension	I	3.0	I	Check the dimension
8.			3.2	U	Open the vice
9.	Remove the work piece	TL		3.4	

Construction of “SIMO” Chart:

The SIMO chart for left hand and right hand analysis sheet inform about the degree of participation of both the hands. The time for each Therblig recorded on the analysis sheet may be shown to scale by means of a SIMO chart. Either the SIMO chart may be prepared independently or the chart may be constructed from the data available on the analysis sheet.

A SIMO chart is beneficial since it allows very accurate and detailed analysis. The work cycle from the film can be studied, easily, peacefully and away from the disturbing surroundings of the actual work station. SIMO chart is critically examined in order to grasp a picture of complete cycle in total details and assists in working out better combination of the desired motions.

SIMO CHART

Department: Film No.....

Operation: Finish hand filling

Charted By.....

Date..... Operator.....

S.No.	Left hand description	Therblig	Time	Therblig	Right hand
1.	Searching and lifting work piece.	SH, H	400		
2.			400	U	Opening the vice
3.	Clamping work piece in vice	PP	800	PP	Clamping work piece in the vice
4.			400	TL	Take the file.
5.	Do the hand filling operation.	U	2000	U	Do the hand filling Operation.
6.			400	TL	Taking the micrometer
7.	Check the dimension	I	1600	I	Check the dimension
8.			400	U	Open the vice
9.	Remove the work piece	TL	400		

3.5 PRINCIPLES OF MOTION ECONOMY

The **principles of motion economy** form a set of rules and suggestions to improve the manual work in manufacturing and reduce fatigue and unnecessary movements by the worker, which can lead to the reduction in the work related trauma.

The principles of motion economy can be classified into four groups:

1. Principles related to the use of *human body*,
2. Principles related to the arrangement of the *work place*,
3. Principles related to the *design of tools and equipment*.
4. Principles related to *time conservation*.

Use of Human Body

- The two hands should begin motions at the same time.
- The two hands should not be idle at the same time except during rest periods.
- Motions of the arms should be made in opposite and symmetrical directions and should be made simultaneously
- Hand motions should be confined to the lowest classification with which it is possible to perform the work satisfactorily:

1. Finger motions
2. Wrist motions

3. Forearm motions
 4. Upper arm motions
 5. Shoulder motions
- Momentum should be employed to assist the worker whenever possible, and it should be reduced to a minimum if it must be overcome by muscular effort.
 - Smooth continuous motions of the hands are preferable to zigzag motions or straight-line motions involving sudden and sharp changes in direction.
 - Ballistic (i.e. free swinging) movements are faster, easier and more accurate than restricted or controlled movements.

Arrangement of the Work Place

- There should be a definite and fixed place for all tools and materials.
- Tools, materials, and controls should be located close in and directly in front of the operator.
- Drop delivers should be used whenever possible.
- Materials and tools should be located to permit the best sequence of motions.
- Arrange the height of the workplace and chair for alternate sitting and standing, when possible.
- Provide a chair of the type and height to permit good posture.

Design of Tools and Equipment

1. Combine tools whenever possible.
2. Preposition tools and materials.
3. Where each finger performs some specific movement, the load should be distributed in accordance with the inherent capacities of the fingers.
4. For light assembly, a screwdriver handle should be smaller at the bottom.
5. Momentum should be used to help the worker in doing their task not to increase their task.

Time Conservation

- Even a temporary delay of work by a man or machine should not be encouraged.
- Machine should not run idle, it is not desirable that a lathe machine is running and its job is rotating but no cut is being taken.
- Two or more jobs should be worked upon at the same time or two or more operations should be carried out on a job simultaneously if possible.
- Number of motion involved in completing a job should be minimized.

3.6 WORK MEASUREMENT

International Labour Organization(ILO) defined work measurement as ‘application of techniques designed to establish the work content of a specified task by determining the time required for carrying it out at a defined standard of performance by a qualified worker’. Conventionally, it is known as time study, which is primarily carried out to determine the standard time required to perform a specific task. Such time standards are used for planning and scheduling work, for cost estimating or for labour cost

control. Otherwise, it may serve as the basis for a wage incentive plan. But we find it has wide application in deciding a wage incentive plan.

Techniques of Work Measurement:

There are different techniques of work measurement. However, the following are the principal techniques:

1. Time study
2. Ratio-delay study (Statistical Sampling Technique)
3. Synthesis from standard data
4. Pre-determined motion time standard
5. Analytic estimating

Out of all these, only the time study technique is widely used because others are complicated in nature. Here also, we will discuss in detail the time study only, while simply defining the other techniques.

Time Study:

ILO defined time study as ‘a technique for determining as accurately as possible from a limited number of observations, the time necessary to carry out a given activity at a defined standard of performance’. For carrying out a time study, equipment’s such as stopwatch, study board, pencils, slide rule, etc. are required.

The different types of stopwatches include:

1. Stopwatches that record one minute per revolution by intervals of one- fifth of a second with a small hand recording 30 minutes.
2. Stopwatches that record one minute per revolution, calibrated in one- hundredth of a minute with a small hand recording 30 minutes.
3. Decimal-hour stopwatches recording one-hundredths of an hour with a small hand recording up to one hour in 100 divisions.

The following steps are necessary for carrying out a time study for the measurement of work:

- a. To collect and complete all available information about the job, which should also include the surrounding conditions and also the attributes of the operators, which are likely to affect the work
- b. To record the details of the methods and also to break down different operations into their elements
- c. To record the time taken by the operators to perform the operation (element-wise) measuring preferably with a timing device such as a stopwatch
- d. To assess the working speed of the operators by comparing the same with a predetermined normal speed
- e. To convert the observed time to normal time
- f. To decide the rate of allowances that may be given over and above the normal time of the operation
- g. To determine the allowed time for operation

Time Study

Time study may be defined as “the art of observing and recording the time required to do each detailed element of an industrial activity/operation.”

The term industrial activity includes mental, manual and machining operation, where:

- (i) Mental time includes time taken by the operator for thinking over some alternative operations.
- (ii) Manual time consists of three types of operations i.e. related with handling of materials, handling of tools and handling of machines.
- (iii) Machining time includes time taken by the machines in performing the requisite operations. Thus time study standardizes the time taken by average worker to perform these operations.

Objectives of Time Study

- (1) Target time for each job can be scientifically estimated. With this estimate realistic schedules and manpower requirements can be prepared.
- (2) Sound comparison of alternative methods is possible by comparing their basic times.
- (3) Useful wage incentive schemes can be formulated on the basis of target times.
- (4) It can lead to proper balancing of the work distribution.
- (5) It can help to analyse the activities for performing a job with the view to eliminate or reduce unnecessary or repetitive operations so that human effort can be minimized.
- (6) To standardise the efficient method of performing operations.
- (7) To standardise conditions for efficient performance.
- (8) To determine man and machines ratio for effective and efficient utilisation of both.
- (9) To provide information and basis for production planning and scheduling activities.

Basic Procedure for Time Study:

The following steps are essential for carrying out the time study of any process in order to determine standard time:

- 1. Select the job for study and define the objective of the study. This needs statement of the use of results, precision required and the desired level of confidence in the estimated time standards.
- 2. Analyse the operation to determine if the standard method and conditions exist and the worker is properly trained. The method study or training of operator should be completed before starting time study if need be.
- 3. Select the operator to be studied if more than one can perform the task.
- 4. Record the information about the standard method, operator, operation, product machine, quality required and working conditions.
- 5. Divide the operation into reasonably small elements.
- 6. Time the operator for each of the elements. Estimate the total number of observations to be taken.
- 7. Collect and record the data of required number of cycles by way of timing and rating the operator.
- 8. For each element of operation note the representative watch time and calculate the normal time as follows: Normal Time = Observed time x Rating Factors

Calculate normal time for the whole job by adding normal time of various elements.

9. Determine standard time by adding allowances to normal time of operation. Some allowances such as personal allowance (20%), fatigue allowance (5%) preparation allowance (5%) are generally used or these can be taken from the company's policy book or by conducting an independent study.

Thus standard time = Normal time + Allowances

Advantages of Time Study

The advantages of time study are the applications to which time standards are put. Generally they are required (a) to plan future production and (b) to evaluate performance of workers or of the production system as a whole.

(a) To plan future production they are required for:

- (1) Estimation of equipment or machinery requirements.
- (2) Estimation of production cost per unit in order to determine/fix selling price.
- (3) Estimating manpower requirements.
- (4) Taking makes or buys decisions.
- (5) Balancing the work of operators constituting a group.
- (6) Estimating delivery dates.
- (7) Estimating times required for a specified inventory level generation.
- (8) Estimation of labour costs.
- (9) Making alternative investment decision in purchasing machines/equipment.
- (10) Developing efficient plant layout and material handling arrangement.

(b) Time standards help in evaluation of performance as in following cases:

- (1) The evaluation of productivity of various workstations.
- (2) The development of individual or group incentive schemes for above average performance.

Limitations of Time Study:

- (1) Standards cannot be established by this technique on jobs which are not well defined.
- (2) In situations where it is difficult to define quality precisely standards and production incentives may cause deterioration in quality levels.
- (3) It is not possible to maintain standards where piece rate system of wage payment exists.
- (4) Labour unions may oppose the application of time study where they are strong.
- (5) Time study is applicable only where the work is visible. So it can be applied only in manual job and not for thinking portions of the job.
- (6) Only specific type of jobs which have identifiable starting and ending points can be timed accurately.

Length of Time Study:

The length of time study will affect the accuracy where bias may result if the period of study is too short to include all the significant variables. Traditional time study practice usually limits the study to observing net cycle time over a relatively short time period to include essential between-cycle variations.

Consideration is usually given to between-hour and between day variations by taking the observations during, say, the middle of the morning work period and during the second or third day of the work week, when it is assumed that the production rate is average as far as these sources of variation are concerned.

Number of Cycles to be Timed:

Time required to perform an element of an operation varies slightly from cycle to cycle. However, this variation can be reduced to some extent by using highly standardized raw materials, good tools and equipment's, good working conditions and a qualified and well trained operator, but the variation cannot be avoided completely.

Time study is a sampling process, therefore, large number of readings should be taken so that the results will be representative of the activity being measured and the desired accuracy may be obtained.

Experience has shown that 95% confidence level and $\pm 5\%$ accuracy is sufficient for time study. From statistics, for this condition, number of cycles to be timed can be found by the following relation.

$$N = \left\{ \frac{40 \sqrt{M \sum X^2 - (\sum X)^2}}{\sum X} \right\}^2$$

where N = Required number of observations to predict the little time with in $\pm 5\%$ accuracy and 95% confidence level.

X = Each stop watch reading.

M = No. of stop watch readings.

3.7 ALLOWANCES AND STANDARD TIME

Allowances in time study can be defined as the extra time figures which are to be added to the basic time of an operation to account for personnel desires, delays, fatigue of operators, any special situation and the policies of the firm or organization. Standard time of a job is obtained by adding various allowances to the basic or normal time of the job.

These allowances are considered or provided to compensate the worker/operator for the production interruptions that may occur due to his personnel legitimate needs or the factors beyond his controls. For example the delay may occur due to operators personnel needs such as drinking water, taking tea, going

to toilet etc., unavoidable delays like waiting for tools, materials or equipment, maintenance of machine and periodic inspection of parts/materials.

The fundamental purpose of allowances is to add enough time to the basic time of the production in order to enable the average worker to meet the standard while performing at a normal pace. The determination of allowances is probably the most controversial part of work study.

Reasons:

1. Individual factors:

A thin, alert and active worker requires a smaller allowance to recover from fatigue than an inept, dull and obese worker. In the similar manner, every worker conducts his work according to the Learning Curve which is unique for everyone.

2. Nature of work:

Allowances calculated or determined for light or medium work are not acceptable for operation involving very hard work and which is done under very difficult conditions. For example, the work involving more eye movement, more physical work (by hands or by legs) or more mental work needs greater allowances than that of light easy and work involving very less physical work or movements.

Some factors inherent to the job such as gloves or masks to be worn while working, constant danger or risk regarding the surface finish or quality of the work etc. varies from job to job. Thus, the determination of allowances becomes more difficult and controversial.

3. Environmental factors:

While determining the relaxation allowance, certain factors like heat, humidity, vibration, dust, light intensity, noise level etc. have to be taken into account. These are called the environmental factors and these factors are of seasonal nature. These factors are more significant for workers conducting the work under open air or where the environmental factors affect the work such as work in a construction company or in shipyards.

Type of Allowances:

1. Relaxation Allowance:

Relaxation allowance is the most essential part of the time added to the basic time. Other allowances like contingency allowances, policy allowances or other special allowances are applied under certain conditions only. Relaxation allowances are added so as to allow the worker or operator to recover from fatigue.

Fatigue can be defined as mental or physical weariness, existing in a person which adversely affects his efficiency in working. Now, this fatigue can be licensed to come extent by some rest breaks, during which

the body part, gets relaxed and recovers from exertion. It can also be lessened by lowering down the rate of working.

Relaxation allowances are added element by element to their basic times so as to obtain the work value of each element separately. After that, the element standard times are added so as to obtain the standard time for the job or operations. Allowances for climatic conditions are applied to the working shift rather than to element or the jobs.

Relaxation allowances itself can be sub-divided into two categories:

1. Fixed Allowances:

Fixed allowances consist of the allowances given for personnel needs or desire. These personnel needs includes going for watching getting a drink etc. It is also agreed that women need more personnel time than man. The usual percentage of the personnel allowances is 5% for male workers and 7% for women.

Fixed allowance also includes the allowances for the basic fatigue. This allowances is given to take account of the energy expended during work and to remove the monotony. Usually, it is taken as 4% a worker who carried out the job while seated, who is engaged in a light work under good working conditions and makes only the normal use of hands, legs and his senses.

2. Variable Allowances:

In calculating the relaxation allowance, variable allowances are added to the fixed allowances which are made, depending upon the circumstances of the job. Variable addition is given be improved.

Following factors are to be taken into account:

1. Standing
2. Weightlifting applications.
3. Light conditions.
4. Abnormal posture or position.
5. Visual strains.
6. Air conditions or availability.
7. Aural strenes.
8. Mental stress.
9. Mental monotony.
10. Physical monotony.



2. Contingency Allowances:

It is also known as delay allowance. It accounts for some other contingencies such as unexpected work as well. “A contingency allowance is an allowances of time that is to be added in normal time or to be included in standard time to meet the legitimate and unexpected items of work or delays, the precise measurement of which is not economical due to their irregular and infrequent nature of occurrences.”

The qualifying statement in the above mentioned definition is 'legitimate.' So, the contingency allowances are given for legitimate contingencies it can be explained with the help of 'delay'. Delays can be avoidable and unavoidable.

These delays which are intentional on the part of worker are avoidable and thus not legitimate. So, no allowance is provided to them. But allowances are given for unavoidable delays which are beyond the control of the worker. For example, breakdowns, of machines, shortage of raw materials, etc.

Some other minor works may come up occasionally. For example, checking the machine setting, rework and measurement. Contingency allowances are to be given for these. Now, the amount of the contingency allowances to be added is expressed as a percentage of basic time.

Usually, it is taken as 5%. These allowances are to be added only when the analyst is completely satisfied that the minor extra work or the unavoidable delays cannot be eliminated. It should never be taken as the loosening factor.

3. Other and Special Allowances:

There are some other and special allowances which are to be added for certain conditions. These allowances are provided for certain reasons and for come specific period of time. Some of their allowances are:

1. Special Allowances:

As the name suggests, these allowances are added for some special conditions. Most of these allowances are temporary and are taken out as soon as conditions normalise.

These include the following:

(a) Start up, shut down, and tooling:

These allowances are given per work period. These allowances compensate for the time loss during cleaning and tooling periods and the shut down at the end of the work period.

(b) Setup and change over:

These types of allowances are provided when a new type of product is started. There include the time taken in set up of machine. Dismantling the previous work piece or tool for new set up is also included in it as the worker is forced to be idle when the new set up is mounted or the previous one is being withdrawn. These allowances are withdrawn as soon as the production starts completely.

(c) Rejection reworks and excess work:

A rejection allowance is provided to compensate for rejection of work. This product rejection is inherent in much process because of come uncontrollable factors. Reworks also present this similar picture. So,

these allowances are added for certain conditions when the work is not being done under standard conditions and some excess work results.

(d) Learning, training and implementation:

And the name suggests, these allowances are given to the trainees before they acquire full ability to perform a task within the actual standard time. Some of the workers are engaged in implant training of new operatives and thus some of their production time is lost in this process. Thus, at this stage, training allowances are added.

Third is the implementation allowance, which is given to the workers when they are encouraged to work on some new processes. It is provided to achieve the whole hearted co-operation from the workers to new methods or processes without any fear of loss of their earning. All the above mentioned allowances are temporary in nature.

2. Policy Allowances:

Policy allowance is not a part of time study. It can be permanent or temporary in nature. It is applied according to the policy of the firm in which the time standard is to be applied. Policy allowance is given to honour the wage agreements that are made by the employers with the trade unions.

A policy allowance can be defined as follows:

“It is an increment (other than the bonus increment) applied to the standard time to achieve a satisfactory level of earnings for a particular level of performance under the exceptional circumstances.”

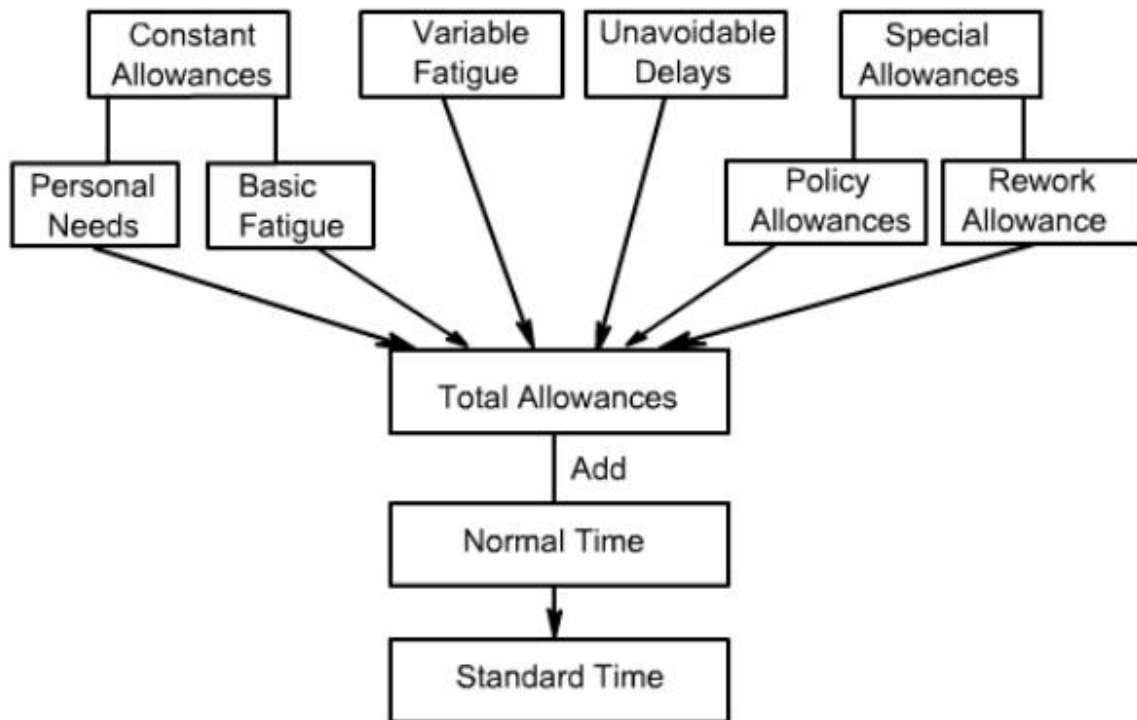
A policy allowance can be added as a temporary factor to compensate for any imperfections in the functioning of a firm, But, in this situation, it should be withdrawn as soon as the condition normalises. So, the policy allowances should be added with due caution and under well defined circumstances.

Standard Time

Standard time is the time required by an average skilled operator, working at a normal pace, to perform a specified task using a prescribed method. It includes appropriate allowances to allow the person to recover from fatigue and, where necessary, an additional allowance to cover contingent elements which may occur but have not been observed.

The standard time can be determined using the following techniques:

1. Time study,
2. Predetermined motion time system aka PMTS or PTS,
3. Standard data system,
4. Work sampling.



Method of Calculation:

The Standard Time is the product of three factors:

1. Observed time: The time measured to complete the task.
2. Performance rating factor: The pace the person is working at. 90% is working slower than normal, 110% is working faster than normal, 100% is normal. This factor is calculated by an experienced worker who is trained to observe and determine the rating.
3. Personal, fatigue, and delay (PFD) allowance.

The standard time can then be calculated by using:

$$\text{Standard Time} = (\text{Observed Time}) (\text{Rating Factor}) (1 + \text{PFD Allowance})$$

Example:

In making a time study of a laboratory technician performing an analysis of processed food in a canning factory, the following times were noted for a particular operation.

Run	1	2	3	4	5	6	7	8	9	10	11	12
Operation time (sec.)	21	21	16	19	20	16	20	19	19	20	40	19
Run	13	14	15	16	17	18	19	20	21	22	23	24
Operation time (sec.)	21	18	23	19	15	18	18	19	21	20	20	19

If the technician's performance has been rated at 120 percent, and the company policy for allowance (personal, fatigue, etc.) stipulates 13 percent,

- Determine the normal time.
- Determine the standard time.

Watch readings falling 50 % above and 25 % below the average may be considered as abnormal.

Ans:

$$T_{av} = \frac{\Sigma \text{cycle time}}{\text{No. of cycles}} = \frac{481}{24} = 20.04 \text{ s.}$$

$$1.5 T_{av} = 30 \text{ s.}$$

$$0.75 T_{av} = 15 \text{ s.}$$

Discarding the time values which are greater than $1.5 T_{av}$ or less than $0.75 T_{av}$,

$$\text{the average observed cycle time} = \frac{441}{23} = 19.2 \text{ s.}$$

$$\text{Normal time} = 19.2 \times \frac{120}{100} = 23.04 \text{ s.}$$

Standard time = normal time + allowances

$$\begin{aligned} &= 23.04 \times \frac{100}{100 - 13} \\ &= 26.5 \text{ seconds.} \end{aligned}$$

3.8 PERFORMANCE RATING

Performance level or a pace is selected as standard to perform certain task. Rating means gauging and comparing the performance or pace rate of a worker against a standard performance level set by the time study analyst. The analyst observes the performance, compares it with other paces and learns to judge pace level as percent of the standard pace.

In actual work measurement situation, the time study engineer/analyst simultaneously observes the actual time taken to perform a particular task and rates the performance.

Performance rating can be calculated as:

$$\text{Performance Rating} = \frac{\text{Observed Performance}}{\text{Normal Performance}} \times 100$$

The context of normal performance must be such that the time standards set from it must be within the working capacity of majority of workers in the unit. It would be of no use if working standards are too high that only few best workers could attain them since the estimates based on their performance would never be fulfilled.

Similarly, low performance standards would lead to high earnings of some workers leading to excessively high labour cost and large difference in earnings of the workers.

Rating Factor:

The time study analyst multiplies actual time or observed time with factor called “Rating Factor” or “Leveling Factor” to set the normal time which a normal worker would take. This is expressed as a percentage of the efficiency of representative worker/operator which shows how efficient an operator is in comparison to some of his fellow operators.

System of Performance Rating:

Various rating techniques are as follows:

- (1) Speed Rating
- (2) Westing-house System of Rating
- (3) Synthetic Rating
- (4) Objective Rating

(1) Speed Rating:

This rating procedure consists of judging the speed of the operator’s movements in relation to a normal pace and is noted as a factor. Rating can be applied to different elements of the job and mathematically

$$\text{Normal Time} = \text{Observed Time} \times \text{Workers Speed/Speed Expected from the Worker}$$

(2) Westing-house System of Rating:

A four factors system performance Rating was developed at Westing house by C.E. Bedaux.

These four factors are:

- (i) Skill
- (ii) Effort
- (ii) Conditions and
- (iv) Consistency.

The numerical values for each of these factors are available in tabular form. The operator was watched while working and given a value from these tables. The observed time is then levelled by applying the values of four rating factors. Table gives the values of these factors.

The observed time for an element of the job was 2 minutes. The ratings were as given below:

Skill excellent B₁
Effort Excessive A₂
Condition Good C



Consistency Good C

Determine the normal time for element under observation.

Performance Rating Table

Skill	Effort	Conditions	Consistency
Super $A_1 = 0.15$ $A_2 = 0.13$ Excellent $B_1 = + 0.11$ $B_2 = + 0.08$ Good $C_1 = + 0.06$ $C_2 = + 0.03$ Average $D = 0.00$ Fair $E_1 = - 0.05$ $E_2 = - 0.10$ Poor $F_1 = - 0.16$ $F_2 = - 0.22$	Excessive $A_1 = 0.13$ $A_2 = 0.12$ Excellent $B_1 = + 0.10$ $B_2 = + 0.08$ Good $C_1 = + 0.05$ $C_2 = + 0.02$ Average $D = 0.00$ Fair $E_1 = - 0.04$ $E_2 = - 0.08$ Poor $F_1 = - 0.12$ $F_2 = - 0.17$	Ideal $A = + 0.06$ Excellent $B = + 0.04$ Good $C = + 0.02$ Average $D = 0.00$ Fair $E = - 0.03$ Poor $F = - 0.07$	Perfect $A = + 0.04$ Excellent $B = + 0.03$ Good $C = + 0.01$ Average $D = 0.00$ Fair $E = - 0.02$ Poor $F = - 0.04$

Solution. Form Table 19.1 the numerical values for the above ratings are

Skill	:	0.11
Effort	:	0.12
Condition	:	+ 0.02
Consistency	:	+ 0.01
Total	=	0.26 i.e. final or overall rating.
The normal time	=	Observed time $\times$ Rating factor
	=	$2 \times \frac{126}{100} = 256 \text{ min.}$ Ans.

(Note: In case the final rating value comes –ve then normal time will be less than the observed time)

(3) Synthetic Rating:

This system of rating was proposed by Morrow. In this system the pace rate of an operator is evaluated with the help of predetermined motion time standards (P.M.T.S.). Thus the time study observer notes down the actual time of performance for the elements as done in the previous method.

Performance times for many such elements are standardised and tabulated. These are known as 'Predetermined Motion Time Values' from this Table the predetermined performance time for the elements are also noted down. The ratio of the latter to the former (average actual time) gives the rating factor. Therefore, symbolically,

$$R = P/A$$

where R = Performance rating factor.

P = Predetermined motion time standard for the element expressed in minutes.

A = Average actual time (observed time) for the same element, in minutes.

Example 2:

The average actual times for 8 elements of a job were measured and tabulated below. The predetermined motion times for two element i.e. 1 and 3 are also given. Calculate the performance rating factor.

Element Number	1	2	3	4	5	6	7	8
Average actual time A.	0.12	0.32	0.20	0.52	0.60	0.22	0.15	0.10

Suppose we chose the element 1 and 3 and determine the predetermined time values by using a P.M.T. system. Suppose these are 0.13 and 0.19 minutes respectively for elements 1 and 3. So the

systematic rating factor for element 1 is $\frac{0.13}{0.12} = 108\%$ and for element 3 is $\frac{0.19}{0.20} = 95\%$

Average R = $108 + 95 = 102\%$

The average rating factor for all the elements = 102%

Element Number	1	2	3	4	5	6	7	8
Average Rating Factor	102	102	102	102	102	102	102	102

(4) Objective Rating:

This system of rating was developed by M.E. Mundel. In this operators tempo of doing a job is taken into account. In addition to this job difficulty or complexity is also taken into consideration.

First of all the time study observer rates the work on the basis of his workplace of doing a work. This is called a pace rating. After the pace rating is done, an allowance is added to the pace rating to take care of the job difficulty.

The following six factors constitute job difficulty:

- (i) Amount of body used
- (ii) Foot pedals
- (iii) Bimanualness
- (iv) Eye-hand coordination
- (v) Handling requirements, and
- (vi) Weight.

Example: 3:

The observed time for an element of job is 0.72 minutes. The pace rating is 80% and the job difficulty factors given below. Find the normal time.

Amount of Body Used	% age Adjustments
Foot pedals	5
Binmanualness	2
Eye hand coordination	5
Handling requirements	3
Weight	5
Total adjustment for job difficulty	20%

Solution. Normal Time = Observed time × Pace rating × Secondary adjustment

$$= 0.72 \times \frac{80}{100} \times \frac{120}{100}$$

$$= 0.68 \text{ minutes} \quad \boxed{\text{Ans.}}$$

Here the total secondary adjustment allowance work out to be 20% but we have multiplied by since due to job complexities a worker is handicapped and put to difficulty. Because of this he cannot work at his normal pace. On account of this he should be allowed little more time than the observed time.

3.9 JOB EVALUATION

Concept of job evaluation:

In simple words, job evaluation is the rating of jobs in an organisation. This is the process of establishing the value or worth of jobs in a job hierarchy. It attempts to compare the relative intrinsic value or worth of jobs within an organisation. Thus, job evaluation is a comparative process.

According to the International Labour Office (ILO) “Job evaluation is an attempt to determine and compare the demands which the normal performance of a particular job makes on normal workers, without taking into account the individual abilities or performance of the workers concerned”.

We may define job evaluation as a process used to establish the relative worth of jobs in a job hierarchy. This is important to note that job evaluation is ranking of job, not job holder. Job holders are rated through performance appraisal. Job evaluation assumes normal performance of the job by a worker. Thus, the process ignores individual abilities of the job holder.

Job evaluation provides basis for developing job hierarchy and fixing a pay structure. It must be remembered that job evaluation is about relationships and not absolutes. That is why job evaluation cannot be the sole determining factor for deciding pay structures.

Objectives of job evaluation:

1. Provide a standard procedure for determining the relative worth of each job in a plant.
2. Determine equitable wage differentials between different jobs in the organisation.
3. Eliminate wage inequalities.
4. Ensure that like wages are paid to all qualified employees for like work.
5. Form a basis for fixing incentives and different bonus plans.
6. Serve as a useful reference for setting individual grievances regarding wage rates.

7. Provide information for work organisation, employees' selection, placement, training and numerous other similar problems.
8. Provide a benchmark for making career planning for the employees in the organisation.

Advantages of job evaluation:

1. Job evaluation being a logical process and objective technique helps in developing an equitable and consistent wage and salary structure based on the relative worth of jobs in an organisation.
2. By eliminating wage differentials within the organisation, job evaluation helps in minimizing conflict between labour unions and management and, in turn, helps in promoting harmonious relations between them.
3. Job evaluation simplifies wage administration by establishing uniformity in wage rates.
4. It provides a logical basis for wage negotiations and collective bargaining.
5. In the case of new jobs, job evaluation facilitates spotting them into the existing wage and salary structure.
6. In the modern times of mechanisation, performance depends much on the machines than on the worker himself/herself. In such cases, job evaluation provides the realistic basis for determination of wages.
7. The information generated by job evaluation may also be used for improvement of selection, transfer and promotion procedures on the basis of comparative job requirements.
8. Job evaluation rates the job, not the workers. Organisations have large number of jobs with specialisations. It is job evaluation here again which helps in rating all these jobs and determining the wages and salary and also removing ambiguity in them.

Drawbacks of job evaluation:

1. Job evaluation is susceptible because of human error and subjective judgment. While there is no standard list of factors to be considered for job evaluation, there are some factors that cannot be measured accurately.
2. There is a variation between wages fixed through job evaluation and market forces. Say Kerr and Fisher, the jobs which tend to rate high as compared with the market are those of junior, nurse and typist, while craft rates are relatively low. Weaker groups are better served by an evaluation plan than by the market, the former places the emphasis not on force but on equity”.
3. When job evaluation is applied for the first time in an organisation, it creates doubts in the minds of workers whose jobs are evaluated and trade unions that it may do away with collective bargaining for fixing wage rates.
4. Job evaluation methods being lacking in scientific basis are often looked upon as suspicious about the efficacy of methods of job evaluation.
5. Job evaluation is a time-consuming process requiring specialised technical personnel to undertake it and, thus, is likely to be costly also.
6. Job evaluation is not found suitable for establishing the relative worth of the managerial jobs which are skill-oriented. But, these skills cannot be measured in quantitative terms.
7. Given the changes in job contents and work conditions, frequent evaluation of jobs is essential. This is not always so easy and simple.
8. Job evaluation leads to frequent and substantial changes in wage and salary structures. This, in turn, creates financial burden on organisation.

Procedure of Job Evaluation:

The procedure of job evaluation involves the comparison of job factors and their relative effectiveness. Job factors are of premier importance's that are taken into consideration as to what degrees of them are present in each job to make comparison meaningful. Skill, efforts, responsibility and working conditions are the factors that are considered for determining the relative worth of job. These factors decide the compensation to be paid for each job.

1. Analyze and prepare job description,
2. Select and prepare a job evaluation plan,
3. Classify jobs,
4. Install the programme,
5. Maintain the programme.

1. Job Analysis and Job Description:

Job evaluation cannot be done without the necessary information in respect of job. This information can be had through job analysis. Job analysis provides the information such as job requirements i.e. machines tools and materials to be used and time for its performance and the activities involved, time for its completion, skill required for its performance etc.

Job description is a by-product of job analysis. The information collected through job analysis is presented in a statement as how it is to be done, what job holder has to do, why it is to be done. This information in the statement form is job description. This is the vital information required for initiation of job evaluation.

2. Job Classification:

Job classification or gradation of jobs is determining the relative levels of different jobs in the organisation. Jobs can be classified into various grades or classes. Technical, non technical, clerical, accounting, supervisory etc. or class I, class II, class III, Class IV etc. according to responsibilities and abilities for performance.

3. Job Pricing:

Pricing the job is the very purpose of job evaluation. The different compensable factors are taken into consideration, factors divided into degrees and money values are assigned. There are four methods of job evaluation. Any method that suits to the need of the organisation can be adopted.

A care is taken to remove inconsistencies from the wage and salary structure. There are two types of inconsistencies, internal and external. Internal inconsistency exists in the wage, structure of the organisation, e.g. a supervisor getting the salary less than his subordinate speaks for internal inconsistency.

External inconsistency refers to the wage differentials between two organisations of the same cadre or between two industries or between two states or regions. As far as possible these inconsistencies should be removed.

Methods of Job Evaluation

1. Non-quantitative Methods:

- (a) Ranking or Job Comparison
- (b) Grading or Job Classification

2. Quantitative Methods:

- (a) Point Rating
- (b) Factor Comparison



The basic difference between these two methods lies in the sense that, under non-quantitative methods, a job is compared as a whole with other jobs in the organisation, whereas in case of quantitative methods, the key factors of a job are selected and, then, measured. The four methods of job evaluation are now discussed one by one.

Ranking Method:

The ranking method is the simplest form of job evaluation. In this method, each job as a whole is compared with other and this comparison of jobs goes on until all the jobs have been evaluated and ranked. All jobs are ranked in the order of their importance from the simplest to the hardest or from the highest to the lowest.

The importance of order of job is judged in terms of duties, responsibilities and demands on the job holder. The jobs are ranked according to “the whole job” rather than a number of compensable factors. The ranking of jobs in a University, based on Ranking Method, may be like this:

Table : Ranking of University Jobs:

Ranking Order	Pay Scale
Professor/Registrar	Rs. 16,40(M50-20,900-500-22,400
Reader/Dy. Registrar	Rs. 12,000-420-18,300
Lecturer/Asst. Registrar	Rs. 8,000-275-13,500

Merits:

- 1. It is the simplest method.
- 2. It is quite economical to put it into effect.
- 3. It is less time consuming and involves little paper work.

Demerits:

1. The main demerit of the ranking method is that there are no definite standards of judgment and also there is no way of measuring the differences between jobs.
2. It suffers from its sheer unmanageability when there are a large number of jobs.

Grading Method:

Grading method is also known as ‘classification method’. This method of job evaluation was made popular by the U.S. Civil Service Commission. Under this method, job grades or classes are established by an authorised body or committee appointed for this purpose. A job grade is defined as a group of different jobs of similar difficulty or requiring similar skills to perform them. Job grades are determined on the basis of information derived from job analysis.

The grades or classes are created by identifying some common denominator such as skills, knowledge and responsibilities. The example of job grades may include, depending on the type of jobs the organisation offers, skilled, unskilled, account clerk, clerk-cum-typist, steno typist, office superintendent, laboratory assistant and so on.

Once the grades are established, each job is then placed into its appropriate grade or class depending on how well its characteristics fit in a grade. In this way, a series of job grades is created. Then, different wage/salary rate is fixed for each grade.

Merits:

1. This method is easy to understand and simple to operate.
2. It is economical and, therefore, suitable for small organisations.
3. The grouping of jobs into classifications makes pay determination problems easy to administer.
4. This method is useful for Government jobs.

Demerits:

1. The method suffers from personal bias of the committee members.
2. It cannot deal with complex jobs which will not fit neatly into one grade.
3. This method is rarely used in an industry.

Points Rating:

This is the most widely used method of job evaluation. Under this method, jobs are broke down based on various identifiable factors such as skill, effort, training, knowledge, hazards, responsibility, etc. Thereafter, points are allocated to each of these factors.

Weights are given to factors depending on their importance to perform the job. Points so allocated to various factors of a job are then summed. Then, the jobs with similar total of points are placed in similar pay grades. The sum of points gives an index of the relative significance of the jobs that are rated.

Merits:

1. It is the most comprehensive and accurate method of job evaluation.
2. Prejudice and human judgment are minimised, i.e. the system cannot be easily manipulated.
3. Being the systematic method, workers of the organisation favour this method.
4. The scales developed in this method can be used for long time.
5. Jobs can be easily placed in distinct categories.

Demerits:

1. It is both time-consuming and expensive method.
2. It is difficult to understand for an average worker.
3. A lot of clerical work is involved in recording rating scales.
4. It is not suitable for managerial jobs wherein the work content is not measurable in quantitative terms.

Factor Comparison Method:

This method is a combination of both ranking and point methods in the sense that it rates jobs by comparing them and makes analysis by breaking jobs into compensable factors. This system is usually used to evaluate white collar, professional and managerial positions.

The first step under this method is the selection of factors to be ‘ utilized for evaluation. The persons preparing the job descriptions should be given the proper explanation of these factors.

The universally used factors are:

- (i) Mental efforts
- (ii) Physical efforts
- (iii) Skill needed
- (iv) Responsibility and
- (v) Working conditions.

Merits:

1. It is more objective method of job evaluation.
2. The method is flexible as there is no upper limit on the rating of a factor.
3. It is fairly easy method to explain to employees.
4. The use of limited number of factors (usually five) ensures less chances of overlapping and over-weighting of factors.
5. It facilitates determining the relative worth of different jobs.

Demerits:

1. It is expensive and time-consuming method.
2. Using the same five factors for evaluating jobs may not always be appropriate because jobs differ across and within organisations.
3. It is difficult to understand and operate.

3.10 MERIT RATING

Merit Rating is also known as performance appraisal or performance evaluation. It is a systematic process for measuring the performance of the employees in terms of job requirements.

It utilizes various rating techniques for comparing individual employees in a work group in terms of personal qualities or deficiencies and the requirements of their respective jobs. It is an established fact that people differ in their abilities and aptitudes. These differences are natural to a great extent and cannot be eliminated even by providing same training and education facilities to them.

There will be some differences in the quality and quantity of work done by different workers even on the same job. Therefore it is essential for the management to know these differentials so that employees having better abilities may be rewarded and the wrong selection and placement maybe restricted or avoided.

“Merit-rating of an employee is the process of evaluating the employee’s performance on the job in terms of the requirements of the job”. It is a technique of assessing the worth of an employee with reference to job requirements.

Objectives of Merit Rating

1. To assess the work of employees in relation to their job requirements.
2. To consider employees/workers for promotions, transfer, layoffs etc.
3. To assess the good and bad points in working of employees and then making suggestions for improvement.
4. To help in wage and salary administrations and taking decisions about incentives and increments to be given to the workers.
5. To evaluate skill and training capabilities of employees and helping in planning suitable training and development programmes for workers.
6. To know the problems faced by workers while doing various jobs.
7. To provide a basis for comparison to segregate efficient and inefficient workers.
8. To help management in placement/transfer to workers according to their capacity, interest, aptitude and qualifications.
9. To help supervisors to know their subordinates more closely for increasing their efficiency and improving productivity.

Advantages of merit rating are as follows:

- (a) It provides a scientific basis for judging the worth of employees. They try to improve upon their performance if it is not up to the satisfaction of the employer.
- (b) It provides a basis for decisions like promotion, demotion, transfer or termination of employees. Better persons are selected for promotion. The systematic evaluation of employees is also kept as a permanent record.
- (c) It helps in distinguishing between efficient and inefficient workers. In this way, it reveals the defects in the selection procedure. Those employees who are misfits may be spotted and appropriate action taken against them.

- (d) Workers may be given increase in pay if their performance is good. It helps the management in avoiding spot judgments and replaces it with planned decisions.
- (e) Objective merit rating develops confidence among the employees if the methods of evaluation are systematic and impartial. Among the workers, a sense of competition is developed resulting into increased output or better performance.
- (f) It helps in creating a congenial working environment in which employer- employee relations are improved. Subordinates get motivated and work hard for getting favorable rating.
- (g) It also helps in stimulating and development of an employee as it points out the weakness of the employees. The training needs of employees can be known and training programmes can be accordingly drawn.
- (h) A systematic evaluation also develops better supervisors and executives. On the basis of merit-rating report, the top management can judge the ability of executives who are writing such reports.

However, formal merit-rating may not take place in case of a small concern since the informal rating can provide all the desired information. In case of a large scale concern, both employer and employee stand benefited from a systematic performance appraisal.

Limitations of merit rating:

1. Halo Effect:

There is a tendency to rate the employee on the basis of one factor only. It is also known as ‘blending tendency’. If the rater finds that the man is good in one factor he may rate him good in all other factors.

2. Clarity in Standards:

Each rater may apply his own standards with the result that final ratings simply cannot be compared. For example, a rater may think that ‘satisfactory’ rating is better than ‘excellent’.

3. Leniency or Strictness:

Lenient raters give high ratings whereas strict raters always give low ratings. Hence, there is a big difference of ratings between two raters. A supervisor may feel that low ratings may reflect his own weakness in dealing with workers.

4. Central Tendency:

Generally the raters evaluate employees by keeping them in the average category though some may be falling in the extreme ends of the scale viz., excellent or poor.

5. Influence of Higher-Paid Jobs:

Usually there is a tendency to give high rating to a person who is doing the highly paid job. Merit rating has nothing to do with the worth of the job (which is the subject matter of job evaluation).

6. Differing Perceptions:

Even if a person tries to be a fairest rater in performance evaluation, he cannot eliminate bias because of differing perceptions. Certain unconscious factors such as race, caste, and creed etc., effect merit rating. If a manager, who is less qualified, evaluates his subordinate who is more or better qualified, merit rating may not be fair. The limitations of merit-rating explained above can be eliminated to some extent by educating and training the raters.

Difference Between Merit Rating and Job Evaluation

1. Rating or job/Employees:

Job evaluation is related with the rating of the job whereas merit rating is a systematic process or evaluation of the capacities and abilities of the man doing the job.

2. Purpose:

The purpose of the job evaluation is to fix wages for the jobs by considering certain factors such as skill, responsibility etc. In case of merit rating, the purpose is to evaluate the employee for the purpose of promotion, training, pay increase or incentives etc.

3. Job difference/Individual differences:

Job evaluation considers the differences in jobs whereas merit rating recognizes the individual differences.

4. Objectives/Subjective Qualities:

The type of factors to be considered are objective in case of job evaluation, whereas in case of merit rating subjective qualities are considered.

5. Performance of individual versus job:

Job evaluation becomes the basis of determining wages whereas merit rating is undertaken to find out the efficiency of the individual for doing specific jobs.

6. Test of worth versus merit:

In job evaluation, an attempt is made to know the worth of a job in terms of certain factors such as responsibility and effort (mental as well as physical). Merit rating identifies the capacity and capability of an individual for doing specific job.

7. Uses:

Job evaluation is meant for determining the wages on an employee. On the other hand, merit rating is utilized for placement and training etc.

Methods of merit-rating

Methods of Performance Appraisal	
<i>Traditional Methods</i>	<i>Modern Methods</i>
1. Ranking method 2. Paired comparison 3. Grading 4. Forced distribution method 5. Forced choice method 6. Checklist method 7. Critical incidents method 8. Graphic scale method 9. Essay method 10. Field review method 11. Confidential report	1. Management by Objectives (MBO) 2. Behaviourally anchored rating scales 3. Assessment centres 4. 360-degree appraisal 5. Cost accounting method

1. Ranking:

Under this method, a man is compared with all others without considering any specific factors. A rank is prepared by placing the best at the top and the poorest in performance at the bottom. This method is simple and is suitable in case of small scale concerns.

However, the method is subject to following limitations:

- (a) It is not only difficult but rather simply impossible to compare a whole man with the whole men.
- (b) When a rank is prepared, it is not possible to know the difference between the two persons listed in the rank order.

The above limitations can be overcome to some extent if paired comparison method is followed. Here an employee's performance is compared with every other employee individually rather than whole men. Under this method the total number of comparisons will be as follows:

$$n(n-1)/2$$

N refers to the number of persons to be compared. The rank prepared on the basis of such comparisons will be definitely better than the simple rank. Suppose there are fifteen employees, the supervisor will have to make 105 comparisons i.e. $15(15-1)/2$. The method is suitable if the number of employees is less.

2. Grading:

Under this method, the performance of the employees is evaluated against certain grades such as poor, good, very good, excellent or outstanding, satisfactory or unsatisfactory. The grade which describes his performance can be allocated to him.

3. Graphic Scales:

Under this method, performance of an employee is evaluated against certain specific factors. Five degrees or scales are established for each factor and each degree is defined. This method is most widely used in merit rating and is also the oldest.

The Factors to be Selected are of Two Types:

- (1) Characteristics of employee viz., initiative, ability to learn, dependability, etc., and
- (2) Contributions of employee's viz., quality and quantity of output, safety record, etc.

For example, four factors selected for merit rating may be a) Quality of output. (b) Quantity of output (c) Dependability (d) Attitudes toward associates and superiors.

The scales or degrees may be constructed as follows:

Quality of Work: Poor - Fair - Good - Very - Good - Excellent

Usually the other factors considered for merit rating in addition to four discussed above are: co-operation, personality, Health, attendance, Knowledge of job, initiative, safety, potential for development.

For each factor we can name five degrees as explained above. The crucial part of this method is, therefore, the determination of factors and their degrees. The following definitions may be given for these degrees:

Poor—Lazy, not interested in his work.

Fair—does his job without any interest in his work.

Good—does his job with interest.

V. Good—Good keen employee.

Excellent — Exceptionally hard worker.

It may be pointed out that this method is just like 'Point System' of Job Evaluation. The rating of an employee may be done by putting a tick or marks from 0 to 20 i.e., for poor 0 marks and for excellent 20 marks.

Sample of Graphic Rating Scale Items and Format

Performance Factor	Performance Rating				
<i>Quality of work is the accuracy, skill, and completeness of work.</i>	☐ Consistently unsatisfactory	☐ Occasionally unsatisfactory	☐ Consistently satisfactory	☐ Sometimes superior	☐ Consistently superior
<i>Quantity of work is the volume of work done in a normal workday.</i>	☐ Consistently unsatisfactory	☐ Consistently unsatisfactory	☐ Consistently satisfactory	☐ Sometimes superior	☐ Consistently superior

Performance Factor	Performance Rating				
Job knowledge is information pertinent to the job that an individual should have for satisfactory job performance.	☐ Poorly informed about work duties	☐ Occasionally unsatisfactory	☐ Can answer most questions about the job	☐ Understands all phases of the job	☐ Has complete mastery of all phases of the job
Dependability in following directions and company policies without supervision.	☐ Required constant supervision	☐ Requires occasional follow-up	☐ Usually can be counted on	☐ Requires very little supervision	☐ Requires absolute minimum of supervision

4. Man-To-Man Comparison:

This method resembles with the factor-comparison method of job- evaluation. Under this method certain Key personnel are selected for each factor (which may include initiative, leadership, dependability, safety etc.) Other employees will be compared with these key personnel by considering one factor at a time.

This method is not very much used in performance appraisal because of difficulty in selecting the key men.

5. Check List:

In order to reduce bias or prejudice of the rater, the merit rating under this method is actually done by the HR Department. The supervisor simply reports the. Performance of the employee by putting ‘Yes’ or ‘No’ against a series of descriptive phrases. The rate is not aware of the weightage of these questions but he can definitely distinguish between positive and negative questions.

There are different checklists for administrative and supervisory personnel. This method is not simple. The main disadvantage of check list method is that it is very difficult to assemble, analyse and weigh a number of phrases applicable to employee’s qualities or shortcomings.

The following is the specimen of check list statements:

		Yes	No
1.	He takes keen interest in his work	☐	☐
2.	He possesses good knowledge of his work	☐	☐
3.	He is good in planning	☐	☐
4.	He is respected by his fellow workers	☐	☐
5.	He volunteers good ideas	☐	☐
6.	He is generally regular	☐	☐

7.	He obeys orders	☐	☐
8.	He always finishes his work in time	☐	☐
9.	He is always co-operative with his colleagues	☐	☐
10.	He maintains his machines properly	☐	☐
11.	He frequently commits mistakes	☐	☐
12.	He tries to evade responsibility	☐	☐

6. Forced Choice Description:

Usually the raters try to rate the employees in the 'average' category because it does not require further explanation. Under this method, the rate is forced to select a statement from a pair of statements though both may be applicable or un applicable to him.

A pair of un favourable statements may be as follows:

- (a) Makes promises that he knows he cannot fulfill.
- (b) Shows favour to certain employees.

He must choose at least one though none may be applicable on the employee.

The Positive Statements May Be as Follows:

- (a) Has a constant flow of new ideas.
- (b) Shows careful judgment.

Even though both these statements may be applicable, the rater must select only one statement. Thus, bias or prejudice (halo effect) of the rater can be eliminated from merit rating under this method.

However, this system is not popular on account of the following limitations:

- (a) It is very difficult to keep secret of the values attached to various statements.
- (b) The rater usually objects to this method as he is forced to make a decision which he does not want to make.
- (c) This method is unsatisfactory. Both the raters and rates dislike this method,

7. Selection of Critical Incidents:

The act or conduct of an employee during crucial or critical circumstances is recorded by the rater. These critical incidents are carefully analysed and the behaviour or qualities of the employees are carefully rated.

The following are some of the examples of such incidents:

- 1. Became upset or nervous when accident took place in the factory.
- 2. Refused promotion or training opportunities.
- 3. Suggested an improvement in the methods of work.
- 4. Did not co-operate with his workers.
- 5. Gave excellent answer to union grievance.
- 6. Persuaded his fellow worker to withdraw his grievance.



8. Forced Distribution Method:

Where there are a large number of employees, the rater may be asked to rate all the employees as follows:

Poor 10%
Average 25%
Fair 35%
Good 20%
Excellent 10%

Generally the rater does not want to rate the employees as poor or excellent because in both the cases, he has to give reasons for justifying the rating. Hence ratings generally cluster around 'average!'. Under this method, the rate is forced to distribute all the employees in certain categories as explained above.

3.11 WAGE PAYMENT SYSTEM

Concept And Meaning of Wage Payment System

The system of wage payment is the method adopted by manufacturing concerns to remunerate workers. It is the way of giving financial compensation to the workers for the time and effort invested by them in converting materials into finished products. It indicates the basis of making payment to the workers, which may be either on time basis or output basis. The selection of the system depends on the type and nature of the concern and its products. The wage payment systems can be divided into two main systems as follows.

1. Piece rate system
2. Time rate system

Importance of Wage Payment System

The amount of wages paid to the workers is one of the major elements of cost. It has a great bearing on the cost of production and profitability of the concern. Hence, every concern is required to adopt a fair system of wage payment.

The importance of wage payment system can be summarized as follows:

- * Wage payment system facilitates the preparation of wage plan for future.
- * Wage payment system helps to determine the cost of production and the profitability of the organization.
- * Wage payment system determines the amount of earning of the workers and their living standards.
- * Wage payment system affects the interest and attitude of the workers.
- * Wage payment system determines the level of satisfaction of the workers and affects the rate of labor turnover.
- * Wage payment system helps in recruiting skilled, experienced and trained workers.
- * Wage payment system helps to increase the productivity and goodwill of the organization.

Essential Characteristics of A Good Wage Payment System

A sound system of wage payment is one that satisfies employer and employee by fulfilling following criteria.

- * Wage payment system should be fair and justifiable to the workers and organization.
- * Wage payment system should help in maximizing workers' satisfaction and minimizing labor

turnover.

- * Wage payment system should assure minimum guaranteed wages to all workers.
- * Wage payment system should assure equal pay for equal work.
- * Wage payment system should provide more wages to efficient and skilled workers.
- * Wage payment system should follow government policy and trade union's norms.
- * Wage payment system should be simple and understandable to all the workers.
- * Wage payment system should help in improving performance and productivity of the workers.
- * Wage payment system should be flexible enough to suit the needs of the organization.

The wage payment is an important factor influencing labour and management relations. Workers are very much concerned with the rates of wages as their standard of living is connected with the amount of remuneration they get. Managements, generally, do not come forward to pay higher wages because cost of production will go up and profits will decrease to the extent.

Following factors influence the determination of wage rate:

1. Ability to Pay:

The ability of an industry to pay will influence wage rate to be paid, if the concern is running into losses, then it may not be able to pay higher wage rates. A profitable enterprise may pay more to attract good workers. During the period of prosperity, workers are paid higher wages because management wants to share the profits with labour.

2. Demand and Supply:

The labour market conditions or demand and supply forces to operate at the national and local levels and determine the wage rates. When the demand for a particular type of skilled labour is more and supply is less than the wages will be more. One the other hand, if supply is more demand on the other hand, is less then persons will be available at lower wage rates also.

According to Mescon,” the supply and demand compensation criterion is very closely related to the prevailing pay comparable wage and on-going wage concepts since, in essence to all these remuneration standards are determined by immediate market forces and factors.

3. Prevailing Market Rates:

No enterprise can ignore prevailing wage rates. The wage rates paid in the industry or other concerns at the same place will form a base for fixing wage rates. If a unit or concern pays low rates then workers leave their jobs whenever they get a job somewhere else. It will not be possible to retain good workers for long periods.

4. Cost of Living:

In many industries wages are linked to enterprise cost of living which ensures a fair wages to workers. The wage rates are directly influenced by cost of living of a place. The workers will accept a wage which may ensure them a minimum standard of living.

Wages will also be adjusted according to price index number. The increase in price index will erode the purchasing power of workers and they will demand higher wages. When the prices are stable, then frequent wage increases may not be required

5. Bargaining of Trade Unions:

The wage rates are also influenced by the bargaining power of trade unions. Stronger the trade union, higher will be the wage rates. The strength of a trade union is judged by its membership, financial position and type of leadership.

6. Productivity:

Productivity is the contribution of the workers in order to increase output. It also measures the contribution of other factors of production like machines, materials, and management .Wage increase is sometimes associated with increase in productivity. Workers may also be offered additional bonus, etc., if productivity increases beyond a certain level. It is common practice to issue productivity bonus in industrial units.

7. Government Regulations:

To improve the working conditions of workers, government may pass a legislation for fixing minimum wages of workers. This may ensure them, a minimum level of living. In under developed countries bargaining power of labour is weak and employers try to exploit workers by paying them low wages. In India, Minimum Wages Act, 1948 was passed empower government to fix minimum wages of workers. Similarly, many other important legislation passed by government help to improve the wage structure.

8. Cost of Training:

In determining, the wages of the workers, in different occupations, allowances must be made for all the exercises incurred on training and time devoted for it.

3.12 PRIMARY WAGE SYSTEM (Time Rate & Piece Rate System)

Concept and Meaning of Piece Rate System of Wage Payment

The piece rate system is that system of wage payment in which the workers are paid on the basis of the units of output produced. Piece rate system does not consider the time spent by the workers. Piece rate system is the method of remunerating the workers according to the number of unit produced or job completed. It is also known as payment by result or output. Piece rate system pays wages at a fixed piece rate for each unit of output produced. The total wages earned by a worker is calculated by using the following formula.

Total Wages Earned= Total units of outputs produced x Wage rate per unit of output.

OR,

Total Wages Earned= Output x Piece Rate

Advantages of Piece Rate System

- * Piece rate system pays wages according to the output produced by the workers. It encourages efficient workers.
- * Piece rate system helps to reduce idle time.
- * Piece rate system gives incentives to the workers to adopt a better method of production for increasing their production and earning.
- * Piece rate system helps the management to determine the exact labor cost per unit for submitting quotation.
- * Piece rate system reduces per unit cost of production due to increased volume of production.
- * Piece rate system requires less supervision cost.

Disadvantages of Piece Rate System

- * Piece rate system does not help in producing quality output as the workers are concentrated more on quantity instead of quality.
- * Piece rate system does not help for a uniform flow of production and makes difficult to regulate the production schedule.
- * It is very difficult to fix an acceptable and reasonable piece rate for each item of output or job.
- * Piece rate system adversely affect the workers' health as well.
- * It requires extra supervision cost for quality output and effective use of materials, tools and equipment.

Concept And Meaning of Time Rate System of Wage Payment

The time rate system is that system of wage payment in which the workers are paid on the basis of time spent by them in the factory. Under this system, the workers and employees are paid wages on the basis of the time they have worked rather than the volume of output they have produced. Hence, according to this system, wages are paid on hourly, weekly or monthly basis. Under time rate system, the wages earned by a worker is determined by using the following formula.

Wages Earned = Time spent(Attended) x Wage rate per hour/day/week/month

Advantages of Time Rate System

The following are some of the important advantages of time rate system of wage payment:

- * Time rate system is simple to understand and easy to calculate.
- * Time rate system is quite useful for organizations that use costly inputs for quality outputs.
- * Time rate system is beneficial for average and below workers.
- * Time rate system assures regular income and creates the feeling of economic security among the workers.
- * Time rate system does not discriminate the workers and is preferred by trade unions.

Disadvantages Of Time Rate System

- * Time rate system does not help in increasing output and improving efficiency as there is no correlation between effort and reward.
 - * Time rate system is not justifiable between efficient and inefficient workers and skilled and unskilled workers.
 - * Time rate system pays for idle time, which increases the cost of production.
- Time rate system encourages a go-slow tendency among workers during working hours and encourages them to work overtime.
- * It is difficult to estimate exact labor cost in advance.
 - * It requires strict supervision to get the required quantity of output.

Differences between Piece Rate System and Time Rate System of Wage Payment

1. Meaning

Piece rate system is a method of wage payment to workers based on the quantity of output they have produced. Time rate system is a method of wage payment to workers based on time spent by them for the production of output.

2. Nature of Payment

Piece rate system pays the workers according to the units of output produced. Time rate system pays the workers according to the time spent in the factory.

3. Emphasis

Piece rate system gives emphasis on larger quantity of output. Time rate system emphasis on better quality of output.

4. Discrimination

Piece rate system discriminates the workers and pays more wages to efficient and skilled workers. Time rate system does not discriminate the workers and pays the same wages to efficient and inefficient workers.

5. Supervision

Piece rate system requires strict supervision to get the required quality output. Time rate system requires strict supervision to get required quantity of output.

6. Determination of Labor Cost

Piece rate system helps to fix per unit labor cost in advance. Time rate system does not help to fix labor cost per unit in advance.

7. Flow of Production

Piece rate system does not bring uniformity in the flow of production and causes an excessive wastage of inputs. Time rate system helps maintain a uniform flow of production and ensures an efficient use of materials, tools and equipments.

3.13 WAGE INCENTIVE PLANS

Wage incentive refers to performance linked compensation paid to improve motivation and productivity. It is the monetary inducements offered to employees to make them perform beyond the acceptance standards.

According to the National Commission of Labour “wage incentives are extra financial motivation. They are designed to stimulate human effort by rewarding the person over and above the time rated remuneration, for improvements in the present or targeted results”.

Scott defines it as “any formal and announced programme under which the income of an individual, a small group, a plant work force or all the employees of a firm are partially or wholly related to some measure of productivity output”.

Human and Nickerson define it in simple terms as “all the plans that provide extra pay for extra performance in addition to regular wages for a job”.

Objectives of Wage Incentive Schemes:

- (i) To use wage incentives as a useful tool for securing a better utilisation of manpower, better productivity scheduling and performance control, and a more effective personnel policy.
- (ii) To improve the profit of a firm through a reduction in the unit costs of labour and materials or both.
- (iii) To increase a worker’s earning without dragging the firm into a higher wage rate structure regardless of productivity.
- (iv) To avoid additional capital investment for the expansions of production capacity.

Principles of a Good Wage and Salary Administration:

- a. Simple and easy to understand.
- b. Union management agreement.
- c. Time standard must be fixed.
- d. Reward must be proportional to the effort.
- e. Complaints and grievances must be properly attended to.
- f. The plans should not change frequently and must be tried out continuously for some length of time.
- g. Equity and fairness.
- h. Workers must be made to understand the plan.
- i. Method study must precede time standard.
- j. There must be a min guaranteed payment.

Advantages of Incisive Plans:

Wage incentive plans benefit not only the employees but also the employers.

- a. Wage incentive plans provide an opportunity for hardworking and ambitious workers to earn more.
- b. It encourages employees to be innovative. They come out with more efficient ways of doing work by overcoming the problems related to productivity and wasteful practice.
- c. Incentive plans help to improve discipline and industrial relations. Effective incentive plan helps in minimizing absenteeism, accidents etc.

- d. The self motivation on the part of the workers to work hard and improve performance so as to earn monetary rewards will reduce the cost of supervision.
- e. The scientific work study undertaken before introducing the incentive plans helps in improving work flow, work methods etc.
- f. The employees are encouraged to work as a team with mutual co-operation as their activities is interdependent, and any obstruction on the part of a worker can affect the output and rewards.
- g. According to the National Commission on Labour, “wage incentive is the cheapest, quickest and surest means of increasing productivity.”

Limitations:

- a. Jealousy and conflicts among workers may arise when some workers earn more than others.
- b. Unless strict check and inspections are maintained, quality may come under stake in the enthusiasm among workers to increase productivity.
- c. In the absence of a ceiling on incentive earnings, some workers may spoil their health.
- d. Strict vigilance becomes necessary to ensure that workers do not disregard safety regulation.
- e. The cost and time of clerical work increases in introducing and administrating the incentive plans.
- f. Whenever production flow is disrupted due to the fault of management, workers insist on compensation.

Requirements of a Good Wage Incentive Plan

(1) Proper Climate:

The management should adopt sound policies regarding recruitment, training, promotion etc. to promote mutual understanding and freedom from suspicion. Good relations among management and workers should prevail for starting any incentive plan to increase productivity.

(2) Scientific Determination of Workload:

The standard workload should be based on scientific basis. Vagueness should have to be avoided because extra payment is made for extra work. To get the benefit of incentive standard work must be fixed on the basis of time and motion studies.

(3) Simplicity:

Incentive plan should be simple and easy to understand by the average worker. Benefits of the plan should be made clear to all the workers so that they can take interest and initiative in work that will enable to lead increased production. It is rather better to discuss the pros and cons of the plan with the workers before its installation.

(4) Beneficial to All:

Plan should benefit majority of the workers and not a section of them. It should cover all workers at all levels. Upper limit for earnings by way of incentives by the workers should not be set. It should promote health and welfare.

(5) Flexibility:

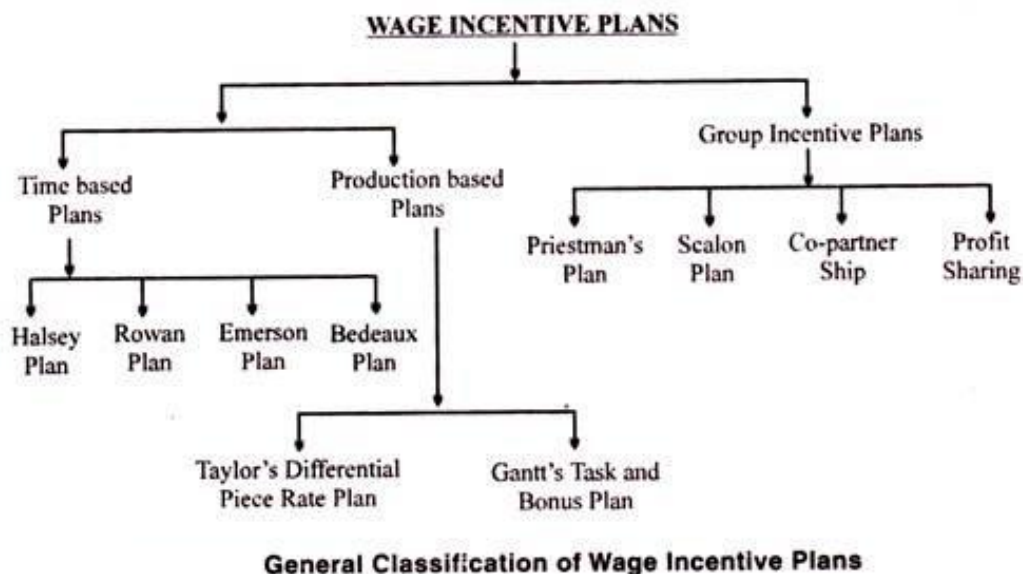
Incentive plan should not be rigid. It should be elastic taking into consideration the social, economic, political conditions in the country and technological changes. It should change accordingly.

(6) Low in Cost:

The cost of operation of the incentive plan should be low. It should not put additional monetary burden on the undertaking. It should not increase the overhead cost. Precisely it should be cost effective.

(7) Guarantee for Basic Rate:

It should guarantee basic rate so that every employee to receive minimum wage irrespective of output he contributes. It is essential to instil confidence and sense of security among the employees. Besides these incentive plans must be fair and just and should satisfy economic and non economic needs of the workers. Standard working conditions should be provided and proper grievance handling machinery is set up to handle problems arising out of execution of incentive plan. For this there is a need for periodical incentive audit so that the deficiencies in the plan be corrected for its smooth functioning.

Classification of Wage Incentive Plans**PRODUCTION BASED PLANS:****1. Straight Piece Rate Plan:**

Under the straight piece rate plan workers are paid based on their output. For example, if the piece rate is Rs. 4 per piece of the product, then a worker who turns out 40 pieces/day earns Rs. 160 (Rs. 4 x 40) as his wage for that day. Whereas another employee who produces 32 pieces/ day earns Rs. 128 (Rs. 4 x 32 pieces). Hence a fast worker earns more compared to the slow worker.

Advantages:

- i. Motivates the workers to increase their output.
- ii. Simple and easy to understand.
- iii. improve productivity.

Disadvantages:

- i. No guaranteed minimum wage. This makes workers insecure.
- ii. Great disparity of earning between slow and fast workers.
- iii. Wastage might increase.
- iv. Quality of production may suffer as the workers concentrate on quantity.
- v. Interpersonal relationship suffers due to jealousy and competition to earn more.
- vi. Enforced idleness like electricity failure or machine breakdown, adversely affect earning of workers.

2. Standard Piece Rate with Guaranteed Minimum Wage:

Here the minimum guaranteed wage is fixed on hourly basis. A worker gets the minimum fixed wage/day plus the incentive for the number of pieces produced. To illustrate this, assume that there is 8 hour's shift the piece rate is Rs 4 and a minimum fixed wage of Rs 16/ hours ($\text{Rs } 16 \times 8 \text{ hours} = \text{Rs. } 128$ per day). The standard time/piece is 15 min.

Now, there are two workers A and B. (If worker A produces 25 prices/day then he earns: Rs. 128 (min. guaranteed wage) + Rs. 100 ($\text{Rs. } 4 \times 25 \text{ pcs}$) = Rs. 228/ day

If worker B produces 40 pieces / day then he earns Rs. 128 (min. guaranteed wage) + Rs. 160 (40 pieces x Rs. 4) = Rs. 228/ day)

Advantages:

- i. Min. guarantee improves sense of security.
- ii. Disparity between slow and faster workers is reduced.

Disadvantages:

- i. Demotivate faster worker.
- ii. Slow workers get higher piece rate viz Rs. 5.12 ($128/ 25$).

3. Taylor's Differential Piece-Rate Plan:

F.W. Taylor started this method as a part of the scheme of scientific management. The underlying principle of this system is to reward an efficient worker and penalise the inefficient person. In Taylor's system, inefficient persons have no place in his organization.

The standard time was fixed for completing a task with the help of time and motion study. If a worker completes the task in the standard time he is paid at higher rate and lower rate is paid if more than the standard time is taken.

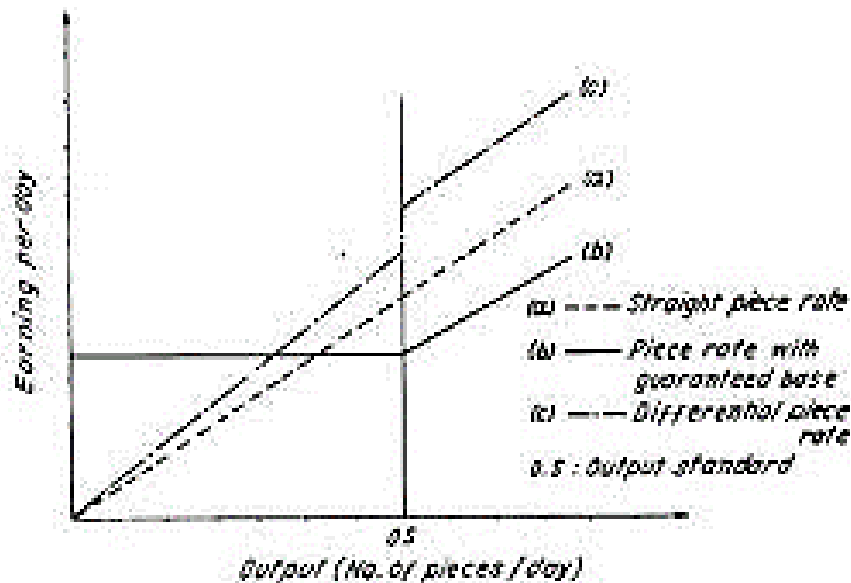
The main features of this system are:

1. Minimum wages are not guaranteed in this plan.



2. A standard time fixed for taking and completing the task.
3. Different rates are fixed for taking standard time or more.
4. Higher rate is given if work is completed in standard or less time and lower rate is offered if more than standard time is taken.

This method can be explained with the help of an example. A standard output of 200 units is fixed in an 8 hours' time. A rate of 45P is paid if the output is 200 or more units and 35P, if production is less than 200 units. Worker A has produced 240 units and B produced 180 units. The wages to be paid to worker A will be Rs. 108 i.e. (240×0.45) and that to B will be Rs. 63 i.e. (180×0.35) .



Comparison of various piece rate plans.

Advantages:

1. This method is simple to understand and wages to be paid to a worker can easily be calculated.
2. It offers good incentives to efficient workers.
3. This method is preferred by employees because it reduces overhead expenses per unit by raising output.

Limitations:

1. This method punishes slow workers very severely by giving them lower rates hence less wages.
2. A seed of disunity is sown among workers. Those producing them will feel jealous of others.
3. Workers are not guaranteed minimum wages and they feel insecure about their earnings.
4. It adversely affects the health of workers because they try to over exert for reaching the standard output.
5. It is difficult to determine labour cost because different rates are paid for production purposes.

4. Gantt's Task and Bonus Plan:

This method is named after H. L. Gantt, a close associate of F.W.T. Taylor. He tried to improve Taylor's method of wage payment. The workers are guaranteed minimum wages for taking standard time or more. A person taking less than the standard time gets time wages plus bonus.

The characteristics of this scheme are as follows:

1. A standard time is fixed for completing the work, and
2. A worker taking standard or more time gets wages on hourly rate.
3. A bonus ranging from 25% to 50% is paid for completing the task in less than standard time.

An example is given to explain payment under this method. A standard time of 10 hours is allowed to complete a task and hourly rate is Rs. 5. A person completing the task in 8 hours will get Rs. 50 as wages. If the same task is completed in 8 hours then wages will be Rs. 12: Rs. 8 will be for time spent and Rs. 4 for bonus (taking 50% as the rate of bonus).

Limitations:

1. Since workers are paid minimum wages, they may not bother to increase their efficiency.
2. The disparity in wages earned by efficient and inefficient workers will be wide and it will create jealousy among them.

TIME BASED PLAN:

(i) Halsey Plan:

This plan was first designed by Halsey in 1890. Under this plan a standard time is fixed for completing a work in advance. A person taking standard or more time is paid for the time taken by him.

A worker completing his task in less than the standard time is paid for some of the time saved. The payments for time saved vary from 33 ⅓% to 66⅔% but generally wages for one half of time saved are paid. The wage of a worker is given by

$$W = T \times R + (S - T/2) R$$

Where S= Standard time or allowed time to complete the job.

T = Time taken

R = Hourly wage rate

Example:

Time allowed for a job Actual Time taken Hourly wage rate Calculate the wage of a worker.

Solution: Given

Standard time S = 18 hours

Time T = 14 hours

Hourly wage rate = Re. 1.0

Using the relation

$$W = T \times R + ((S-T)/2) R$$

$$= 14 \times 1 + (18-14/2) \times 1$$

$$= 14 + 2 = \text{Rs. } 16.$$

In this equation a worker gets Rs. 2.00 as incentive for saving 4 hours i.e. it is a 50-50 Halsey plan. In this case the total saving of 4 hours amounts to Rs. 4 the 50% saving goes to management and 50% to worker who has saved time.

Advantages of Halsey Plan:

1. It guarantees minimum wages to all workers, whether efficient or inefficient.
2. Efficient workers are induced to show better results by offering them additional wages as incentive.
3. The employer also gains under this system because workers are not paid for full time saved by them and fifty percent goes to management.
4. This method is very simple. Workers can make their calculations very easily.
5. There is no need to over speed because incentive is not for more production but for time saved as is clear from above example.

Limitations of Halsey Plan:

1. The workers are not given full reward to their efforts. Normally they are paid for half of the time saved and not for full time. This may discourage workers.
2. It may be difficult to fix standard time for completing a job.
3. The quality of products suffers because workers try to complete the work in shortest possible time.

(ii) Rowan Plan:

This system is similar to that of Halsey plan. A worker is guaranteed minimum wages for time spent on the job. He gets bonus for completing the job in less than the standard time. The only difference between Halsey and Rowan Plans is the method of calculating bonus is that proportion of the wages of the time taken which he saved bears to the standard time allowed.

Wages are calculated by the following relations:

$$W = T \times R + (S-T/S) \times T \times R$$

Example. 1:

Standard Time: 32 hours

Actual Time taken: 26 hours

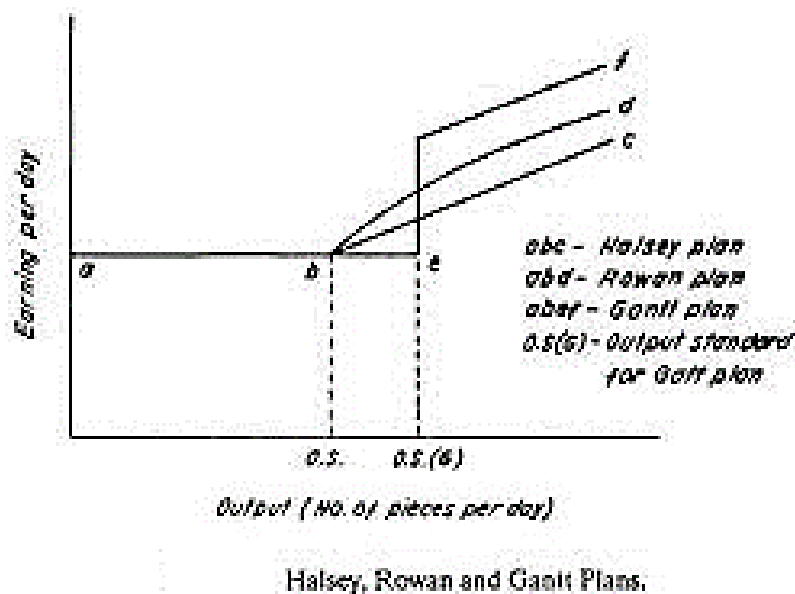
Hourly Rate: 4

Solution:

$$26 \times 4 + (32-26) = 104 + 6/32 \times 26 \times 4$$

$$= 104 + 19.50 = \text{Rs. } 123.50$$

The additional bonus, a worker will get is Rs. 19.50 in this case. However, the total saving of 6 hours amounts to Rs. 24 so Rs. 4.50 goes to management and Rs. 19.50 to worker.



Advantages of Rowan Plan:

1. This method provides minimum wages to workers.
2. The worker is not induced to rush through the work because bonus increases at a decreasing rate at higher levels of efficiency. Thus the quality of goods under this system will not suffer.
3. Labour cost per unit is reduced because time saved is shared by the worker and management both.
4. The increase in production will reduce overhead cost per unit produced.

Disadvantages of Rowan Plan:

1. The calculation of bonus under this system is complicated. In Halsey plan workers know that he will get additional wages for half of the time saved. In this method a certain proportion of time saved is paid as incentive. The calculation involved is difficult for workers to understand.
2. This method is unjust for efficient workers since bonus is paid at decreasing rate.
3. Labour cost is generally higher in this method.

(iii) Emerson Plan:

Emerson, an associate of F.W. Taylor, developed this efficiency plan in 1910. A standard output is fixed for determining the efficiency of workers. A worker reaching up to 66⅔% of efficiency is paid only

minimum wages and bonus is paid only when his efficiency crosses this limit. The rate or bonus increases with the increase in efficiency.

For example, 2000 units are fixed as a standard production, a person producing 2000 units achieves 100%-efficiency, another person producing 1600 units reaches 80% efficiency and so on.

Under this plan bonus is 20% of wages earned at 100% efficiency and increases by 1% with every percent increase in efficiency. If efficiency is 110% then bonus will be 30% at this level. Efficiency of workers is well acknowledged in this system.

Benefits of Emerson Plan:

1. It is simple easily understandable by workers.
2. Workers get security because minimum wages are paid if efficiency is upto $66\frac{2}{3}\%$.
3. It provides stimulus to workers for increasing their efficiency. The rate of bonus increases progressively so provides encouragement for improving efficiency.
4. It provides incentive even to beginners and less efficient persons.

Limitations of Emerson Plan:

1. Standards may be set fairly high and workers may not be able to achieve them.
2. Workers may not be encouraged to increase their output beyond the standard level because benefits may be nominal after that level.
3. The records of standards will have to be kept separately for different categories of workers. It increases clerical work.

(iv) Bedeaux Plan:

This plan was devised by Charles E. Bedeaux in 1911. It provides comparable standards for all workers. The benefit of time saved goes both to the worker and his supervisor in the ratio of 3/4 and 1/4th respectively. A supervisor also helps a worker in saving his time so he is also given some benefit in this method.

The standard time for each job is determined in terms of minutes which are called Bedeaux points or B's. Each B represents one minute through time and motion study. A worker is paid time wages up-to standard B's or 100% performance. Bonus is paid when actual performance exceeds standard performance in terms of B's.

Advantages of Bedeaux Plan:

1. It ensures minimum wages to all workers.
2. This method is very simple and is easy to understand.
3. The supervisor is motivated to co-operate with the workers for increasing their efficiency.

Limitations of Bedeaux Plan:

1. Workers are tempted to hurry up with the job and strict supervision will be necessary for maintaining proper quality control.
2. Workers resent sharing of their efforts with supervisors or superior.

3 The standard task may be too difficult to perform within a specified time.

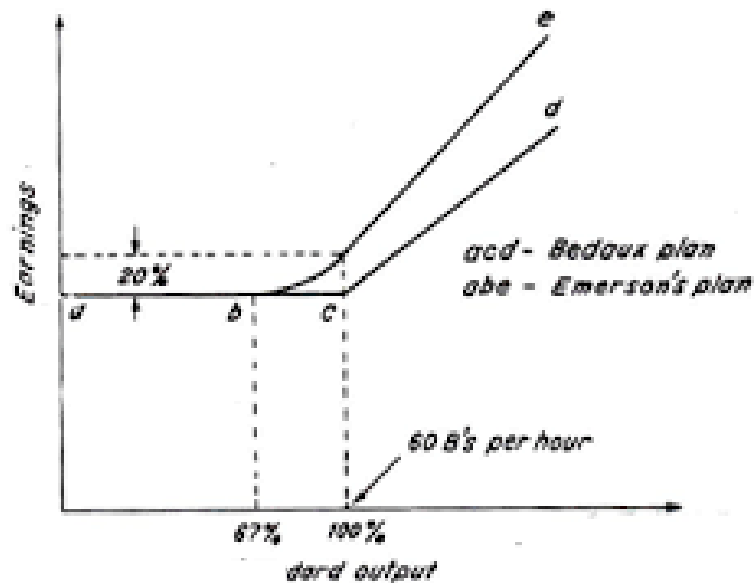
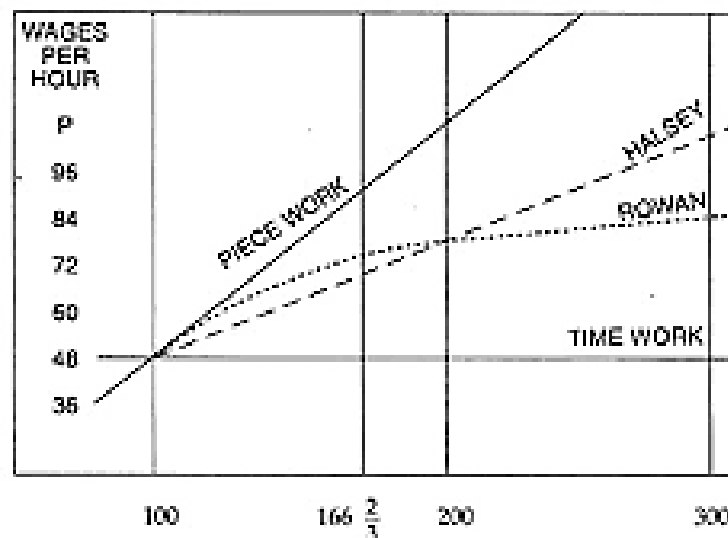


Fig. 30.4. Bedaux and Emerson plans.

GRAPH OF COMPARATIVE EARNINGS UNDER ALTERNATIVE BONUS SCHEMES.



GROUP INCENTIVE PLANS:

Under individual incentive system workers are paid on the basis of their personal performance. Their wages will be directly linked to their efforts. A worker may improve his remuneration by raising the level of output.

There may be circumstances when individual performance may not be measurable. A number of persons may be associated in completing a task. The work of one person may be influenced by the work of the other. Under such conditions, incentives may be offered for raising group performance.

The industry engaged on assembly type of work as in computers, washing machines and Televisions etc. group incentive plan may be practicable. The performance depends upon the group effort rather than on individual initiative. The incentives are given for raising output beyond a certain level of output. The amount of bonus is divided among all persons associated in completing the task.

Suitability:

Group incentive schemes are suitable under the following situations:

1. When individual performance cannot be measured precisely.
2. The workers comprising a group possess the same type of skill or ability.
3. The completion of the task is linked with the collective efforts of the group.
4. The aim is to provide incentive to indirect workers rather than direct workers.
5. The number of persons constituting a group is not large.

Methods of Distributing Group Bonus:

Many methods are used for distributing bonus some of the commonly used criterion are as follows:

1. If all the persons in the group possess the same type of ability or skill then bonus may be distributed equally among them.
2. When group members are paid wages according to same time scale, bonus may also be divided according to that time scale.
3. If workers earn different amounts of wages then bonus may be distributed in proportion to wages earned by them.
4. Bonus may also be paid on the basis of certain percentage, fixed on the basis of the experience, skill and wages earned by a worker.

Types of Group Incentive Plans:

There may be different schemes for paying group bonus some of these are discussed as follows:

1. Priest-man's Plan:

A standard production is fixed for the whole enterprise under this plan. If productivity exceeds the standard then bonus is paid in accordance with the increase. In case production does not reach the standard then workers get maximum wages only. For example, a standard production of 200,000 units is fixed for the year.

Actual production during the year is 240,000 units since production has gone up by 20% workers will get 20% higher wages as bonus. The workers get sufficient incentive to raise their performance. A team spirit is visible among the workers because production will increase with the collective efforts of various limbs of the organization.

This method does not offer incentive to individual workers. Inefficient workers share the efforts of efficient workers because increased production benefits all the manpower in the organization.

2. Scalon Plan:

This method is named after Joseph Scalon. There is a payment of one percent participating bonus for every one percent increase in productivity under this plan. The bonus is available to all workers except top management.

The entire amount of bonus is not paid every month. A reserve fund of one-half of first fifteen percent is created for off-setting any change in labour cost. In case, this reserve remains unused at the end of the year then this amount is also distributed among workers in the last month of the year and a fresh reserve is created in the year.

3. Co-Partnership:

The employees are offered shares of the enterprise at reduced rates in this plan. The payment is also collected in instalments. The employees share profits of the enterprise as its members.

The underlying idea of this method is to make workers feel as a part of the organization and understand view point of the management. As co-partners they will behave in a responsible manner and will try to make the concern more and more profitable and successful.

4. Profit Sharing:

When shareholders share profits for contributing towards capital then workers should also get a part of profits for contributing their labour. The workers are an integral part of any organization and their contribution to its prosperity should also be rewarded by making them the recipients of profits. This realisation that employees/workers contribute significantly to increase profit has encouraged the adoption of this system.

Profit sharing is a method of remuneration under which an employer undertakes to pay his employees a share in the net profits of an enterprise, in addition to regular wages.

Benefits of Group Incentive Plans:

(1) Easy to implement:

It is easy to implement since measurement of group output is easy than the individual's output.

(2) Low Overhead Cost:

The overhead cost is reduced because of reduced paper work.

In general, individual incentive plans tend to motivate the workers to a larger extent than group incentive plans. With the increased rate of production the unit production cost is reduced.

Limitations of Group Incentive Plans:

(i) Tend to lower the overall productivity.

(ii) Due to uniformity of pay irrespective of individual's lower or higher contribution in a group effort personnel problem arise.

IMPORTANT QUESTIONS

1. Explain different process charts and flow diagrams used in method study.
2. What is the difference between method study and work measurement?
3. How will you calculate the standard time for doing a work?
4. Explain different types of allowances.
5. Explain different wage incentive plans.
6. What is piece rate and time rate system.
7. Explain different Job evaluation methods.
8. Explain different Merit rating methods.
9. What is meant by Critical examination?
10. Explain Micromotion study and SIMO chart.
11. What are Therbligs?
12. Explain different principles of Motion economy.
13. Explain different performance rating methods.
14. Differentiate Job evaluation and Merit rating.
15. What are the requirements of a good wage incentive plan.
16. What is the need for workstudy?
17. List the objectives of method study and work measurements.
18. Explain the procedure of method study.
19. What is the purpose of micro motion study? List its advantages.
20. Explain the basic procedure of time study.
21. List the objectives of time study. Mention the advantages also.
22. What are the elements in a standard time?
23. List the advantages and drawbacks of job evaluation.
24. List the advantages, limitations and objectives of merit rating.
25. Explain group incentive plan.



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MODULE 4

4.1 INDUSTRIAL RELATIONS

Concept of IR:

According to Dale Yoder, IR is a designation of a whole field of relationship that exists because of the necessary collaboration of men and women in the employment processes of Industry”.

Armstrong has defined IR as “IR is concerned with the systems and procedures used by unions and employers to determine the reward for effort and other conditions of employment, to protect the interests of the employed and their employers and to regulate the ways in which employers treat their employees”

In the opinion of V. B. Singh “Industrial relations are an integral aspect of social relations arising out of employer-employee interaction in modern industries which are regulated by the State in varying degrees, in conjunction with organised social forces and influenced by the existing institutions. This involves a study of the State, the legal system, and the workers’ and employers’ organizations at the institutional level; and of the patterns of industrial organisation (including management), capital structure (including technology), compensation of the labour force, and a study of market forces all at the economic level”.

Objectives of IR:

The primary objective of industrial relations is to maintain and develop good and healthy relations between employees and employers or operatives and management. The same is sub- divided into other objectives.

Thus, the objectives of IR are designed to:

1. Establish and foster sound relationship between workers and management by safeguarding their interests.
2. Avoid industrial conflicts and strikes by developing mutuality among the interests of concerned parties.
3. Keep, as far as possible, strikes, lockouts and gheraos at bay by enhancing the economic status of workers.
4. Provide an opportunity to the workers to participate in management and decision making process.
5. Raise productivity in the organisation to curb the employee turnover and absenteeism.
6. Avoid unnecessary interference of the government, as far as possible and practicable, in the matters of relationship between workers and management.
7. Establish and nurse industrial democracy based on labour partnership in the sharing of profits and of managerial decisions.
8. Socialise industrial activity by involving the government participation as an employer.

According to Krikaldy, industrial relations in a country are influenced, to a large extent, by the form of the political government it has. Therefore, the objectives of industrial relations are likely to change with change in the political government across the countries.

4.2 PSYCHOLOGICAL ATTITUDES TO WORK AND WORKING CONDITIONS

Defining Worker Stress

- Hans Selye (1976) : Stress is primarily a psychological reaction to certain threatening environmental events. Worker stress would simply refer to the stress caused by events in the work environment.
- Stressor : an environmental event that is perceived by an individual to be threatening.
- Worker stress : the psychological and/or psychological reactions to events that are perceived to be threatening or taxing.

Sources of Worker Stress

Situational stress : stress arising from certain conditions that exist in the work environment, or in the worker's personal life.

- Stressful Occupation : It is generally believed that certain occupations, such as air traffic controllers, physicians, and other health care providers, police officers, and firefighters, are particularly stressful.

Organizational Sources of Work Stress:

a. Situational Stressors

A great deal of worker stress is caused by the work organization. Some of this organizational stress is caused by the work tasks themselves-the physical and psychological demands of performing a job.

b. Work Task Stressors

- Work overload : a common source of stress resulting when a job requires excessive speed, output, or concentration.
- Underutilization: a source of stress resulting from workers feeling that their knowledge, skills or energy are not being fully used.

c. Work Role Stressors

- Job ambiguity : a source of stress resulting from a lack of clearly defined jobs and/or work tasks.
- Lack of Control : a feeling of having little input or effect on the job and/or work environment; typically result in stress
- Physical work conditions : physical conditions in the work environment are another organizational source contributing to worker stress. Jobs that must be performed under extreme temperatures, loud noise, or poor lighting or ventilation can be quite stressful.
- Interpersonal Stress : stress arising from difficulties with others in the workplace.
- Harassment : All forms of harassment, including sexual harassment, harassment due to group membership (e.g., gender, race, sexual orientation), and being singled out by an abusive supervisor or colleague are all extremely stressful.
- Organizational change : A final organizational is change. People tend to grow accustomed to certain work procedures and certain work structures and they resist change. Most of us prefer things to remain stable and predictable.

d. Individual Sources of Work Stress : Dispositional Stressors

- Type A behavior pattern : a personality characterized by excessive drive, competitiveness, impatience, and hostility that has been linked to greater incidence of coronary heart disease.
- Susceptibility/resistance to stress :
 - o Hardiness: the notion that some people may be more resistant to the health-damaging effects of stress.
 - o Self-efficacy: an individual's belief's in his or her abilities to engage in courses of action that will lead to desired outcomes.

4.3 FATIGUE

Human capacity to work is limited. Every work requires care, attention and concentration. A person cannot work continuously for longer periods. The work output will be more in the morning and it goes on decreasing with passage of time as a worker will feel tired, mentally and physically, in the evening. The decrease in efficiency of working due to longer working periods is known as fatigue.

It is one of the most significant problems before industrial engineers. Fatigue has been defined as a negative appetite for work activity. According to Vitals, "The tension, the subjection to specialization of work and a rhythm uncontrolled by the individual, promote fatigue characterized by decrease in the power to work, a decrease in pressure taken in work, and an increase in the enjoyment of the hours spent away from work".

Fatigue can precisely be explained as follows:

1. It leads to loss of efficiency.
2. It results in more or less complete loss of responsiveness of tissue.
3. It may be termed as the reduced capacity of work resulting from work itself.
4. It is considered as state of 'lack of interest' in not only work but in other activities too.

Fatigue is related to the mental status of person. One person may not be tired even after long hours of work whereas the other person may feel fatigued or tired even after few hours of work. Some persons may feel fresh after a short relaxation during working periods while others may not regain energy "even after longer rests." A number of factors such as aptitude, interest, nature of job, work environment or working conditions etc., influence the mind of persons on jobs.

Types of Fatigue:

Fatigue may be of the following types:

1. Physical fatigue:

Physical fatigue is caused either by continuous work for long periods or the nature of work like the work may be heavy and requiring lot of physical exertion. The physical capacities of workers are limited and they may feel tired after a continuous work for longer periods.

2. Mental fatigue:

A worker utilizes his brain for doing a job for a longer period. The doing of same work again and again will result in mental fatigue of the worker.

3. Nervous fatigue:

When work needs the use of mental and physical capabilities for a continuous period, it will result in nervous fatigue.

4. Industrial fatigue:

When a worker continues to work the job for longer periods, his efficiency and productivity will result in nervous fatigue.

Causes of Fatigue:

1. Longer hours of continuous work without rest pauses may cause fatigue because muscles will be tired after long work.
2. Unsatisfactory working environment like insufficient light insanitation, congestion, noise elevated temperatures etc.
3. Fatigue may also be caused by personal reasons such as family tension, bad health etc.
4. There may be faulty design of machines and equipment which leads to wastage of energy of workers on them.
5. The complexity or production process may also put more load on workers resulting into fatigue.
6. The workers may not be properly placed on jobs as per their skill levels.
7. The harsh attitude of supervisor may also result in fatigue of workers.
8. Inconvenient and awkward posture of worker (required for some particular jobs) i.e., standing or bending for longer periods may also be the cause of fatigue.

4.4 METHODS OF REDUCING FATIGUE

A fatigued worker will not be able to work with his original rhythm. Every industrial enterprise is expected to devise ways and means of reducing fatigue so that production does not suffer for longer periods.

Following methods may help in reducing fatigue and providing freshness to workers:

1. Rest pauses:

When workers continue to work for longer periods then they; feel tired and need some rest. Rest pauses are of great importance in reducing fatigue as well as monotony. Rest pauses should be provided in such a way that workers feel relieved after working for some hours. A well planned rest pauses schedule will be of .great help in reducing fatigue.

2. Less hours of work:

Longer hours of work are the main cause of fatigue. The hours of work should be reduced to that optimum level where workers are able to keep up their working speed. The factories act in India allows only 48 hours a week for adult workers and it should be followed strictly.

3. Proper lighting:

Poor illumination is an important cause of disturbance and fatigue. The work place should be properly lighted so that workers are able to work without burdening their eye sight and brain.

4. Improving environmental conditions:

The humidity, temperature and ventilation influence workers at work. A proper balance of temperature, humidity should be maintained to make the work place comfortable and worth working so that fatigue is reduced.

5. Reduction of noise:

Undesirable noise will cause fatigue. It may also cause muscular tension. Unnecessary noise should be curbed to its, minimum level. Lower noise levels increase productivity of workers by reducing fatigue, removing cause of irritation.

6. Proper selection of employees:

A wrong selection of employees and their placement may also be a reason for fatigue and monotony. A job may require more physical input than worker has. In such situations the worker will feel fatigued after short duration of work. A square peg in a square hole will help in reducing fatigue and boredom so proper selection shall help in this regard.

7. Job rotations:

Sometimes workers start feeling bore and tired while working repeatedly on the same job. If job rotation a is provided it will avoid boredom and fatigue.

8. Counseling of employees:

Anxieties and worries may be the main cause of fatigue with some workers. There may be job dissatisfaction domestic problem etc. the officers of personnel department should counsel the employees and help them in solving their problems.

4.5 EFFECT OF COMMUNICATION IN INDUSTRY

Communication can be defined as transmission or exchange of ideas, views, message, information or instructions between two or more persons by different means.

The common means and ways for transmission of ideas are:

- (a) Spoken words;
- (b) Written words;
- (c) Diagrams, pictures or graphs; and
- (d) Gestures.



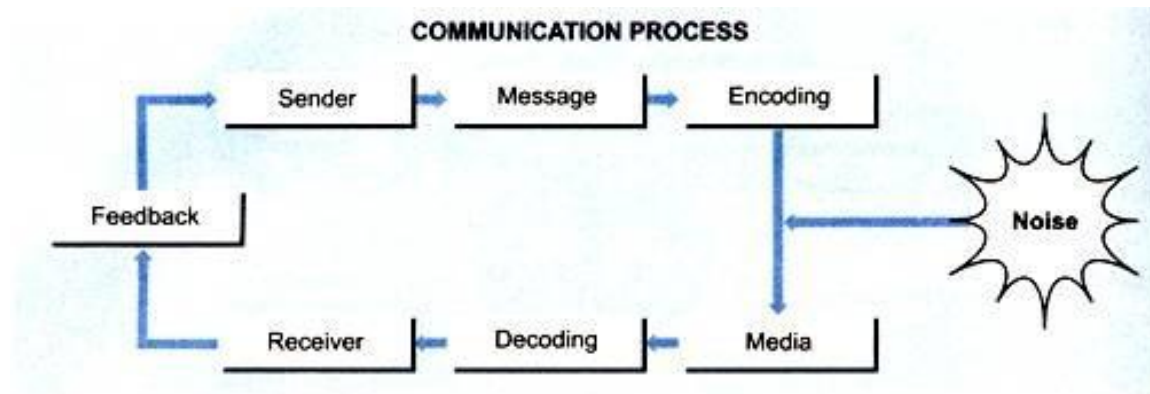
The communication is a two way process as it begins with sender a comes from receiver to sender. Minimum two parties are involved in the process of communication, i.e., the sender and the receiver.

Communication Process/Elements of Communication Process:

The communication process begins when sender thinks of an idea or message to be conveyed to other person. Then the sender encodes the message, i.e., plans out the words or the language in which the

message must be sent to the other party. After encoding the message the sender transmits or transfers the message by using different ways and means. After transmission the message reaches to receiver.

The receiver decodes the message to understand the message and give his response or feedback to the sender. When the feedback reaches back to sender then only the communication process ends which means the communication is a circular process which starts with sender and ends with sender.



1. Sender:

Sender is the person who conveys the message. The communication process begins immediately when the idea comes in the mind of sender.

2. Message:

The contents of ideas or message or suggestion which sender wants to share with receiver is called message.

3. Encoding:

Receiver cannot read the mind of sender. So receiver converts the idea into a language or other communication symbols such as pictures, gestures, etc. This is known as encoding.

4. Media:

It is the way or means through which encoded message has to be transmitted to receiver. The common ways of transmission are phone, letter, internet, message, etc.

5. Decoding:

Decoding refers to converting the encoded message into language and understanding the message.

6. Receiver:

Receiver is the person who receives the communication and understands the message.

7. Feedback:

After understanding the message the receiver sends his response on that idea or message to sender. When response reaches the sender then only communication process ends.

8. Noise:

Noise is any element which results in disturbance, distraction or interruption in the flow of information. The noise can be a loud sound made by any vehicle, disturbance in the telephone line, disconnection of line, losing of letter or documents on the way, poor connectivity in the e-mail or internet. Due to noise the message is not conveyed to receiver and receiver is not able to decode the message in the same manner as expected by the sender.

Importance of Communication:

The communication is an important element of directing because of following points:

1. Act as basis of Coordination and Cooperation:

Generally the organisational objectives are set up at planning stage and these objectives have to be communicated to management at all levels. That is why through communication only the objectives and goals of the organisation are made clear to every employee.

While setting up the personal targets and objectives of employees the managers must get the accurate information which reaches manager through communication only. Through communication the top level management is able to interact with lower level management and all the employees. This interaction helps in getting the commitment and co-operation and coordination of people.

2. Act as basis for Decision Making:

For taking any major decision or solving any problem in the organisation there is a need to get the most accurate information and information moves in organisation through communication only. Whenever the managers are taking decisions they keep in mind the 'pros' and 'cons' or positive or negative aspects. The accurate information regarding the positive and negative aspect comes only through communication.

3. Increase managerial efficiency:

Every individual in the organisation is assigned a job or task. He is made responsible for some activities. He is granted authority to carry on those responsibilities. This classification of task, responsibility and authority is possible only when the information reaches accurately to the employees.

The employees must know clearly who has to report to whom, what part of total job they are expected to perform and what are their decision making powers. The clarity about these questions comes only with smooth flow of communication.

4. Establish effective leadership:

If there is two way information flows between the superiors and subordinates then there will be definitely positive reaction of employees. Generally rules are framed by the top level authority but these are applied on all the employees in the organisation. So it is always advisable to interact clearly with all the levels of employees before framing the rigid rules and regulations. The rules are more effective when they are formed with the interaction of employees.

5. Helps in Process of Motivation and Morale Development:

Motivation is a psychological process of developing willingness to work. In the motivation process the superiors try to analyse the needs of subordinates and the needs can be recognised only when there is smooth flow of information and exchange of views between the superiors and subordinates.

With the communication it becomes more convenient for the superiors to offer financial and non-financial incentives. The job satisfaction and the moral of the employee depend on the communication between superior and subordinates.

6. Helps in Smooth Working of an Enterprise:

All interactions in organisation depend upon communication. Smooth working of an enterprise is possible only when there is no communication gap. Right from establishing of enterprise till its survival communication is essential.

Through communication managers give directions to subordinates for smooth functioning of organisation.

7. Promotes Cooperation and Peace:

Through two way communication process managers try to develop mutual understanding between management and workers. Through smooth flow of communication subordinates and superior can discuss their problems, grievances and aspirations. Cooperation brings peace in the organisation.



Forms of Organisational Communication:

There are two forms of organisational communication:

- i. Formal Communication; and
- ii. Informal Communication.

1. Formal communication:

Formal communication refers to official communication taking place in the organisation. Whenever there is exchange of views or message or information related to official matter such as assignment of task,

fixing of responsibilities, granting authority or setting up of targets, objectives etc. then it is known as formal communication.

Formal communication generally takes place in the written form such as issue of notice, letter, memo, document, etc. Verbal or oral channels are avoided in formal communication as there is no record or proof of such communication.

According to direction of flow formal communication can be divided into four types:

i. Downward Communication:

The flow of information from the top level or from the superiors to lower level or subordinates is known as downward communication. In this communication the manager passes instructions to his subordinates and the success of downward communication depends upon communication skill of the managers. The common ways of having downward communication are:

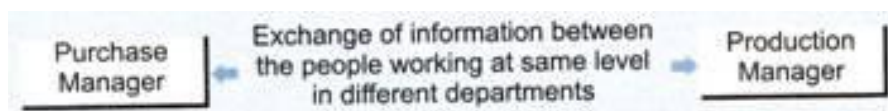
ii. Upward Communication:

When the subordinates inform or pass any information to superiors then it is known as upward communication. This communication flows from subordinates to superiors. Upward communication generally consists of work performance, opinion, grievances or problems of the employees.

In the recent years upward communication is gaining more importance because the subordinates prefer to involve themselves in the decision making and they have started demanding a better working condition. The upward communication may take place in the following manner:

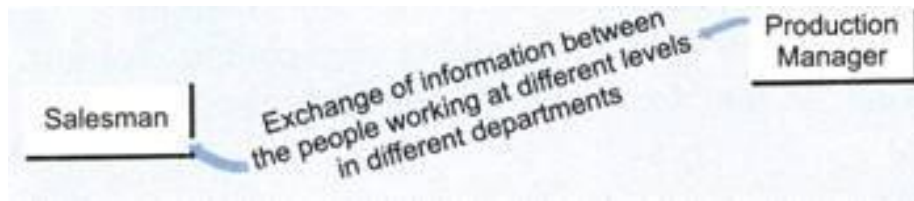
iii. Horizontal Communication:

Horizontal communication is the communication between the two or more persons working at the same level of authority. Generally different departmental heads discuss the policy of their department with each other. That will be considered as the horizontal communication. Discussion between two managers of same rank is also a horizontal communication. For example, the marketing manager may discuss from time to time the quality and packing of product with the production manager.



iv. Diagonal Communications:

Diagonal communication is a flow of information between the persons working in different departments and holding different levels of authority. For example, the salesman may request the production manager to modify the product design according to the requirement of the customer. The diagonal communication prevents delay in flow of information and helps in taking fast decisions as subordinates of one department can directly communicate with the managers of other department.



Advantages of Formal Communication:

1. The formal communication is very systematic and ensures orderly flow of information.
2. The source of information can easily be located.
3. In formal communication it is easy to fix the responsibilities of different employees as there is proof for the information.
4. Through formal communication it is easy to exercise control over the work performance of different employees.

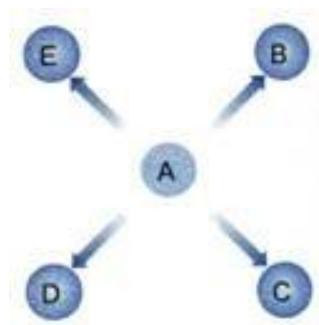
Disadvantages:

1. The information is delayed as generally scalar chain is followed while passing of information.
2. In formal communication the information is conveyed or passed in impersonal manner.
3. The information may not be transmitted accurately to avoid the unfavourable effect of communication.

The common networks of formal communication are:

1. Wheel Pattern:

It is the most centralised way of communication. Under wheel pattern all the information flows from one person only who is generally leader of the group. The other members have no communication link with each other. It is a common pattern of centralised organisation.



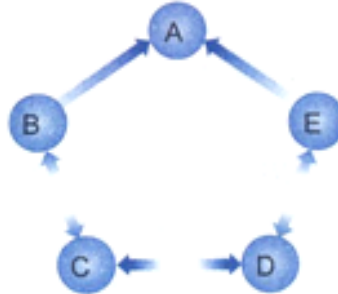
2. Chain Pattern:

Under the chain pattern each person gets the information from one person that is their immediate boss and passes the information to one person who is their immediate subordinate. So, every member is attached to one person.



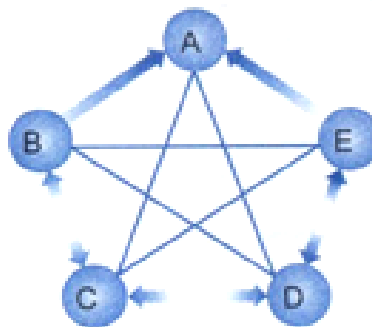
3. Circle Pattern:

In the circle pattern each person communicates with two more persons in a group. The person may receive or give information to two more persons in the organisation. In this type of organisation every member participates equally in flow of information. That is why it is more decentralised.



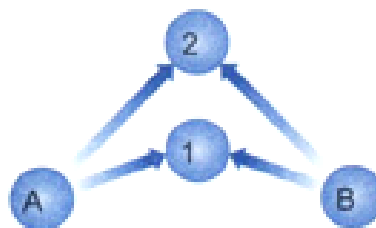
4. All Source (Channel) Pattern:

Under all channel pattern there is no fixed pattern of communication. All the members in the organisation are linked to each other and information can flow from anywhere in the organisation. It is most decentralised organisation.



5. Inverted V:

In this network a subordinate is allowed to communicate with his immediate superior and also with the superior of his superior. However in latter case only limited communication can take place.



2. Informal Communication:

Informal communication between different members of organisation who are not officially attached to each other is known as informal communication. Generally, the social interactions, friendly talks and non-official matters are discussed in the informal communication.

There is no fixed direction or path for the flow of information under informal communication. The information moves in a very vague, confusing and zig-zag manner. That is why the network of informal communication is known as Grape Vine.

There is no fixed source of information under informal communication and information can move in any direction. There are several types of informal systems.

But the common networks of informal communication are:

1. Gossip:

Under gossip pattern one person tells many ^4 f which means information is with one member of organisation and he shares the information with many other people in his social group.

2. Clusters:

Under this pattern or method the information is shared between two persons who trust each other, then one of the persons from that group passes the information to members of other group and one person from other group may pass it to another and so on. So, under cluster method one person tells the information to select other persons.

3. Single Strand:

In single strand pattern each individual communicates to the other in sequence.

4. Probability:

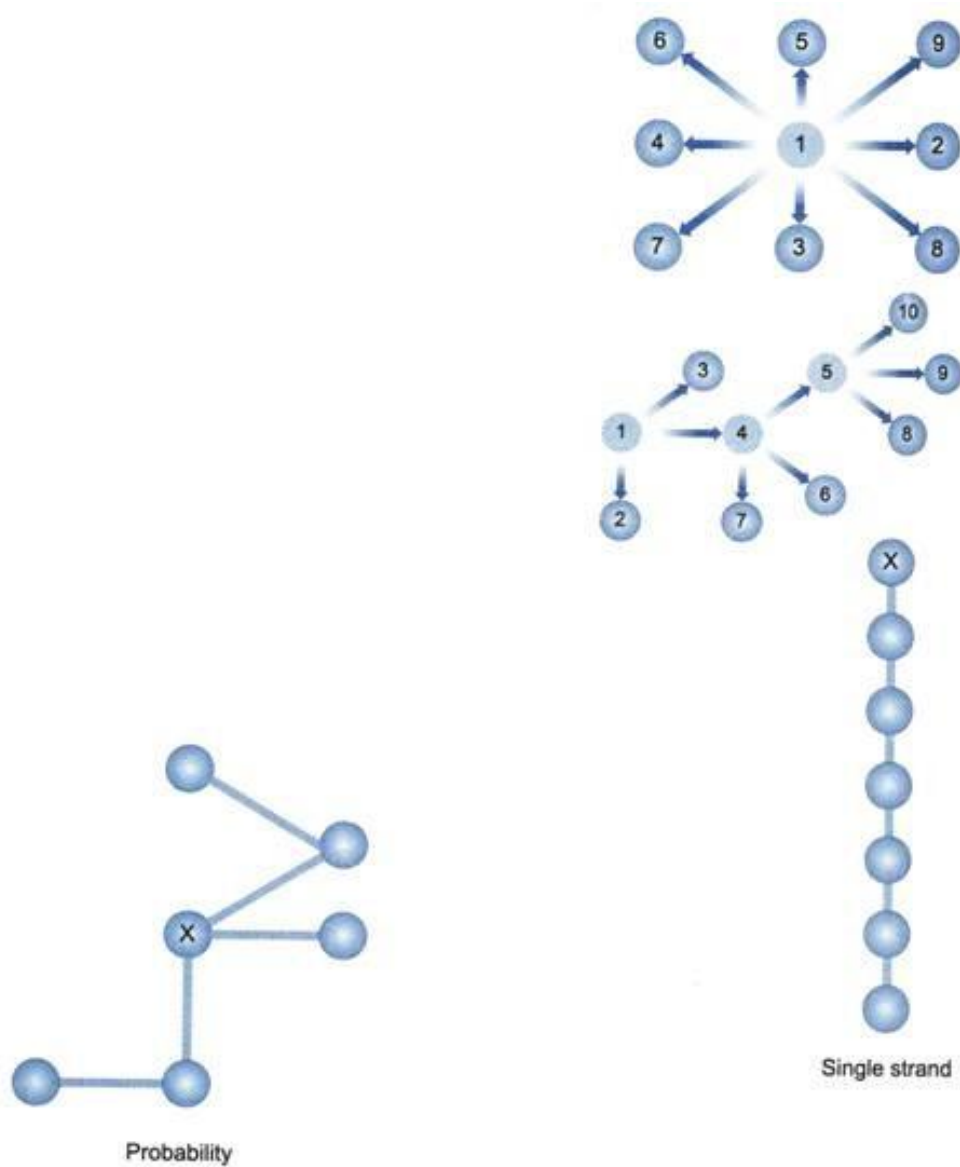
In probability network the individual communicates randomly with other individuals.

Advantages of Informal Communication (benefits from employees' point of view):

1. The employees can develop friendly relationship and get social satisfaction.
2. Through informal communication the matters can be discussed which cannot be discussed through official channels.
3. It provides information to all the members cutting across the official channels.

Benefits from management point of view:

1. The managers can fill up communication gap if there is any through informal communication.
2. Informal communication travels very fast. So urgent matters can be communicated informally.
3. The managers can get true and accurate response of subordinates on various policy matters.
4. Informal communication provides emotional relief to employees and results in reducing union and management problems.



Disadvantages of Informal Communication:

1. The information travels in a very unsystematic manner and gets distorted while passing through various channels.
2. Informal communication helps in spreading rumors and false facts.
3. Responsibilities cannot be fixed as the source of information cannot be known.
4. The confidential information may get leaked under informal communication.

Improving Communication Effectiveness:

All organizations face barriers to effective communication. By using following measures they can overcome such barriers.

1. Clarify the idea before communication:

In the first place we must be clear about what we want to communicate. The message can be conveyed properly only if it is clearly formulated in the mind of the communicator. The message should be encoded in direct and simple language so that the receiver is able to understand it without much difficulty.

2. Communication according to the need of the receiver:

Whenever we communicate we must keep in mind the needs of the receiver of the message or it should be our effort to see that whatever message or information we send across must be of value to receiver. This will make receiver more receptive. Sender must select the words according to the education level of receiver.

3. Consult others before communicating:

Before communicating the message it is advisable to consult others. Effective communication is the responsibility of all persons in the organisation as all have to work towards a common goal. If plans are developed with consultation and involvement of subordinates, they will accept it with full cooperation.

4. Use of proper language, tone and contents of message:

For an effective communication the words, tone and symbols used in message must be selected very carefully. The language used should not offend the sentiments of listener. The symbols used in message must be known to the listener.

5. Proper feedback:

Feedback helps to know the effect or success of communication given by sender. Communication is complete only when the message is understood by the receiver.

We can never know whether receiver has understood the message or not unless we get the feedback. Feedback also provides opportunity for suggestions and criticism.

6. Communication for the present as well as for future:

Communication must meet the need of present organisation as well as for future organisation. There must be consistency in the past, present and future communication.

7. Follow up communication:

There should be proper follow up of the information given by manager to subordinate. This follow up helps to remove hurdles, misunderstanding of instructions given by manager to subordinates.

8. Good listener:

The sender must listen to receiver's words alternatively; on the other hand receiver must also listen with due attention. Patient and attentive listening solve many problems.

9. Open mind:

The parties to communication must have open mind. They should not try to withhold information for their personal interest. They should not react before receiving and listening the full message.

10. Completeness of message:

A message is effective only when it is given completely. The receiver should not be left guessing. It may lead to misunderstanding. A complete message carries all necessary facts and figures.



4.6 INDUSTRIAL DISPUTES

In common parlance, dispute means difference or disagreement or strife over some issues between the parties. As regards industrial dispute, since its settlement proceeds as per the legal provisions contained in the 'Industrial Disputes' Act, 1947, hence it seems pertinent to study the concept of industrial disputes from a legalistic angle.

According to Section 2 (k) of the Industrial Disputes Act, 1947, the term 'industrial dispute' means "any dispute or difference between employers and employers or between employers and workmen, or between workmen and workmen, which is connected with the employment or non-employment or the terms of employment and conditions of employment of any person".

The above definition is too broad and includes differences even between groups of workmen and employers engaged in an industry. However, in practice, industrial disputes mainly relate to the difference between the workmen and the employers.

Dispute differs from discipline and grievance. While discipline and grievance focus on individuals, dispute focuses on collectivity of individuals. In other words, the test of industrial dispute is that the interest of all or majority of workmen is involved in it.

The following principles judge the nature of an industrial dispute:

1. The dispute must affect a large number of workmen who have a community of interest and the rights of these workmen must be affected as a class.
2. The dispute must be taken up either by the industry union or by a substantial number of workmen.
3. The grievance turns from individual complaint into a general complaint.

4. There must be some nexus between the union and the dispute.
5. According to Section 2A of the Industrial Disputes Act, 1947, a workman has a right to raise an industrial dispute with regard to termination, discharge, dismissal, or retrenchment of his or her service, even though no other workman or any trade union of workman or any trade union of workmen raises it or is a party to the dispute.

Forms of Industrial Disputes (Effects):

The industrial disputes are manifested in the following forms:

Strikes: Strike is the most important form of industrial disputes. A strike is a spontaneous and concerted withdrawal of labour from production. The Industrial Disputes Act, 1947 defines a strike as “suspension or cessation of work by a group of persons employed in any industry, acting in combination or a concerted refusal or a refusal under a common understanding of any number of persons who are or have been so employed to continue to work or accept employment”.

According to Patterson “Strikes constitute militant and organised protest against existing industrial relations. They are symptoms of industrial unrest in the same way that boils symptoms of disordered system”.

Depending on the purpose, Mamoria et. al. have classified strikes into two types: primary strikes and secondary strikes.

(i) Primary Strikes:

These strikes are generally aimed against the employers with whom the dispute exists. They may include the form of a stay-away strike, stay-in, sit-down, pen-down or tools- down, go-slow and work-to-rule, token or protest strike, cat-call strike, picketing or boycott.

(ii) Secondary Strikes:

These strikes are also called the ‘sympathy strikes’. In this form of strike, the pressure is applied not against the employer with whom the workmen have a dispute, but against the third person who has good trade relations with the employer.

However, these relations are severed and the employer incurs losses. This form of strike is popular in the USA but not in India. The reason being, in India, the third person is not believed to have any locus standi so far the dispute between workers and employer is concerned.

General and political strikes and bandhs come under the category of other strikes:

Lock-Outs:

Lock-out is the counter-part of strikes. While a ‘strike’ is an organised or concerted withdrawal of the supply of labour, ‘lock-out’ is withholding demand for it. Lock-out is the weapon available to the employer to shut-down the place of work till the workers agree to resume work on the conditions laid

down by the employer. The Industrial Disputes Act, 1947 defined lock-out as “the temporary shutting down or closing of a place of business by the employer”.

Lock-out is common in educational institutions also like a University. If the University authority finds it impossible to resolve the dispute raised by the students, it decides to close-down (or say, lockout) the University till the students agree to resume to their studies on the conditions laid down by the University authority. Recall, your own University might also have declared closure sometimes for indefinite period on the eve of some unrest / dispute erupted in the campus.

Gherao:

Gherao means to surround. It is a physical blockade of managers by encirclement aimed at preventing the egress and ingress from and to a particular office or place. This can happen outside the organisational premises too. The managers / persons who are gheraoed are not allowed to move for a long time.

Sometimes, the blockade or confinements are cruel and inhuman like confinement in a small place without light or fans and for long periods without food and water. The persons confined are humiliated with abuses and are not allowed even to answer “calls of nature”.

The object of gherao is to compel the gheraoed persons to accept the workers’ demands without recourse to the machinery provided by law. The National Commission on Labour has refused to accept ‘gherao’ as a form of industrial protest on the ground that it tends to inflict physical duress (as against economic press) on the persons gheraoed and endangers not only industrial harmony but also creates problems of law and order.

Picketing and Boycott:

Picketing is a method designed to request workers to withdraw cooperation to the employer. In picketing, workers through display signs, banners and play-cards drew the attention of the public that there is a dispute between workers and employer.

Workers prevent their colleagues from entering the place of work and persuade them to join the strike. For this, some of the union workers are posted at the factory gate to persuade others not to enter the premises but to join the strike.

Boycott, on the other hand, aims at disrupting the normal functioning of the organisation. The striking workers appeal to others for voluntary withdrawal of co-operation with the employer. Instances of boycotting classes and examinations are seen in the Universities also.

Types of Industrial Disputes:

The ILO’ has classified the industrial disputes into two main types.

1. Interest Disputes
2. Grievance or Right Disputes.



1. Interest Disputes:

These disputes are also called 'economic disputes'. Such types of disputes arise out of terms and conditions of employment either out of the claims made by the employees or offers given by the employers. Such demands or offers are generally made with a view to arrive at a collective agreement. Examples of interest disputes are lay-offs, claims for wages and bonus, job security, fringe benefits, etc.

2. Grievance or Right Disputes:

As the name itself suggests, grievance or right disputes arise out of application or interpretation of existing agreements or contracts between the employees and the management. They relate either to individual worker or a group of workers in the same group.

That's way in some countries; such disputes are also called 'individual disputes'. Payment of wages and other fringe benefits, working time, over-time, seniority, promotion, demotion, dismissal, discipline, transfer, etc. are the examples of grievance or right disputes.

If these grievances are not settled as per the procedure laid down for this purpose, these then result in embitterment of the working relationship and a climate for industrial strife and unrest. Such grievances are often settled through laid down standard procedures like the provisions of the collective agreement, employment contract, works rule or law, or customs /usage in this regard. Besides, Labour Courts or Tribunals also adjudicate over grievance or interest disputes.

Generally, industrial disputes are considered as 'dysfunctional' and 'unhealthy'. These are manifested in the forms of strikes and lock-outs, loss of production and property, sufferings to workers and consumers and so on. But, sometimes industrial disputes are beneficial as well.

It is the dispute mainly which opens up the minds of employers who then provide better working conditions and emoluments to the workers. At times, disputes bring out the causes to the knowledge of the public where their opinion helps resolve them.

Causes of Industrial Disputes:

Psychological Causes:

- (i) Authoritarian leadership (nature of administration).
- (ii) Clash of personalities.
- (iii) Difficulty in adjusting in given conditions or with each other (employee and employer).
- (iv) Strict discipline.
- (v) Demand for self-respect and recognition by workers.

Institutional Causes:

- (i) Non recognition of trade/labour union by the management.
- (ii) Matters of collective bargaining.
- (iii) Unfair conditions and practices.
- (iv) Pressure on workers to avoid participation in trade unions.

Economic Causes:

- (a) Terms and conditions of employment.
 - (i) More work hours.
 - (ii) Working in night shifts.
 - (iii) Disputes on promotions, layoff, retrenchment and dismissal etc.
- (b) Working conditions.
 - (i) Working conditions such as too hot, too cold, dusty, noisy etc.
 - (ii) Improper plant and work place layout.
 - (iii) Frequent product design changes etc.
- (c) Wages and other benefits.
 - (i) Inadequate wages.
 - (ii) Poor fringe benefits.
 - (iii) No bonus or other incentives etc.

Denial of Legal and other Right of Workers:

- (i) Proceeding against labour laws and regulations.
- (ii) Violation of already made agreements i.e., between employees and employers.

4.7 COLLECTIVE BARGAINING

Collective bargaining is a process of negotiating between management and workers represented by their representatives for determining mutually agreed terms and conditions of work which protect the interest of both workers and the management. According to Dale Yoder, “Collective bargaining is essentially a process in which employees act as a group in seeking to shape conditions and relationships in their employment”.

Michael J. Jucious has defined collective bargaining as “a process by which employers, on the one hand, and representatives of employees, on the other, attempt to arrive at agreements covering the conditions under which employees will contribute and be compensated for their services”.

Thus, collective bargaining can simply be defined as an agreement collectively arrived at by the representatives of the employees and the employers. By collective bargaining we mean the ‘good faith bargaining’. It means that proposals are matched with counter proposals and that both parties make every reasonable effort to arrive at an agreement’ It does not mean either party is compelled to agree to a proposal. Nor does it require that either party make any specific concessions.

Why is it called collective bargaining? It is called “collective” because both the employer and the employee act collectively and not individually in arriving at an agreement. It is known as ‘bargaining’ because the process of reaching an agreement involves proposals and counter proposals, offers and counter offers.

Objectives:

The basic objective of collective bargaining is to arrive at an agreement between the management and the employees determining mutually beneficial terms and conditions of employment.

1. To foster and maintain cordial and harmonious relations between the employer/management and the employees.
2. To protect the interests of both the employer and the employees.
3. To keep the outside, i.e., the government interventions at bay.
4. To promote industrial democracy.

Importance:

The need for and importance of collective bargaining is felt due to the advantages it offers to an organisation.

1. Collective bargaining develops better understanding between the employer and the employees:

It provides a platform to the management and the employees to be at par on negotiation table. As such, while the management gains a better and deep insight into the problems and the aspirations of the employees, on the one hand, the employees do also become better informed about the organisational problems and limitations, on the other. This, in turn, develops better understanding between the two parties.

2. It promotes industrial democracy:

Both the employer and the employees who best know their problems, participate in the negotiation process. Such participation breeds the democratic process in the organisation.

3. It benefits the both-employer and employees:

The negotiation arrived at is acceptable to both parties—the employer and the employees.

4. It is adjustable to the changing conditions:

A dynamic environment leads to changes in employment conditions. This requires changes in organisational processes to match with the changed conditions. Among other alternatives available, collective bargaining is found as a better approach to bring changes more amicably.

5. It facilitates the speedy implementation of decisions arrived at collective negotiation:

The direct participation of both parties—the employer and the employees—in collective decision making process provides an in-built mechanism for speedy implementation of decisions arrived at collective bargaining.

4.8 TRADE UNION

Trade unions are associations of workers and are formed with the intention of protecting the workers against exploitation of the employers and also to improve the workers' conditions. The industrial revolution in England and in other countries and the advent of the factory system of production are greatly responsible for the emergence of trade unions.

The modern industrial establishments employ the workers in conditions which make them helpless in bargaining individually. The workers are unorganised and are subject to exploitation by the employer. This exploitation of the employers is resented by the working class. Hence they form unions to protect themselves against the exploitations of the employers.

Thus, the trade union movement arose for the purpose of defending their rights, for improving their living and working conditions and for protecting their interests. From a small beginning, this movement has now grown into worldwide movement for achieving social justice, economic equality and preservation of democratic values of life.

Objectives of Trade Union:

- (1) To improve the economic lot of workers by securing them better wages.
- (2) To secure for workers better working conditions.
- (3) To secure bonus for the workers from the profits of the enterprise/organization.
- (4) To ensure stable employment for workers and resist the schemes of management which reduce employment opportunities.
- (5) To provide legal assistance to workers in connection with disputes regarding work and payment of wages.
- (6) To protect the jobs of labour against retrenchment and layoff etc.
- (7) To ensure that workers get as per rules provident fund, pension and other benefits.
- (8) To secure for the workers better safety and health welfare schemes.
- (9) To secure workers participation in management.
- (10) To inculcate discipline, self-respect and dignity among workers.
- (11) To ensure opportunities for promotion and training.
- (12) To secure organizational efficiency and high productivity.
- (13) To generate a committed industrial work force for improving productivity of the system.

Functions of Trade Unions:

- (1) Collective bargaining with the management for securing better work environment for the workers/employees.
- (2) Providing security to the workers and keeping check over the hiring and firing of workers.
- (3) Helping the management in redressal of grievances of workers at appropriate level.
- (4) If any dispute/matter remains unsettled referring the matter for arbitration.
- (5) To negotiate with management certain matters like hours of work, fringe benefits, wages and medical facilities and other welfare schemes.
- (6) To develop cooperation with employers.
- (7) To arouse public opinion in favour of labour/workers.

Benefits of Trade Union:

Workers join trade union because of a number of reasons as given below:

1. A worker feels very weak when he is alone. Union provides him an opportunity to achieve his objectives with the support of his fellow colleagues.
2. Union protects the economic interest of the workers and ensures a reasonable wage rates and wage plans for them.
3. Union helps the workers in getting certain amenities for them in addition to higher wages.
4. Union also provides in certain cases cash assistance at the time of sickness or some other emergencies.
5. Union organize negotiation between workers and management and are instruments for settlement of disputes.
6. Trade union is also beneficial to employer as it organizes the workers under one banner and encourages them follow to peaceful means for getting their demands accepted.
7. Trade union imparts self-confidence to the workers and they feel that they are an important part of the organization.
8. It provides for promotion and training and also helps the workers to go to higher positions.
9. It ensures stable employment for the workers and opposes the motive of management to replace the workers by automatic machines.
10. Workers get an opportunity to take part in the management and oppose any decision which adversely effects them.

4.9 WORKERS PARTICIPATION IN MANAGEMENT

Management considers participation as a joint consultation prior to decision-making, while workers mean co-determination by the term 'participation'. Government looks at 'participation' as an association of labour with management without the final authority or responsibility in decision-making. But, generally conceived, it means a way of reducing power difference and, therefore, equality is stressed.

Frankly speaking, this idea of reduction of power inequality, between management and labour through workers' participation cannot be accepted since there is the obvious difference in formal power position between the labourer and the director.

However, certain elements of participation are very much clear though the more precise conceptual properties of participation are not so clear. The elements are: the amount of verbal interaction between a supervisor and his subordinates, the extent to which a supervisor performs his role in a group-centred way and the degree to which a supervisor is active in making decisions and setting goals for his work units.

Whatever might be the interpretations of the term participation, the fact remains that it means sharing, in an appropriate manner, the decision-making power with the lower ranks of the organisation. There are levels of participation in decision-making which range on a scale from minimum to maximum.

Characteristics:

1. Participation implies practices which increase the scope for employees' share of influence in decision-making process with the assumption of responsibility.
2. Participation presupposes willing acceptance of responsibility by workers.
3. Workers participate in management not as individuals but as a group through their representatives.
4. Worker's participation in management differs from collective bargaining in the sense that while the former is based on mutual trust, information sharing and mutual problem solving; the latter is essentially based on power play, pressure tactics, and negotiations.
5. The basic rationale for worker's participation in management is that workers invest their labour and their fates to their place of work. Thus, they contribute to the outcomes of organization. Hence, they have a legitimate right to share in decision-making activities of organisation.

Objectives:

1. Promote mutual understanding between management and workers, i.e., industrial harmony.
2. Establish and encourage good communication system at all levels.
3. Create and promote a sense of belongingness among workers.
4. Help handle resistance to change.
5. Induce a sense among workers to contribute their best for the cause of organisation.
6. Create a sense of commitment to decisions to which they were a party.

Levels of Participation:

Having known the objectives of WPM, the question then is to what extent workers can participate in decision-making process. In other words, it is important to know the extents/levels of co-determination in an organisation.

Viewed from this angle, Mehtras has suggested five levels of workers' participation ranging from the minimum to the maximum. Since these levels of workers' influence the process and quality of decision making in an organisation. We are therefore highlighting here these levels briefly ranking them from the minimum to the maximum level of participation.

Informative Participation:

This refers to management's information sharing with workers on such items those are concerned with workers. Balance Sheet, production, economic conditions of the plant etc., are the examples of such items. It is important to note that here workers have no right of close scrutiny of the information provided and management has its prerogative to make decisions on issues concerned with workers.

Consultative Participation:

In this type of participation, workers are consulted in those matters which relate to them. Here, the role of workers is restricted to give their views only. However the acceptance and non-acceptance of these

views depends on management. Nonetheless, it provides an opportunity to the workers to express their views on matters involving their interest.

Associative Participation:

Here, the role of the workers' council is not just advisory unlike consultative participation. In a way, this is an advanced and improved form of consultative participation. Now, the management is under a moral obligation to acknowledge, accept and implement the unanimous decision of the council.

Administrative Participation:

In the administrative participation, decisions already taken are implemented by the workers. Compared to the former three levels of participation, the degree of sharing authority and responsibility by the workers is definitely more in this participation.

Decisive Participation:

Here, the decisions are taken jointly by the management and the workers of an organisation. In fact, this is the ultimate level of workers' participation in management.

Benefits

- (1) Enhancement of the sense of responsibility of workers,
- (2) Change of mental attitude of workers to accept any change,
- (3) Encouragement for better decisions,
- (4) Improvement of morale and team work,
- (5) Use of the creative employees,
- (6) Encouragement for self-expression and self-advancement,
- (7) Fostering dignity to workers,
- (8) Improvement of product quality and quantity,
- (9) Reduction of workers' grievances and industrial conflicts.

4.10 INDUSTRIAL SAFETY

The importance of industrial safety was realized because of the fact that every year millions occupational/ industrial accidents occur which result in loss of production time equivalent to millions of man hours, machine hours etc.

Of these about one-fifth production time is lost by those actually injured due to temporary and permanent disablement and the remaining production time is lost by fellow operators/ people in helping the injured, in taking care of the damage caused by accident etc. the loss to the industrial unit would appear much more alarming when death cases due to accidents are considered.

It is therefore essential to identify/examine the causes of industrial accidents and take steps to control them. Many disciplines are concerned with this safety approach. Industrial engineering is one field which

deals with design of efficient work place, equipment and industrial layout design. Other disciplines which can contribute to safe working environment are psychology, sociology and Medicare science.

The following steps may be taken to effectively and efficiently eliminate an unsafe working environment:

- (1) Elimination if possible of the causes of accidents.
- (2) If it is not possible to eliminate the cause of accidents, make arrangements to shield the hazardous place by guards, enclosures or similar arrangements.

Need for Safety:

In view of above discussion, need and concern for safety is therefore need of the hour. There are some direct costs/ effects of an accident but there are certain indirect costs involved in it also e.g. machine down time, damage to machine, ideal time of nearby equipment and horror created among workers, loss of time etc. in aid cost compensation, legal implications and allied costs etc. So safety measures would not only eliminate/ avoid above cost but would mean performing their moral responsibility towards workmen/operators also.

An accident is by virtue of unsafe factor he results of an unsafe condition it may be the combined effect of two. An unsafeact results in the form of operator/people doing thing without proper authority, misuse of safety devices, ignoring warnings and precautions etc.

An unsafe condition may be present in various forms e.g. faulty or defective electrical fittings, inadequate maintenance of gang way. Use of defective tools etc. So to prevent the occurrence of accidents, unsafe acts have to be avoided/ eliminated or checked.

Unsafe acts:

For rectification of the causes because of unsafe acts attention must be paid to following factors:

(1) Personnel adjustment:

If a foreman/supervisor identifies that a worker is unfit either physically or mentally or a job/ task, he should be quickly taken off the work in consultation with the personnel department.

(2) Method/technique used:

Some techniques requiring change should be replaced by safe methods.

(3) Operator training:

Job method may be safe or unsafe but the operator must be trained to perform the job.

(4) Publicity and education about accident prevention:

The workers/ people are led by the skill, energy and leadership of foreman/supervisor. So it is the duty of these people to educate the workmen about prevention of accidents. The aim is to teach them to become safety conscious so that they are able to recognize an unsafe act or situation and act in such a manner that accident is avoided.

The unsafe conditions:

To avoid accidents due to unsafe conditions, various provisions have been discussed in the “Factories Act” these may be concerned with moving parts of prime movers, electrical generators and transmission machinery: fire protection devices, control of dangerous fumes, lifting of excessive weights and safe guards over lighting machines, chains and ropes etc.

Thus safety in industry helps:

- (i) Increasing the production rate.
- (ii) Reducing the cost of production.
- (iii) Reducing damage to machinery and equipment.
- (iv) Preventing unwanted suffering and pain to employees of the organization.
- (v) Preventing premature/untimely death of talented workers who may be an asset to the enterprise and society.

Safety Measures for Workers Working in Industries

The term safety measures means the provision of devices, equipment or other methods which ensure safety e.g. presence of guard endorsing a rotating part of a machine, and fire extinguishers etc. Safety procedure relate to the methods to be followed for the minimization of accidents e.g. provision of double controls. While designing an electrical installation i.e. one local and other remote so as to ensure double safety.

Safety Measures:

(1) Design of factory buildings:

These should include:

- (i) Proper lighting and ventilation provisions.
- (ii) Proper control of overhead cranes.
- (iii) Due consideration to be given to sharp bends, heavy slopes and undue steps.
- (iv) Availability of emergency exits.
- (v) Adequate control on electrical installation must be available.
- (vi) First aid arrangement in different shops.

(2) Safety against machine accidents:

The following provisions should be made:

- (i) Dangerous parts should be out of reach during operation.
- (ii) Some guards are built into a permanent casing, while some are attached after wards.
- (iii) Machines or their parts should be fenced when it is not possible to provide safe guards.



- (iv) All boilers and other pressure vessels must be kept in good/ proper condition. Safety valves, pressure gauges and water gauges etc. must be thoroughly examined at regular intervals.
- (v) Cranes, hoists and lifts etc. must be of sound construction. They should be periodically tested.
- (vi) Repair work on machines to be avoided during running of machines.
- (vii) All belts, gears, fly wheels and pulleys etc. should be covered with adequate guards.

(3) Safety against fire:

Following measures should be taken.

- (i) Adequate provision of passages and exits should be made.
- (ii) The doors should be open out wards.
- (iii) Adequate warning signals should be provided.
- (iv) Fire extinguishers should kept at suitable places.
- (v) Storing of explosives (if required to store).
- (vi) Training in fire fighting should be given to all employees.

(4) Prevention of Electrical accidents:

To prevent electrical accidents following measures should be taken:

- (i) Insulate all electrical equipment, wiring appliances and controls and these should be periodically tested.
- (ii) All metallic covers over electrical machinery or control equipment should be properly earthed.
- (iii) Underground cables/wires should be safe from mechanical damage or exposure to excessive moisture.
- (iv) Every live conductor should begin with a safety fuse.
- (v) The C.T.C. extinguishers should be clearly marked.
- (vi) Repair work should be started after switching the power off.
- (vii) Use safety equipment's such as insulated tools, rubber gloves etc. whenever required.

4.11 PERSONAL PROTECTIVE DEVICES FOR THE EMPLOYEES

1. Head protection

Head is to be protected against –

- i) Objects falling from a height particularly in construction sites;
- ii) Heat near the boilers and furnaces;
- iii) Splashing of chemicals.

Various types of industrial helmets are available and they are to be selected and used by the employees according to the requirements. Women workers have to use hairnets and safety caps if there is a risk of hair becoming caught in gears, drills or spinning shafts of machine.

2. Ear protection:

To guard against deafness, ear-plugs, ear-muffs and helmets incorporating ear-muffs plus communication devices are to be used.

3. Eye protection

Eyes are to be protected against –

- i) Glare and radiation as in the case of welding;
- ii) Flying fragments as in the case of grinding, turning on a lathe or a milling machine;
- iii) Dust as in the case of carpentry and foundry;
- iv) Chemical splashes in chemical industries;
- v) Molten metal in foundries.

Lenses, eye-shields, goggles and safety spectacles are the chief means of eye protection.

4. Body protection Body is to be protected against –

- i) Minor mishaps;
- ii) Chemical hazards and
- iii) Fire.

Apron, boiler-suits, asbestos suits, hot entry suits are some of the body protective devices. Body protection against falling from heights is provided by safety-belts.

5. Hand protection

Gloves, pads, arm shields and wrist bands provide protection for the arms and hands against any hazards. Hand protection (skin protection) is also provided by barrier creams of various types and special hand cleaners.

6. Lung protection

Light, simple face-masks are sufficient to protect workers against dust, non-toxic sprays and fumes. In dangerous conditions, it is necessary to use either respirator which purifies the air being breathed or breathing apparatus which provides oxygen or air from non-contaminated source. Breathing apparatus must be used where the surrounding is contaminated.

7. Foot protection

In foundries, the workers must wear safety footwear which will protect their feet from injuries due to handling of molten metal. In some cases, special types of safety shoes are to be used to avoid injuries due to falling weight on toes.

In addition to the above personal protective devices, the employer should provide:

- a. Suitable containers for flammable liquids.
- b. Safety nets.
- c. Non-slippery floors.

- d. Safety signs.
- e. Gas detectors.
- f. Fire fighting and warning equipment.

Personal protective equipment should be used as a last resort. The primary thing is to eliminate the hazards so that a careless employee will not be in danger even if he does not wear a hat or an apron.

IMPORTANT QUESTIONS

1. What are the psychological factors which affects the attitude of workers towards the work?
2. Explain different methods for eliminating fatigue.
3. Explain the causes and effects of industrial disputes.
4. What is collective bargaining?
5. Explain worker's participation in management.
6. Explain the importance of having effective communication in an industry.
7. Explain different types of industrial disputes.
8. Explain different personal protective devices used in a production plant.
9. Explain different types of fatigue.
10. Explain formal and informal communication.
11. List the objectives, functions and benefits of trade unions.
12. Explain Industrial safety.



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MODULE 5

5.1 PRODUCTION PLANNING AND CONTROL

Production planning and control is a device that regulates the movements of materials, performance of machines and operation of labour in the best technical and economical manner; so as to obtain right quantity of production of required quality – at a time which is promised for delivery of goods to customers.

“Production control is the process of planning production in advance of operations, establishing the exact route of each individual item, part or assembly; setting starting and finishing dates for each important item, assembly and the finished products, and releasing the necessary orders as well as initiating the required follow-up to effectuate the smooth functioning of the enterprise.”

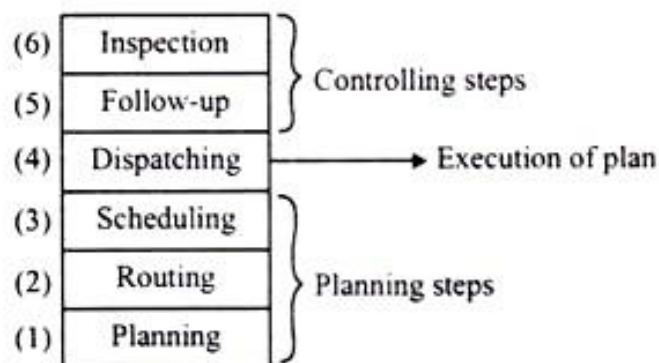
Steps in Production Planning and Control:

Production control involves the following steps:

- (i) Planning
- (ii) Routing
- (iii) Scheduling
- (iv) Dispatching
- (v) Follow-up or checking the progress
- (vi) Inspection

Out of these six steps involved in production control, the first three steps relate to planning; the fourth relates to execution of plan and the last two refer to the control aspect of planning.

The above idea is depicted by means of the following diagram:



(1) Planning:

For planning of productive operations in detail, the planning department will receive full information from management about the quantity to be produced and the dates when delivery has been promised to customers. The planning department will also get the necessary engineering and drawing specifications from the engineering department.

Broadly, at the stage of planning the following issues are considered on which bases charts and written plans are prepared:

- (a) What work should be done?
- (b) How shall the work be done?
- (c) Where shall the work be done?
- (d) When shall the work be done?

(2) Routing:

Routing involves the determination of the path that work shall follow and the order in which various operations will be carried out. The objective of routing is to find out the best and the cheapest sequence of operations. While preparing the route card, it must be kept in mind that machines in the plant are operated at their full capacity; and manpower and other facilities are best utilized.

(3) Scheduling:

Scheduling is the determination of the time that should be required to perform each operation and also the time necessary to perform the entire series, as routed, making allowance for factors concerned. It involves the preparation of a time-table, indicating the total time needed for the manufacture of a product as also the time expected to be spent at each machine and process.

In preparing schedules, the persons concerned will have to take into consideration the various types of orders on hand and the dates by which their completion has been promised. Some orders may be such as will require over-time work; because completion is not possible according to the delivery dates set for them, in the regular course of production.

(4) Dispatching:

Dispatching literally means sending something towards a particular destination. Here, it means taking all such steps, as are necessary to implement the programme of production chalked out as per routing and scheduling steps.

In particular, dispatching refers to:

1. Procurement of necessary tools, jigs and fixtures etc.; before they are actually required by the workmen.
2. Giving workers the necessary work orders, instructions, drawings etc. for initiating the work.

(5) Follow-Up (or Checking the Progress):

Follow-up is the control aspect of production planning and control. It involves taking steps to check up whether work proceeds according to plans and how far there are variances from standards; and also taking necessary corrective steps to set things in order.

(6) Inspection:

Inspection is the quality control aspect of production planning and control. It ensures that goods produced are of the right quality. The inspectors may inspect materials, semi-finished and finished products either at the work bench or in special laboratories or testing rooms.

To ensure maintenance of high standards of quality, a programme of SQC (Statistical Quality Control) may be fused with a system of production planning and control.

Objectives/Advantages of Production Planning and Control:

(i) Continuous Production:

Production control ensures continuous production with least possible interruptions; as it eliminates all sources of interruptions in production like non-availability of materials, tools, poor maintenance of machines etc.

(ii) Cost Control and Profit Maximization:

Production control helps in cost control (and thus in profit maximization) by optimizing use of productive resources and eliminating waste and spoilage.

(iii) Customer Satisfaction:

Production control ensures better service to customers due to timely delivery of goods and qualitative products. It, thus, leads to customer satisfaction and better business relations with customers.

(iv) Planning of Resource Requirements and Inventory Control:

Production control seeks to assess in advance requirements of manpower, machinery and other facilities to meet the desired targets of production. It also helps to maintain regular supply of raw-materials, work-in-progress and finished goods with minimum investment in inventories.

(v) Minimum Material Handling and Storage Costs:

Production control helps in minimization of material handling and storage costs.

(vi) Economy in Production Time:

Production control reduces the loss of time by the workers waiting for materials, and causes improvement in plant morale.

(vii) Equipment Utilization:

Production control makes for the most effective use of equipment.

Objectives/advantages of production control – at a glance

1. Continuous production
2. Cost-control and profit-maximisation
3. Customer satisfaction
4. Planning of resource requirements and inventory control
5. Minimum material handling and storage costs
6. Economy in production time
7. Equipment utilization

5.2 IMPORTANCE OF PRODUCTION PLANNING

Production planning involves the means by which a manufacturing plan is determined, information issued for its execution, data collected and recorded, which will enable the plant to be controlled through all its stages. A few definitions are given here in order to have clear understanding of the term ‘Production Planning’.

“The planning of industrial operations involves four considerations, namely, what work shall be done, how the work shall be done and lastly, when the work shall be done”.

“The technique of forecasting or picturing ahead every step in a long series of separate operations, each step to be taken in the right place of the right degree and at the right time and each operation to be done at maximum efficiency.”

“Production planning is a series of related and co-ordinated activities performed by not one but a number of different departmental groups, each activity being to systematise in advance the manufacturing efforts in its area. ”

Objectives of Production Planning:

1. To achieve coordination among various departments relating to production.
2. To make adequate arrangement of men, money, materials, machines tools, implements and equipment relating to production.
3. To decide about the production targets to be achieved by keeping in view the sales forecast.
4. To keep production operation continuous.
5. To achieve desired share of the market.
6. To fix right type of man for right type of job.
7. To achieve the desired level of profit.
8. To make all arrangements to remove possible obstacles in the way of smooth production.
9. To achieve economy in production cost and time.
10. To initiate production on modern lines.
11. To operate the plant at planned level of efficiency.
12. To develop alternative plans in order to meet any emergency or contingency.

5.3 TYPES OF PRODUCTION

Some of the most important types of production are: (i) Job Production (ii) Batch production and (iii) Mass or flow production! The final decision regarding any particular method of production is very much affected by the nature of the products and the quantity to be produced. Production methods may be broadly classified as Job Production, Batch production and Mass or Flow Production.

(i) Job Production:

Under this method peculiar, special or non-standardized products are produced in accordance with the orders received from the customers. As each product is non- standardized varying in size and nature, it requires separate job for production. The machines and equipment's are adjusted in such a manner so as to suit the requirements of a particular job.

Job production involves intermittent process as the work is carried as and when the order is received. It consists of bringing together of material, parts and components in order to assemble and commission a single piece of equipment or product.

Ship building, dam construction, bridge building, book printing are some of the examples of job production. Third method of plant layout viz., Stationery Material Layout is suitable for job production.

Characteristics:

The job production possesses the following characteristics.

1. A large number of general purpose machines are required.
2. A large number of workers conversant with different jobs will have to be employed.
3. There can be some variations in production.
4. Some flexibility in financing is required because of variations in work load.
5. A large inventory of materials, parts and tools will be required.
6. The machines and equipment setting will have to be adjusted and readjusted to the manufacturing requirements.
7. The movement of materials through the process is intermittent.

Limitations:

1. The economies of large scale production may not be attained because production is done in short-runs.
2. The demand is irregular for some products.
3. The use of labour and equipment may be an inefficient.
4. The scientific assessment of costs is difficult.

(ii) Batch production:

Batch production pertains to repetitive production. It refers to the production of goods, the quantity of which is known in advance. It is that form of production where identical products are produced in batches on the basis of demand of customers' or of expected demand for products.

This method is generally similar to job production except the quantity of production. Instead of making one single product as in case of job production, a batch or group of products are produced at one time. It should be remembered here that one batch of products may not resemble with the next batch.

Under batch system of production the work is divided into operations and one operation is done at a time. After completing the work on one operation it is passed on to the second operation and so on till the product is completed. Batch production can be explained with the help of an illustration. An enterprise wants to manufacture 20 electric motors.

The work will be divided into different operations. The first operation on all the motors will be completed in the first batch and then it will pass on to the next operation. The second group of operators will complete the second operation before the next and so on. Under job production the same operators will manufacture full machine and not one operation only.

Batch production can fetch the benefits of repetitive production to a large extent, if the batch is of a sufficient quantity. Thus batch production may be defined as the manufacture of a product in small or large batches or lots by series of operations, each operation being carried on the whole batch before any subsequent operation is operated. This method is generally adopted in case of biscuit and confectionery and motor manufacturing, medicines, tinned food and hardware's like nuts and bolts etc.

Characteristics:

1. The work is of repetitive nature.
2. There is a functional layout of various manufacturing processes.
3. One operation is carried out on whole batch and then is passed on to the next operation and so on.
4. Same type of machines is arranged at one place.
5. It is generally chosen where trade is seasonal or there is a need to produce great variety of goods.

(iii) Mass or flow production:

This method involves a continuous production of standardized products on a large scale. Under this method, production remains continuous in anticipation of future demand. Standardization is the basis of mass production. Standardized products are produced under this method by using standardized materials and equipment. There is a continuous or uninterrupted flow of production obtained by arranging the machines in a proper sequence of operations. Process layout is best suited method for mass production units.

Flow production is the manufacture of a product by a series of operations, each article going on to a succeeding operation as soon as possible. The manufacturing process is broken into separate operations.

The product completed at one operation is automatically passed on to the next till its completion. There is no time gap between the work done at one process and the starting at the next. The flow of production is continuous and progressive.

Characteristics:

The mass or flow production possesses the following characteristics.

1. The units flow from one operation point to another throughout the whole process.
2. There will be one type of machine for each process.
3. The products, tools, materials and methods are standardised.
4. Production is done in anticipation of demand.
5. Production volume is usually high.
6. Machine set ups remain unchanged for a considerable long period.
7. Any fault in flow of production is immediately corrected otherwise it will stop the whole production process.

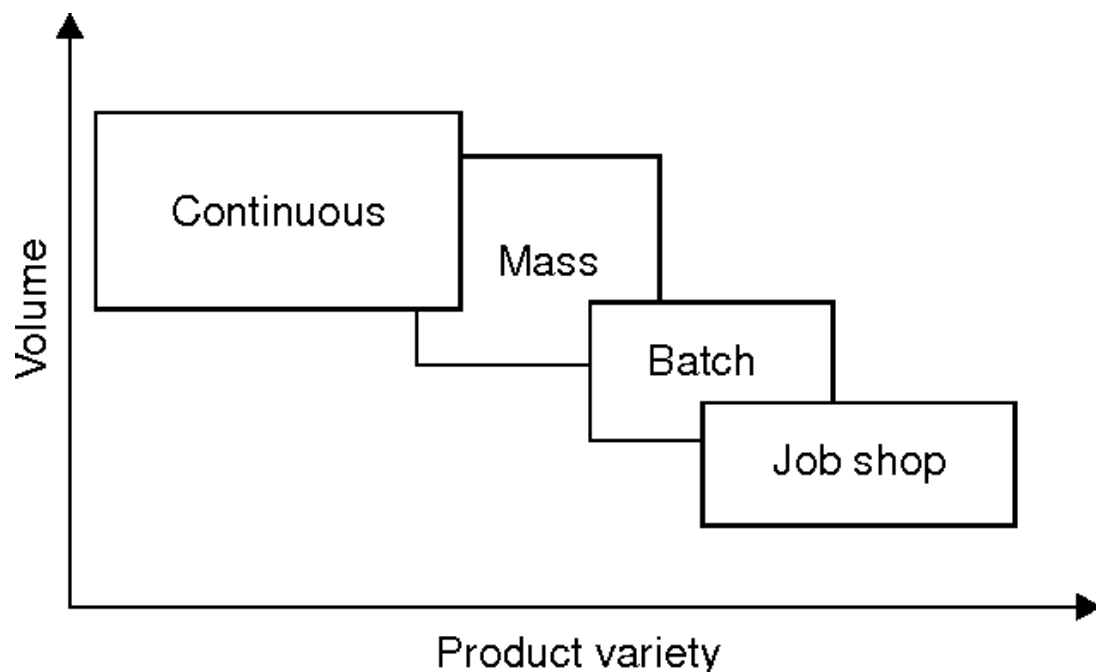


Suitability of flow/mass production:

1. There must be continuity in demand for the product.
2. The products, materials and equipments must be standardised because the flow of line is inflexible.
3. The operations should be well defined.
4. It should be possible to maintain certain quality standards.
5. It should be possible to find time taken at each operation so that flow of work is standardised.
6. The process of stages of production should be continuous.

Advantages of mass production:

1. The product is standardised and any deviation in quality etc. is detected at the spot.
2. There will be accuracy in product design and quality.
3. It will help in reducing direct labour cost.
4. There will be no need of work-in-progress because products will automatically pass on from operation to operation.
5. Since flow of work is simplified there will be lesser need for control.
6. A weakness in any operation comes to the notice immediately.
7. There may not be any need of keeping work-in-progress, hence storage cost is reduced.



5.4 INTRODUCTION AND NEED FOR A NEW PRODUCT

Introducing new products is rather difficult as it involves long-range planning. Customer's need should be identified. Competing and substitutes products should be evaluated and above all the strength of the company should be examined before deciding to produce a new product.

Guidelines for New Product Development

1. Market requirements:

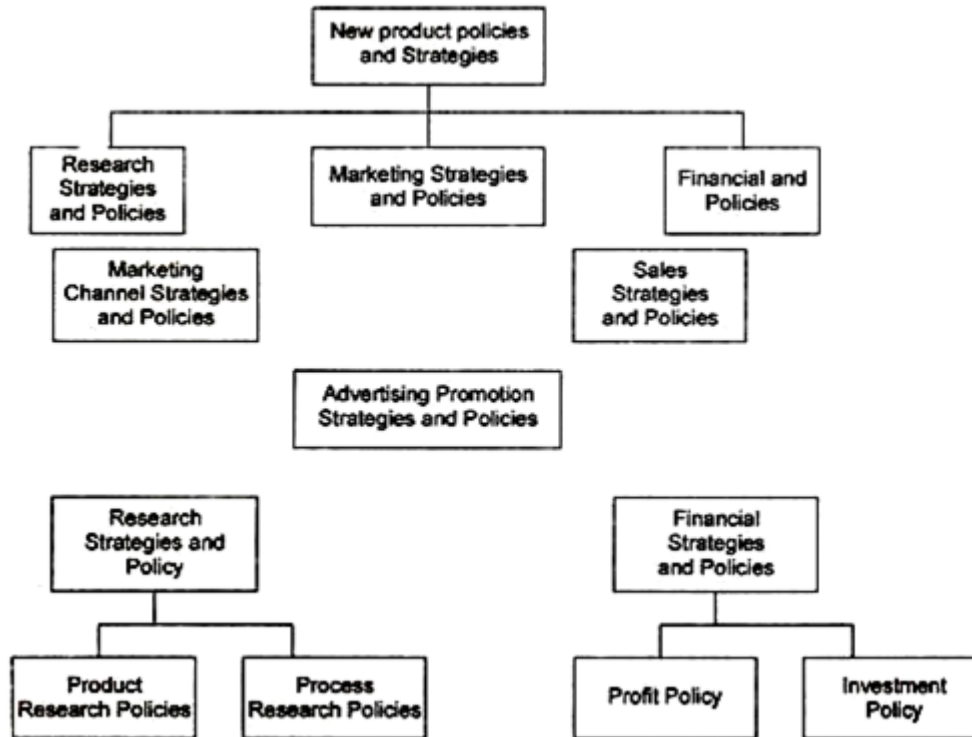
Successful innovators pay close attention to the requirements of the market. They undertake extensive and intensive market research to identify market wants and needs and maintain close liaison with potential users during the development.

2. Diligent development work:

Successful innovators tend to perform their development work diligently and efficiently rather than more quickly. They eliminate the possible technical defects from the product before launch and they have bigger development teams and spend more time and treasure on development.

3. Top management support:

Successful innovators have the wholehearted support of the people at the top in the corporate hierarchy. These 'product champions' are more likely to have the power to overcome organizational inertia and other impeding factors of new product development.



4. Good external communications:

Successful innovators have an open mind and good external communications. That is, they keep in good and constant contact with institutions in science and technology. These institutions can be polytechnics, research associations and so on. These act as both good source of original ideas for new products and easy and effective solutions of technical problems in the course of development.

Need for Developing a New Product

A new product, may be called innovation, seems to be necessary due to an enormous number of factors. The dynamic nature of business environment offers only two options – be innovative, or die. Marketer has to search for superior ways to meet increasing expectations.

Those companies fail to respect or respond the need of time have to withdraw from the market. Innovation or new product development directly concerns with firm's survival, growth, and development. A marketer continuously tries to offer superior product to his consumers.

He desires to offer more satisfaction than competitors. He also attempts to match goods and services with changing needs and wants of the target market. As per change in the fashion, preference, habits, and attitudes of consumers, a company must change its offer/product. New products are primarily developed to meet market expectations. However, there are some other dominants factors forcing a marketer to go for developing new products.

1. Changes in Market:

Today's market is much dynamic as compared to the past. Due to increased education, borderless marketing, severe competition, and availability of a number of substitutes have posed tremendous challenges for today's marketers. Market fashion, preference, and habits are constantly changing and marketer finds no option except to respect such market changes, by positive response in terms of innovation. Thus, consumer behaviour is one of the dominant reasons for innovation.

2. Changes in Technology:

Due to continuous technological development, new production methods are invented. Old technology and production methods are replaced by newer ones. A company spends a large amount of money for technological research. To match the technological changes, new products are developed.

3. Increasing Competition:

Increasing competition is one of significant reasons leading to go for innovation. Every company struggles to attract and maintain consumers by offering superior products. To offer more competitive advantages and to satisfy consumer more effectively and efficiently, the product innovation seems to be necessary.

4. Diversification of Risk:

In many cases, a company develops new products just to diversify risk. Existing products may not be capable to match with market needs and wants. By offering more varieties, a company can minimize the degree of obsolescence. Thus, the need for continuous innovation arises because older products are thrown out of market.

5. Reputation and Goodwill:

To create image and reputation as an innovative and dynamic firm, the innovation is adopted. Company wishes to convince the market that it tries seriously to meet consumer's expectations. Obviously, a company developing new products periodically has more reputation, and can attract consumers easily.

6. Utilization of Excess Capacity:

Excess capacity may be in form of production capacity or human skills. To utilize maximum plant and material capacity, a company may go for developing a new product. Sometimes, excess managerial or human capacity may also tempt the company to opt for innovation.

7. Seasonal Fluctuations:

Sometimes, new products are developed just to minimize seasonal fluctuations in demand. By producing new product, a company can meet seasonal requirements of market. Market is satisfied due to matching products in each of the seasons, and company can get attractive business.

8. Growth and Development:

Innovation is an effective way to win more market share or sales. Marketer can exploit emerging opportunities by innovative products. When it is not possible to accelerate growth rate by the existing products, a company prefers to develop new products to expand its market, maximize sales, and earn more profits.

Steps in new Product development

1. Generation of New Product Ideas:

The first step in product planning and development is generation of ideas for the development of new/innovative products.

Ideas may come from internal sources like company's own Research and Development (R&D) department, managers, sales-force personnel etc.; or from external sources like, customers, dealers, competitors, consultants, scientists etc.

At this stage, the intention of management is to generate more and more new and better product ideas; so that the most practical and profitable ideas may be screened subsequently.

2. Screening of Ideas:

Screening of ideas means a close and detailed examination of ideas, to determine which of the ideas have potential and are capable of making significant contribution to marketing objectives. In fact, generation of ideas is not that significant as the system for screening the generated ideas.

The ideas should be screened properly; as any idea passing this stage would cost the firm in terms of time, money and efforts, at subsequent stages in product planning and development.

3. Product Concept Development:

Those product ideas which clear the screening stage must be developed into a product concept – identifying physical features, benefits, price etc. of the product. At this stage product idea is transformed into a product concept i.e. a product which target market will accept.

4. Commercial Feasibility:

At this stage, the purpose is to determine whether the proposed product idea is commercially feasible, in terms of demand potential and the costs of production and marketing. Management must also ensure that product concept is compatible with the resources of the organization technological, human and financial.

5. Product Development:

Product development encompasses the technical activities of engineering and design. At this stage, the engineering department converts the product concept into a concert form of product in view of the required size, shape, design, weight, colour etc. of the product concept.

A model or prototype of the product is manufactured on a limited scale. Decisions are also made with regard to packaging, brand name, label etc. of the product.

6. Test Marketing:

A sample of the product is tested in a well-chosen and authentic sales environment; to find out consumers' reaction. In view of consumers' reactions, the product may be improved further.

7. Commercialisation:

After the management is satisfied with the results of test marketing, steps are taken to launch a full-fledged programme for the production, promotion and marketing of the product. It is the stage where the new product is born; and it enters its life cycle process.

5.5 PRODUCT LIFE-CYCLE

Like human beings, products also have a limited life-cycle and they pass through several stages in their life-cycle. A typical product moves through five stages, namely—introduction, growth, maturity, decline and abandonment. These stages in the life of a product are collectively known as product life-cycle.

The length of the cycle and the duration of each stage may vary from product-to-product, depending on the rate of market acceptance, rate of technical change, nature of the product and ease of entry. Every stage creates unique problems and opportunities and, therefore, requires a special marketing strategy.

Stages of Product Life Cycle

Product life cycle comprises of four steps/stages. Each stage of product life cycle can be characterized in terms of at least four aspects – sales volume, amount of profits, level of promotional efforts and expenses, and degree of competition. Each stage demands the unique marketing strategy. Let us briefly describe each of the stages of the PLC.

Introduction Stage

Introduction stage starts when a new product is, for the very first time, made available for purchase. Consumers are not aware of product, or they may not have general opinion and experience regarding product. Moreover, a new product has to face the existing products. So, the sales remain limited.

In the very initial stage, there is loss or negligible profit. During this period, the direct competition is almost absent. Company has not mastered production and selling problems. Price is normally high to recover/offset costs of development, production, and marketing with minimum sales. So, sales rise at gradually.

Characteristics of introduction stage include:

- (i) Huge selling and promotional costs are required to increase awareness of customers.
- (ii) Price is kept high to recover high development, production, and marketing costs.
- (iii) Marketer has to tackle technical and production problems.
- (iv) Sale is low and increasing at a lower rate.
- (v) There is loss or negligible profit.
- (vi) There is no competition

Growth Stage

This is the stage of a rapid market acceptance. Due to increased awareness, the product gets positive response from market. This stage is marked by a rapid climb in sales. Sales rise at the increasing rate. Profits follow the sales. Seller shifts his promotional attempts from “try-my-brand” to “buy-my-brand.”

Company tries to develop effective distribution network. Here, the most of production and marketing problems are mastered. Due to rise in profits, competitors are attracted. At a right time, price may be reduced to attract the price-sensitive buyers.

Company continues, even increases, its selling and promotional efforts to educate and convince the market and meet competition. At the end of growth stage, sales start increasing at decelerated rate, consequently, profits starts to decline.

Characteristics of growth stage include:

- (i) Sales increase rapidly (or at increasing rate) as a result of consumer acceptance of the products.
- (ii) Company can earn maximum profits.
- (iii) Competitors enter the market due to attractive profits.
- (iv) Price is reduced to attract more consumers.
- (v) Distribution network is widened and improved.
- (vi) Necessary primary changes are made in product to remove defects.
- (vii) Company enters the new segments and new channels are selected.

Maturity Stage

This stage is marked with slow down of sales growth. Sales continue to rise but at decreasing rate. Competitors have entered the market and existing products face severe competition. Sales curve is pushed downward. It is just like an inverse “U.” During this stage, for certain period of time, sales remain stable. This level is called the Saturation. Profits also decline. Normally, this stage lasts longer and marketers face formidable challenges.

The stages may be divided into three phases:

- i. Growth Maturity:
Sales-growth rate starts to decline.
- ii. Stable Maturity:
Sales remain stable (i.e., saturation stage).
- iii. Decline Maturity:

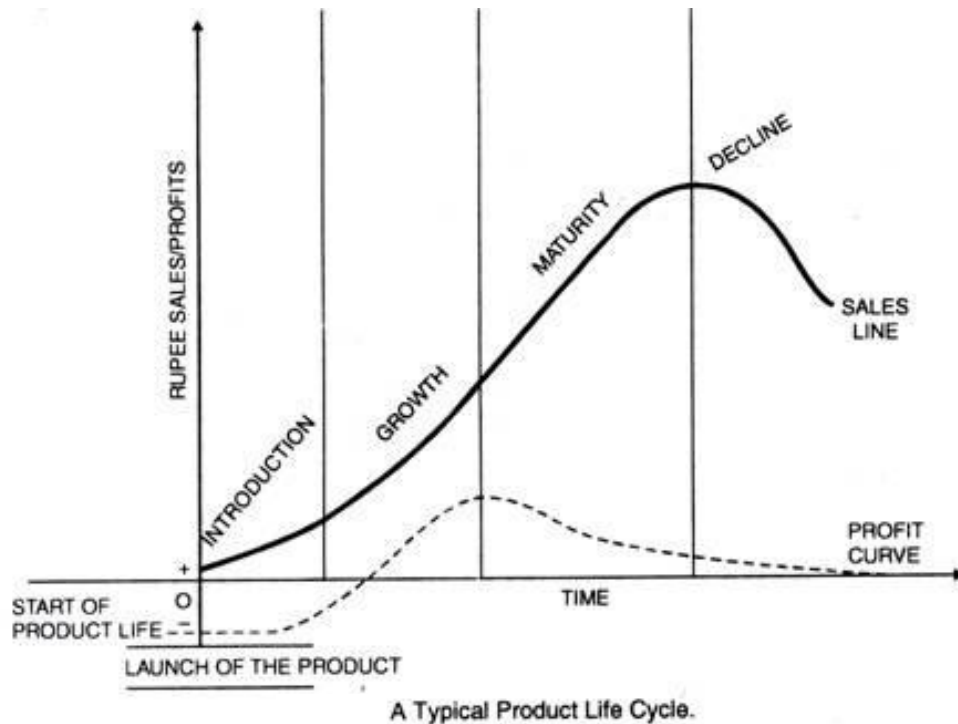
Sales now start to decline.

Marginal producers are forced to drop out the products. Those who operate formulate various strategies to extend the stage. Market, products, and marketing programme are to be modified to sustain the stage.

Characteristics of maturity stage include:

- i. Sales increase at decreasing rate.
- ii. Profits start to decline.
- iii. Marginal competitors leave the market.
- iv. Customer retention is given more emphasis.
- v. Product, market, and marketing mix modifications are undertaken.





Decline Stage

This is the last stage of product life cycle. Here, sales start declining rapidly. Profits also start erasing. There is a minimum profit or even a little loss. Advertising and selling expenses are reduced to realize some profits. This stage is faced by only those who survived in maturity stage.

Most products obsolete as new products enter the market. All products have to face the stage earlier or later. New products start their own life cycle and replace old ones. A number of competitors withdraw from the market. Those who remain in the market prefer to drop smaller segments, make minor changes in products, and continue selling the products in profitable segments and channels.

Here, logic has its own role. Management continues with the same product with expectation that sales improve when economy improves; marketing strategy is revised expecting that competitors will leave the market; or product is improved to attract new market segments.

However, unless a strong reason exists, it is costly and risky to continue with the same products. Later on it is difficult to manage selling and promotional efforts. Marketer must check every possibility before dropping the product completely.

Characteristics of decline stage include:

- i. Sales fall rapidly.
- ii. Profits fall more rapidly than sales.
- iii. Product modification is adopted.
- iv. Gradually, the company prefers to shift resources to new products.
- v. Most of sellers withdraw from the market.
- vi. Promotional expenses are reduced to realize a little profit.

5.6 PRODUCTION CONTROL

There is lot of disagreement between different experts of management regarding the meaning of production control. The term itself appears to be quite confusing and misleading. In literary sense control means action to check/regulate.

In the opinion of Mary Gushing Niles, “Control is maintaining a balance in activities towards a goal or set of goals evolved during production planning.” Planning only outlines some course of action whereas control is an execution process involving standardisation, evaluation and corrective functions.

According to Fayol, “Control consists in verifying whether everything occurs in conformity with the adopted plan and established principles. The objective of control is to point out weaknesses and shortcomings, if any, in order to rectify them and prevent recurrence. It operates on everything viz. material, equipment, men, operations etc. For control to be effective, it must be applied within reasonable time and be followed-up sanctions.”

Thus production control is some scientific procedure to regulate an orderly flow of material and co-ordinate various production operations to accomplish the objective of producing desired item. In right quantity of desired quality at the required time by the best and the cheapest method i.e., to attain highest efficiency in production.

Objectives of Production Control:

The success of an enterprise greatly depends on the performance of its production control department.

The production control department generally has to perform the following functions:

- (i) Provision of raw material, equipment, machines and labour.
- (ii) To organise production schedule in conformity with the demand forecast.
- (iii) The resources are used in the best possible manner in such a way that the cost of production is minimised and delivery date is maintained.
- (iv) Determination of economic production runs with a view to reduce setup costs.
- (v) Proper co-ordination of the operations of various sections/departments responsible for production.
- (vi) To ensure regular and timely supply of raw material at the desired place and of prescribed quality and quantity to avoid delays in production.
- (vii) To perform inspection of semi-finished and finished goods and use quality control techniques to ascertain that the produced items are of required specifications.
- (viii) It is also responsible for product design and development.

Thus, the fundamental objective of production control is to regulate and control the various operations of production process in such a way that orderly flow of material is ensured at different stages of the production and the items are produced of right quality in right quantity at the right time with minimum efforts and cost.

Objectives and Advantages of Production Control:

The following are the objectives and benefits under the four sub-headings of Direction, Co-ordination, Control and Innovation:

1. Direction:

- (i) Efforts can be directed to those areas of production that will contribute towards accomplishment of a given objective.
- (ii) Programme can be closely geared as consumer requirements.
- (iii) Manufacturing cycles are shortened resulting in reduction of in process inventory costs and consumer satisfaction.
- (iv) Work must be performed as per pre-planned schedules through distribution of the jobs/work according to the objective needs of the schedules.
- (v) Supervisory staff is directed to take corrective measure whenever and wherever required.

2. Co-ordination:

- (i) The workers/employees of the enterprise can be provided awareness regarding their role in the business.
- (ii) Customer can be provided information about the status of his order.
- (iii) Overall cost will go down in view of systematization and reduction in paper work involved.
- (iv) By the greater utilization of resources through sound loading & scheduling production is increased.

3. Control:

- (i) Less time required for analysis of reports by the management.
- (ii) Necessary information is available for the determination of when and where preventive/corrective measures are required.
- (iii) It provides a yardstick to the management for knowing the work progress.

4. Innovation:

- (i) Time becomes available to work out details that would otherwise be left to advance planning.
- (ii) It becomes a necessity that all activities of production are time phased.
- (iii) More flexibility in the manufacturing operations is obtained so that we can accept and deliver the occasional use order for the odd product which gives us a new consumer or additional business from an old consumer.

Production Control Procedure:

This procedure consists of following four functions:

(i) Routing:

Connected with where the work is to be done.

(ii) Scheduling:

Connected with when the work is to be done.

(iii) Dispatching:

Connected with starting the work in the plant.

(iv) Follow up or Progress Reporting:

Collection of information on the progress of work in the unit.

5.7 ELEMENTS OF PRODUCTION PLANNING AND CONTROL IN AN ORGANIZATION

Some of the important elements involved in the process of production planning and control in organization are: (a) Planning; (b) Routing; (c) Scheduling; (d) Despatching; (e) Checking the progress or follow-up

(a) Planning:

This is the first and the most important element of production planning and control. Planning refers to deciding in advance what is to be done in future. A separate planning department is established in the organisation which is responsible for the preparation of policies and plans with regard to production to be undertaken in due course.

While explaining the concept of scientific management, F.W. Taylor emphasised the need of separating planning function from the function of actual operation in an organisation. For successful implementation of production control, production planning is of utmost importance. The planning department prepares various charts, manuals production budgets etc., on the basis of information received from management.

These plans and charts or production budgets are given practical shape by carrying various elements under production control. If production planning is defective, production control is bound to be adversely affected. For achieving the production targets, production planning provides sound basis for production control.

(b) Routing:

Production routing is a process concerned with determining exact route or path, a product has to follow right from raw material till its transformation into finished product. A few definitions of routing can be cited here:—

“Routing may be defined as the selection of paths or routes over which each piece is to travel in being transformed from raw material into finished product”.

“Production routing involves the planning of the exact sequence of work stations to be used in processing a part of product. Once a layout has been established the routing of an item is the determination of the path that item should follow as it is manufactured”. —James C. Lundy

“Routing is the specification of the flow or sequence of operations and processes to be followed in producing a particular manufacturing lot”.

“Routing includes the planning of where and by whom work shall be done, the determination of the path that work shall follow and the necessary sequence of operations; it forms a groundwork for most of the scheduling and dispatching functions of planning department.”

The above mentioned definitions clearly lay down that routing is concerned with the selection of the most economical and appropriate path for the product in the process of final completion from raw material to finished product.

Objects of routing:

The main objective of routing is to lay down the best and the most economical sequence of operations to be undertaken in the process of production. Another objective of routing is to determine proper tools and equipments and the required number of workers required for doing or carrying total production processes in an organisation.

Routing becomes automatic and continuous in case of continuous manufacturing units where standardized products are produced by undertaking standardized production operations.

On the other hand, in case of job order units or intermittent- process industries such as ship building, every product requires different designs and varying sequences of operations.

Procedure followed in routing:

In case where a new product is going to be produced, different steps are involved in a total routing procedure. These steps are:

- (a) Complete analysis or study of the product as to decide what parts of the product are to be manufactured and what may be purchased from the market.
- (b) Analysis of the article so as to know what sorts of materials are needed for producing the article or product. This includes the complete study with regard to quality, quantity, kind and grade of materials required.
- (c) To determine different manufacturing operations and their sequence. This can be worked out by knowing accurately about the machines and their layout. This also necessitates the knowledge of allied equipment, jigs, tools and implements needed for efficient production.
- (d) Determining lot to proper size in relation to order placed by the customers.
- (e) Possibility of scrap in manufacturing a product must be properly determined. Anticipated scrap should be compared with actual scrap. Steps should be undertaken to control excess scrap.
- (f) Determination of the cost of the article or the product produced must be properly worked out. Calculation of total cost and per unit cost production is primarily the job of costing department, but still cost estimates pertaining to direct material, direct labour, direct expenses and indirect expenses and overheads must be prepared by the production department. These estimates will be greatly helpful for the costing department.

(g) Complete information pertaining to different types of production control forms viz., time and job cards, inspection cards and tool tickets, etc. must be kept by the works manager. This will be very helpful in carrying planned and systematic production.

(h) Preparation of route sheets is another important step in routing procedure. Route sheets relate to specific production orders. One sheet is prepared for each part or component of the product.

Route sheets also indicate the sequence of operations to be undertaken and also contain various requirements of production viz., men, materials and machinery etc. Route sheets also indicate total number of pieces to be produced and number of pieces to be included in each lot where production is carried in lots.

It must be remembered that routing is a complex and tedious process as such it should be entrusted to an expert who knows all the intricacies and complexities of production operations.

A number of factors viz. human considerations, plant layout, type of production undertaken and processes employed and type of equipment being undertaken must be kept in mind before selecting a proper route for production.

(c) Scheduling:

Scheduling in simple words means fixation of time and date when each operation is to be commenced and completed. It is an important part of production control as all future process of production is based on it. Scheduling lays down ground work for all subsequent steps in production process.

A few definitions of scheduling are given as under:

“The determination of the time that should be required to perform each operation and also the time necessary to perform the entire series as routed, making allowance for all factors concerned.”

“Scheduling involves establishing the amount of work to be done and the time when each element of the work will start, or the order of work. This includes allocating the quantity and rate of output of the plant or department and also the date or order of starting each unit of work at each station along the route prescribed.”

From the above mentioned definitions, it is clear that scheduling is concerned with allocating time for each operation of production and finally total time in the completion of production.

Types of scheduling:

Scheduling is of three types viz:

- (i) Master scheduling;
- (ii) Manufacturing or operation scheduling;
- (iii) Retail operation scheduling.

(i) Master scheduling:

It relates to a specified period; say a month, a week or a fortnight. It contains production requirements of a single product or different products during the specified period of time. It is easier to prepare master schedule for a single product, but difficulty arises where the number of products are more. It is also known as over-all schedule.

The preparation of master schedule varies from industry to industry according to type of production undertaken by them. Master schedule usually contains information pertaining to direct material requirements, estimated requirements in man-hours per product at various work centres and estimated overhead expenses etc.

(ii) Manufacturing or operation scheduling:

Manufacturing schedules are prepared in case of process or continuous type of industries. In case of mass production industries, where uniform products of same size, colour and design etc., are produced, manufacturing schedules can be easily prepared.

But in case where a product is produced in different sizes, quantity, colour and design, it is bit difficult to prepare manufacturing schedule. The important information contained in this schedule relates to name, number of the product, quantity to be produced each day, week or any other stipulated time.

(iii) Detail operation scheduling:

This type of schedule relates to allocation of time for each production operation within each machine and manufacturing process in the organisation.

Both routing and scheduling are important elements in the process of product control. They are interdependent on each other. Proper route cannot be assigned to a product without proper schedule, at the same time schedules cannot be prepared properly without the knowledge of exact route of production.

(d) Dispatching:

Dispatching relates to the process of initiating production in accordance with pre-conceived production plan. It is concerned with giving practical shape to the production plan. This includes issuing necessary orders and instructions and other important guidelines and information pertaining to work.

Some important definitions of dispatching are enumerated here: “A good definition of dispatching is the routine of setting productive activities in motion through the release of orders and instructions, in accordance with previously planned times and sequences, embodied in route sheet and schedule charts.”

“Despatches put production in effect by releasing and guiding manufacturing orders in the sequence previously determined by route sheets and schedules.”

“The despatching function involves the actual granting of permission to proceed according to plans already-laid down. This is similar in case of the traveller, to his employer finally approving his vacation leave.”

Procedure or steps followed in dispatching:

Following steps are undertaken in discharging the function of dispatching’:

1. Issuing materials from stores to different production processes.
2. Assignment of work to various machines and work places.
3. Procuring necessary tools, equipment and fixtures to be issued to workmen as and when needed.
4. Issuing necessary work orders, giving instructions and other information with regard to work to the workers.
5. To record and maintain the time taken from starting to completion of each job and also recording the total production time.
6. After the completion of work, all tools, implements, drawings and charts etc., to be returned to respective issuing departments.
7. Recording idle time of machines and workers.

(e) Checking the progress or Follow-up:

Follow-up or expediting function relates to evaluation and appraisal of work performed. If goods are to be produced as planned, proper follow up or expediting must be undertaken. A properly planned follow up procedure is helpful in finding errors and defects in the work and it also suggests remedial measures.

“Follow up or expediting is that branch of production control procedure which regulates the progress of materials and part through the production process” The function of follow-up is carried by ‘follow-up men’. These men act as intermediaries between various departments bringing about co-ordination between them. ‘Follow-up men’ are also referred as expeditors, ‘go-betweens’, ‘stock chasers’ and ‘progress-men’ etc.

Follow-up function can be applied in accordance with product or process layout. Follow up under product layout is easier to undertake as the follow-up men are responsible for the progress of a single product from inspection to packing.

On the other hand, follow-up under process layout is difficult to carry on account of scattered departments. Different follow-up men are appointed in different department’s viz., lathe department, welding department and finishing department etc.

In brief the element of follow up is concerned with the following three steps viz.,

- (a) To review the present situation with regard to materials, work-in-progress and finished products.
- (b) Expediting the performance of those departments which lag behind.
- (c) Removing obstacles in the way of production for smooth and uninterrupted flow of production.

5.8 GANTT CHARTS

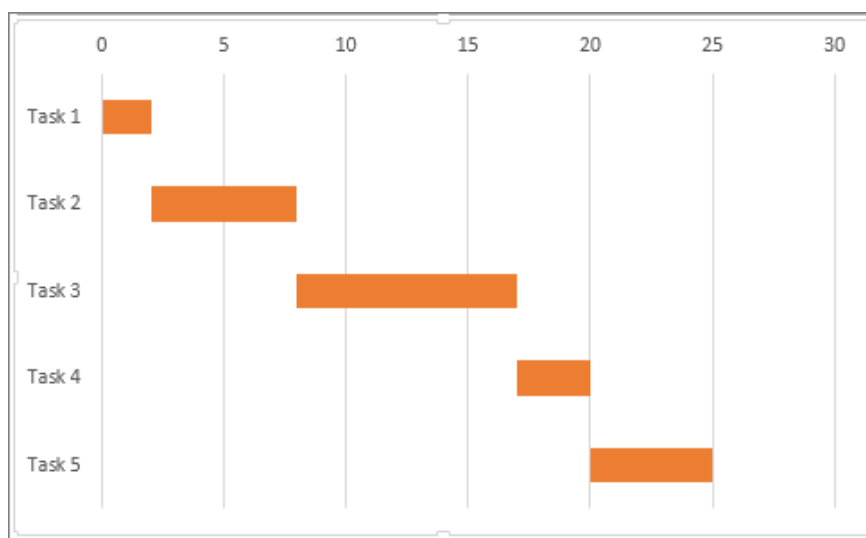
A Gantt chart is a useful graphical tool which shows activities or tasks performed against time. It is also known as visual presentation of a project where the activities are broken down and displayed on a chart which makes it is easy to understand and interpret.

A Gantt chart is a popular tool in project management. It basically drills down activities which need to be done by a fixed time period. It is commonly used for tracking project schedules. On the chart, tasks are shown on the vertical axis while the scheduled time-spend is laid out on the horizontal axis. Each task is represented by a bar that shows the time required for the project. The bar then represents or shows percentage of tasks that have been completed. It also shows dependencies, which simply means the interlinkages between various activities in the project.

Understanding the interlinkage between activities is very important to monitor and Gantt charts help the project manager to do just that. It conveys the information about the completion of other activities in the project. This information is important because of the interlinkages between various activities and if one activity gets delayed it will have an impact on others.

Gantt chart is a useful tool in planning and scheduling the projects. It keeps the management updated as to when the project will get completed. It also keeps the management informed about any additional resources that are required, and manage dependencies between tasks.

They are commonly used in scheduling production processes, employee roster or scheduling, events scheduling, production processes, etc. Microsoft Excel can also be used to create Gantt charts apart from other independent software available in the market.



Limitations of Gantt chart:

- (i) More time will be required for calculations and plotting of these values on chart so as to keep
- (ii) The errors committed by people in providing the necessary information and plotting may be difficult to check.
- (iii) It is a cumbersome job to collect the necessary information from various departments.

(iv) The relationship between various operations cannot be revealed in this chart.

(v) With any modifications in schedule, it is very difficult to incorporate the corresponding changes in Gantt chart.

5.9 INVENTORY CONTROL

An inventory is a stock of goods maintained for the purpose of future production or sales. In broad sense, the term inventory refers to all materials, parts, supplies, tools, in-process or finished products recorded in the books by an organisation and kept in its stocks, warehouse or plant for some period of time. It is a list or schedule of materials held on behalf of an enterprise. The quantity and value of every item is also mentioned in such list.

According to R.L. Ackoff and M.W. Sasieni, “Inventory consists of usable but idle resources. The resources may be of any type; for example, men, materials, machines or money. When the resources involved are materials or goods in any stage of completion, inventory is referred to as stock. ”

In a nutshell, the term inventory may be defined as “the stock of goods, commodities or other economic resources that are stored or reserved at any given period for future production or for meeting future demand.

Types/Classification of Inventory:

1. Direct Inventories:

Direct inventories are those inventories that play a major role in the production and constitute a vital part of finished goods. These inventories can be easily assigned to specific physical units. Direct inventories may be categorised into four groups.

(i) Raw materials:

Raw materials are the physical resources to be used in the manufacture of finished products. They include materials that are in their natural or raw form. For example, cotton in the case of textile mill, sugarcane in the case of sugar factory, oil seeds in the case of an oil mill etc. The chief objective of keeping raw material is to ensure uninterrupted production in the event of delays in delivery and also to enjoy the economies of large scale buying.

(ii) Semi-finished Goods:

Semi-finished goods are those materials which are not cent per cent (100%) complete in all respects i.e., some processing still remains to be done before the product can be sold. For example, a person who is engaged in the manufacture of furniture, may purchase unpolished furniture from market and sell it after polishing the same.

(iii) Finished Goods:

Finished goods are complete products that are ready for sale or distribution. For instance, in case of a hosiery factory, sweaters, shawls etc. are finished products.

(iv) Spare Parts:

Spare parts means duplicate parts of a machine. Usually, almost all the industrial concerns maintain spare parts of various machines which they use for manufacture. This will enable them to ensure smooth running of machines which in turn provide for uninterrupted production.

2. Indirect Inventories:

Indirect inventories include those items which are necessary for manufacturing but do not become component of the finished goods. They normally include petrol, maintenance materials, office materials, grease, oil lubricants etc. These inventories are used for ancillary purposes to the business and cannot be assigned to specific, physical units. These inventories may be used in the factory, the office or the selling and distribution divisions.

Objectives:

- (i) To minimise capital investment in inventory by eliminating excessive stocks;
- (ii) To ensure availability of needed inventory for uninterrupted production and for meeting consumer demand;
- (iii) To provide a scientific basis for planning of inventory needs;
- (iv) To tide over the demand fluctuations by maintaining reasonable safety stock;
- (v) To minimise risk of loss due to obsolescence, deterioration, etc.;
- (vi) To maintain necessary records for protecting against thefts, wastes leakages of inventories and to decide timely replenishment of stocks.

Advantages of Inventory Control:

- 1. It improves the liquidity position of the firm by reducing unnecessary tying up of capital in excess inventories.
- 2. It ensures smooth production operations by maintaining reasonable stocks of materials.
- 3. It facilitates regular and timely supply to customers through adequate stocks of finished products.
- 4. It protects the firm against variations in raw materials delivery time.
- 5. It facilitates production scheduling, avoids shortage of materials and duplicate ordering.
- 6. It helps to minimise loss by obsolescence, deterioration, damage, etc.
- 7. It enables the firms to take advantage of price fluctuations through economic lot buying when prices are low.

Limitations of Inventory Control:

- (i) Efficient inventory control methods can reduce but cannot eliminate business risk.
- (ii) The objectives of better sales through improved service to customer; reduction in inventories to reduce size of investment and reducing cost of production by smoother production operations are conflicting with each other.
- (iii) The control of inventories is complex because of the many functions it performs. It should be viewed as shared responsibilities.

Inventory Costs:

(i) Set up Cost:

These costs include clerical cost on orders and discount rates on quantity of goods purchased. The costs are included in the cost of material at two stages. Firstly, when material is purchased and stored, and secondly, when goods manufactured are stored from the said material. Every company has to store its goods that it wants to be sold. Storage function is necessary because production and consumption cycles rarely match.

(ii) Cost of Spoilage and Obsolescence:

The next is the cost of spoilage and obsolescence. It refers to loss of goods while in stock. Any product or material is bound to spoil if stored for a long time. The risk of spoilage is an open risk. The cost of spoilage is bound to be taken into account. Similarly the cost of obsolescence, some spare parts and machine components may become obsolete if they are stored for a long time. This is true when there are rapid technological changes. As a result, the cost of spoilage and obsolescence gives rise to the accountability of inventory cost.

(iii) Cost of Placing an Order:

This cost may be for placing order on outside suppliers for procuring raw goods to be manufactured inside the firm.

Depending upon the type of stock, this cost may vary. Cost of placing an order includes the following:

- (i) Set up cost of machines
- (ii) Cost involved in follow-up
- (iii) Cost involved in receiving the order
- (iv) Paper work costs.



(iv) Cost of Carrying Stock

This is the cost which a firm actually incurs for carrying the stock. Cost of carrying stock is calculated by taking into consideration the following items:

- (i) Interest on capital
- (ii) Tax and insurance charges
- (iii) Storage cost
- (iv) Allowance for spoilage
- (v) Obsolescence.

(v) Cost of Running out of Stock

Whenever stock exhausts for any item, this cost is incurred. These costs are different in nature. The cost of running out of stock for a raw material or spare part is made up of plant down time and possible special delivery costs. For a finished good, such costs are known as dissatisfaction to customers or lost customers.

Importance of Inventory Control:

The aim of holding inventories is to allow the firm to separate the process of purchasing, manufacturing, and marketing of its primary products. Inventories are a component of the firm's working capital and as such represent a current account. Inventories are also viewed as a source of near all cash.

The purpose is to achieve efficiencies in areas where costs are involved. The scientific inventory control results in the reduction of stocks on the one hand and substantial decline in critical shortages on the other. In the following paragraph we can specify the various importances that accrue from holding inventories.

(i) Reducing Risk of Production Shortages:

Firms mostly manufacture goods with hundreds of components. The entire production operation can be halted if any of these are missing. To avoid the shortage of raw' material the firm can maintain larger inventories.

(ii) Reducing Order Cost:

Where a firm places an order, it incurs certain expenses. Different forms have to be completed. Approvals have to be obtained, and goods that arrive must be accepted, inspected and counted. These costs will vary with the number of orders placed. Smaller the inventories lesser the capital needed to carry inventories.

(iii) Minimise the Blockage of Financial Resources:

The importance of inventory control is to minimise the blockage of financial resources. It reduces the unnecessary tying up of capital in excess inventories. It also improves the liquidity position of the firm.

(iv) Avoiding Lost Sales:

Most firms would lose business without goods on hand. Generally a firm must be prepared to deliver goods on demand. By ensuring timely availability of adequate supply of goods, inventory control helps the firm as well as consumers.

(v) Achieving Efficient Production Scheduling:

The manufacturing process can occur in sufficiently long production runs and with pre-planned schedules to achieve efficiencies and economies. By maintaining reasonable level of inventory production scheduling becomes easier for the management.

(vi) Gaining Quantity Discounts:

While making bulk purchases many suppliers will reduce the price of supplies and component supplies will reduce the price of supplies and component parts. The large orders may allow the firm to achieve discounts on regular basis. These discounts in turn reduce the cost of goods and increase the profits.

(vii) Taking the Advantage of Price Fluctuations:

When the prices of the raw materials are low the firm makes purchases in economic lots and maintains continuity of operations. By reducing the cost of raw materials and procuring high prices for its goods the firm maximises profit. This with the help of inventory control the firm takes advantage of price fluctuations.

(viii) Tiding over Demand Fluctuations:

Inventory control also helps the firm in tiding over the demand fluctuation. These are taken care of by keeping a safety stock by the firm. Safety stock refers inventories carried to protect against variations in sales rate, production rate and procurement time. Inventory control aims at keeping the cost of maintaining safety stock minimum.

(ix) Deciding timely Replenishment of Stocks:

Inventory control results in the maintenance of necessary records, which can help in maintaining the stocks within the desired limits. With the help of adequate records the firm can protect itself against thefts, wastes and leakages of inventories. These records also help in deciding about timely replenishment of stocks.

Terms and Definition in Inventory Control:

(i) Demand:

It is the number of items required per period. The demand is the most critical, yet an uncontrollable component, without demand there would be no need for maintaining inventory.

(ii) Lead Time:

It is the time between placing an order and its receipt in stock. It may be assumed deterministic or probabilistic depending upon the reputation of supplier or his past behaviour.

It is the period that elapses between placing an order and receiving the same. These can be classified as administrative lead time, transporting lead time and inspection lead time. Consideration of lead time is one of the important factors to be taken into consideration in inventory control.

(iii) Quantity Discount:

It is an allowance granted by the vendor to the purchaser of the materials for encouraging large size orders. Sometimes there is an agreement between the vendor and the purchaser that quantity discount will be allowed by the vendor on purchase of certain specified quantity of items.

This is allowed by a supplier on account of the savings in his cost which arise from production on a mass scale and distribution on to the purchaser by means of a quantity discount.

(iv) Safety Stock:

It is also known as minimum stock level of material/items below which the actual stock should not be allowed to fall. This much quantity of material pertaining to a particular item must be kept in stores all times. The fixation of this level acts as safety measure and hence, it is also known as “Safety Stock” or ‘Buffer Stock’.

In case the actual stock falls below this level, there is a danger of interruption in production and the management has to give top priority to the acquisition of its fresh supplies. The main objective of fixing the minimum level of materials is to ensure that required quantity of various input materials are available in stores at all times.

The main factors which are taken into account in fixing the level are:

- (a) The average rate of consumption of materials during production.
- (b) The time required to obtain fresh supplies under top priority conditions.
- (c) Reorder level/point.
- (d) The production requirements of materials.
- (e) The minimum quantity of materials which could be produced advantageously.

For calculating minimum or safely stock level, the following formula is applied.

$$\text{Minimum Stock Level} = \text{Reorder Level} - (\text{Normal Consumption} \times \text{Normal delivery time})$$

For example. Normal Consumption = 300 units per week.

Normal delivery time = 7 weeks. Reorder level = 2800 units per week

By putting the values in the above formula

$$\text{Minimum stock level} = 2800 - (300 \times 7)$$

$$= 2800 - 2100 = 700 \text{ units.}$$

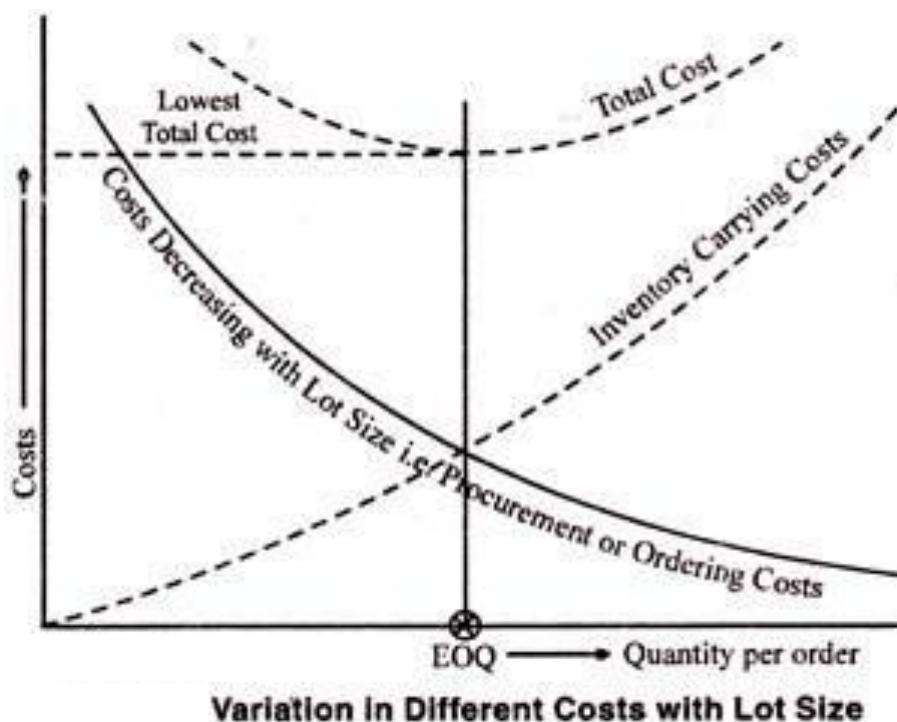
(v) Lot Size & Economic Ordering Quantity (EOQ) or Economic Lot Size (E.L.S.):

The amount of material procured or quantity produced during one production run by an enterprise is known as lot size. The quantity to be ordered depends upon a number of factors. It is evident that with increase in inventory size, expenditure on storage, deterioration and spoilage of goods obsolescence etc. is likely to increase whereas expenditure on setting up of the plant, procurement of materials etc. will decrease.

Thus with lot size, there are two sets of factors having opposite contributions towards the expenditure i.e. one encourages the lot size and other discourages. The total cost related with particular lot size is a combination of expenditures on all these factors as is illustrated in Fig.

Total Cost = Materials Cost + Cost due to factors whose cost increases with lot size + Cost due to factors whose cost decreases with lot size.

The lot size for which the total cost per period is minimum is known as economic lot size or economic order quantity i.e. E.L.S. or EOQ.

**(vi) Re-ordering Level:**

When the quantity of materials reaches a certain level then fresh order is sent to procure materials again. The order is sent before the materials reach minimum stock level.

Reordering level is fixed between minimum level and maximum level. The rate of consumption, number of days required to replenish the stocks, and maximum quantity of materials required on any day are taken into consideration while fixing reordering level.

Re-ordering level is fixed with following formula:

Reordering Level = Maximum Consumption Rate x Maximum Reorder period.

(vii) Maximum Level:

It is the quantity of materials beyond which a firm should not exceed its stocks. If the quantity exceeds maximum level limit then it will be termed as overstocking. A firm avoids overstocking because it will result in high material costs. Overstocking will lead to the requirement of more capital, more space for storing the materials, and more charges of losses from obsolescence.

Maximum stock level will depend upon the following factors:

1. The availability of capital for the purchase of materials in the firm.
2. The maximum requirements of materials at any point of time.
3. The availability of space for storing the materials as inventory.
4. The rate of consumption of materials during lead time.
5. The cost of maintaining the stores.
6. The possibility of fluctuations in prices of various materials.
7. The nature of materials. If the materials are perishable in nature, then they cannot be stored for long periods.
8. Availability of materials. If the materials are available only during seasons then they will have to be stored for the future period.
9. Restrictions imposed by the government. Sometimes, government fixes the maximum quantity of materials which a concern can store. The limit fixed by the government will become the deciding factor and maximum level cannot be fixed more than that limit.
10. The possibility of changes in fashions will also affect the maximum level.

Wheldon has suggested the following formula for calculating maximum stock level:

Maximum Stock Level = Reordering Level + Reordering Quantity – (Minimum Consumption x Minimum Reordering period)

(viii) Danger Level:

It is the level below which stocks should not fall in any case. If danger level approaches then immediate steps should taken to replenish the stocks even if more cost is incurred in arranging the materials. Danger level can be determined with the following formula:

Danger Level = Average Consumption x Maximum reorder period for emergency purchases.

(ix) Average Stock Level:

The Average stock level is calculated such as:

$$\text{Average Stock Level} = \text{Minimum stock Level} + 1/2 \text{ of Reorder Quantity}$$

Example:

From the following information, calculate minimum stock level, maximum stock level and re-ordering level:

- (i) Maximum Consumption = 200 units per day
- (ii) Minimum Consumption = 120 units per day
- (iii) Normal Consumption = 160 units per day
- (iv) Reorder period = 10-15 days
- (v) Reorder quantity = 1,600 units
- (vi) Normal reorder period = 10 days.



Solution:

$$\text{Reordering Level} = \text{Maximum Consumption} \times \text{Maximum Reorder period}$$

$$= 200 \text{ units} \times 15 = 3,000 \text{ units}$$
$$\text{Minimum Stock Value} = \text{Reordering Level} - (\text{Normal Consumption} \times \text{Nominal Reordering Period})$$

$$= 3,000 - (160 \times 10) = 3,000 - 1,600 = 1,400 \text{ units}$$

$$\text{Maximum Stock Level} = \text{Reordering Level} + \text{Reorder Quantity} - (\text{Minimum Consumption} \times \text{Reorder period})$$
$$= 3,000 + 1,600 - (120 \times 10) = 3,000 + 1,600 - 1,200 = 3,400 \text{ units.}$$

5.10 INVENTORY MODELS

Inventory model is a mathematical model that helps business in determining the optimum level of inventories that should be maintained in a production process, managing frequency of ordering, deciding on quantity of goods or raw materials to be stored, tracking flow of supply of raw materials and goods to provide uninterrupted service to customers without any delay in delivery.

There are two types of Inventory model widely used in business.

1. **Fixed Reorder Quantity System**
2. **Fixed Reorder Period System.**

Fixed Reorder Quantity System

Fixed Reorder Quantity System is an Inventory Model, where an alarm is raised immediately when the inventory level drops below a **fixed quantity** and new orders are raised to replenish the inventory to an

optimum level based on the demand. The point at which the inventory is ordered for replenishment is termed as **Reorder Point**. The inventory quantity at Reorder Point is termed as **Reorder Level** and the quantity of new inventory ordered is referred as **Order Quantity**.

Average Demand (DAv): It is the average number of order requests made per day.

Average Lead Time (TL): The time required to manufacture goods or product.

Average Lead Time Demand (DL): Average number of orders requested during the Lead Time

Average Lead Time Demand (DL) = Average Demand (DAv) X Average Lead Time (TL)

Safety Stock (S): It is the extra stock that is always maintained to mitigate any future risks arising due to stock-outs because of shortfall of raw materials or supply, breakdown in machine or plant, accidents, natural calamity or disaster, labour strike or any other crisis that may stall the production process.

The quantity of safety stock is often derived by analysing historical data and is set to an optimized level by evaluating carefully the current cost of inventory and losses that may be incurred due to future risk.

Reorder Level (RL): Reorder level is the inventory level, at which an alarm is triggered immediately to replenish that particular inventory stock. Reorder level is defined, keeping into consideration the **Safety Stock** to avoid any stock-out and **Average Lead Time Demand** because even after raising the alarm, it would take one complete process cycle (**Lead Time**) till the new inventories arrive to replenish the existing inventory.

Reorder Level (RL) = Safety Stock (S) + Average Lead Time Demand (DL)

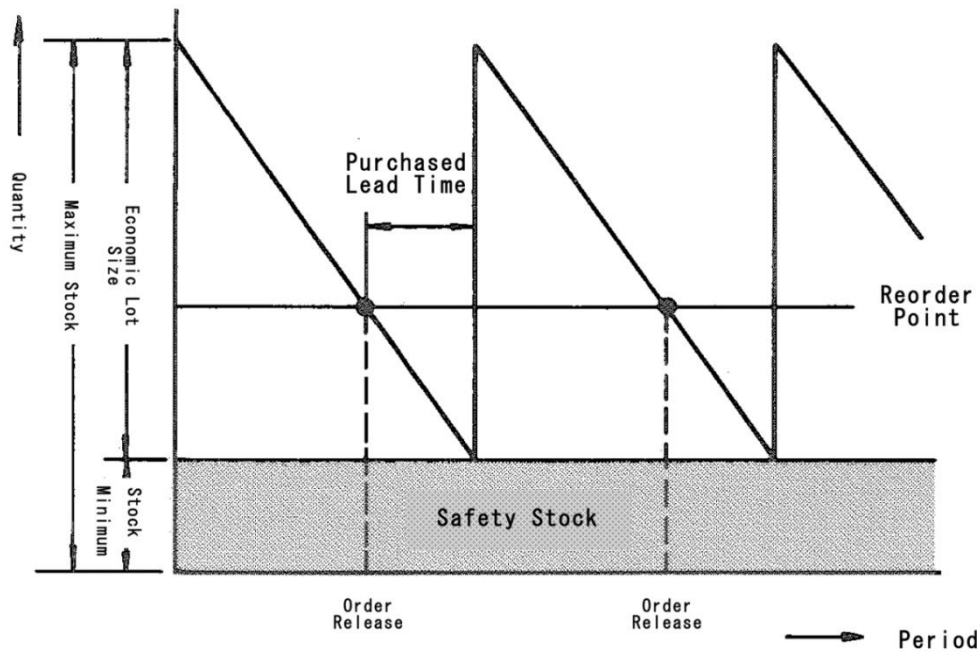
Order Quantity (O): Order quantity is the Demand (Order requests) that needs to be delivered to the customer.

Minimum Level: At least Safety Stock has to be always maintained to avoid any future stock-outs as per the standard practices of inventory management.

Minimum Level (LMin) = Safety Stock (S)

Maximum Level: The maximum level that can be kept in stock is safety stock and the demand (the quantity ordered).

Maximum Level (LMax) = Safety Stock (S) + Order Quantity (O)



Inventory Model: Fixed Reorder Quantity System

Example: The order quantity of an Item is 600 Units. The safety Stock is 200 Units. The Average Lead Time is 5 Days and average consumption per days is 40 units.

Order Quantity (O) = 600 Units

Safety Stock (S) = 200 Units

Average Lead Time (TL) = 5 Days

Average Demand (DAV) = 40 Units

Average Lead Time Demand (DL) = Demand (DAV) X Lead Time (TL) = 200 Units

Reorder Level (RL) = Safety Stock (S) + Average Lead Time Demand (DL) = 400 Units

Minimum Level (LMin) = Safety Stock (S) = 200 Units

Maximum Level (LMax) = Safety Stock (S) + Order Quantity (O) = 800 Units

Fixed Reorder Period System

Fixed Reorder Period System is an Inventory Model of managing inventories, where an alarm is raised after every **fixed period of time** and orders are raised to replenish the inventory to an optimum level based on the demand. In this case replenishment of inventory is a continuous process done after every fixed interval of time.

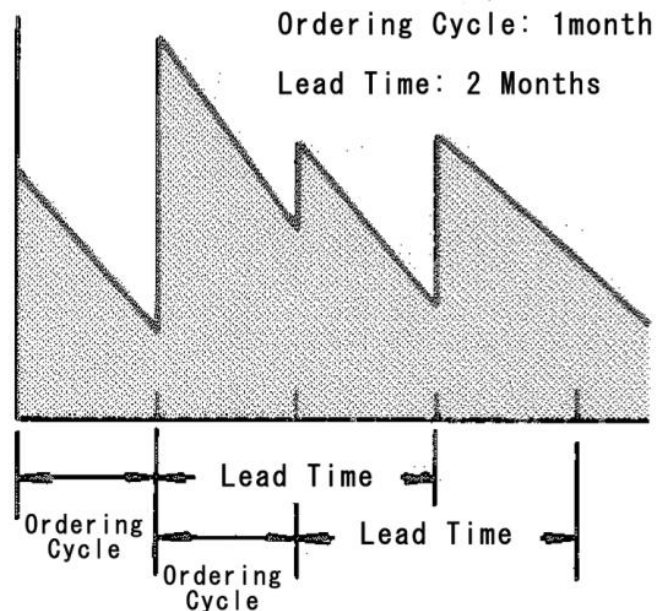
Regular Intervals (R): Regular Interval is the fixed time interval at the end of which the inventories would be reviewed and orders would be raised to replenish the inventory

Inventory on Hand (It): Inventory on hand is the Inventory level measured at any given point of time.

Maximum Level (M): It is the maximum level of inventory allowed as per the production guidelines. The maximum level is derived by analysing historical data.

Order Quantity: In this system, inventory is reviewed at regular intervals (R), inventory on hand (It) is noted at the time of review and order quantity is placed for a quantity of $(M) - (It)$.

Order Quantity (O) = (M) – (It).



Inventory Model: Fixed Reorder Period System

Example: Inventory is replenished at every regular interval of 5 days. The maximum allowable inventory is 800 Units. The inventory reviewed on Day-5, Day-10, Day -15 and Day -20 were 387 Units, 201 Units, 498 Units and 127 Units respectively.

Regular Intervals (R) = 5 Days

Maximum Level (M) = 800 Units

Inventory on Hand: $I_5 = 387$ Units, $I_{10} = 201$ Units, $I_{15} = 498$ Units and $I_{20} = 127$ Units

Order Quantity (O) = $(M) - (It)$.

Order Quantity (O_5) = $800 - 387 = 413$ Units

Order Quantity (O_{10}) = $800 - 201 = 599$ Units

Order Quantity (O_{15}) = $800 - 498 = 302$ Units

Order Quantity (O_{20}) = $800 - 127 = 673$ Units

1. Economic Ordering Quantity (EOQ) Model:

One of the important decisions to be taken by a firm in inventory management is how much inventory to buy at a time.

This is called 'Economic Ordering Quantity (EOQ)'. EOQ also gives solutions to other problems like:

- (i) How frequently to buy?
- (ii) When to buy?
- (iii) What should be the reserve stock?

Assumptions

1. That the firm knows with certainty how much items of particular inventories will be used or demanded for within a specific period of time.
2. That the use of inventories or sales made by the firm remains constant or unchanged throughout the period.
3. That the moment inventories reach to the zero level, the order of the replenishment of inventory is placed without delay.

Determination of EOQ:

EOQ Model is based on Baumol's cash management model. How much to buy at a time, or say, how much will be EOQ is to be decided on the basis of the two costs:

- (i) Ordering Costs, and
- (ii) Carrying Costs.

The various components of ordering costs and carrying costs are shown in the following:

Components of Ordering Costs and Carrying Costs:

Ordering Costs	Carrying Costs
Requisitioning	Warehousing
Order Placing	Handling
Transportation	Administrative
Storing	Insurance
Administrative	Deterioration and Obsolescence

EOQ can be determined by applying the following commonly used formula:

$$EOQ = \sqrt{\frac{2C_o D}{C_c}}$$

Where:

Q = Economic Ordering Quantity (EOQ)

D = Quantity purchased in a year or month or Demand of a product annually.

C_o = Cost of placing an order.

C_c = Annual or monthly cost of storage of one unit known as 'carrying cost'.

Let us illustrate this with an imaginary example.

Let us assume the following data for a firm:

Annual requirements 800 units

Ordering cost (per order) Rs. 50

Carrying cost (per unit) Rs. 2

Purchasing cost (per unit) Rs. 100

Now, using the EOQ formula, EOQ quantity will be as follows:

$$EOQ = \sqrt{\frac{2C_oD}{C_c}} = \sqrt{\frac{2*50*800}{2}}$$

= 200 Units

5.11 SELECTIVE INVENTORY CONTROL TECHNIQUES

Selective Inventory Control is an essential part of Materials Management. Selective control emphasizes on variations in methods of control from item to item based on selective basis. We can not apply uniform control since it's expensive and gives diffused effect. For this purpose we can use some criterion such as lead time, consumption, criticality, cost of the items, procurement difficulties etc. The following classification can be used for selective treatment of various types of materials.

Classification of Inventory Control

No.	Classification	Full Form	Criterion Employed
1	ABC Analysis	Always Better Control	Usage Value (i.e. Consumption per period x price per unit)
2	VED Analysis	Vital Essential Desirable	Loss of Production or Criticality of the item
3	HML Analysis	High Medium Low	Unit Price i.e. does not take consumption into account
4	SDE Analysis	Scarce Difficult Easy	Procurement Difficulties
5	GOLF Analysis	Government Ordinary Local Foreign	Source of procurement
6	SOS Analysis	Seasonal Off Seasonal	Seasonality
7	FSN Analysis	Fast Slow Non Moving	Issues from stores
8	XYZ Analysis		Inventory Investment

1. ABC Method

In order to exercise effective control over materials, A.B.C. (Always Better Control) method is of immense use. Under this method materials are classified into three categories in accordance with their respective values. Group 'A' constitutes costly items which may be only 10 to 20% of the total items but account for about 50% of the total value of the stores.

A greater degree of control is exercised to preserve these items. Group 'B' consists of items which constitutes 20 to 30% of the store items and represent about 30% of the total value of stores.

A reasonable degree of care may be taken in order to control these items. In the last category i.e. group 'C' about 70 to 80% of the items is covered costing about 20% of the total value. This can be referred to as residuary category. A routine type of care may be taken in the case of third category.

Type	Account for (Quantity)	Account for (Expenditure)
A Items	5-10%	70-75%
B Items	10-20%	10-20%
C Items	70-80%	5-10%

Steps in ABC analysis:

The steps involved in ABC analysis are as follows:

- Calculate the annual usage value of each item by multiplying the number used with the price of the item.
- Arrange the items in descending order according to the usage value.
- Compute percentage of total usage value for each item.
- Find out the average inventory of each item by dividing the usage value with 2 and number of orders.

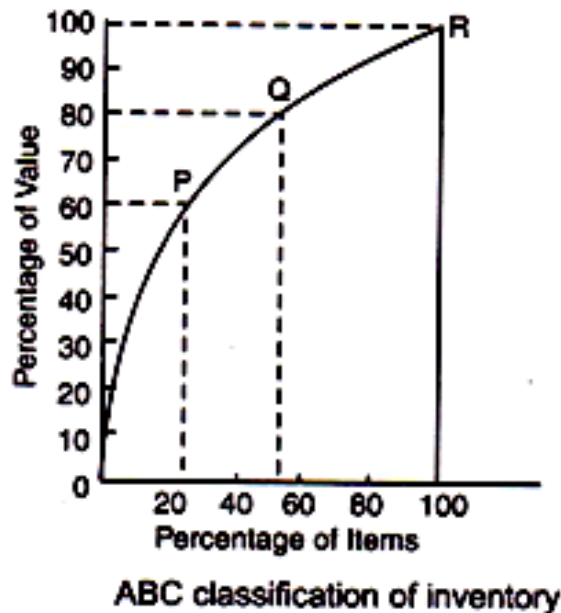
The ABC analysis can be shown on a graph too. For this purpose the cumulative percentages of the number of items are shown on X-axis and percentage of values on F-axis. Where the plotted curve takes a sharp turn a point is marked.

Each such point indicates one category of items. For example, in Fig. point P on the curve indicates 20% of item; with 60% usage value. This category can be called A. Point Q shows 30% items with 20% usage value. This category is B. Point R reflects 50% items with 20% usage value which is category C.

Advantages of ABC Analysis:

- It ensures better control over the costly items in which a large amount of capital is invested.
- It helps in developing scientific method of controlling inventories. Clerical costs are considerably reduced and stock is maintained at optimum level.

3. It helps in maintaining stock turnover rate at comparatively higher level through scientific control of inventories.
4. It ensures considerable reduction in the storage expenses. It results in stock carrying stock.
5. It helps in maintaining enough safety stock for C category of items.



Limitations of ABC analysis:

1. This technique can be successfully employed only, if there is proper standardisation of materials in the store.
2. A good system of codification of materials should be in operation for the success of this analysis.
3. The analysis is based on monetary value of the items in use. Other important factors one ignored.

2. VED Analysis

Classification of items based on their criticality

Initial	Stands For	Description
V	Vital	Those items for want of which production would come to stop
E	Essential	items whose stockouts cost is very high
D	Desirable	items which do not cause any immediate loss of production or stockouts entail nominal expenditure & cause minor disruptions

An item may be vital due to following reasons

- Non availability of item may cause serious production losses
- Procurement lead time is very high

- It's non standard item and is procured to buyer's design
- Single source of supply and located far off from buyers location

3. FSN Analysis:

This classification works like this:

F = Fast Moving

S = Slow Moving

N = Non-moving



FSN analysis is based on the assumption that all items of inventory are not required all the time in stores. Some items are required on regular basis and some once in a while. Therefore, Fast moving items must be kept nearer to the point of issue and similarly Non- moving items can be kept in a remote place as they are required occasionally.

Therefore for the purpose of controlling items under FSN analysis, 'F' type items need to be reviewed on regular basis while 'S' type items may be examined further and their disposal can be considered.

To conduct FSN analysis, the date of receipt or the last date of issue, whichever is later, is considered to determine the number of months, which have lapsed since the last transaction.

4. HML Analysis:

This classification works like this:

H = High Cost Items

M = Medium Cost Items

L = Low Cost Items

Likewise ABC analysis, items are classified on the basis of cost of the items. The point of difference between these two techniques is that under HML analysis, for the purpose of classifying inventories into various categories, only cost of the items is considered while their annual consumption value is totally ignored.

5. SDE Analysis:

This classification is carried out based on the lead time required to procure the items.

The classification is as follows:

a. Scarce (S):

Items which are imported and those items which require more than 6 months' lead time.

b. Difficult (D):

Items which require more than a fortnight but less than 6 months' lead time.

c. Easily Available (E):

Items which are easily available; mostly local items, i.e. less than a fortnights' lead time.

This classification helps in reducing the lead time required at least in case of vital items. Ultimately, this will reduce stock-out costs in case of stock-outs.

Inventory Models

Model I:

Derivation of Harris formula for instantaneous supply, continuous consumption and zero buffer stock:
This relationship shall be valid for purchase as well as manufacturer of goods:

Case I. Purchase Case:

Using the Notations

Q = Economic order quantity EOQ (for input materials)

D = Annual demand or Consumption of input materials = yearly usage of items

C = Unit Cost i.e. cost of unit material

P = Procurement cost or ordering cost per quantity Q and it may include.

- (i) Cost of processing the requisitions.
- (ii) Cost of receiving the material or transportation cost upto the plant
- (iii) Cost of quality control & inspection involved.

I = rate of interest charged per unit per year

H = Inventory holding or carrying cost

Which may include the following:

- (i) Cost of storage and handling the materials.
- (ii) Taxes and insurance on inventories during storage.
- (iii) Deterioration and obsolescence of inventories during storage.
- (iv) Cost of maintaining records of input material inventories.

(H is taken as percent of unit material cost)

Case II. Production Case:

Q = Economic lot size to be manufactured.

D = Annual production.

C = Unit cost of production which includes:

- (i) Raw material cost per unit.
- (ii) Labour cost per unit.
- (iii) Over heads cost per unit.

P = Preparation or set up for quantity

Q = It includes the following:

- (i) Labour cost for set up of machines/equipment's, tools, jigs and fixtures and accessories.
- (ii) Paper work cost involved in production planning and control, tool room and input materials stores.
- (iii) Cost of scrap caused due to initial trial runs during production process and machine idle time during this process.

H = Inventory carrying rate or holding cost. It is taken as percent of unit cost, it may also be expressed as rupees per unit cost.

To derive the relationship for EOQ and ELS i.e. Economic order quantity or Economic lot size,

Let Y_c = Total yearly cost (total annual investment)

Y_c = Material cost + Annual Inventory Carrying or Holding Cost + Procurement or Preparation or set up cost.

Material Cost = Unit Cost x Annual Production/usage = $C \times D$ Annual inventory carrying cost/Holding cost.

$$= (\text{Holding cost}) \times \text{Average Inventories} = \frac{HQ}{2} \times C$$

$$\text{Preparation cost or set up cost} = \text{Preparation cost per unit} \times D = \frac{P}{Q} \times D$$

$$\text{So } Y_c = C \times D + (HC) \frac{Q}{2} + \frac{P}{Q} \times D$$

For economic ordering quantity or lot size differentiating Y_c with respect to, Q and equating it to zero.

$$\text{We have } \frac{dY_c}{dQ} = 0 = \frac{d}{dQ} \left[CD + (HC) \frac{Q}{2} + \frac{P}{Q} \cdot D \right] = 0 + \left(\frac{HC}{2} \right) - \frac{PD}{Q^2}$$

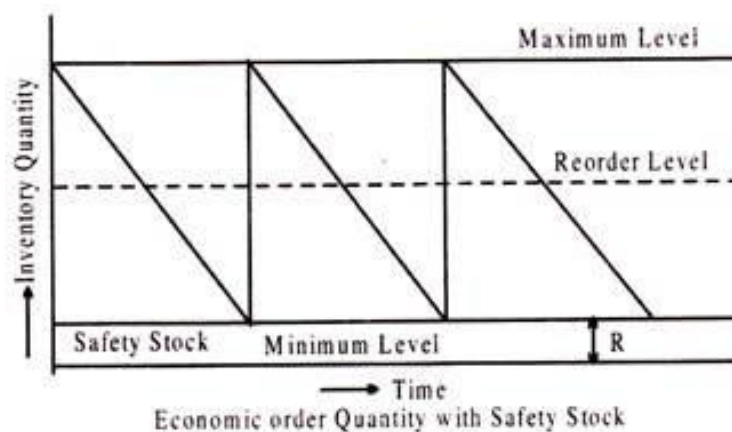
$$\text{or } Q = \sqrt{\frac{2PD}{H.C}} \quad \dots 1$$

In this derivation no reserve stock has been kept.

Model II:

Economic Order Quantity or Economic lot size with reserve stock:

When the supply is instantaneous and consumption is continuous and there is f reserve stock and for this model the pattern of inventory is shown in the adjoining fig



Following the previous notations and R as reserve stock, we have the total annual cost.

$$Y_c = \text{Material cost} + \text{Inventory carrying or Holding cost} + \text{or Setup cost.}$$

$$\text{Material cost} = C \cdot D$$

$$\text{Inventory carrying cost} = H \times \text{cost of inventory} = H(R + \frac{Q}{2})C$$

Here we take the average of Q and zero plus safety stock

$$\text{Setup cost} = \frac{P}{Q} \times D$$

$$Y_c = C \cdot D + H \cdot C \cdot (R + \frac{Q}{2}) + \frac{P}{Q} \times D$$

$$= C \cdot D + \frac{PD}{Q} + HRC + H \cdot \frac{Q}{2} \cdot C$$

$$\text{Let } \frac{R}{Q} = f \text{ i.e. let } R \text{ be taken as a fraction of } Q \text{ or } R = fQ$$

$$\therefore Y_c = C \cdot D + \frac{P}{Q} \cdot D + H \cdot Q \cdot f \cdot C + H \cdot \frac{Q}{2} \cdot C$$

For economic order, quantity or economic lot size differentiating with respect to Q and putting equal to zero, we have

$$\frac{dY_c}{dQ} = \frac{d}{dQ} \left[C \cdot D + \frac{P}{Q} \cdot D + f \cdot Q \cdot H \cdot C + \frac{HQ}{2} \cdot C \right]$$

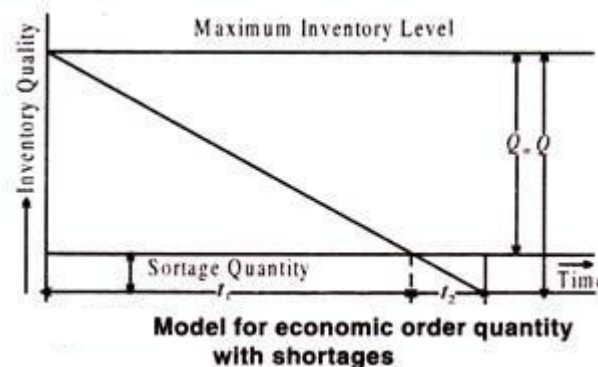
$$\frac{P}{Q^2} \cdot D + H \cdot f \cdot C + \frac{HC}{2} = 0 \quad \text{or} \quad \frac{P}{Q^2} D = HC \left[f + \frac{1}{2} \right] = \frac{CH(2f+1)}{2}$$

$$\text{or} \quad Q^2 = \frac{2PD}{CH(1+2f)}$$

$$\therefore Q = \sqrt{\frac{2PD}{C \cdot H \cdot (1+2f)}} \quad \dots 2$$

Model III:

Economic order quantity or economic lot size when there is shortage as shown in Fig.



We have considered the cases where all demands were met from stock and the system was never out of stock.

In the present case all demand must be met ultimately but the system may be short of stock for some time. For shortages penalty is charged to the supplier in order to discourage such practice. In case of shortages back orders are to be permitted.

With usual notations as followed in previous models with

C_1 = Penalty per unit shortages cost per unit time.

Q = Order Quantity or lot size.

Q_m = Maximum inventory level.

n = Number of order placed per year = D/Q

Total time of the cycle $t = t_1 + t_2$

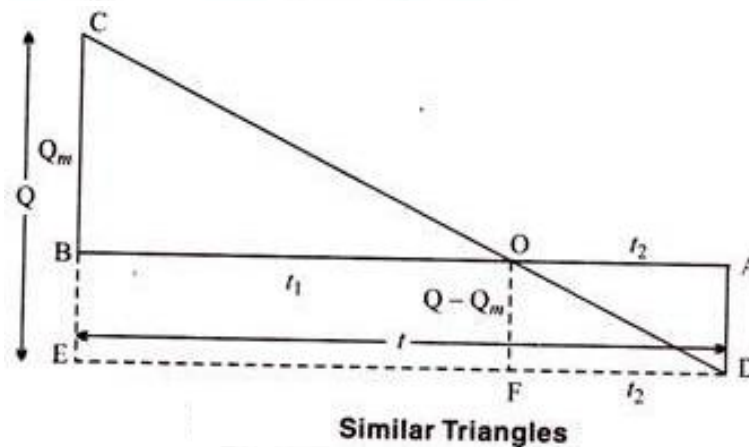


t_1 is the time during which demand is met from inventory stock t_2 is the time when shortage occur. Using geometrical relationship from these similar triangles as shown in Fig.

We have $\frac{t_1}{t} = \frac{Q_m}{Q}$ from $\Delta S \ COB$ & CED

or $t_1 = \frac{Q_m}{Q} t$... (1)

and $t_2 = \frac{Q - Q_m}{Q} t$ from $\Delta S \ COB$ & OFD or OAD ... (2)



Now the average number of units in inventory during time $t_1 = \frac{Q_m}{2}$

Average inventory cost during time $t_1 = \frac{Q_m}{2} C \cdot t_1$

Average inventory carrying cost during this period = $H \cdot \frac{Q_m}{2} C \cdot t_1$

Since n is the number of orders placed annually the annual inventory carrying cost will be

$$= H \cdot \frac{Q_m}{2} C \cdot t_1 \times n$$

Similarly the average number of units short during time $t_2 = \frac{Q - Q_m}{2}$

The average shortage cost during time $t_2 = \frac{Q - Q_m}{2} C_1 \cdot t_2$

Annual shortage cost = $\frac{Q - Q_m}{2} \cdot C_1 \cdot t_2 \cdot n$

So total yearly cost

$$Y_c = \text{Material cost} + \text{inventory carrying cost} + \text{set up cost} + \text{Shortage cost}$$

$$= C \cdot H \frac{Q_m}{2} t_1 \cdot n + CD + \frac{P}{Q} \cdot D + \frac{Q - Q_m}{2} C_1 t_2 \cdot n$$

Substituting the value of t_1 and t_2 from equations (1) and (2)

$$Y_c = CD + \frac{P}{Q} D + C \cdot H \frac{Q_m}{2} \frac{Q_m}{Q} t \cdot n + \frac{Q - Q_m}{2} C_1 \frac{(Q - Q_m)}{Q} t \cdot n$$

Put $n \cdot t = 1$ year.

$$Y_c = CD + \frac{P}{Q} D + C \cdot H \frac{Q_m^2}{2Q} + \frac{(Q - Q_m)^2}{2Q} C_1$$

$$= CD + \frac{P}{Q} D + C \cdot H \frac{Q_m^2}{2Q} + \frac{(Q - Q_m)^2}{2Q} C_1$$

Case I . To get optimum Q_m , differentiating Y_c with respect to Q_m and putting it equal to zero

$$\therefore \frac{d}{dQ_m} \left(CD + \frac{P}{Q} D + CH \frac{Q_m^2}{2Q} + \frac{(Q - Q_m)^2}{2Q} C_1 \right)$$

$$\text{or} \quad \left(0 + 0 + CH \frac{2Q_m}{2Q} + 2 \frac{(Q - Q_m)}{2Q} (-1) C_1 \right) = 0$$

$$CHQ_m - QC_1 + Q_m C_1 = 0$$

$$Q_m (CH + C_1) = QC_1$$

$$Q_m = \frac{QC_1}{CH + C_1}$$

...3

Since the factor $\frac{C_1}{CH + C_1}$ is always less than 1 so the optimum Q_m will always be less than Q .

Case II. To get optimum Q

$$\text{Put} \quad \frac{dY_c}{dQ} = 0 \quad \text{and} \quad Q_m = Q \frac{C_1}{CH + C_1} \quad \text{in the result}$$

$$\frac{dY_c}{dQ} = \frac{d}{dQ} \left[CD + \frac{P}{Q} D + \frac{CHQ_m^2}{2Q} + \frac{(Q - Q_m)^2}{2Q} C_1 \right] = 0$$

On simplification we get Q_{opt}

$$Q_{opt} = \sqrt{\frac{2PD}{CH}} \sqrt{\frac{C_1 + CH}{C_1}}$$

...4

$$\text{Since factor } \sqrt{\frac{C_1 + CH}{C_1}} \text{ is greater than 1, so } Q > \sqrt{\frac{2PD}{CH}}$$

Model IV:

Economic lot size when supply is continuous (not instantaneous) consumption is continuous, no shortages and no buffer stock. The supply rate is greater than the rate of consumption p so there is a gradual build-up of inventory at the rate or $(S - p)$.

$\therefore$ The rate of inventory build-up = $(S - p)$ Model for economic lot sizes with supply and consumption rates varying, no shortages, no buffer stock, rate of supply = n rate of consumption = p .

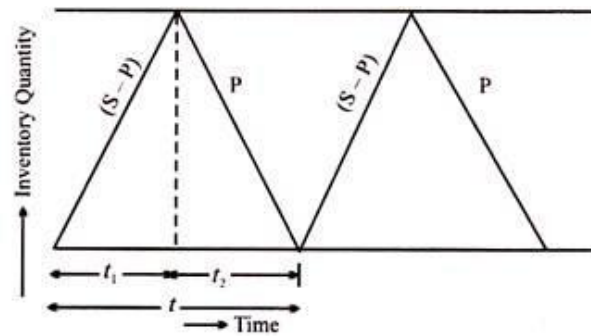


Fig. 12.8: Model for economic lot sizes with supply and consumption rates varying, no shortages, no buffer stock, rate of supply = S , rate of consumption = p .

Inventory after time $t_1 = (S - p)t_1$

$$= St_1 \left(1 - \frac{p}{S} \right) = Q(1 - p/S)$$

Since $Q = St_1$

$$\text{Average inventory} = \frac{Q}{2} (1 - p/S)$$

The annual cost

$$Y_c = \text{Material cost} + \text{set up cost} + \text{Inventory holding cost}$$

$$\text{or } Y_c = C \cdot D + \frac{P}{Q} \cdot D + \frac{Q}{2} (1 - p/S) CH$$

For minimum total annual cost,

$$\text{Put } \frac{dY_c}{dQ} = 0 = \frac{d}{dQ} \left[CD + \frac{PD}{Q} + \frac{Q}{2} (1 - p/S) CH \right]$$

$$\text{or } 0 = \frac{P}{Q^2} D + \frac{CH}{2} (1 - p/S)$$

$$\text{or } Q = \sqrt{\frac{2PD}{CH(1 - p/S)}}$$

...5

IMPORTANT QUESTIONS

1. Explain the importance of Planning in a production system.
2. Explain different techniques of Production planning.
3. Explain Product Life Cycle (PLC).
4. Explain the function of Production control.
5. What is Gantt chart.
6. Explain Routing, scheduling, dispatching and followup.
7. Explain EOQ model.
8. Explain different Inventory models.
9. Explain different selective inventory control techniques.
10. Explain different steps in production control.
11. Explain different types of production.
12. Explain different types of inventory.
13. Explain different inventory costs.
14. Explain different steps in new product development.
15. Explain different types of scheduling.



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MODULE 6

6.1 QUALITY CONTROL & INSPECTION

Quality Control is a systematic control of various factors that affect the quality of the product. The various factors include material, tools, machines, type of labour, working conditions, measuring instruments, etc.

Quality Control can be defined as the entire collection of activities which ensures that the operation will produce the optimum Quality products at minimum cost.

As per A.Y.Feigorbaum Total Quality Control is: “An effective system for integrating the quality development, Quality maintenance and Quality improvement efforts of the various groups in an organization, so as to enable production and services at the most economical levels which allow full customer satisfaction.

In short, we can say that quality control is a technique of management for achieving required standards of products.

Factors Affecting Quality:

In addition to men, materials, machines and manufacturing conditions there are some other factors which affect the product quality. These are:

- Market Research i.e. demand of purchaser.
- Money i.e. capability to invest.
- Management i.e. Management policies for quality level.
- Production methods and product design.

Modern quality control begins with an evaluation of the customer's requirements and has a part to play at every stage from goods manufactured right through sales to a customer, who remains satisfied.

Objective of Quality Control:

- To decide about the standard of quality of a product that is easily acceptable to the customer and at the same time this standard should be economical to maintain.
- To take different measures to improve the standard of quality of product.
- To take various steps to solve any kind of deviations in the quality of the product during manufacturing.

Functions of Quality Control Department:

- Only the products of uniform and standard quality are allowed to be sold.
- To suggest method and ways to prevent the manufacturing difficulties.
- To reject the defective goods so that the products of poor quality may not reach to the customers.
- To find out the points where the control is breaking down and investigate the causes of it.
- To correct the rejected goods, if it is possible. This procedure is known as rehabilitation of defective goods.

Advantages of Quality Control:

- Quality of product is improved which in turn increases sales.
- Scrap rejection and rework are minimized thus reducing wastage. So the cost of manufacturing reduces.
- Good quality product improves reputation.
- Inspection cost reduces to a great extent.
- Uniformity in quality can be achieved.
- Improvement in manufacturer and consumer relations.

Consequences of Poor Quality

Some of the major ways that quality affects an organization are:

- i) Loss of business,
- ii) Liability,
- iii) Productivity,
- iv) Costs



Techniques of Quality Control:

(I) Inspection

(II) Statistical quality control (SQC)

(I) Inspection:

Inspection is that component of quality control programme which is concerned with checking on the performance of items to the specifications set for it. It involves periodic checking and measuring – before, during and after the production process. Because of the numerous variables that enter into manufacturing, inspection is a never ending process.

Inspection may be ‘Centralised’ or ‘Floor Inspection.’:

Under centralized inspection, all the work from a department is sent to the Inspection Department, before passing on to the next operation. Floor inspection, on the other hand, follows the practice of sending inspectors to the floor and inspects work at the machines of operatives. It is also called patrolling or travelling inspection.

Advantages of centralized inspection:

- (i) Centralised inspection ensures impartial supervision; because the inspector is not under the strain of not rejecting the work of a person with whom he has good personal relations.
- (ii) Under centralized inspection, it is easier to keep records of items/parts which are approved or rejected.
- (iii) Production work is liable to less interruption, under centralized inspection.

Advantages of floor inspection:

- (i) Since work is inspected on the floor; delay in sending work to next station is avoided.
- (ii) Inspector can immediately locate the fault and suggest rectification.
- (iii) It involves minimum material handling.

(II) Statistical Quality Control (SQC):

SQC is based upon the laws of probability. It is a system for controlling the quality of production within specified limits (tolerance limits) by means of a sample procedure and continuing analysis of inspection results.

“SQC is a simple statistic method for determining the extent to which quality goods are being met without necessarily checking every item produced and for indicating whether or not the variations which occur are exceeding normal expectations. It enables us to decide whether to reject or accept a particular product.”

SQC does not produce a quality product. It merely informs management that things are not going as they should. Management must take necessary action to remove the causes of variations and ensure production of quality products.

Inspections vs. SQC:

It is an interesting academic exercise to compare inspection and SQC.

The two techniques of quality control may be compared as follows:

- (i) The result of inspection is acceptance or rejection of production; while SQC enables management to take action so that products will meet specifications. As such inspection enables one “to be wiser after the event” whereas SQC enables one “to get wiser before the event.”
- (ii) Inspection can be cent per cent; while SQC always involves sampling.

Techniques of SQC:

Techniques of SQC can be divided into two parts:

- (1) Process control
- (2) Acceptance sampling

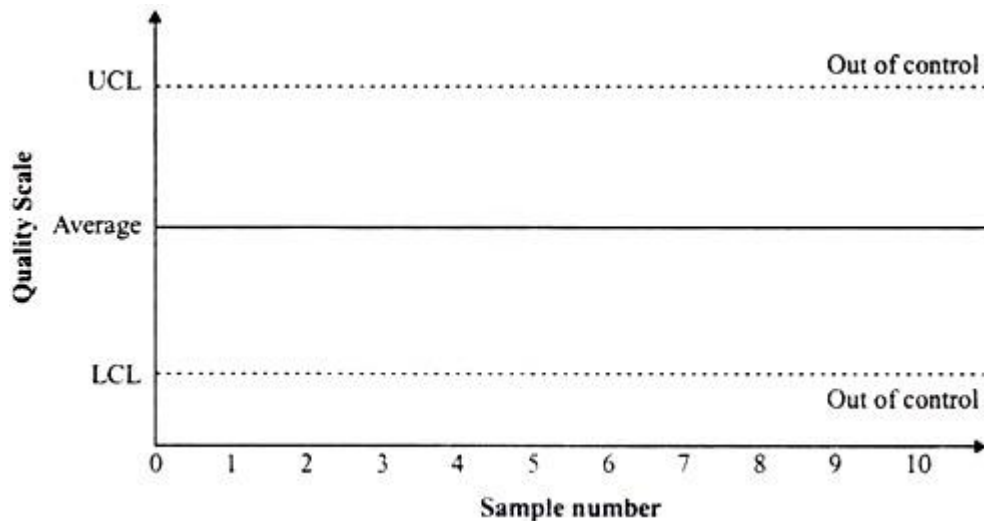
Following is a brief account of these techniques of SQC:

(1) Process Control:

The checking up of quality characteristics under process control is done with the help of charts. There may be many types of charts like ‘X-Chart’, ‘R-Chart’, ‘C-Chart’ and ‘P-Chart’. All types of charts are

similar in composition and structure. All of them represent how quality- characteristic is changing from one sample to another.

A control chart when prepared would appear as follows:



Note: UCL = Upper Control Limit

LCL = lower control Limit

A process is considered out of control and an action to check and correct the process is taken; when a plotted point falls outside the control limits.

Advantages of control charts:

1. They provide visual aids
2. They are easy to prepare.
3. They give early warning of trouble

(2) Acceptance Sampling:

Control charts are useful for process control. In case of receipt of materials and dispatch of finished goods; a different method is used, that of acceptance sampling. Acceptance sampling plans are of utmost value when the nature of the process used to manufacture products remains unchanged.

In acceptance sampling, decisions [e.g. whether acceptable/not acceptable (rejection)] about the quality of batches or lots are made after inspection of only a portion i.e. a sample. If the sample of items conforms to requisite quality levels; then the whole batch from which the sample is taken is accepted. If the sample does not conform to the requisite quality level; then the whole batch is rejected.

An acceptance sampling is defined as:

Lot size (N)

Sample size (n)

Acceptance number (C)

Suppose $N = 9000$; $n = 300$ and $C = 7$; then this sampling plan means that a lot of 9000 items has 300 units (sample size) inspected. If seven or less defectives are found in 300 units sample; the lot is accepted. If eight or more defectives are found in the sample; the lot is rejected.

A close study of acceptance sampling technique would reveal that there is likelihood that a lot of satisfactory quality is rejected on the basis of sample result. This is technically called producer's risk. Similarly, the consumer (or buyer) has the risk of accepting a lot of unsatisfactory quality, on the basis of sample results. This risk is called consumer's risk.

Advantages of acceptance sampling:

- (i) Less expensive than 100% inspection
- (ii) Used where 100% inspection is not possible.
- (iii) Useful when inspection may cause damage or complete destruction.

Advantages of SQC:

(i) Reduced Cost:

Since only a fraction of output is inspected; costs of inspection are greatly reduced.

(ii) Early Warning of Defects:

SQC gives an early warning of defects in the production process; so that these defects can be detected and corrected at inception.

(iii) Simple Technique:

SQC techniques are simple and can be operated by semi-skilled operators.

(iv) Continuous Inspection:

SQC is a technique which provides a continuous inspection of the product at various stages of the manufacturing process.

(v) Adherence to Specifications:

SQC enables a process to be held in a state of statistical control i.e. a state in which variability is the result of chance causes alone.

Advantages of SQC – at a glance

1. Reduced cost
2. Early warning of defects
3. Simple technique
4. Continuous inspection
5. Adherence to specifications.

6.2 DESTRUCTIVE & NON-DESTRUCTIVE TESTING METHODS

Non Destructive Testing

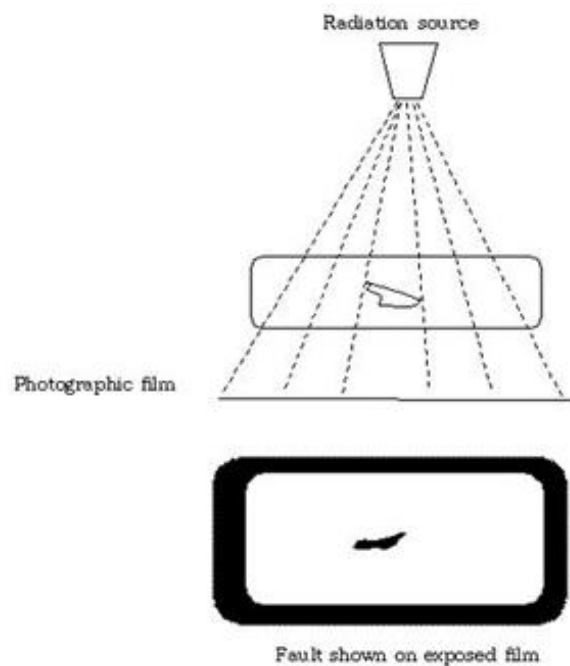
Nondestructive testing (NDT) is the process of inspecting or testing materials, components or assemblies for discontinuities, or differences in characteristics without destroying the workpiece or specimen. In other words, when the inspection or test is completed the part or the workpiece can still be used.

1. Visual and Optical testing

This method involves visually checking the work piece for surface defects, geometrical errors etc. Inspection is carried out using naked eye or by using magnifying glass. Instruments like steel rule, screw gauge, vernier caliper etc. can be used. Corrosion, misalignment of parts, physical damage and cracks are just some of the discontinuities that may be detected by visual examinations.

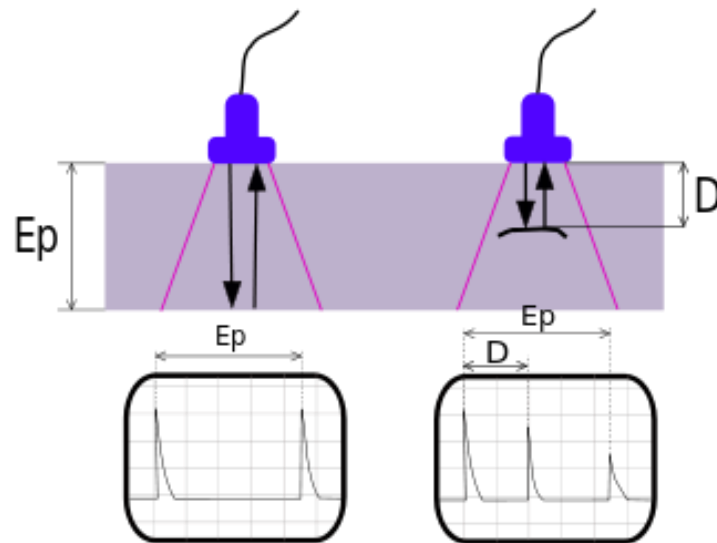
2. Radiography testing

In this method, mostly X rays are used for finding the internal defects in the components. X rays are passed through the workpiece and reaches the X ray film placed below the work piece as shown in figure. If any internal cracks or defects are present in the workpiece, it is shown in the image produced in the photographic film (X ray film). Instead of X rays, gamma rays are also used which has more penetrating power.



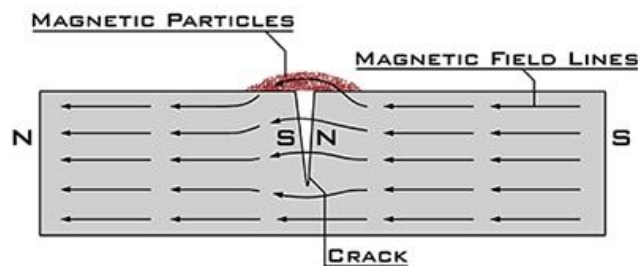
3. Ultrasonic testing

In this testing method, high frequency sound waves are used to find out the internal defects. Sound waves are passed through the component and the reflection of these sound waves are captured using sensors. If any internal cracks or defects are present, the sound waves will get reflected from these and the sensors records this variation in reflection. And these are converted into digital signals and visually showed in LCD or CRT displays. By checking the graphs, the presence of internal defects are easily detected.



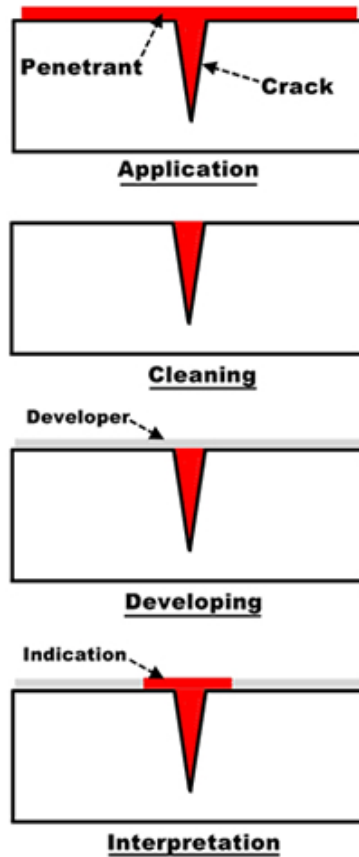
4. Magnetic Particle testing

In this method, electric current is made to pass through the workpiece. The workpiece becomes an electro magnet and magnetic flux lines are created in the workpiece. If any surface defects or cracks are present in the workpiece, small magnetic poles are created at these cracks as shown in the figure. While the workpiece is being magnetized, magnetic particles in the form of powder (iron powder) is sprinkled over this workpiece. Then the excess powder is removed. But some of the powder remains in the workpiece where small magnetic poles are present. By the presence of these remaining magnetic particles in the workpiece, the location of surface defects are spotted.



5. Liquid Penetrant testing

Liquid penetrant testing is one of the fastest and easiest methods to find surface defects. There is no special equipment used in this method. A low viscous liquid called penetrant or pigment is applied on the surface of the workpiece. The liquid will be having a vibrant colour. This liquid penetrant will penetrate into the cracks in the surface. After sometime, excess penetrant is removed from the surface but the liquid penetrated in the cracks will remain there. A light coating of a powder known as developer is applied over the surface of the workpiece and the function of this powder is to absorb the penetrant entrapped in the cracks. While absorbing the penetrant, colour of developer changes and due to this colour change, the presence of surface cracks are spotted easily.



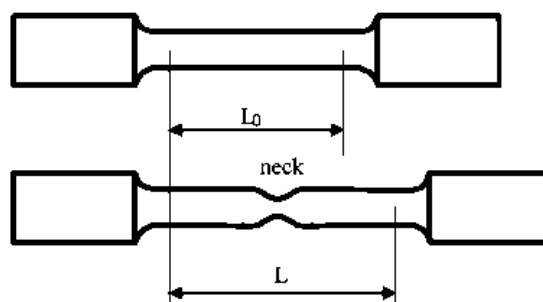
Destructive Testing

Destructive testing methods are used to find out the tensile strength, compressive strength, impact strength, hardness etc. In this testing methods, certain load is applied to the work piece or the specimen which physically damages it. So these testing methods are named destructive testing.

1. Tensile test

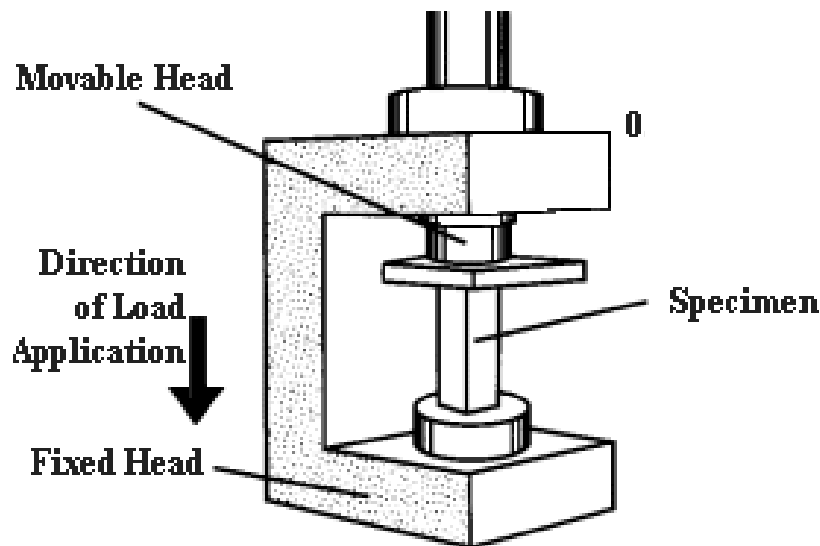
Tensile test is conducted to measure the tensile strength, ductility etc. of the material. The specimen is gripped in the tensile testing machine and tensile load is applied. Due to this pulling load, the specimen elongates and after a certain load, it stops elongating and the specimen breaks. The load at this point is measured using the machine and it is read from the dial. The initial diameter of the specimen, diameter at the point of fracture, initial length of the specimen, length after the fracture etc. are measured and substituted in the formulas for finding the tensile strength, % elongation etc. The machine used for tensile test is known as Universal Testing Machine (UTM).

Standard tensile test specimen



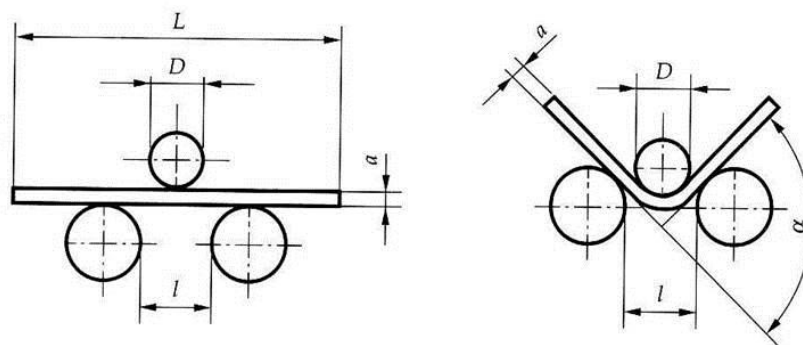
2. Compression test

Compression test is conducted to find out the compressive strength of the material. It is conducted similar to tensile test but the force is compressive. Usually compression test is conducted on brittle materials like cast iron and in concrete and brick. The testing is done using Universal testing machine (UTM). The specimen used in compression test is usually made of cylindrical. When the load is applied on the specimen, it compresses and after a certain load, the specimen breaks. This load is read from the dial of UTM. The diameters of the specimen before and after fracture and the load at fracture are substituted in equations to find the compressive strength of the material.



3. Bend test

Bend test is conducted to find the bending strength, ductility etc. of the material. This test can also be done using a UTM but need certain attachments for gripping the specimen. The test is conducted on both ductile and brittle materials. The load at the time of start of bending and at fracture is measured using the machine and these values are used to find the bending strength of the material.



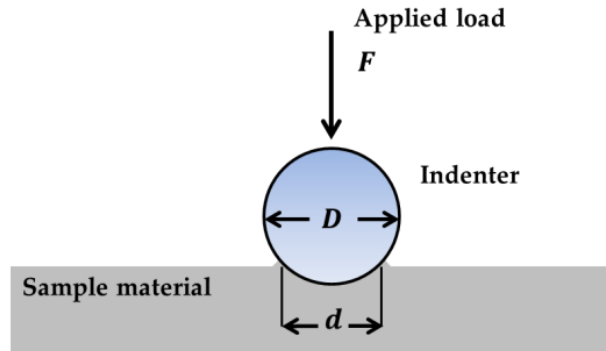
4. Hardness test

This testing is used to find out the resistance to wear of the specimen. Based on the machine used, hardness testing is divided into 3.

a) Brinell Hardness Test

In Brinell hardness test, a hardened steel ball is pressed into the specimen. Heavy load is applied on the specimen using this hardened steel ball which causes an indentation on the surface of the workpiece/specimen. After removing the load, diameter of this indent is measured using brinell

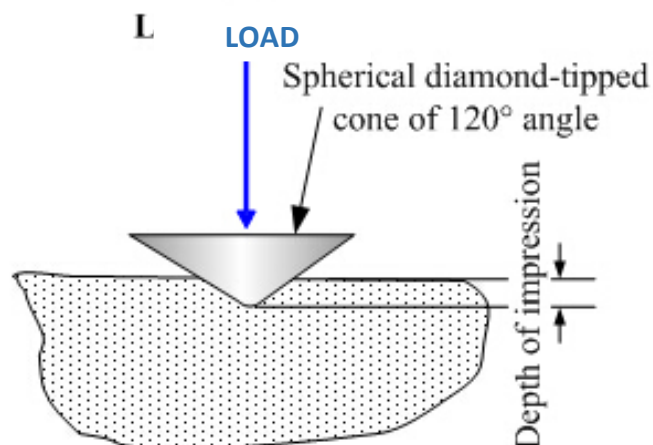
microscope. Usually the diameter of the steel ball is 10mm. Applied load is ranging between 100kg to 3000kg. During the test, the load is maintained constant for 10-15 seconds. The diameter of the steel ball, diameter of the indent, load applied etc. are substituted in the theoretical formula for finding the Brinell hardness number (BHN) of that material. BHN is used to represent the hardness of the materials. If BHN is higher, the material is harder. If BHN is less, material is soft.



The disadvantages of brinell hardness testing is, (a) it produces large indent on soft materials, (b) it cannot be used to find the hardness of materials harder than the steel ball used, (c) the test is not suitable on thin work pieces.

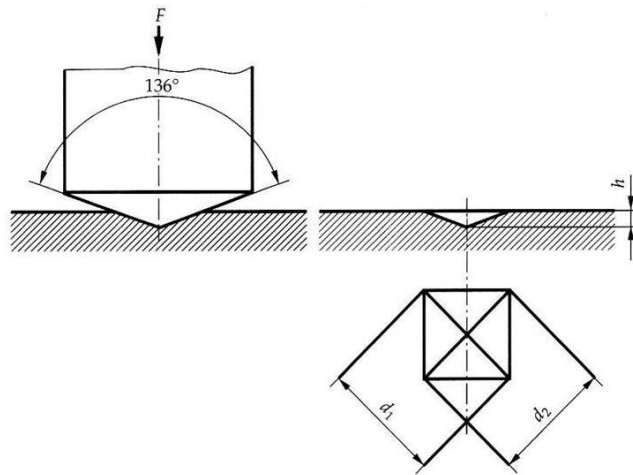
b) Rockwell Hardness Test

The procedure of Rockwell hardness testing is similar to brinell. But instead of hardened steel ball, a conical shaped diamond is used. Load is applied on the work piece using the diamond and due to the heavy load, an indent is produced in the workpiece. Due to the use of diamond instead of steel, Rockwell testing is suited for finding hardness of harder materials which are beyond the range of brinell. And also the depth of indent created in the specimen is also small compared to brinell. Another advantage is that, there is no need to measure the depth of the indent. The Rockwell hardness number (RHN) is readily read from the dial of the machine. So the time for measuring the diameter and applying in the equation is saved. But the limitation is that, the accuracy of the testing is very less compared to brinell.



c) Vicker's Hardness Test

Vickers hardness test is the most accurate hardness test among the three. In this, a pyramid shaped diamond is used as the indenter. This test is suitable for very hard materials. Load is applied on the workpiece using the diamond and due to the heavy load, an indent is produced in the work piece. The diagonal length of the indent is accurately measured using the microscope. The diameter of the indent and the applied load is substituted in the theoretical equation for finding the Vicker Hardness number (VHN).

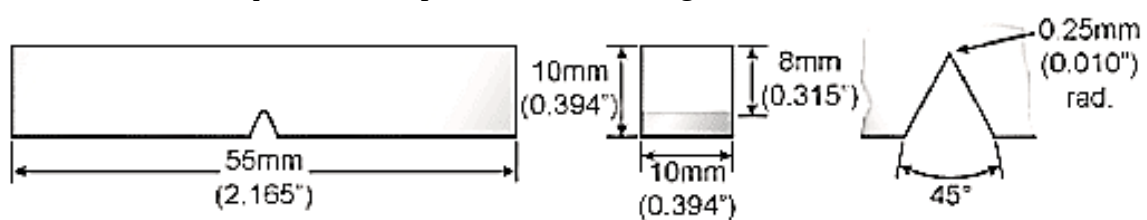


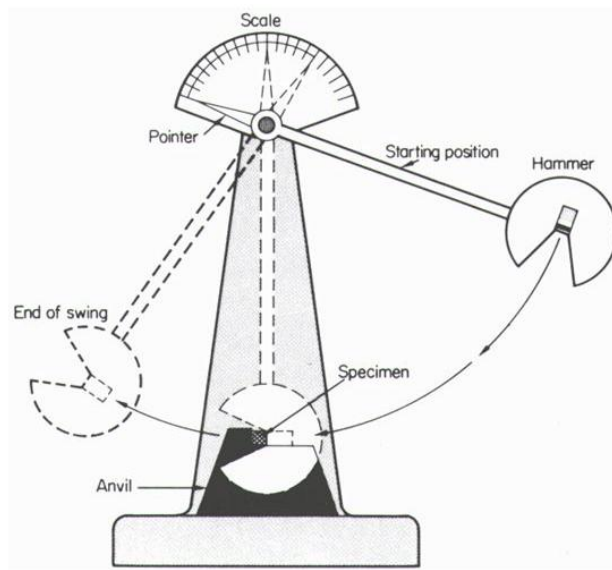
5. Impact Test

Materials may fail when they are suddenly subjected to loads or stresses. Impact tests are performed in order to assess their capacity to withstand such impact loads. This property of the material to withstand such sudden loads are called impact strength or impact resistance. In impact test, the specimen is struck and broken by a single blow in a specially designed testing machine. Based on these machines, impact tests are divided into two. Charpy impact test and Izod impact test.

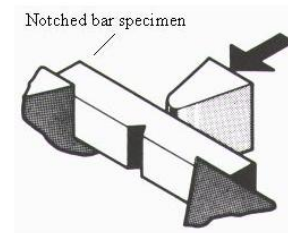
a) Charpy Impact Test

The specimen used in a Charpy impact test is a 10mm square cross section having a length of 55mm. A 2mm depth notch is made at the center with an angle of 45° . This specimen is placed horizontally in the testing machine in such a way that the striking hammer hits the specimen just opposite to the notch. The Charpy impact testing machine consists of a body which supports a striking hammer (pendulum), and a support to place the specimen. The hammer is placed at the end of a component hinged at the other end, it act as a pendulum. A dial is also attached at the hinge to measure the angle of movement of the pendulum. The hammer is raised to a known angle and is released. The hammer comes down and strikes the specimen, breaks it and moves forward to a certain height. This height is measured in the dial in terms of angle. The difference between the initial angle of the pendulum and the angle reached after breaking the specimen, gives the energy absorbed by the specimen. This is the energy that was taken by the specimen to break and is equal to the impact resistance/strength.

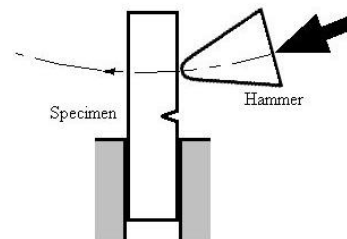




Apparatus to measure impact strength



Charpy test configuration

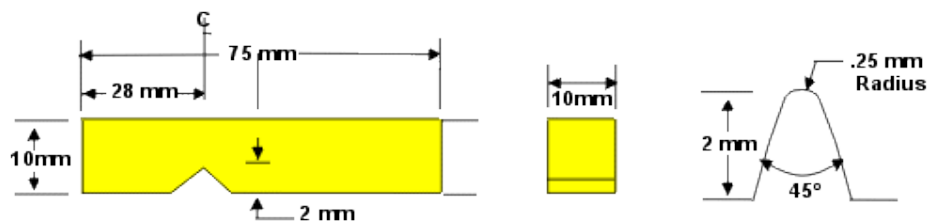


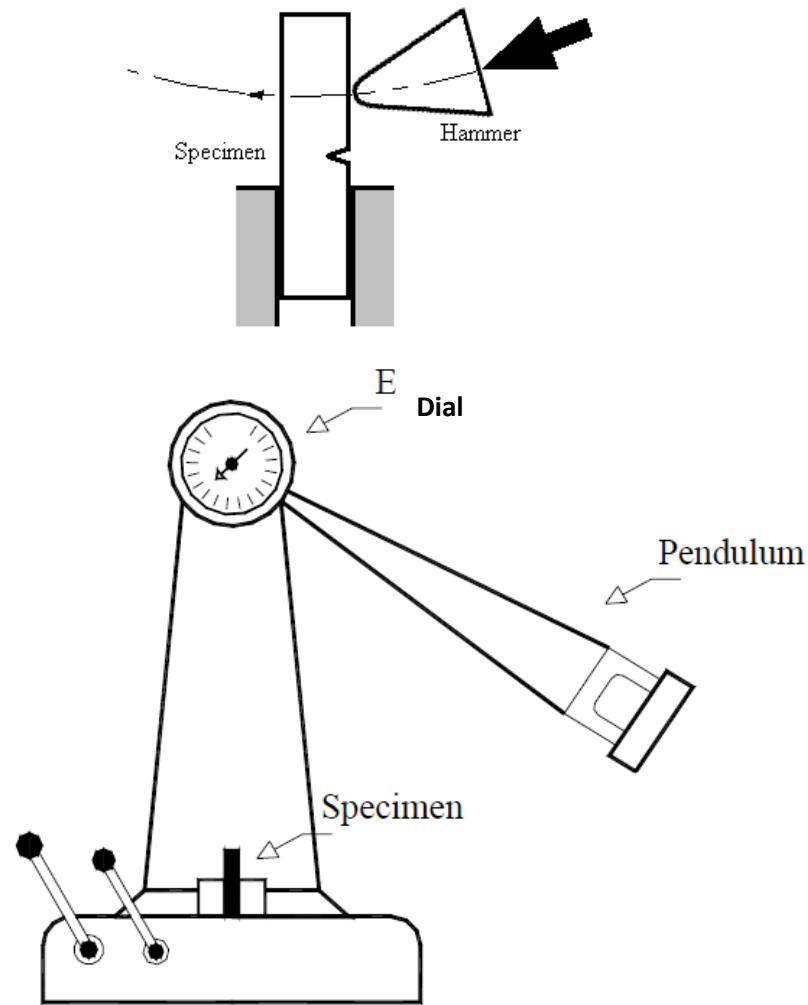
Izod test configuration

b) Izod Impact Test

Izod impact testing is similar to Charpy impact test. There are mainly two differences. The length of the specimen is different. The specimen is fixed vertically in the testing machine.

The specimen used in an Izod impact test is a 10mm square cross section having a length of 75mm. A 2mm depth notch is made with an angle of 45° , 28mm away from the end. This specimen is fixed vertically in the testing machine in such a way that the striking hammer hits the specimen at the same side of the notch. The Izod impact testing machine consists of a body which supports a striking hammer (pendulum), and a support to fix the specimen. The hammer is placed at the end of a component hinged at the other end, it acts as a pendulum. A dial is also attached at the hinge to measure the angle of movement of the pendulum. The hammer is raised to a known angle and is released. The hammer comes down and strikes the specimen, breaks it and moves forward to a certain height. This height is measured in the dial in terms of angle. The difference between the initial angle of the pendulum and the angle reached after breaking the specimen, gives the energy absorbed by the specimen. This is the energy that was taken by the specimen to break and is equal to the impact resistance/strength.





Impact Testing Machine.

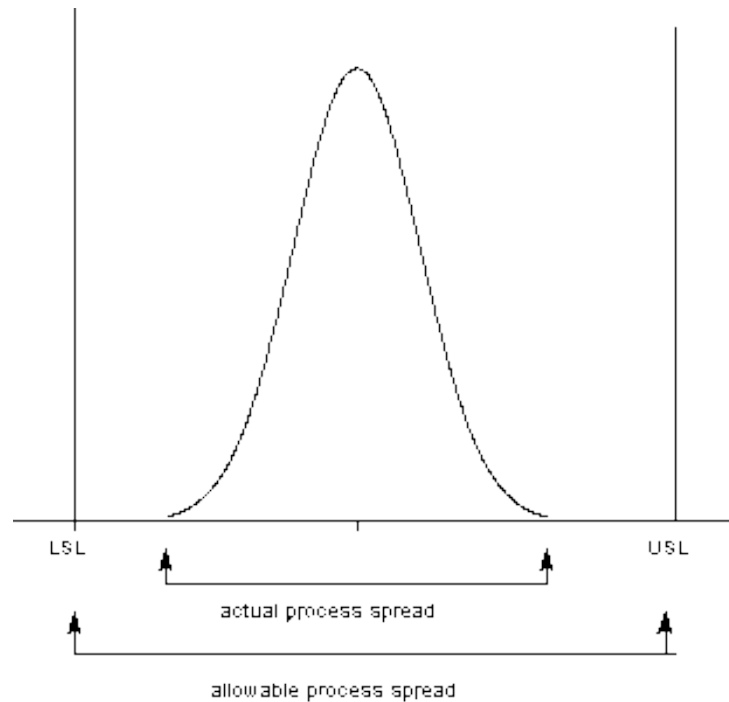
6.3 PROCESS CAPABILITY

Process capability compares the output of an *in-control* process to the specification limits by using *capability indices*. The comparison is made by forming the ratio of the spread between the process specifications (the specification "width") to the spread of the process values, as measured by 6 process standard deviation units (the process "width").

Process Capability Indices

We are often required to compare the output of a stable process with the process specifications and make a statement about how well the process meets specification. To do this we compare the natural variability of a stable process with the process specification limits.

A process where almost all the measurements fall inside the specification limits is a capable process. This can be represented pictorially by the plot below:



6.4 STATISTICAL QUALITY CONTROL

Statistical quality control refers to the use of statistical methods in the monitoring and maintaining of the quality of products and services. One method, referred to as acceptance sampling, can be used when a decision must be made to accept or reject a group of parts or items based on the quality found in a sample. A second method, referred to as statistical process control, uses graphical displays known as control charts to determine whether a process should be continued or should be adjusted to achieve the desired quality.

Acceptance sampling

Assume that a consumer receives a shipment of parts called a lot from a producer. A sample of parts will be taken and the number of defective items counted. If the number of defective items is low, the entire lot will be accepted. If the number of defective items is high, the entire lot will be rejected. Correct decisions correspond to accepting a good-quality lot and rejecting a poor-quality lot. Because sampling is being used, the probabilities of erroneous decisions need to be considered. The error of rejecting a good-quality lot creates a problem for the producer; the probability of this error is called the producer's risk. On the other hand, the error of accepting a poor-quality lot creates a problem for the purchaser or consumer; the probability of this error is called the consumer's risk.

The design of an acceptance sampling plan consists of determining a sample size n and an acceptance criterion c , where c is the maximum number of defective items that can be found in the sample and the lot still be accepted. The key to understanding both the producer's risk and the consumer's risk is to assume that a lot has some known percentage of defective items and compute the probability of accepting the lot for a given sampling plan. By varying the assumed percentage of defective items in a lot, several different sampling plans can be evaluated and a sampling plan selected such that both the producer's and consumer's risks are reasonably low.

Statistical process control

Statistical process control uses sampling and statistical methods to monitor the quality of an ongoing process such as a production operation. A graphical display referred to as a control chart provides a basis for deciding whether the variation in the output of a process is due to common causes (randomly occurring variations) or to out-of-the-ordinary assignable causes. Whenever assignable causes are identified, a decision can be made to adjust the process in order to bring the output back to acceptable quality levels.

Control charts can be classified by the type of data they contain. For instance, an $\bar{x}$ -chart is employed in situations where a sample mean is used to measure the quality of the output. Quantitative data such as length, weight, and temperature can be monitored with an $\bar{x}$ -chart. Process variability can be monitored using a range or R -chart. In cases in which the quality of output is measured in terms of the number of defectives or the proportion of defectives in the sample, an np -chart or a p -chart can be used.

All control charts are constructed in a similar fashion. For example, the centre line of an $\bar{x}$ -chart corresponds to the mean of the process when the process is in control and producing output of acceptable quality. The vertical axis of the control chart identifies the scale of measurement for the variable of interest. The upper horizontal line of the control chart, referred to as the upper control limit, and the lower horizontal line, referred to as the lower control limit, are chosen so that when the process is in control there will be a high probability that the value of a sample mean will fall between the two control limits. Standard practice is to set the control limits at three standard deviations above and below the process mean. The process can be sampled periodically. As each sample is selected, the value of the sample mean is plotted on the control chart. If the value of a sample mean is within the control limits, the process can be continued under the assumption that the quality standards are being maintained. If the value of the sample mean is outside the control limits, an out-of-control conclusion points to the need for corrective action in order to return the process to acceptable quality levels.

6.5 CONTROL CHARTS

Since variations in manufacturing process are unavoidable, the control chart tells when to leave a process alone and thus prevent unnecessary frequent adjustments. Control charts are graphical representation and are based on statistical sampling theory, according to which an adequate sized random sample is drawn from each lot. Control charts detect variations in the processing and warn if there is any departure from the specified tolerance limits. These control charts immediately tell the undesired variations and help in detecting the cause and its removal.

In control charts, where both upper and lower values are specified for a quality characteristic, as soon as some products show variation outside the tolerances, a review of situation is taken and corrective step is immediately taken.

If analysis of the control chart indicates that the process is currently under control (i.e. is stable, with variation only coming from sources common to the process) then data from the process can be used to predict the future performance of the process. If the chart indicates that the process being monitored is not in control, analysis of the chart can help determine the sources of variation, which can then be eliminated to bring the process back into control. A control chart is a specific kind of run chart that allows significant change to be differentiated from the natural variability of the process.

The control chart can be seen as part of an objective and disciplined approach that enables correct decisions regarding control of the process, including whether or not to change process control parameters. Process parameters should never be adjusted for a process that is in control, as this will result in degraded process performance.

In other words, control chart is:

- A device which specifies the state of statistical control,
- A device for attaining statistical control,
- A device to judge whether statistical control has been attained or not.

Purpose and Advantages:

1. A control charts indicates whether the process is in control or out of control.
2. It determines process variability and detects unusual variations taking place in a process.
3. It ensures product quality level.
4. It warns in time, and if the process is rectified at that time, scrap or percentage rejection can be reduced.
5. It provides information about the selection of process and setting of tolerance limits.
6. Control charts build up the reputation of the organization through customer's satisfaction.

A control chart consists of:

- Points representing a statistic (e.g., a mean, range, proportion) of measurements of a quality characteristic in samples taken from the process at different times [the data]
- The mean of this statistic using all the samples is calculated (e.g., the mean of the means, mean of the ranges, mean of the proportions)
- A center line is drawn at the value of the mean of the statistic
- The standard error (e.g., standard deviation/sqrt(n) for the mean) of the statistic is also calculated using all the samples
- Upper and lower control limits (sometimes called "natural process limits") that indicate the threshold at which the process output is considered statistically 'unlikely' are drawn typically at 3 standard errors from the center line

The chart may have other optional features, including:

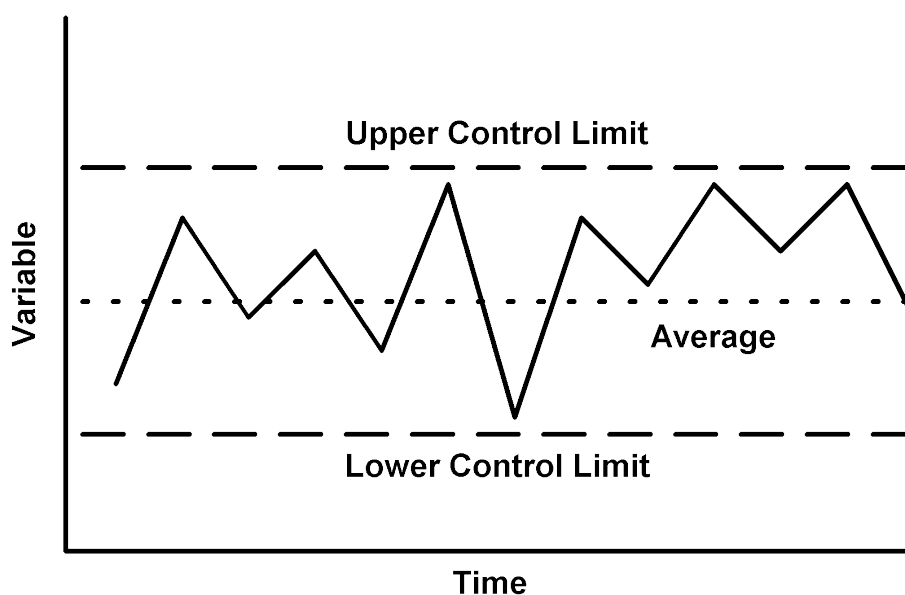
- Upper and lower warning limits, drawn as separate lines, typically two standard errors above and below the center line
- Division into zones, with the addition of rules governing frequencies of observations in each zone
- Annotation with events of interest, as determined by the Quality Engineer in charge of the process's quality.

Types of Control Charts

Variables	$\bar{x}$ Chart R Chart σ Chart
Attribute	p Chart C Chart np Chart U Chart

Control charts can be used to measure any characteristic of a product, such as the weight of a cereal box, the number of chocolates in a box, or the volume of bottled water. The different characteristics that can be measured by control charts can be divided into two groups: variables and attributes.

- A control chart for variables is used to monitor characteristics that can be measured and have a continuum of values, such as height, weight, or volume. A soft drink bottling operation is an example of a variable measure, since the amount of liquid in the bottles is measured and can take on a number of different values. Other examples are the weight of a bag of sugar, the temperature of a baking oven, or the diameter of plastic tubing.
- A control chart for attributes, on the other hand, is used to monitor characteristics that have discrete values and can be counted. Often they can be evaluated with a simple yes or no decision. Examples include color, taste, or smell. The monitoring of attributes usually takes less time than that of variables because a variable needs to be measured (e.g., the bottle of soft drink contains 15.9 ounces of liquid). An attribute requires only a single decision, such as yes or no, good or bad, acceptable or unacceptable (e.g., the apple is good or rotten, the meat is good or stale, the shoes have a defect or do not have a defect, the lightbulb works or it does not work) or counting the number of defects (e.g., the number of broken cookies in the box, the number of dents in the car, the number of barnacles on the bottom of a boat).



Mean (X-Bar) ($\bar{x}$) Charts

A mean control chart is often referred to as an x-bar chart. It is used to monitor changes in the mean of a process. To construct a mean chart we first need to construct the center line of the chart. To do this we take multiple samples and compute their means. Usually these samples are small, with about four or five observations. Each sample has its own mean. The center line of the chart is then computed as the mean of all sample means, where n is the number of samples:

1. It shows changes in process average and is affected by changes in process variability.
2. It is a chart for the measure of central tendency.
3. It shows erratic or cyclic shifts in the process.
4. It detects steady progress changes, like tool wear.
5. It is the most commonly used variables chart.
6. When used along with R chart:
 - a. It tells when to leave the process alone and when to chase and go for the causes leading to variation;
 - b. It secures information in establishing or modifying processes, specifications or inspection procedures;
 - c. It controls the quality of incoming material.
7. X-Bar and R charts when used together form a powerful instrument for diagnosing quality problems.

Range (R) charts

These are another type of control chart for variables. Whereas x-bar charts measure shift in the central tendency of the process, range charts monitor the dispersion or variability of the process. The method for developing and using R-charts are the same as that for x-bar charts. The center line of the control chart is the average range, and the upper and lower control limits are computed. The R chart is used to monitor process variability when sample sizes are small ($n < 10$), or to simplify the calculations made by process operators. This chart is called the R chart because the statistic being plotted is the sample range.

1. It controls general variability of the process and is affected by changes in process variability.
2. It is a chart for measure of spread.
3. It is generally used along with X-bar chart.

Plotting of $\bar{x}$ and R charts:

A number of samples of component coming out of the process are taken over a period of time. Each sample must be taken at random and the size of sample is generally kept as 5 but 10 to 15 units can be taken for sensitive control charts. For each sample, the average value $\bar{X}$ of all the measurements and the range R are calculated. The grand average $\bar{\bar{X}}$ (equal to the average value of all the average $\bar{X}$) and $\bar{R}$ ($\bar{R}$ is equal to the average of all the ranges R) are found and from these we can calculate the control limits for the $\bar{X}$ and R charts.

$$\bar{\bar{X}} = \frac{\bar{x}_1 + \bar{x}_2 + \cdots + \bar{x}_m}{m}$$

$$\bar{R} = \frac{R_1 + R_2 + \cdots + R_m}{m}$$

Variables Data ($\bar{x}$ and R Control Charts)

$\bar{x}$ Control Chart

$$UCL = \bar{\bar{x}} + A_2 \bar{R}$$

$$LCL = \bar{\bar{x}} - A_2 \bar{R}$$

$$CL = \bar{\bar{x}}$$

R Control Chart

$$UCL = \bar{R} D_4$$

$$LCL = \bar{R} D_3$$

$$CL = \bar{R}$$



Here the factors A_2 , D_4 and D_3 depend on the number of units per sample. Larger the number, the closer the limits. The value of the factors A_2 , D_4 , and D_3 can be obtained from S.Q.C tables. However for ready reference these are given below in tabular form:

n	A_2	D_3	D_4	d_2
2	1.880	0.000	3.267	1.128
3	1.023	0.000	2.574	1.693
4	0.729	0.000	2.282	2.059
5	0.577	0.000	2.114	2.326
6	0.483	0.000	2.004	2.534
7	0.419	0.076	1.924	2.704
8	0.373	0.136	1.864	2.847
9	0.337	0.184	1.816	2.970
10	0.308	0.223	1.777	3.078

6.6 RELIABILITY

Reliability is the probability that a product or equipment will perform satisfactorily for a given time under normal conditions of use. So reliability is related to quality but it is something more than that. Quality is concerned with the initial performance of a product or a service whereas reliability is related to the continuation of performance over a period of time.

A product having better performance initially may fail to provide the same performance later on. In such a case a product is not considered reliable. Hence, the manufacturers should produce not only quality products but also reliable products.

Measurement of Reliability:

Reliability can be measured only if it is expressed in quantitative terms. Probability and time require consideration for measurement of reliability. Whether a product is reliable or not depends on the probability of its failure during a given period and the time span for which it provides the rated performance.

To improve the reliability of products; data/information related to failures during a given period of time is collected; analysed and corrective actions are taken. The measurement of reliability depends on the type of products.

For example:

- (i) Certain products function only once such as fuse in a electrical or electronic circuit.
- (ii) Certain products possess a long life but if they fail cannot be maintained/repared.
- (iii) Certain products possess a long life but if they fail, can be maintained and then again can be put to use.

The conditions of use of the product are also to be taken into consideration while measuring the reliability of a product. A product if not used properly or if it is not well maintained its useful life may be shortened and its reliability may decrease.

Reliability information available in the form of curves or charts is very useful in taking certain important decisions.

The following questions can be easily answered with the help of charts:

- (i) What changes should be incorporated in the design of a product to improve its reliability,
- (ii) Which item should be inspected and when?
- (iii) Which component or machine should be inspected and when?
- (iv) How far a particular investment proposal would prove to be financially viable?
- (v) How far a product or a machine or process or a component is reliable and how can its reliability be improved?
- (vi) How should the plant and machinery or equipment etc. be maintained for its maximum?

Reliability Improvement:

Reliability is measured to see to what extent a product or process is reliable. In case the reliability shows decreasing trend, corrective actions to improve the reliability are taken.

Reliability of a product is directly related to the reliability of materials used for manufacturing as well as reliability of tools, machines and equipment and production processes etc. used for the production of that product.

Hence to improve the reliability of a certain product adopts the following:

- (i) More precise and perfect components should be used for assembly of product.
- (ii) Standard quality input materials procured from reliable sources should be utilized.

- (iii) Plant/machinery must be properly repaired and maintained.
- (iv) Suitable and standard equipment/machines should be used.
- (v) Machines or tools should be replaced before they become unsuitable.

6.7 BATH TUB CURVE

The bathtub curve is widely used in reliability engineering. It describes a particular form of the hazard function which comprises three parts:

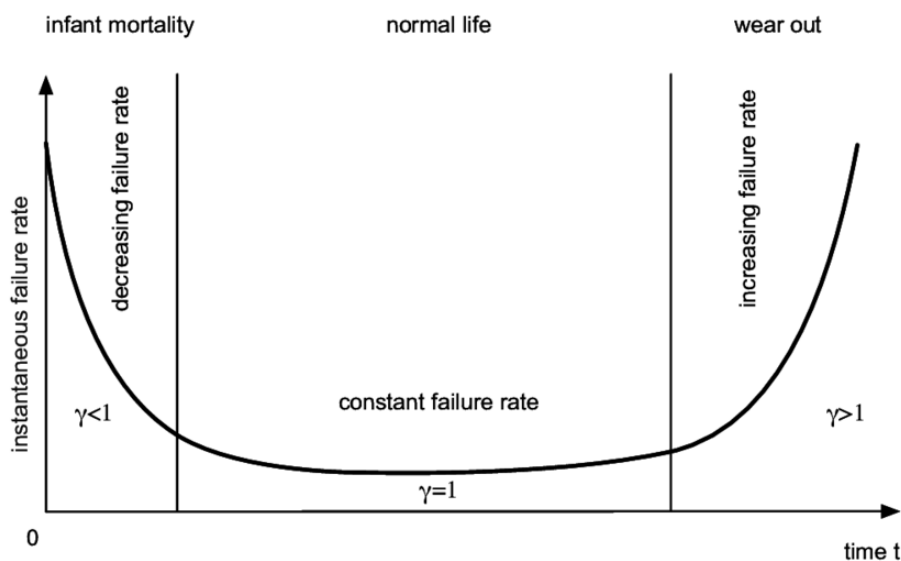
- The first part is a decreasing failure rate, known as early failures.
- The second part is a constant failure rate, known as random failures.
- The third part is an increasing failure rate, known as wear-out failures.

The name is derived from the cross-sectional shape of a bathtub: steep sides and a flat bottom.

The bathtub curve is generated by mapping the rate of early "infant mortality" failures when first introduced, the rate of random failures with constant failure rate during its "useful life", and finally the rate of "wear out" failures as the product exceeds its design lifetime.

In less technical terms, in the early life of a product adhering to the bathtub curve, the failure rate is high but rapidly decreasing as defective products are identified and discarded, and early sources of potential failure such as handling and installation error are surmounted. In the mid-life of a product—generally, once it reaches consumers—the failure rate is low and constant. In the late life of the product, the failure rate increases, as age and wear take their toll on the product. Many consumer product life cycles strongly exhibit the bathtub curve.

While the bathtub curve is useful, not every product or system follows a bathtub curve hazard function, for example if units are retired or have decreased use during or before the onset of the wear-out period, they will show fewer failures per unit calendar time (not per unit use time) than the bathtub curve.



Bath Tub Curve

6.8 TOTAL QUALITY MANAGEMENT

Total quality management (TQM) is the continual process of detecting and reducing or eliminating errors in manufacturing, streamlining supply chain management, improving the customer experience, and ensuring that employees are up to speed with training. Total quality management aims to hold all parties involved in the production process accountable for the overall quality of the final product or service. Total quality management (TQM) is a structured approach to overall organizational management. The focus of the process is to improve the quality of an organization's outputs, including goods and services, through continual improvement of internal practices. The standards set as part of the TQM approach can reflect both internal priorities and any industry standards currently in place.

A core definition of total quality management (TQM) describes a management approach to long-term success through customer satisfaction. In a TQM effort, all members of an organization participate in improving processes, products, services, and the culture in which they work.

Advantages of TQM:

(i) Sharpens Competitive Edge of the Enterprise:

TQM helps an organisation to reduce costs through elimination of waste, rework etc. It increases profitability and competitiveness of the enterprise; and helps to sharpen the organisation's competitive edge, in the globalized economy of today.

(ii) Excellent Customer Satisfaction:

By focusing on customer requirements, TQM makes for excellent customer satisfaction. This leads to more and more sales, and excellent relations with customers.

(iii) Improvement in Organisational Performance:

Through promoting quality culture in the organisation, TQM lead to improvements in managerial and operative personnel's performance.

(iv) Good Public Image of the Enterprise:

TQM helps to build an image of the enterprise in the minds of people in society. This is due to stress on total quality system and customers' requirements, under the philosophy of TQM.

(v) Better Personnel Relations:

TQM aims at promoting mutual trust and openness among employees, at all levels in the organisation. This leads to better personnel relations in the enterprise.

Limitations of TQM:

(i) Waiting for a Long Time:

TQM requires significant change in organisation; consisting of:

1. Change in methods, processes etc. of organisation.
2. Change in attitude, behaviour etc. of people

Launching of TQM and acceptance of the philosophy of TQM requires a long waiting for the organisation. It is not possible to accept and implement TQM overnight.

(ii) Problem of Labour Management Relations:

Success of TQM depends on the relationships between labour and management; because participation of people at all levels is a pre-requisite for TQM programme implementation. In many organisations, here and abroad, labour-management relations are quite tense. As such, launching, acceptance and implementation of TQM programme is nothing more than a dream for such organisations.

Elements of TQM

(i) Management's commitment to quality:

If an organisation is serious about implementing TQM, the lead has to be taken by the top management with full commitment.

It must initiate quality improvement programmes. The top management should continue all the efforts and provide the resources to continue quality improvement programmes. This is provided by collecting, reporting and use of quality related cost information.

(ii) Customer satisfaction:

TQM is designed in such a manner so as to meet the expectations of customers. In the present era, customer is the king. It must be recognised that customers are the most important persons for any business. The very existence of an organisation depends on them. They are the life blood of a business and deserve the most courteous and affectionate treatment.

(iii) Preventing rather than detecting defects:

TQM checks the poor quality products or services rather than simply to detect and sort out defects. "Prevention rather than detection" is the main characteristic of TQM. Some of the important techniques of TQM which aim at the prevention of defects rather than the detection of the defects are statistical process control, continuous process improvement and problem solving and system failure analysis etc.

(iv) Measurement of Quality:

Quality is a measurable entity and we must know what current quality levels are i.e. Where we are or where we stand in respect of the quality and what quality levels we are aspiring for or where we are going

(v) Continuous improvement:

TQM comprises of a continuous process of improvement covering people, equipment, suppliers, materials and procedures. It includes every aspect of an operation in an organisation. In Japan the word "Kaizen" is used to describe the continuous process of improvement. In USA, TQM zero defects and six-sigma are used to describe such efforts.

(vi) Corrective action for root cause:

TQM aims at preventing repetition of problems by identifying the root causes for their occurrence and developing means and corrective actions to solve the problems of the root level. Failure analysis and problem solving skills are very useful techniques in this regard.

(vii) Training:

Proper training programmes have to be undertaken to train the employees for the use of TQM concepts and techniques. Employees have to be provided regular training for continuous improvement.

(viii) Recognition of high quality:

TQM aims at developing long term relationships with a few high quality suppliers rather than those suppliers who supply the inferior goods at the low cost.

(ix) Involvement of Employees:

Involvement of employees means that every employee is completely involved at every step of production process which plays an active role in helping the organisation to meet its targets. Employee involvement and empowerment can be assured by enlarging the employee's job so that responsibility and authority is moved to the lowest level possible in the organisation.

(x) Benchmarking:

Benchmarking is a systematic method by which organisations can measure themselves against the best industry practices. Benchmarking aims at developing best practices that will lead to better performance. It helps a company to learn and incorporate the best practices into its own operations. Benchmarking is a technique of distinguishing an organization's efforts with the best performance in the field and also to suggest how the gap between the two performances can be removed. Thus, benchmarking is a technique of continuous improvement.

6.9 QUALITY CIRCLES

Quality circle may be defined as a small group of workers (5 to 10) who do the same/similar works voluntarily meeting together regularly during their normal working time, usually under the leadership of their own superior to

- Identity
- Analyse, and
- Solve work related problem.

The group present the solution to the management and whenever possible implement the solution themselves. The QC concept was first originated in Japan in 1960.

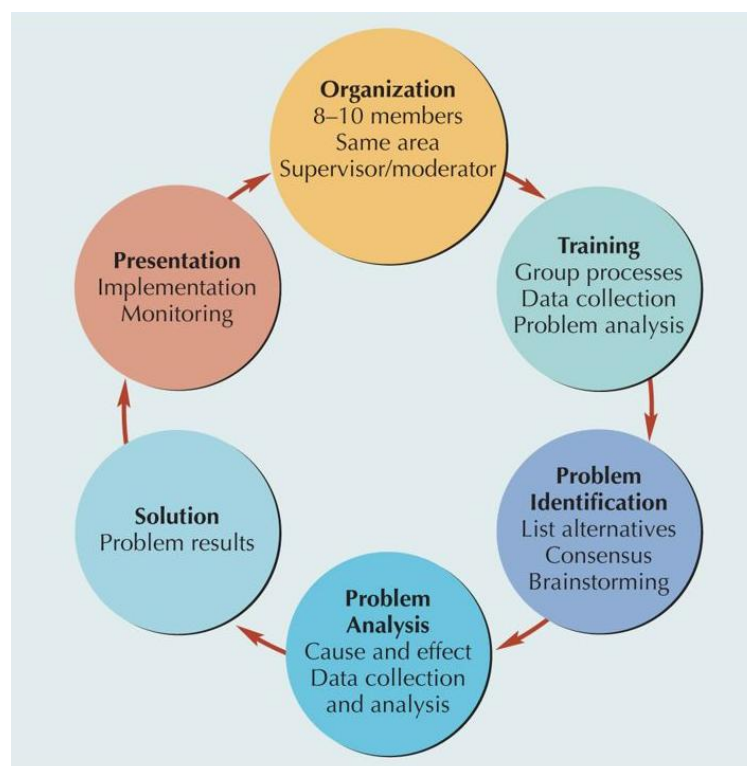
Philosophical basis of QC

- 1) It is a belief that people will take pride and interest in their work if they get autonomy and take part in the decision making process.
- 2) It develops a sense of belongingness in the employees towards a particular organization.
- 3) It is also a belief that each employee desires to participate in making the organization a better place.
- 4) It is a mean/method for the development of human resources through the process of training, work experience and participate in problem solving.

- 5) A willingness to allow people to volunteer their time and effort for improvement of performance of organization.
- 6) The importance of each members role in meeting organizational goal.

Characteristics of Quality Circle

- 1) Quality circles are small primary groups of employees/workers whose lower limit is 3 and upper limit is 12.
- 2) Membership is voluntary. The interested employees in some areas may come together to form a quality circle.
- 3) Each quality circle is led by area supervisor.
- 4) The members meet regularly every week/as per agreeable schedule.
- 5) The QC members are specially trained in technique of analysis and problem solving in order to play their role effectively.
- 6) The basic role of QC is to identify work related problems for improving quality and productivity.
- 7) QC enable the members to exercise their hidden talents, creative skills etc.
- 8) It promotes the mutual development of their member through co- operative participation.
- 9) It gives job satisfaction because of identifying and solving challenging problems while performing job.
- 10) Members receives public recognition.
- 11) Members receives recognition from company's management in the form of memento, certificate etc.
- 12) As a result of above recognition there is development of self steem and self confidence of employees.



Objectives of QC

- 1) To improve the quality and productivity.
- 2) To reduce cost of products/services by waste reduction. Effective utilization of resources eliminating errors/defects.
- 3) To utilize the hidden creative intelligence of the employee.
- 4) To identify and solve work related problem.
- 5) To motivate people for solving challenging tasks.
- 6) To improve communication within the organization.
- 7) To increase employees loyalty and commitment to organizational goal.
- 8) To enrich human capability, confidence, morale, attitude and relationship.
- 9) To pay respect to humanity and create a happy bright workplace.
- 10) To satisfy the human needs of recognition and self-development.

6.10 ISO

Quality standards may be designed by individual enterprises; but enterprises may find it difficult to convince customers about the value of these individual standards. However, if an external agency is involved in developing and certifying quality standards of enterprises; confidence of customers may be easily won over.

ISO 9000 series deals with this aspect of quality standards.

The International Standards Organisation (ISO) established in 1946 at Geneva, functions as an apex body of 'standard organizations' of various member countries including India., It introduced ISO 9000 series of quality standards in 1987, as a set of guidelines for developing and demonstrating a documented quality system.

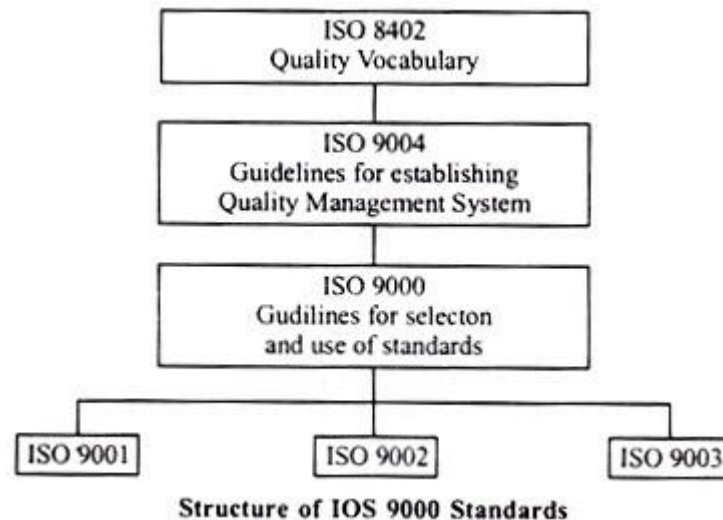
ISO 9000 series includes certificates in different areas; and a business enterprise can obtain relevant certificate under this series, after fulfilling the required conditionality's.

Structure of ISO-9000 Standards:

ISO-9000 series is a set of five international standards and supplemented by a vocabulary standard, defining the various terms used in the series.

These standards are shown through the following chart:





Selection and Use of Standards:

An organisation which intends to adopt ISO 9000 quality management system has to select one of the three quality system models – ISO 9001, ISO 9002 or ISO 9003.

ISO 9001 System:

Model for quality assurance in design/development, production installation and servicing; for use, when conformance to specifications is to be assumed by the supplier during production and installation.

ISO 9002 System:

Model for quality assurance in production, installation and servicing; for use, when conformance to specified requirements is to be assured by the supplier during production and servicing.

ISO 9003 System:

Model for quality assurance in final selection and testing; for use, when conformance to required specifications is to be assured by the supplier solely to final inspection.

Implementation of ISO 9000 Quality Management System:

The various phases in the installation of ISO 9000 quality management system are:

Phase I: Preparation Phase:

- (i) Management commitment to ISO quality management system.
- (ii) Setting up of implementation committee
- (iii) Quality awareness programme
- (iv) Data collection of existing system and preparation plan

Phase II: Implementation:

- (v) Develop and implement documented quality system
- (vi) Establish internal quality audit system and monitor and stabilize the quality system.

Phase III: Registration and Certification:

- (vii) Apply for registration
- (viii) Adequacy audit by certifying agency.
- (ix) Competence audit by certifying agency
- (x) Certification

Benefits of ISO 9000 Certification:

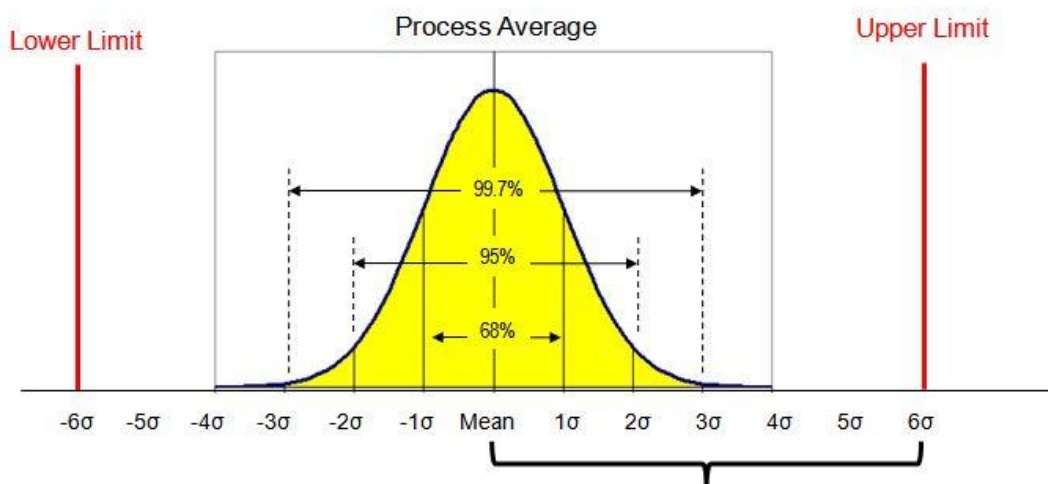
ISO 9000 certification entitles an organisation to following advantages:

- (i) It brings about discipline in system, procedures and documentation.
 - (ii) It improves the confidence of customers in company's product. Hence, the image of the company is enhanced.
 - (iii) It helps an enterprise to sell products in international markets. European Common Market countries prefer exports from ISO 9000 certified companies.
- Hundreds of Indian companies have adopted ISO standards and have been certified under ISO 9000. Some well-known ISO 9000 certified companies are: Hindustan Lever Ltd., Mafatlal Industries Ltd., Hero Cycles Ltd., Escorts Tractors Ltd., Hindustan Aeronautics Ltd. etc. etc.
- (iv) It helps an enterprise to avail of governmental concessions on imports.

6.11 SIX SIGMA

Six Sigma is a disciplined, statistical-based, data-driven approach and continuous improvement methodology for eliminating defects in a product, process or service. It was developed by Motorola and Bill Smith in the early 1980's based on quality management fundamentals, then became a popular management approach at General Electric (GE) with Jack Welch in the early 1990's. Hundreds of companies around the world have adopted Six Sigma as a way of doing business.

Sigma represents the population standard deviation, which is a measure of the variation in a data set collected about the process. If a defect is defined by specification limits separating good from bad outcomes of a process, then a six sigma process has a process mean (average) that is six standard deviations from the nearest specification limit. This provides enough buffer between the process natural variation and the specification limits.



For example, if a product must have a thickness between 10.32 and 10.38 inches to meet customer requirements, then the process mean should be around 10.35, with a standard deviation less than 0.005 (10.38 would be 6 standard deviations away from 10.35), assuming a normal distribution.

Six Sigma can also be thought of as a measure of process performance, with Six Sigma being the goal, based on the defects per million. Once the current performance of the process is measured, the goal is to continually improve the sigma level striving towards 6 sigma. Even if the improvements do not reach 6

sigma, the improvements made from 3 sigma to 4 sigma to 5 sigma will still reduce costs and increase customer satisfaction.

Sigma Level	Defects per Million	Yield
6	3.4	99.99966%
5	230	99.977%
4	6,210	99.38%
3	66,800	93.32%
2	308,000	69.15%
1	690,000	30.85%

IMPORTANT QUESTIONS

1. Explain different non-destructive testing methods
2. Explain Bath tub curve.
3. What is Process Capability.
4. Explain X chart and R chart.
5. Explain different elements of TQM.
6. What is Quality circles?
7. What is Six sigma quality.
8. Explain the ISO standard.
9. Explain different destructive testing methods.
10. What is Statistical quality control?
11. List the factors affecting quality. What are the functions of quality control dept?
12. Explain different techniques of quality control.
13. Explain reliability.



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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY EIGHTH SEMESTER B.TECH DEGREE EXAMINATION

Course Code: ME404

Course Name: INDUSTRIAL ENGINEERING

Max. Marks: 100

Duration: 3 hours

PART A

Answer any three full questions, each carries 10 marks

1. a) Explain Cost-Volume-Profit analysis. What is Break-Even chart. (6)
- b) List the factors affecting Manufacturing and Purchase decisions (4)
2. a) Explain different steps in Product development process. (8)
- b) List different applications of Industrial Engineering. (2)
3. a) Explain the principles of Material Handling. (6)
- b) Compare Preventive and Breakdown maintenance (4)
4. a) Explain different methods of Replacement (6)
- b) How will you determine the economic life of a product? (4)

PART B

Answer any three full questions, each carries 10 marks

5. a) Explain different wage incentive plans. (10)
6. a) Explain different Job evaluation methods. (6)
- b) What is meant by Critical examination? (4)
7. a) What is collective bargaining? (4)
- b) Explain worker's participation in management. (4)
- c) List different methods for eliminating fatigue. (2)
8. a) What are the psychological factors which affects the attitude of workers towards the work? (3)
- b) Explain different personal protective devices used in a production plant. (7)

PART C

Answer any three full questions, each carries 10 marks

- 9. a) Explain Product Life Cycle (PLC) (7)
 - b) What is Gantt chart. (3)
 - 10. a) Explain Routing, scheduling, dispatching and followup (8)
 - b) What is EOQ? (2)
 - 11. a) Explain different selective inventory control techniques (10)
 - 12. a) What is Process Capability. (3)
 - b) What is Six sigma quality (3)
 - c) What is Statistical quality control? (4)
 - 13. a) Explain different destructive testing methods. (10)
 - 14. a) Explain different elements of TQM. (5)
 - b) Explain X chart and R chart. (5)
-

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PRINCIPLES OF MANAGEMENT

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